

STANWELL POWER STATION

OFFICE REFURBISHMENT, NEW BUILDING & ASSOCIATED WORKS



SPECIFICATION

Proprietor

STANWELL CORPORATION LIMITED

Location

SWITCHYARD ROAD, STANWELL, QLD, 4702

Project No

25_003

- A1 – Registry Refurbishment
- A2 – Workshop Refurbishment
- A3 – Warehouse Refurbishment
- A4 – New Build

Revision	Date	Approved by
T1	12.03.2026	Tender Issue



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1.1 GENERAL

General: Refer to the Stanwell Corporation Tender Documents for all details relating to the Tender.

PROJECT INFORMATION

Contact person

Inquiries: Refer inquiries to the following:

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- Telephone: +61 423 479 607
- Email: Kaitlyn.Feeney@stanwell.com

GENERAL

General conditions

Contract: AS 4000-2025

Cross reference: The clause **GENERAL, INTERPRETATION**, in *0171 General requirements*, also applies.

1.2 DESCRIPTION OF WORKS

Stanwell Corporation Limited (Stanwell) is undertaking the renovation of three existing office spaces and the development of a new workplace facility for staff at Stanwell Power Station.

The project includes the construction of a new single-storey workplace building with an approximate Gross Floor Area (GFA) of 818 m², together with approximately 134 m² of surrounding verandah.

In addition, the works comprise refurbishment of three existing areas, including:

- Registry refurbishment – approximately 415 m²
- Workshop refurbishment – approximately 715 m²
- Main workshop refurbishment – approximately 340 m²

The refurbishment works generally involve the stripping out of existing finishes and fittings, followed by comprehensive fitout and upgrade works.

1.3 THE SITE

The refurbishment & new building will be located at Stanwell Power Station, 22km west of Rockhampton in Central Queensland. See Figure 1.

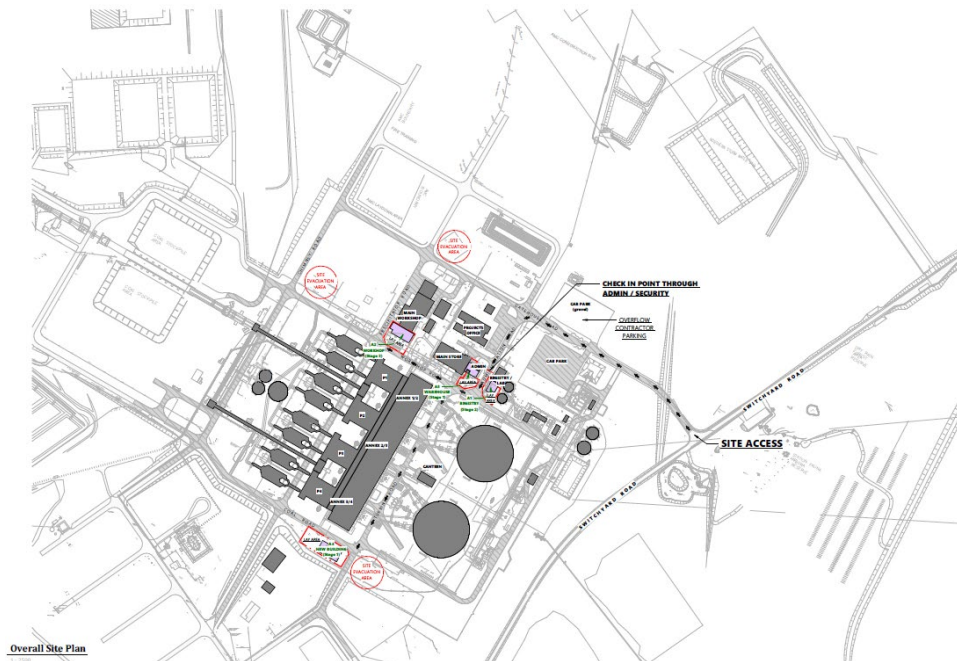


Figure 1 – to be read in conjunction with A0-000 – site plans

Site Conditions

Minimum temperature (039083)	°C	-1.0
Maximum temperature (039083)	°C	45.3
Mean relative humidity (039083)	%	40 – 73
Site elevation (Site of work)	M (ASL)	60
Wind speed region (according to AS/NZS 1170.2:2021)	-	Region C
Corrosion region (to AS 4312-2019)		C3 – Very High (Industrial)
Importance level		2
Terrain category		2

Site limitations: Comply with the following restrictions on the use of the site:

- Liaise with client before any services are cut off
- You must have your security access card on your person at all times while on site and must swipe in on entry and swipe out when leaving. Never loan another person your card.
- Ensure that all relevant parties complete all necessary inductions/paperwork and gain proper authorisations before commencing work
- Traffic management plan or WMS is required to be developed and communicated appropriately if there is any work that affects the normal traffic flow or impacts on pedestrian footpaths.
- If you are authorised to drive a vehicle on Site, you must exercise due care and attention when driving and adhere to all applicable road signs (including maximum speed signs). Seat belts must be worn in moving vehicles where they have been fitted. Qld government road rules apply.
- Areas of the Site may be monitored by cameras.
- You must wear the correct Personal Protective Equipment while on site. (Apart from in exemption zones) Minimum PPE requirements include:
 - A 100% cotton, long-sleeved, high visibility shirt – with reflective striping and 100% cotton trousers OR overalls. (shirt sleeves must be rolled down and buttoned when on plant)
 - safety boots or shoes (closed in shoes are required, at all times, for visitors if they don't have safety boots)
 - safety helmet (apart from in workshops, unless a risk assessment identifies the need to wear a helmet – (orange helmet are not allowed as these are specific for Emergency Response Team.members)
 - safety glasses (Clear or indoor-outdoor light mirror safety glasses are only allowed when working indoors).
 - the carrying of gloves (in operational areas)
 - Raincoats, as required - need to be high visibility with reflectors.
- Contractor to park in designated parking area

Integration with Stanwell Power Station

As the Works will be situated at SPS, the Contractor must comply with all relevant policies, procedures and rules of the Principal in relation to the Site or the Works. This is to ensure that the Works do not have the potential to compromise the SPS Operations.

Site access and deliveries to the Site will utilise shared private roads between Power Station Road (at entry to SPS), and the proposed location of the Works.

Access: All site access to go through security as entering site.

All personnel required to enter SPS site will be required to be inducted to Stanwell Power Station. To be inducted involves the completion of 5 onlines modules, 3-4hours to complete. On successful completion, a security card will be issued that will permit entry into SPS. If work is required to be undertaken under Stanwell's Safe System of works, additional work party member training will be required.

The Contractor shall be the *Work Health and Safety* (WHS) Principal Contractor (PC) of the Site with battery limits defined in the Architectural Drawings in the Site Precinct Plan: A0

SCOPE OF WORKS

The project comprises a single-storey new build & three refurbishment office building.

- New single storey office building – Approximaetly 818sqm & 134 m² verandah
- Registry refurbishment – approximately 415 m²
- Workshop refurbishment – approximately 715 m²
- Main workshop refurbishment – approximately 340 m²

New Building Facility

- 2 x Open-plan office space for up to 56 staff
- Meeting rooms and associated breakout areas
- Kitchen and dining facilities
- Lockers and amenities
- Storage areas
- Covered outdoor area
- Associated support services

External and Site Works:

- Landscaping works
- Civil New Carparking & driveway
- External concrete and paths
- External mechanical plant
- Hydrant and stormwater works

Three Refurbishment Buildings

- Open-plan office spaces
- Meeting rooms and associated breakout areas refurbishment
- Kitchen and dining facilities refurbishments
- Lockers and amenities refurbishments
- Storage areas refurbishments
- Associated support services

Acoustic Design Considerations:

- Acoustic treatments to manage external ambient noise
- Internal acoustic privacy measures for meeting rooms, training spaces, and open-plan work areas
- Compliance with sound rating requirements to ensure a productive and comfortable workplace environment

Preferred suppliers

Nominated works: Select a subcontractor from the **Preferred suppliers schedule**.

Preferred suppliers schedule

Works	Supplier	Supplier contact
Master keying	Windsx (swip card)	Bravis
Fire	Firelec	
AV	Programmed	

Separate contracts

Work concurrent to this Contract: The following works on Site, to be carried out by persons engaged by the Principal, are not included in this Contract: Audio Visual Scope of Works, including auxiliary cabling. Contractor to coordinate timing for rough-in and fit-off for concurrent contract. Allow a minimum 3 weeks notice for those works to be conducted on Site

1.4 PROTECTION OF PEOPLE AND PROPERTY

Occupied premises

General: For the parts of the Site Documented in the **Occupied premises schedule**:

- Allow occupants to continue in secure possession and occupancy of the premises for the required period.
- Maintain safe access for occupants.
- Arrange work to minimise nuisance to occupants and for their safety.
- Protect occupants against weather, dust, dirt, water or other nuisance. Proposals: Submit details of proposed methods.
- Purpose of submission: For information.
- Timing of submission: Before commencement of WUC.

Occupied premises schedule

Occupants	Occupied premises	Period of occupancy
Stanwell Staff and Sub Contractors	Stanwell Power Station	Duration of the Works

Safety

Accidents: Promptly notify the Superintendent of the occurrence of the following:

- Accidents involving death or personal injury.
- Accidents involving loss of time.
- Incidents with accident potential such as equipment failure, slides and cave-ins. Accident reports:

Submit reports of accidents.

- Purpose of submission: For information.

PROTECTIVE CLOTHING

Requirement: Make available protective clothing for the use of visitors, as follows:

- Safety helmets: Type 1 to AS/NZS 1801 (2024).
- High visibility safety vests: To AS 4602.1 (2024). Certification:

Required.

- Certification provider: An organisation accredited by the Joint Accreditation System of Australia and New Zealand (JASANZ).

Number of helmets: Six

Number of high visibility safety vests: Six

of the Site: The Site is limited to the area identified in the Documents. Access is to be maintained to the footpath allowing access to the bus drop off areas, unless approved in advance by the Superintendent and the Principal.

1.5 THE SITE

Major works to adjoining Site/s: Major works for power generation unit outages and maintenance are scheduled for:

1. Unit 3 outage: 30/05/26 to 24/07/26
2. Unit 4 outage: 03/07/27 to 27/08/27

During these times the total population on the Power Station location will increase. This may impact coordination of access to and around the construction Site.

Access area restrictions: Access onto and within the Site, and use of the Site for Temporary Works and constructional plant, including working and storage areas, location of offices, workshops, sheds, roads and parking, is restricted to the following areas: as Documented; the Construction Area and Lay Down Area, Lay Down Area: The exact extent of the Lay Down Area is to be determined in consultation with the Principal on commencement of the WUC on-site. Limitations to Delivery Times: Deliveries should be avoided during shift start and end periods; 06:30 to 07:30 and 16:00 to 17:00, Monday to Friday.

Secure area

Conditions of entry: Security clearance and inductions are required for access to the Secure Area. For Works within the Secure Area, a secondary temporary line of security is to be established in coordination with the Principal.

Entry permits: Application is to be made to the Principal for entry to the Secure Area.

Personnel: Submit the full name, address, and date and place of birth of persons required to enter designated secure areas.

- Purpose of submission: For review.

- Timing of submission: At least 14 days before entry is required.

1.6 SETTING OUT THE WORKS

Surveys

Setting out: as Documented. Refer also to: Additional Detail Survey of Part of Stanwell Power Station: by Capricorn Survey Group CQ.

Check surveys: required; particularly for all services. Final survey:
Required.

1.7 CLEANING UP

Final cleaning

General: Before the Date for Practical Completion, clean throughout, including interior and exterior surfaces exposed to view. Vacuum carpeted and soft surfaces. Clean debris from the Site, roofs, gutters, downpipes and drainage systems. Remove waste and surplus materials.

1.8 MATERIALS, LABOUR AND CONSTRUCTION PLANT

General

Temporary Works: Provide and maintain required hoardings, barricades, guards, fencing, shoring, temporary roadways, footpaths, signs, lighting and traffic management.

Apprentice employment policy Policy statement:

[complete/delete] **Access roads**

Temporary roads: none.

Principal's existing roads: Use only designated roads.

Location: Access by Gatehouse Road and part of Administration Road.

Parking

Principal's existing parking areas: Use only designated parking areas. Location: Contractor to establish within the Construction Area.

Overflow: Carpark B at the approval of the Principal. No access:
Carpark A.

Temporary fence Requirement: by the

Contractor. **Use of existing services**

General: Existing services may be used as temporary services for the performance of the Contract.

Existing services schedule

Service	Conditions of use
Electricity	To be isolated for demolition. Contractor to provide temporary connection infrastructure as required.
Potable water.	To be isolated for demolition. Contractor to provide temporary connection infrastructure as required.
Raw water	Accessible within secure part of the Site, to be accessed by water truck. Security induction required.

Note: the Contractor is required to provide their own telecommunications and internet services. No ablution facilities will be made available for the Contractor on Site.

Project signboards

General: Provide project specific signboards and as follows:

- Locate where directed.
- Maintain in good condition for duration of the WUC.
- Obtain permission for removal.
- Remove on completion.

Other signboards: Obtain approval before display of advertisements or provision of other signboards.

Contractor's representative

General: Must be accessible, and fluent in English and technical terminology.

Contacts: Submit names and telephone numbers of responsible persons who may be contacted after hours during the course of the Contract.

- Purpose of submission: For information.
- Timing of submission: At the first Site meeting.

Items supplied by Principal

General: Materials and other items supplied free of charge to the Contractor for installation in the execution of the Works.

Unload and take delivery, inspect for Defects and take care of the items. If Defects are found, advise. Return unused items to the Principal.

1.9 WORKING HOURS

General

Working hours that can be accommodated by the Principal: 6am to 6pm Working days that can be accommodated by the Principal: 7 days per week. Work to be undertaken outside of these times is to be approved by the Principal

PROGRAMMING

Program of Works

Construction Program: Submit a construction Program showing the following:

- Sequence of WUC.
- Critical paths of activities related to the WUC.
- Allowance for holidays.
- Activity inter-relationships.
- External dependencies including provision of access, Document approvals and work by others.
- Periods within which various stages or parts of the WUC are to be executed. Time scale:

Working days.

Updated Program: Submit revised construction Programs as required by the progress of the WUC. Identify changes since the previous issue, and show the estimated percentage of completion for each item of WUC.

Purpose of submission: For information.

Timing of submission: Within 14 days after the date for possession of the Site and with each progress claim.

Program chart: Display in the Contractor's Site office an up-to-date bar chart and network diagram based on the construction Program.

Site meetings

General: Hold and attend Site meetings throughout the Contract and arrange for the attendance of appropriate subcontractors, Superintendent and appropriate consultants.

Frequency: weekly.

Minutes: Make a record of Site meetings. Distribute a copy of the minutes to each party.

- Purpose of submission: For review.
- Timing of submission: Within 5 working days after each meeting.

Progress photographs

General: Take colour progress photographs within 7 days before each Site meeting. Submit 2 sets of prints and the digital files. Identify the project, date, time, location and orientation.

- Purpose of submission: For information.
- Timing of submission: At each Site meeting. Format: jpeg, png or pdf.

1.11 PAYMENT

Progress claims

Progress claim breakdown: Submit a statement of amounts claimed in respect of each worksection or trade heading designated in the specification, together with Variations included in the claim.

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide a project quality management system, as Documented.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 *General requirements*.

1.3 STANDARDS

General

Standard: To AS/NZS ISO 9001 (2016).

1.4 INTERPRETATION

Definitions

General: For the purposes of this worksection, the definitions given in AS/NZS ISO 9000 (2016) and the following apply:

- Quality package: A designated part of the Works, which may include the whole Works, for which an individual quality system is required.
- Service: After sales service, repairs, maintenance.

1.5 SUBMISSIONS

Quality plan

Quality package: Submit a quality plan for each quality package, at least 10 Business Days before work on that package commences. Keep a copy of each approved quality plan on Site.

Calculations

Statistical techniques: Provide the methodology for statistical evaluation.

Execution details

Requirement: Provide the procedure for sign-off and audit.

1.6 INSPECTION

Notice

Inspection: Give notice so inspection may be made of the following:

- As determined by the Quality Plan

2 PROJECT QUALITY MANAGEMENT SYSTEM REQUIREMENTS

2.1 DOCUMENTATION REQUIREMENTS

Quality plan

Standard: Conform to the recommendations of AS ISO 10005 (2018). Include inspection and Test plans.

Documented procedures

Review: Provide evidence of revision(s) including dates, approval and status of each procedure.

Register: Maintain a register of Documented procedures including the title, identifier and revision status.

2.2 CONTROL OF EXTERNALLY PROVIDED PROCESSES, PRODUCTS AND SERVICES

General

External audits: Perform pre-tender surveys and audits of subcontractors and suppliers, as necessary with an approved auditor. Include audit and surveillance proposals in the quality plan along with results of pre-tender surveys.

Verification: The Superintendent may verify at source, or upon receipt, that purchased products conform to Documented requirements.

2.3 PRODUCTION AND SERVICE PROVISION

Product identification and traceability

Requirement: As Documented.

Identification: Identify all items of Work, samples and Site records by lot/number/barcode. Traceability:

Provide and maintain records of components for audit.

Service

Requirement: As Documented.

3 MONITORING AND MEASUREMENT

3.1 DOCUMENT CONTROL

Changes to Documents

Revision: Review and approve changes to Documents using the same functions of the organisations that performed the original review and approval of the Documents.

Retention: As Documented.

Evaluation: Make quality records available to the Superintendent for evaluation, within 2 days of the initial request.

Period of evaluation: Agree for each request, but at least 5 Business Days.

3.2 INSPECTION AND TEST PLANS

Content

Plan: Include the following:

- Detail all inspections and Tests required including Hold points.
- Identify acceptance criteria and frequency of sampling and Testing.
- Identify responsibilities for inspection and Testing and product/service approval criteria.

Control of non-conforming product

Acceptance of concession: Before the provision or repair of a non-conforming product, obtain permission to use the product.

Hold points

Stages: Hold points during the construction/manufacturing process require release by the Superintendent.

Release: Requirements for release of a Hold point may include the following:

- Provision of information required by the technical specifications.
- Certification of design/construction or installation.
- Submission of any checklists or non-conformance forms as required.
- Inspection/demonstration of Works.

Frequency of Testing

General: Conform to the Test requirements of the relevant standards and worksections, as Documented.

3.3 AUDITING

General

Audit plan: Conform to the recommendations of AS/NZS ISO 19011 (2019) clauses 6.4 and 6.5. Initial systems audit:

Carry out before date of Site possession.

3.4 CORRECTIVE ACTION

General

Review: Provide a procedure to review the various control methods to minimise non-conformance. Record amendments to the project quality management system resulting from corrective action.

Non-conforming Works: Include in the quality plan the procedure for reporting any non-conforming Works to the Superintendent and any corrective action requests.

4 SELECTIONS

4.1 SCHEDULES

Project quality management system schedule

Quality package	Activities included	Standard	Options
The whole of the Works	All	AS/NZS ISO 9001 (2016)	See below

1 GENERAL

1.1 PRECEDENCE

General

Responsibility: All obligations within this specification are the responsibility of the Contractor, unless expressly stated otherwise.

Order of precedence: If there is conflict or inconsistency between the worksections of this specification, the requirements of worksections take the following order of precedence:

- All worksections other than those listed below.
- *0701 Mechanical systems, 0801 Hydraulic systems, 0901 Electrical systems and 1001 Fire services systems.*
- *0171 General requirements.*

1.2 SUPPORTING SPECIFICATIONS

General

Cross reference: Documents to be referenced as supporting this specification include:

- *Structural Engineering Specification.*
- *Electrical Engineering Specification.*
- *Mechanical Engineering Specification.*
- *Hydraulic Engineering Specification.*

1.3 CROSS REFERENCES

Common requirements

Requirement: Conform to the following worksections:

- *0181 Adhesives, sealants and fasteners.*
- *0183 Metals and prefinishes.*
- *0184 Termite management.*
- *0185 Timber products, finishes and treatment.*

1.4 REFERENCED DOCUMENTS

General

Precedence: The requirements of worksections override conflicting requirements of their referenced Documents. The requirements of the referenced Documents are minimum requirements.

Contractual relationships: Responsibilities and duties of the Principal, Contractor and Superintendent are not altered by requirements in the Documents referenced in this specification.

Current editions: All referenced Documents are the editions, with amendments, current on 1st September 2025.

Exception to current editions: If statutory requirements reference other editions or standards, conform to those other editions or standards. If the NCC (2022) references editions other than the current edition, the same editions cited in the NCC (2022) are referenced in each worksection.

Maintenance and repair works: If statutory requirements applicable to the maintenance or repair works reference other editions or standards, conform to those other editions or standards.

European standards: Any national European Standard (e.g. IS EN or DIN EN) may be used in place of the equivalent referenced European Standard (EN).

1.5 CONTRACT DOCUMENTS

Services diagrammatic layouts

General: Layouts of service lines, plant and equipment shown on the drawings are diagrammatic only, except where figured dimensions are provided or calculable.

Before commencing WUC on-Site:

- Obtain measurements and other necessary information.
- Coordinate the design and installation in conjunction with all trades.

Levels

General: Spot levels take precedence over contour lines and ground profile lines.

Drawings and manuals for existing services

Subsurface services: All underground services are to be confirmed by the Contractor before commencing WUC on-Site.

Underground services shown on Documents have not been verified. Warranty: No warranty is given as to the completeness or accuracy of drawings and/or manuals of existing services.

1.6 COORDINATION OF TRADES

General: The Contractor is responsible for coordinating the worksections of this specification with the relevant trades, including the determination of responsibility at trade interfaces.

1.7 INTERPRETATION Documentation conventions

Imperative mood and streamlined language: The words shall or shall be are implied where a colon is used following a keyword or within a sentence or sentence fragment.

Subject of sentences and phrases: Specification requirements are to be performed by the contractor, unless stated otherwise.

Abbreviations

General: For the purposes of this specification the following abbreviations apply:

- AS: Australian Standard.
- BCA: National Construction Code Series Volume One: Building Code of Australia Class 2 to 9 Buildings and Volume Two: Building Code of Australia Class 1 and Class 10 Buildings.
- EN: European Norm (European Standard).
- GRP: Glass Reinforced Plastic.
- IP: Ingress protection.
- NATA: National Association of Testing Authorities.
- NCC: National Construction Code.
- NZS: New Zealand Standard.
- PCA: National Construction Code Series Volume 3: Plumbing Code of Australia.
- PVC: Polyvinyl Chloride.
- PVC-U: Unplasticised Polyvinyl Chloride. Also known as UPVC.
- SDS: Safety data sheets.
- VOC: Volatile Organic Compound.
- WHS: Work Health and Safety.

Definitions

AS4000(2024) definitions: terms defined in the AS4000(2024) General Conditions of Contract are Capitalised in this specification.

General: For the purposes of this specification, the following definitions apply:

- Access for maintenance: Includes access for maintenance, inspection, measurement, operation, adjustment, repair, replacement and other maintenance related tasks.
- Accessible, readily: Readily accessible, easily accessible, easy access and similar terms mean capable of being reached quickly and without the use of a tool, without hazard, climbing over or removing obstructions, using a movable ladder and in any case not more than 2.0 m above the ground, floor or platform.
- Accredited Testing Laboratory: One of the following:
 - . An organisation accredited by the National Association of Testing Authorities (NATA) to undertake the relevant Tests.
 - . An organisation outside Australia accredited to undertake the relevant Tests by an authority recognised by NATA through a mutual recognition agreement.
 - . An organisation recognised as being an Accredited Testing Laboratory under legislation at the time the Test was undertaken.
 - . An organisation accredited for compliance with AS ISO/IEC 17025 (2018) to undertake the relevant Tests.
- Adjust: To align, support, straighten, reposition, secure, recalibrate, reset, tighten, loosen or otherwise perform work to restore to proper working order.
- Appropriately qualified person: To NCC (2022) Schedule 1.
- Attendance: Attendance, provide attendance and similar expressions mean give assistance for examination and Testing.
- Baseline data: Data derived from the final design, installation and commissioning, which serve as a basis for verification of results of routine servicing.
- Check: To examine to determine accuracy, quality or condition.
- Commissioning: Advancement of an installation from static completion to full working order, including verification that the systems, subsystems, and their components meet the project requirements. This includes all work described as commissioning in referenced Documents, even if carried out before static completion.
- Concealed: Obstructed from view, including below ground, in walls, cavities or roof spaces.
- Consumable: Materials or components intended to be depleted, worn out, replaced or replenished within the service life of the associated systems, plant or equipment as part of normal operation and maintenance.
- Contract administrator: Has the same meaning as architect, Superintendent or Principal's authorised person and is the person appointed by the owner or Principal under the Contract.
- Contractor: Has the same meaning as builder and is the person or organisation bound to carry out and complete the WUC.
- Default: Specified value, product or installation method that is to be provided unless otherwise Documented.
- Design life: The period of time for which it is assumed, in the design, that an asset will be able to perform its intended purpose with only anticipated maintenance but no major repair or replacement being necessary.
- Design parameters: Information used as the basis for design. It includes design requirements, performance criteria, performance parameters and similar terms.
- Disconnect: Includes all preparation required to safely detach, isolate, unfasten, unplug or break the connection between a supply of electricity, gas or water and a device or an appliance.
- Documented: Documented, as Documented and similar terms mean contained in the Contract Documents. (Where specific references are provided they are to assist with locating the primary reference information, but are not an exhaustive list of locations of reference information within the Contract Documents. The Contractor is to become familiar with the full extent of Contract Documents).

- Economic life: The period of time from the acquisition of an asset to the time when the asset, while still physically capable of fulfilling its function and with only anticipated maintenance, ceases to be the lowest cost alternative for satisfying that function.
- Electricity distributor: Any person or organisation that provides electricity from an electricity distribution system to one or more electrical installations. Includes distributor, supply authority, network operator, local network service provider, electricity retailer or electricity entity, as may be appropriate in the relevant jurisdiction.
- Errors and omissions: For the design prepared by the Contractor, errors and omissions have the same meaning as Defects.
- Fastener: Device used for fixing an item to another item, base material or substrate, by mechanical means.
- Fastening: Assembly used to secure an item to another item, base material or substrate, comprising one or more fasteners and accessories.
- Fire hazard properties: To NCC (2022) Schedule 1.
- Fixing: The action of securing an item to another item, base material or substrate.
- Gas Network Operator: Has the same meaning as network operator in AS/NZS 5601.1 (2022).
- Geotechnical Site investigation: The process of evaluating the geotechnical characteristics of the Site in the context of existing or proposed construction.
- Give notice: Give notice, submit, advise, inform and similar expressions mean give notice (submit, advise, inform) in writing to the Superintendent.
- High level interface: Systems transfer information in a digital format using an open system interface.
- Hold point: The activity cannot proceed without the approval of the Superintendent.
- Hot dip galvanized: Zinc coated or zinc iron alloy coated to AS/NZS 4680 (2025) after fabrication with coating thickness and mass conforming to AS/NZS 4680 (2025) Tables 3 and 4.
- Include: Include or including means to contain or comprise and are not words of limitation.
- Ingress protection: IP, IP code, IP rating and similar expression have the same meaning as IP Code in AS 60529 (2025).
- Joints:
 - . Construction joint: A joint with continuous reinforcement provided to suit construction sequence.
 - . Contraction joint: An opening control joint with a bond breaking coating separating the joint surfaces to allow independent and controlled contraction of different parts or components, induced by shrinkage, temperature changes or other causes. It may include unbound dowels to assist vertical deflection control.
 - . Control joint: An unreinforced joint between or within discrete elements of construction that allows for relative movement of the elements.
 - . Expansion joint: A closing control joint with the joint surfaces separated by a compressible filler to allow axial movement due to thermal expansion or contraction with changes in temperature or creep. It may include unbound dowels to assist vertical deflection control.
 - . Sealant joint: A joint filled with a flexible synthetic compound that adheres to surfaces within the joint to prevent the passage of dust, moisture and gases.
 - . Structural control joint: A control joint (contraction, expansion and isolation) in structural elements when used with applied material and finishes.
 - . Substrate joint: A joint in the substrate, which includes construction joints and joints between different materials.
 - . Weakened plane joint: A contraction joint created by forming a groove, extending at least one-quarter the depth of the section, either by using a grooving tool, by sawing or by inserting a premoulded strip.
- Local authority (local council): A body established for the purposes of local government by or under a law applying in a state or territory.
- Low level interface: Systems transfer information via terminals and voltage free contacts.
- Maintenance: Work that is carried out to preserve an asset, to allow for its continued use and function over its designed service life.

- . Corrective maintenance: Maintenance initiated as a result of plant, equipment, systems, elements, before or after functional failure. This work can be planned or unplanned and includes repairs and replacement.
- . Preventive maintenance: Planned maintenance of plant, equipment and other systems or elements, including cyclical or periodic maintenance, fire safety measures and statutory requirements.
- Manufacturer's and supplier's recommendations: Recommendations, instructions, requirements, specifications (and similar expressions) provided in written or other form by the manufacturer and/or supplier relating to the suitability, use, installation, storage and/or handling of a product.
- Metallic-coated: Steel coated with zinc or aluminium-zinc alloy as follows:
 - . Metallic-coated steel sheet: To AS 1397 (2021). Metal thicknesses specified are base metal thicknesses.
 - . Ferrous open sections zinc coated by an in-line process: To AS/NZS 4791 (2006).
 - . Ferrous hollow sections zinc coated by a continuous or specialised process: To AS/NZS 4792 (2006).
- Network Utility Operator: To NCC (2022) Schedule 1. A person who undertakes the piped distribution of drinking water or non-drinking water for supply; or is the operator of a sewerage system or a stormwater drainage system.
- Obtain: Obtain, seek and similar expressions mean obtain (seek) in writing from the Superintendent.
- Pipe: Includes pipe and tube.
- Practical completion or Defects free completion: refer to the General Conditions of Contract.
- Pre-commissioning: Verifying that the installation of a system is complete and ready for commissioning.
- Principal: Principal has the same meaning as owner, client and proprietor and is the party to whom the Contractor is legally bound to construct the Works.
- Professional engineer: To NCC (2022) Schedule 1.
- Proprietary: Identifiable by naming the manufacturer, supplier, installer, trade name, brand name, catalogue or reference number.
- Prototype: A full size mock-up of components, systems or elements to demonstrate or Test construction methods, junctions and finishes, and to define the level of quality.
- Provide: Provide and similar expressions mean supply and install and include development of the design beyond that Documented.
- Record drawings: Record drawings has the same meaning as as-installed drawings, as-built drawings and work-as-executed drawings. It includes technical drawings and digital models.
- Recovered/reclaimed materials: Material previously used in a building or project that is then re-used in another project. The material may be altered, re-sized, refinished, or adapted, but is not reprocessed in any way, and remains in its original form.
- Referenced Documents: Standards and other Documents whose requirements are included in this specification by reference.
- Required: Required by the Contract Documents, the local or statutory authorities.
 - . If required: A conditional specification term for Work that may be shown in the Documents or is a Legislative Requirement.
- Sample: A physical example that illustrates workmanship, materials or equipment, and establishes standards by which the Work will be judged. It includes samples and sample panels.
- Static completion: The state of a system when installation works are complete but have not been commissioned.
- Statutory authority: A public sector entity created by legislation, that is, a specific law of the Commonwealth, State or Territory.
- Subcontractor: A person or organisation, other than the principal, having a contract with the Contractor for the provision of part of the Works.
- Supply: Supply, furnish and similar expressions mean supply only.
- Tests - integrated system: Tests conducted on the project as a complete, integrated system to verify successful integration, interaction, and operation of all interrelated systems to the project requirements.

- Tests - production: Tests carried out on an item, before delivery to the Site.
- Tests - Site: Tests carried out on Site.
- Tests - type: Tests carried out on an item identical with a production item, including with respect to materials, material suppliers, manufacturing processes, dimensions and marking.
- Tolerance: The permitted difference between the upper limit and the lower limit of dimension, value or quantity.
- Utility service provider: Includes Electricity distributor, Network Utility Operator, Gas Network Operator and organisations providing other reticulated utilities including data and telecommunications services.
- Verification: Provision of evidence or proof that a performance requirement has been met or a default exists.

2 SUBMISSIONS AND INSPECTIONS

2.1 SUBMISSIONS

General

Requirement: Make submissions, as Documented.

Submit to: Superintendent or Superintendent's Representative.

Contractor review: Before submitting, review each submission item, and check for coordination with other Work of the Contract and conformance to Contract Documents.

Submission times

Default timing: Submit information or other material for information, comment or approval at least 5 Business Days before ordering products or starting installation of the respective portion of the Works.

Submission response times: Allow in the construction Program for at least the following times:

- Shop drawings: 10 Business Days
- Samples and prototypes: 10 Business Days
- Manufacturer's or supplier's recommendations: 10 Business Days
- Product data: 10 Business Days
- Product/design substitution or modification: 10 Business Days

Proposed products schedules: Submit a schedule of proposed products that have not been specified as proprietary items within 3 weeks of starting WUC on Site.

Identification

Requirement: Identify the project, Contractor, subcontractor or supplier, manufacturer, applicable product, model number and options, as appropriate and include relevant Contract Document references. If the submission covers more than one item, identify the item in the Contract Documents the submitted items relate to.

Non-conformance: Identify proposals that do not conform with project requirements, and characteristics that may be detrimental to successful performance of the completed WUC.

Errors

Requirement: If a submission contains errors, make a new or amended submission as appropriate, indicating changes made since the previous submission.

Electronic submissions

Electronic copies file format: .pdf

CAD file format: .dwg (version 2022 compatible) Quantity: 1

Transmission medium: electronic file transfer or email

Project requirements

General: Submit the following, as Documented:

- Authority approvals: Notes of meetings with regulatory authorities and utility service providers whose requirements apply to the WUC and evidence that notices, fees and permits have been sought and paid, that utility service provider connections are complete and that statutory approvals by the authorities whose requirements apply to the Work have been received.
- Baseline data: To **BASELINE DATA**.
- Building penetrations: Details of the methods to maintain the required structural, fire and other properties to **BUILDING PENETRATIONS**.
- Certification: Certificates of conformance to Documented and statutory requirements.
- Commissioning plan: For the whole of the Work to **COMMISSIONING**.
- Commissioning program: For the whole of the Work to **COMMISSIONING**.
- Design Documentation: Drawings, calculations and specifications as Documented.
- Electronic facility and asset management information: For the whole of the Work to **ELECTRONIC FACILITY AND ASSET MANAGEMENT INFORMATION**.
- Execution details: Execution programs, schedules and details of proposed methods and equipment. For building services include the following:
 - Embedded services: Proposed method for embedding services in concrete walls or floors or chasing into concrete or masonry walls.
 - Fixing of services: Typical details of locations, types and methods of fixing services to the building structure.
 - . Inaccessible services: If services will be enclosed and not accessible after completion, submit proposals for location of service runs and fittings.
- Fire performance: Evidence of conformity to requirement for non-combustibility, fire hazard properties and fire-resistance of building elements.
- Marking and labelling: Samples and schedules of proposed marking and labels to **MARKING AND LABELLING**.
- Operation and maintenance manuals: For the whole of the Work to **OPERATION AND MAINTENANCE MANUALS**.
- Products and materials: Products and materials data, including manufacturer's technical specifications and drawings, product data sheets, type Tests results, evidence of conformity to Documented requirements, product certification, performance and rating tables, service connection requirements and installation and maintenance recommendations.
- Prototypes: Prototypes of components, systems or elements.
- Records: Record drawings to **RECORD DRAWINGS**, reports of Testing or work, photographs, system diagrams, schedules and logbooks.
- Safe Work Method Statement: For high risk construction works.
- Safety in design report: For the proposed work to **DESIGN DEVELOPMENT, Safety in design**.
- Samples: Representative of proposed products and materials and including proposals to incorporate samples into the Works, if any to **SAMPLES AND PROTOTYPES**.
- Shop drawings: To **SHOP DRAWINGS**.
- Substitutions: To **SUBSTITUTIONS**.
- Tests: Test reports for Testing performed under the Contract.
- Warranties: To **WARRANTIES**.

2.2 INSPECTION

Notice

Concealment: If notice of inspection is required for parts of the Works that are to be concealed, give notice when the inspection can be made before concealment.

Notification times

Minimum notice: As Documented.

Light levels

Lighting levels for inspection: To AS/NZS 1680.2.4 (2017).

Attendance

General: Provide attendance for Documented inspections and Tests

PERFORMANCE

3.1 CORROSION RESISTANCE

Atmospheric corrosivity category

General: Atmospheric corrosivity category as defined in AS 4312 (2019):

- Exterior atmospheric corrosivity category: C3, as determined by Dulux Protective Coatings
- Interior atmospheric corrosivity category: C2, as determined by Dulux Protective Coatings

Galvanizing

Requirement: Galvanize mild steel components (including fasteners) to AS/NZS 1214 (2016) or AS/NZS 4680 (2025) as appropriate, if:

- Exposed to weather.
- Embedded in masonry.
- Exposed to or in air spaces behind the external leaf of masonry walls.
- In contact with chemically treated timber, other than copper chrome arsenate (CCA).

3.2 NOISE LEVELS

General

Requirement: Install systems to operate within the noise level limits, as Documented for the Contract design and Documented equipment performance.

3.3 STRUCTURE

General

Requirement: If provision of the Works requires structural design, provide structures, installations and components as follows:

- Fixed accessways: To AS 1657 (2018).
- Structural design actions: To the AS/NZS 1170 series. Importance level: 2

4 DESIGN

4.1 DESIGN DEVELOPMENT

General

Requirement: Complete the design of the Work, including development of the design beyond that Documented.

Conflict with the Documents: If it is believed that a conflict exists between statutory requirements and the Documents, notify the Superintendent immediately and provide a recommendation to resolve the conflict.

Stanwell Engineering Requirements

Design provided by the Contractor is to satisfy the following Stanwell Engineering Requirements,

Appendix H – Principal Supplied Documents:

- ASM-STD-ENG-MAN-01 - Documentation Drawings and Manuals.
- T-3640 - Earthing.
- T-3643[2] - General Electrical & Engineering Standards.
- T-3645 - Lighting & Small Power.
- T-3646[3] - Low Voltage Switchgear.
- T-3655 - Equipment Identification Labelling Signage & Marking Physical Hazards.
- T-3685 - Noise and Vibration.
- T-3686 - Painting & Corrosion Prevention.
- HSE-PROC-18 - Principal Contractor HSE Requirements.

Where a conflict exists between the Stanwell Engineering Requirements and the project Documented design, the Superintendent shall be notified immediately.

Building Approval and Certificate of Occupation

Form 15 submissions: Complete the design of the Work, including development of the design beyond that Documented, and provide a Form 15 from an RPEQ registered engineer for:

- Roof Water Drainage Performance Solution Report; refer to Hydraulic Engineer's Documents.
- Timber Trusses; refer to Structural Engineer's Documents.
- Verified Mechanical design following coordination with final truss design; refer to Mechanical Engineer's Documents.
- Light steel framing systems such as walls, ceilings and the like.
- Seismic restraint for non-structural elements such as suspended services and the like.

The Form 15 is to be submitted prior to commencement of the Works, at the time of shop drawing submission. Refer also to *6.4 Support of Plant and Equipment* and *6.5 Restraint of Non-Structural Parts and Components*.

Form 12 submission: submit all Form 12s required to certify the installation of the design elements to the satisfaction of the certifying authority.

Local Authority Charges: Contractor is responsible for payment of all local authority inspections and charge associated with the project including plumbing and drainage.

Traditional Delivery of Design Provided

If the Contractor chooses to deliver the project as Documented, including the project's verified design elements, then the remaining verification of supply items, systems, and installation to the design parameters, required by the certifier for the certificate of classification remain as Documented.

Time and Cost for Amended Design, Design Verification and Alternative Delivery

The extent of any amended process, including any additional time, fees, or services (including for the Principal's certifier) are deemed to be included in the Contractor's lump sum tender offer.

4.2 DESIGNER

General

Design by Contractor: If the Contractor provides design, use only appropriately qualified and registered persons to undertake design work. If requested, provide Documents verifying the qualification and experience.

Engineering design by Contractor: If the Contractor provides engineering design: Provide design carried out by an RPEQ registered engineer.

Evidence: Signature of a nominated professional engineer in the respective discipline to each design Document submitted.

Insurance: Provide evidence of currency of insurance of the engineering designer for professional indemnity insurance and public liability insurance.

ALTERNATIVE DESIGN SOLUTIONS

Conforming alternative design solutions

General: Proposals for alternative designs that conform to Documented requirements are permitted.

Requirement: If proposing alternative design solutions, submit the proposed alternative solution with a comparison to the original. Include the following factors in the comparison:

Safety, health and amenity of building occupants, including maintenance personnel. Cost.

Anticipated life.

Energy efficiency.

Environmental impact, including greenhouse gas emissions. Effect on construction Program.

Effect on other trades. Availability of products. Certification requirements. Maintenance requirements. Access for maintenance.

Flexibility for future changes. Safety of equipment.

Sustainability.

Other information required for submission in **SUBSTITUTIONS. Non-conforming alternative design solutions**

Requirement: If the proposed design solution does not conform to the Documented requirements, conform to **SUBSTITUTIONS. Costs**

Requirement: Pay the cost of submissions and evaluation and/or Testing of proposed alternatives, whether subsequently adopted or not. The costs will be calculated at the current charge-out rates of the relevant consultant.

DESIGN DOCUMENTS

General

Requirement: Provide calculations, drawing, shop drawings, digital models and specifications to Document a design conforming to the Documented design parameters and statutory requirements, as required.

Drawings

Requirement: To **SHOP DRAWINGS**.

Drawing registers: Submit drawing registers with the drawings showing current and previous drawing issues.

Specifications

Requirement: Provide project specific specification using the NATSPEC National Master Specification and incorporating this specification by reference.

Variations from this specification: If it is proposed to depart from the wording of this specification, submit the proposed changes as Documented in **ALTERNATIVE DESIGN SOLUTIONS**.

Licence: Provide specifications prepared by a current subscriber to the applicable NATSPEC package. Worksections: Use worksections from the current update of NATSPEC and customise for the project.

Additional material: If the specification requires material not included in this specification, use worksections from the current update of NATSPEC and customise for the project.

Additional content: If additional content is required for the project specification not covered in either format.

Redundant content: Delete redundant content from the NATSPEC worksection *Templates*.

ERRORS IN AND OMISSIONS FROM THE DESIGN

General

Requirement: If the design contains errors and/or omissions, correct the errors and make a new or amended submission, as appropriate, indicating changes made since the previous submission.

Notice: Immediately on detecting a design error and/or omission, submit notification in writing describing the error and/or omission, anticipated consequences and proposed corrective action.

Consequences of errors and/or omissions: Pay the costs associated with correcting design errors and/or omissions and minimise the effects of the errors.

PRODUCT AND MATERIALS

5.1 GENERAL

Sources policy

General: Locally sourced materials and suppliers are preferred.

Consistency

General: For each material or product use the same source or manufacturer and provide consistent type, size, quality and appearance.

Low VOC emitting paints

Paint types: To the recommendations of AS/NZS 2311 (2017) Table 4.2.

Prohibited materials

General: Do not provide the following:

- Materials exceeding the limits of those listed in the Safe Work Australia *Hazardous Chemical Information System* (HCIS) Workplace exposure standards.
- Blowing agents:
 - . Materials that use chlorofluorocarbon (CFC) or hydrochlorofluorocarbon (HCFC) in the manufacturing process.
 - . A blowing agent with a Global Warming Potential (GWP) not less than 700.

5.2 PROPRIETARY ITEMS

Manufacturer's or supplier's recommendations

General: Provide manufactured items to the manufacturer's or supplier's recommendations.

Proprietary items/systems/assemblies: Deliver, handle, store, assemble, install or fix to substrate to the manufacturer's or supplier's recommendations.

Project modifications: Advise of activities that supplement, or are contrary to the manufacturer's or supplier's recommendations.

Identification of proprietary items

Sealed containers: If items are supplied by the manufacturer in closed or sealed containers or packages, bring them to point of use in the original containers or packages.

Other items: Marked to show the following, as applicable:

- Manufacturer's identification.
- Brand name.
- Product type.
- Quantity.
- Reference code and batch number.
- Date of manufacture.

5.3 SUBSTITUTIONS

Substitution Request Form

Identified proprietary items: Identification of a proprietary item does not necessarily imply exclusive preference for the identified item, but indicates the necessary properties of the item.

Alternatives: If alternatives to the Documented products, methods or systems are proposed, submit a completed Substitution Request. The following conditions must be met:

1. Contractor must submit a fully completed Substitution Request Form at least 5 Business Days before ordering products for, or starting installation of, the respective portion of the Works. Any substitution requests made without submitting a completed form will not be considered. A copy of the required form is enclosed within the appendices of this specification.
2. State whether the use of the substitution will require alteration to any part of the Works. If the substitution is adopted, carry out any such alteration without extra charge. In the case where the affected Works are part of another subcontract, obtain cost from the other subcontractor for such alteration and include it in the cost of the substitution.
3. No claim shall arise from any rejection, nor, unless otherwise agreed, shall adoption of a substitution be grounds for any claim for Variation to cost or time.

Conditions: Contractor's request for substitutions will be received and considered when extensive revisions to Contract Documents are not required and changes are in keeping with general intent of Contract Documents, and when timely, fully Documented and properly submitted, all as judged by the Architect.

Approval: No work involving any request for substitution shall be commenced, or materials ordered, until the Contractor has received written evidence of the approval of the substitution provided by the Architect.

Costs: Pay the cost of submissions and of evaluations and Tests of proposed alternatives, whether subsequently adopted or not. The costs will be calculated at the current charge-out rates of the relevant consultant(s).

5.4 SAMPLES AND PROTOTYPES

General

Incorporation into Works: Only incorporate samples and prototypes that have been endorsed for inclusion in the Works. Do not incorporate other samples or prototypes.

Retention: Keep endorsed samples and prototypes in good condition, until the Date for Practical Completion.

Unincorporated samples and prototypes: Remove on completion.

5.5 SHOP DRAWINGS

General

Standard: To AS 1100.101 (1992), AS 1100.201 (1992), AS 1100.301 (2008), AS 1100.401 (1984) and AS/NZS 1100.501 (2002) as applicable.

Documentation: Include dimensioned drawings showing details of the fabrication and installation of structural elements, building components, services and equipment, including relationship to building structure and other services, cable type and size, and marking details.

Diagrammatic layouts: Coordinate work shown diagrammatically in the Contract Documents, and prepare dimensioned set-out drawings.

Services coordination: Coordinate with other building and service elements. Show adjusted positions on the shop drawings.

Space requirements: Check space and access for maintenance requirements of equipment and services indicated diagrammatically in the Contract Documents.

Commissioning requirements: Show provisions for Testing and commissioning on the drawings. Access for maintenance: Show space and provisions for access for maintenance.

ANCILLARY BUILDING WORK

6.1 WALL CHASING

Holes and chases

General: If holes and chases are required in masonry walls, make sure structural integrity of the wall is maintained. Do not chase walls with a fire-resistance level or an acoustic rating.

Parallel chases or recesses on opposite faces of a wall: Not closer than 600 mm to each other. Chasing blockwork: Only chase core-filled hollow blocks or solid blocks that are not structural. **Concrete blockwork chasing table**

Block thickness (mm)	Maximum depth of chase (mm)
190	35
140	25
90	20

6.2 FIXING

General

Suitability: If equipment is not suitable for fixing to non-structural building elements, fix directly to structure and trim around penetrations in non-structural elements.

Fasteners

General: Use proprietary fasteners capable of transmitting the loads imposed, and sufficient for the rigidity of the assembly

6.3 BUILDING PENETRATIONS

Penetrations

Requirement: Maintain the required structural integrity, fire performance, waterproofing performance and other properties when penetrating or fixing to the following:

- Structural building elements including external walls, fire walls, fire doors and access panels, other Tested and rated assemblies or elements, floor slabs and beams.
- Membrane elements including damp-proof courses, waterproofing membranes and roof coverings. If penetrating membranes, provide a waterproof seal between the membrane and the penetrating component.

Sealing

Fire-resisting building elements: Seal penetrations with a system conforming to AS 4072.1 (2005).

Non fire-resisting building elements: Seal penetrations around conduits and sleeves. Seal around cables within sleeves. If the building element is acoustically rated, maintain the rating.

Sleeves

General: If piping, cables or conduits penetrate building elements, provide metal or PVC-U sleeves formed from pipe sections as follows:

- Movement: Arrange to permit normal pipe or conduit movement.
- Diameter (for non fire-resisting building elements): Sufficient to provide a ring shaped space around the pipe or pipe insulation of at least 12 mm.
- Ferrous surfaces: Prime paint.
- Sealing: Seal between pipes or conduits and sleeves to prevent the entry of vermin.
- Terminations:
 - . Cover plates fitted: Flush with the finished building surface.
 - . Fire-resisting and acoustic rated building elements: 50 mm beyond finished building surface.
 - . Floors draining to floor wastes: 50 mm above finished floor.
 - . Other locations: 5 mm beyond finished building surface.
 - . Termite management: To AS 3660.1 (2014).
- Thickness:
 - . Metal: 1 mm or greater.
 - . PVC-U: 3 mm or greater.

SUPPORT OF PLANT AND EQUIPMENT

Concrete plinths

General: Provide concrete plinths as Documented and under all equipment located on concrete floor slabs as follows:

- Surround: Zinc (hot-dipped) coated steel, at least 75 mm high and 1.6 mm thick. Fix to the floor with masonry anchors. Fill with concrete.
- Height: 75 mm or greater, as Documented.
- Reinforcement: Single layer of F62 fabric.
- Concrete: Grade N20.
- Finish: Steel float, flush with top edge of the surround. **Support of ground level plant and equipment** Ground level: Conform to the following:
 - If the ground slope is 15° or more, or the area of the plant and equipment is extensive, obtain the advice of a professional engineer for the Documentation of a suitable slab or platform.
 - In all other cases, provide proprietary plastic or concrete supports installed with falls that achieve a raised, impervious and water shedding bearing surface.

Support of plant and equipment mounted on roofs or elevated platforms

Platforms: If a platform is required, or the area of the plant and equipment mounted on roofs or elevated platforms is extensive, obtain the advice of a professional engineer for the Documentation of a suitable platform.

Balustrades: If balustrades or screening are required, obtain the advice of a registered architect.

Roof level support: If any of the following apply to roof level support, obtain the advice of a professional engineer:

- The total load from any unit of plant or equipment exceeds 500 kg.
- The load from a unit of plant or equipment to any single support point exceeds 100 kg.
- The average loading of plant and equipment over the area extending 1 m on all sides beyond the plant and equipment exceeds 25 kg/m².

6.4 RESTRAINT OF NON-STRUCTURAL PARTS AND COMPONENTS

Seismic restraint

Standard: To the NCC cited AS 1170.4 (2007). Earthquake design category: II

Requirement: Conform to the following:

Use parts or components that will not be damaged by earthquake actions.

Protect systems against the adverse effects of parts or components such as tilt switches, that although may not be damaged in an earthquake, may malfunction.

Engineering services: To the recommendations of *SA Gov G172 (2021) Seismic restraint of engineering services* and the following:

Securely fix all plant and equipment to the building structure. Do not rely on gravity and/or friction to resist seismic forces.

Anti-vibration mounts: Use horizontally restrained type.

Wind restraint

General: Provide restraints to resist the effect of ultimate limit state wind pressures. Wind pressure: To AS/NZS 1170.2 (2021).

7 BUILDING SERVICES

7.1 SERVICES CONNECTIONS

Connections

General: Connect to utility service provider services or service points. Excavate to locate and expose connection points.

Reinstate the surfaces and facilities that have been disturbed.

Utility service provider requirements

General: If the utility service provider elects to perform or supply part of the Works, make the necessary arrangements. Install equipment supplied, but not installed, by the utility service provider.

7.2 SERVICES INSTALLATION

General

Installation: Install equipment and services as follows:

- Plumb and securely fixed.
- Allow for movement in both structure and services.
- Arrange services running together, parallel to each other and adjacent building elements.

Concealment: Conceal all cables, ducts, trays and pipes except where installed in plant spaces, ceiling spaces and riser cupboards or Documented to be exposed. If alternative routes are available, do not locate on external walls.

Lifting: Provide heavy items of equipment with permanent fixtures for lifting to the manufacturer's recommendations.

Suspended ground floors: Keep all parts of services suspended under ground floors at least 150 mm clear of the ground surface. Make sure services do not impede access.

Dissimilar metals

Jointing: Join dissimilar metals with fittings of electrolytically compatible material.

Temporary capping

Pipe ends: During construction, protect open ends of pipe with metal or plastic covers or caps.

Piping

General: Install piping in straight lines at uniform grades without sags. Arrange to prevent air locks. Provide sufficient unions, flanges and isolating valves to allow removal of piping and fittings for maintenance or replacement of plant.

Spacing: Provide at least 25 mm clear between pipes and between pipes and building elements, additional to insulation.

Changes of direction: Provide as follows:

- If practicable, long radius elbows or bends and sets, and swept branch connections.
- If pipes are led up or along walls and then through to fixtures, provide elbows or short radius bends.
- Do not provide mitred fittings.

Vibration: Arrange and support piping to prevent vibration whilst permitting necessary movement. Minimise the number of joints.

Embedded pipes: Do not embed pipes that operate under pressure in concrete or surfacing material.

Embedded pipes: If pipes that operate under pressure are to be embedded in concrete or surfacing material conform to AS 2896 (2021) clause 4.3.3.4. Pressure Test and rectify leaks before the concrete is poured.

Valve groupings: If possible, locate valves in groups.

Pressure Testing precautions: Isolate items not rated for the Test pressure. Restrain pipes and equipment to prevent movement during pressure Testing.

Support and structure

Requirement: Provide incidental supports and structures to suit the services.

Pipe support systems

Standard: To AS 4041 (2006) clause 3.28.

General: Provide hangers, brackets, saddles, clips, and support system components to resist live and dead loads and to control pipe movement caused by thermal and water pressure effects. Incorporate provisions for adjustment of spacing, alignment, grading and load distribution. Support pipework from associated equipment or building structure. Support valves, strainers and major line fittings so that no load is placed on connected piping or transmitted to it during operation and maintenance.

Fastenings: Provide fastenings to the associated equipment or building structure designed to withstand the loads imposed by the pipe supports.

Channel section supports: Proprietary channel section with clamps and hangers sized to match external diameter of pipe being supported. Provide all components from the same manufacturer.

Channel and fastening material: Metallic-coated steel.

Vertical pipes: Provide anchors and guides to maintain long pipes in position, and supports designed for the mass of the pipe and its contents.

Saddles: Do not use saddle type supports for pipes larger than DN 20.

Dissimilar metals: If pipe and support materials are dissimilar, provide industrial grade electrically non-conductive material securely bonded to the pipe to separate them. Provide fasteners of electrolytically compatible material.

Fixing to masonry and concrete: Provide metallic-coated steel or non-ferrous metal bolts or screws into chemical or expanding metal masonry anchors.

Uninsulated pipes: Clamp piping supports directly to pipes. Provide electrical isolation of dissimilar metals.

Insulated pipes:

- Spacers: Provide spacers at least as thick as the insulation between piping supports and pipes. Extend either side of the support by at least 20 mm.
- Spacer material: Rigid insulation material of sufficient strength to support the piping and suitable for the temperature application.

- Vapour barriers: For cold pipes, apply aluminium foil tape over the circumference of the spacer to form a vapour barrier. Fit to spacer before installation of the bracket on the pipe.
- Metal sheathing: If metal sheathing is Documented, provide a band of the Documented sheathing materials between the aluminium foil tape and the support for the full width of the spacer.

Hanger sizes: Conform to the following:

- Gas installations: To AS/NZS 5601.1 (2022) Table 5.8.3.
- Other pipes: Provide hangers sized to the manufacturer's recommendations to suit operating conditions and regulatory requirements including the loads due to valves and other attached components, pipe material, pipe contents and temperature and seismic loads.

Support spacing: Provide supports at no greater spacing than the following:

- Cold and heated water: To the NCC cited AS/NZS 3500.1 (2021) Table 5.7.4.
- Sanitary plumbing: To the NCC cited AS/NZS 3500.2 (2021) Table 10.2.1.
- Stormwater: To the NCC cited AS/NZS 3500.2 (2021) clause 4.9.
- Fuel gas: To AS/NZS 5601.1 (2022) Table 5.8.2.
- Fire sprinklers and combined wet suppression systems: To AS 2118.9 (1995) Table 2.6.1.
- Fire hydrants:
- . Metal piping: To AS 2419.1 (2021) clause 10.6.
- . Plastic piping: To the NCC cited AS/NZS 3500.1 (2021).
- Gaseous fire suppression systems:
- . General gaseous fire suppression systems: To AS 4214 (2018) clause 6.3.4.
- . Carbon dioxide fire suppression systems: To AS 6183 (2025) clause 6.3.4.
- Medical gases: To AS 2896 (2021) Table 4.1.
- Refrigerant: To AS/NZS 5149.2 (2016) Tables 5 and 6.
- Other ferrous pipes under pressure: To AS 4041 (2006) Table 3.28.2.
- Other copper pipes: To AS 4809 (2017) Table 6.2.
- ABS pipes: To AS/NZS 3690 (2009) Table 6.2.
- PVC pipes: To AS/NZS 2032 (2006) Table 6.3.
- Polyolefin including PE and PP pipes: To AS/NZS 2033 (2024) clause 7.3.2.
- Other non-ferrous pipe carrying liquids: To the NCC cited AS/NZS 3500.1 (2021) Table 5.7.4.
- Other pipes carrying air or gases: To AS/NZS 5601.1 (2022) Table 5.8.2.
- Proprietary grooved piping systems: To the manufacturer's recommendations.

Additional supports: Provide additional supports as follows:

- Proprietary grooved piping systems: To the manufacturer's recommendations.
- Valves and other heavy pipe mounted components: Adjacent to the valve or component.
- Adjacent pipe mounted components requiring regular maintenance.
- At changes of direction and adjacent to wall or floor penetrations.
- Where required to anchor piping or control thermal or other movement.

Differential movement

General: If the geotechnical Site investigation report predicts differential movements between buildings and the ground in which pipes or conduits are buried, provide control joints in the pipes or conduits, as follows:

- Arrangement: Arrange pipes and conduits to minimise the number of control joints.
- Magnitude: Accommodate the predicted movements.

7.3 PLANT AND EQUIPMENT

General

Location: Locate so failure of plant and equipment (including leaks) does not create a hazard for the building occupants and causes a minimum or no damage to the building, its finishes and contents including water sensitive equipment or finishes.

Safe tray and an overflow pipe: Provide to each tank, hot water heater and storage vessel.

7.4 ACCESS FOR MAINTENANCE

General

Requirement: Provide access for maintenance of all items requiring inspection, measurement, operation, adjustment, repair, replacement and other maintenance-related tasks.

Standards: Conform to the relevant requirements of AS 1657 (2018), AS 1892.1 (2018), AS 2865 (2009) and AS/NZS 3666.1 (2011).

Work Health and Safety: Conform to the requirements of the applicable *Work Health and Safety* regulations.

Access safety systems: Provide access safety systems to *0193 Building access safety systems*.

Refrigerated or cooling plant: If the space is a refrigerated or cooling chamber inside a duct, air handling plant or similar, provided with an access door or personnel access panel and of sufficient size for a person to enter, provide the following to BCA (2022) G1D3:

- An access door.
- Internal lighting with external indicator lamp.
- An alarm.

Protection from injury: Protect personnel from injury caused by contact with objects including those that are sharp, hot or protrude at low level.

Plant room flooring surfaces: R10 Slip resistance classification to AS 4586 (2013).

Trip hazards: Do not run small services including drains and conduits across floors where they may be a trip hazard.

Manufacturer's standard equipment: If necessary, modify manufacturer's standard equipment to provide the plant access Documented.

Clearances

Minimum clearances for access: Conform to the following:

- Vertical clearance: Not less than 2100 mm, vertically above horizontal floors, ground and platforms.
- Horizontal clearance: Preferably not less than 750 mm clear, but in no case less than 600 mm between equipment or between equipment and building features including walls.
- If tools are required to operate, adjust or remove equipment, provide sufficient space so the tools can be used in their normal manner and without requiring the user to employ undue or awkward force.
- Hinged or removable components: To the manufacturer's recommendations.
- Within plant items: Conform to the preceding requirements, and not less than the clearances recommended in BS 8313 (1997).
-

Elevated services other than in occupied areas

Access classifications:

- Access class A: Readily accessible. Provide clear and immediate access to and around plant items. If plant or equipment is located more than 2.0 m above the ground, floor or platform, provide a platform with handrails accessible by a stair, all to AS 1657 (2018).

- Access class B: If the plant item requiring access is located more than 2.0 m above the ground, floor or platform, provide a platform with handrails accessible by a non-vertical ladder, all to

AS 1657 (2018).

- Access class C: Locate plant so temporary means of access conforming to *Work Health and Safety* regulations can be provided.

Temporary means of access: Make sure there is adequate provision in place, which is safe and effective.

Areas in which access is restricted to authorised maintenance personnel: Provide access as follows:

- Instruments, gauges and indicators (including warning and indicating lights) requiring inspection at any frequency: Readily accessible.
- Access required monthly or more frequently: Access class A.
- Access required between monthly and six monthly: Access class A or B.
- Access required less frequently than six monthly: Access class A, B or C. Other areas: Provide access as follows:
- Locate to minimise inconvenience and disruption to building occupants or damage to the building structure or finishes.
- In suspended ceilings, locate items of equipment that require inspection and/or maintenance above tiled parts. If not possible, provide access panels where located above set plaster or other inaccessible ceilings. Arrange services and plant locations to reduce the number of access panels. Coordinate with other trades to use common access panels where feasible.
- Do not locate equipment requiring access above partitions.
- Instruments, gauges and other items requiring inspection at any frequency: Readily accessible.
- Labelling: If equipment is concealed in ceilings, provide marking to **MARKING AND LABELLING**, equipment concealed in ceilings.

Facilities for equipment removal and replacement

Requirement: Provide facilities to permit removal from the building and replacement of plant and equipment, including space large enough to accommodate it and any required lifting and/or transportation equipment. Arrange plant so large and/or heavy items can be moved with the minimum changes of direction.

Removal of components: Allow sufficient space for removal and replacement of equipment components including air filters, tubes of shell and tube heat exchangers, removable heat exchanger bundles, coils and fan shafts. Provide access panels or doors large enough to permit the safe removal and replacement of components within air handling units.

Facilities for access

Equipment behind hinged doors: Provide doors opening at least 150°.

Equipment behind removable panels: Provide panels with quick release fasteners or captive metal thread screws.

Removable panels: Provide handles to permit easy and safe removal and replacement.

Insulated plant and services: If insulation must be removed to access plant and services for maintenance, arrange it to allow for removal and replacement without damage.

Piping

Requirement: Conform to the following:

- Provide access and clearance at fittings that require maintenance, inspection or servicing, including control valves and joints intended to permit pipe removal.
- Arrange piping so it does not interfere with the removal or servicing of associated equipment or valves or block access or ventilation openings.
- Preferably run piping, conduits, cable trays and ducts at high level and drop vertically to equipment.

Electrical equipment and controls

Electrical equipment: Provide clearances and access space to AS/NZS 3000 (2018).

Switchboards and electrical control equipment: Locate near the main entrance to plant space and with switchboards visible from the plant being operated.

Control panels: Locate near and visible from the plant being controlled.

7.5 VIBRATION SUPPRESSION

General

Requirement: Minimise the transmission of vibration from rotating or reciprocating equipment to other building elements.

Standard

Machinery noise and vibration: Vibration severity in Zone A to ISO 20816-1 (2016) and ISO 20816-3 (2022).

Speeds

General: If no maximum speed is prescribed, do not exceed 1500 r/min for direct driven equipment.

Connections

General: Provide flexible connections to rotating machinery and assemblies containing rotating machinery. Isolate pipes by incorporating sufficient flexibility into the pipework or by use of proprietary flexible pipe connections installed to prevent placing stress on pipes due to end reaction.

Inertia bases

General: If necessary to achieve the required level of vibration isolation, provide inertia bases having appropriate mass and to the following:

- Construction: Steel or steel-framed reinforced concrete with reinforcing bars welded between base sections. Position foundation bolts for equipment before pouring concrete.
- Supports: Support on vibration isolation mountings using height saving support brackets.

Vibration isolation mountings

General: Except for external equipment that is not connected to the structure of any building, support rotating or reciprocating equipment on mountings as follows:

- Static deflections less than 15 mm: Single or double deflection neoprene in-shear mountings incorporating steel top and base plates and a tapped hole for bolting to equipment.
- Static deflections not less than 15 mm: Spring mountings.

Selection: Provide mountings selected to achieve 95% isolation efficiency at the normal operating speeds of the equipment.

Installation: Set and adjust vibration isolation mounting supports to give clearance for free movement of the supports.

Spring mountings: Provide freestanding laterally stable springs as follows:

- Clearances: Not less than 12 mm between springs and other members such as bolts and housing.
- High frequency isolation: 5 mm neoprene acoustic isolation pads between base plate and support.
- Levelling: Provide bolts and lock nuts.
- Minimum travel to solid: Not less than 150% of the designated minimum static deflection.
- Ratio of mean coil diameter to compressed length at the designated minimum static deflection: Not less than 0.8:1.
- Snubbing: Snub the springs to prevent bounce at start-up.
- Vertical resilient limit stops: To prevent spring extension when unloaded, to serve as blocking during erection and which remain out of contact during normal operation.

7.6 FINISHES TO BUILDING SERVICES

General

Requirement: If exposed to view (including in plant rooms), paint building services and equipment. Surfaces painted or finished off Site: Conform to *0183 Metals and prefinishes*.

Exceptions: Do not paint chromium or nickel plating, anodised aluminium, glass-reinforced plastic, stainless steel, non-metallic flexible materials and normally lubricated machined surfaces. Surfaces with finishes applied off Site need not be re-painted on Site provided the corrosion resistance of the finish is not less than that of the respective finish Documented.

Standard: Conform to the recommendations of AS/NZS 2311 (2017) Sections 3, 6 and 7 or

AS 2312.1 (2014) Sections 6, 7 and 8, as applicable.

Inaccessible surfaces: If surfaces are inaccessible after installation, complete finish before installation.

Painting systems

New unpainted interior surfaces: To AS/NZS 2311 (2017) Table 5.1. New unpainted exterior surfaces: To AS/NZS 2311 (2017) Table 5.2. **Paint application**

Coats: Apply the first coat immediately after substrate preparation and before contamination of the substrate can occur. Make sure each coat of paint or clear finish is uniform in colour, gloss, thickness and texture and free of runs, sags, blisters or other discontinuities.

Combinations: Do not combine paints from different manufacturers in a paint system.

Protection: Remove fixtures before starting to paint and refix in position undamaged when painting is complete.

Underground metal piping

Requirement: Provide corrosion protection for the following:

- Underground ferrous piping.
- Underground non-ferrous metal piping in chemically aggressive soils and environments. Corrosion protection: Select from the following:
 - Cathodic protection: Sacrificial anodes or impressed current. Incorporate a facility for periodic Testing. Conform to the recommendations of AS 2832.1 (2015).
 - Continuous wrapping using proprietary petroleum taping material.
 - Impermeable flexible plastic coating.
 - Sealed polyethylene sleeve.

Aggressive soils: If metallic piping or components are installed in chemically aggressive soil, provide additional protection as follows:

- Material: Continuous polyethylene sleeve to ASTM D1248 (2016) with a minimum thickness of 0.25 mm.
- Installation: Wrap or sleeve pipes and components. Tape joints between sections of polyethylene and between polyethylene and piping.

Repairs to finishes

Requirement: Repair damaged finishes to restore their corrosion protection, appearance and service life.

Painting of pipe threads: After pipe installation and before other finishes or insulation are applied, paint exposed threads in metallic-coated steel pipe with zinc-rich paint.

7.7 MARKING AND LABELLING

General

Requirement: Mark and label services and equipment for identification purposes as follows:

- Locations exposed to weather: Provide durable materials.
- Pipes, conduits and ducts: To AS 1345 (1995) throughout its length, including in concealed spaces.
- Cables: Label to indicate the origin and destination of the cable.

Consistency: Label and mark equipment using a consistent scheme across all services elements of the project.

Label samples and schedules

Requirement: For each item or type of item, prepare a schedule of marking and labelling, including the following:

- A description of the item or type of item for identification.
- The proposed text for marking or labelling.
- The proposed location of the marking and labelling. Submission timing: Before marking or labelling. **Electrical**

accessories

Circuit identification: Label isolating switches and outlets to identify circuit origin.

Operable devices

Requirement: Mark to identify the following:

- Controls.
- Indicators, gauges, meters.
- Isolating switches.
- Wall mounted equipment in occupied areas
- Location: Provide labels on wall mounted items in occupied areas including the following:
 - Services control switches.
 - Temperature and humidity sensors.
 - Points lists
- Automatic control points: Provide plasticised, fade-free points lists for each automatic control panel and include terminal numbers, point addresses, short and long descriptors in the lists. Store in a pocket on the door of the panel.
- Pressure vessels
- General: Mount manufacturer's certificates in glazed frames on a wall next to the vessel.
- Valves and pumps
 - General: Label to associate pumps with their starters and valves. Screw fix labels to body or attach label to valve handwheels with a key ring.
- Underground services
 - Survey: Accurately record the routes of underground cables and pipes before backfilling. Include on the record drawings.
 - Records: Provide digital photographic records of underground cable and pipe routes before backfilling. Include in operation and maintenance manual.
 - Location marking: Accurately mark the location of underground cables and pipes with route markers consisting of a marker plate set flush in a concrete base, engraved to show the direction of the line and the name of the service.
 - Markers: Place markers at ground level at each joint, route junction, change of direction, termination and building entry point and in straight runs at intervals of not more than 100 m.

Marker bases: 200 mm diameter x 200 mm deep, minimum concrete.

Direction marking: Show the direction of the cable and pipe run by means of direction arrows on the marker plate. Indicate distance to the next marker.

Plates: Brass, aluminium or stainless steel with black filled engraved lettering, minimum size 75 x 75 x 1 mm thick.

Plate fastening: Waterproof adhesive and 4 brass or stainless steel countersunk screws.

Marker height: Set the marker plate flush with paved surfaces, and 25 mm above other surfaces.

Marker tape: Where electric bricks or covers are not provided over underground wiring, provide a 150 mm wide yellow or orange marker tape bearing the words WARNING – electric cable buried below, laid in the trench 150 mm below ground level.

Plastic pipe: Provide a detectable marker tape with trace wire to identify the route of buried piping. Terminate with 1000 mm coil in a readily accessible location. Tag to match the record drawings.

Labels and notices

Materials: Select from the following:

- Cast metal.
- For indoor applications only, engraved two-colour laminated plastic.
- Proprietary pre-printed self-adhesive flexible plastic labels with machine printed black lettering.
- Stainless steel or brass minimum 1 mm thick with black filled engraved lettering. Emergency functions: To AS 1319 (1994).

Colours: Generally to AS 1345 (1995) as appropriate, otherwise black lettering on white background except as follows:

- Danger, warning labels: White lettering on red background.
- Main switch and caution labels: Red lettering on white background.

Edges: If labels exceed 1.5 mm thickness, radius or bevel the edges.

Labelling text and marking: To correspond to terminology and identifying number of the respective item as Documented on the record drawings and in operating and maintenance manuals.

Lettering heights:

- Danger, warning and caution notices: Minimum 10 mm for main heading, minimum 5 mm for remainder.
- Equipment labels within cabinets: Minimum 5 mm.
- Equipment nameplates: Minimum 40 mm.
- Identifying labels on outside of cabinets: Minimum 5 mm.
- Isolating switches: Minimum 5 mm.
- Switchboards, main assembly designation: Minimum 25 mm.
- Switchboards, outgoing functional units: Minimum 10 mm.
- Switchboards, sub assembly designations: Minimum 15 mm.
- Valves:
 - . Not less than DN 65: Minimum 25 mm.
 - . Less than DN 65: Minimum 10 mm.
- Self-adhesive flexible plastic labels:
 - . Labels less than 2000 mm above floor: 5 mm.
 - . Labels minimum 2000 mm above floor: 10 mm.
 - . Other locations: Minimum 5 mm.

Label locations: Locate labels so they are easily seen and are either attached to, below or next to the item being marked.

Fixing: Fix labels securely using screws, rivets, proprietary self-adhesive labels or double-sided adhesive tape and as follows:

- If labels are mounted in extruded aluminium sections, use rivets or countersunk screws to fix the extrusions.
- Use aluminium or monel rivets for aluminium labels. Vapour barriers: Do not penetrate vapour barriers.

8 COMPLETION

8.1 TOOLS AND SPARE PARTS

Spare parts

General: Provide spare parts listed as Documented.

Replacement: Replace spare parts used during the maintenance period.

Tools and spare parts schedule

Submission timing: At least 8 weeks before the Date for Practical Completion.

Requirement: Prepare a schedule of tools, portable instruments and spare parts necessary for maintenance of the installation.

For each item state the recommended quantity and the manufacturer's current price. Include the following in the prices:

- Checking receipt, marking and numbering in conformance with the spare parts schedule.
- Packaging and delivery to Site.
- Painting, greasing and packing to prevent deterioration during storage.
- Referencing equipment schedules in the operation and maintenance manuals.
- Suitable means of identifying, storing and securing the tools and instruments. Include instructions for use.

8.2 TRAINING

General

Standard: To SA TS 5342 (2021).

Duration: Instruction to be available for the whole of the commissioning and running-in periods.

Format: Conduct training at agreed times, at system or equipment location. Also provide seminar instruction to cover all major components.

Operation and maintenance manuals: Use items and procedures listed in the final draft operation and maintenance manuals as the basis for instruction. Review contents in detail with the Principal's staff.

Certification: Provide written certification of attendance and participation in training for each attendee. Provide register of certificates issued.

Demonstrators

General: Use only qualified manufacturer's representatives who are knowledgeable about the installations.

Operation

General: Explain and demonstrate to the Principal's staff the purpose, function and operation of the installations.

Maintenance

General: Explain and demonstrate to the Principal's staff the purpose, function and maintenance of the installations.

Seasonal operation

General: For equipment requiring seasonal operation, demonstrate during the appropriate season.

8.3 CLEANING

Removal of material

General: Dispose of building waste material off Site to the requirements of the relevant authorities.

Final cleaning

General: Before the Date for Practical Completion, clean throughout, including all exterior and interior surfaces except those totally and permanently concealed from view.

Labels: Remove all visible labels not required for maintenance.

8.4 WARRANTIES

General

Requirement: If a warranty is Documented, name the Principal as warrantee. Register with manufacturers as necessary. Retain copies delivered with components and equipment.

Approval of applicator or installer: If the warranty is conditional on the manufacturer's approval of the applicator or installer, submit the manufacturer's written approval of the installing company, and authorised personnel, with evidence of qualifications and experience in the specific use of the product, material or system.

Principal's responsibilities: Submit details of responsibilities of the Principal required to keep warranties in force.

Warranty types

Manufacturer's warranty: Warranty to cover manufacturing Defects and Defects with products and materials delivered to Site.

Manufacturer and applicator's/installer's interlocking warranty: Interlocking warranty to cover manufacturing Defects and Defects with products and materials delivered to Site, including their application or installation.

Supplier's warranty: Warranty to Defects in materials delivered to Site.

9 TESTING AND COMMISSIONING

9.1 TESTING - GENERALLY

Inspection and Testing plan

Requirement: Provide inspection and Testing plan consistent with the construction Program including details of Test stages and procedures.

Notice

Site Tests: Give notice of the time and place of Documented Tests.

Inspection: Give sufficient notice for inspection to be made of the commissioning, Testing and verification Tests on completion of commissioning.

Attendance

General: Provide attendance at Tests.

Suppliers: If necessary to carry out Documented Tests, arrange equipment suppliers to assist.

Testing authorities

Requirement: Have Tests carried out by an Accredited Testing Laboratory, accredited for the Documented Test method, except for Site Tests or Test methods that do not have an Accredited Testing Laboratory.

Test equipment

Accuracy: Use Testing equipment designed to Test and/or measure system performance within the Documented tolerances.

Calibration: Use only instruments that have current calibration certificates issued by an Accredited Testing Laboratory. Tag or label instruments with calibration date and calibration authority name. Provide copies of certification if requested.

Maximum period since last calibration: As recommended by the manufacturer but less than 12 months, except as Documented.

Recalibration: If dropped or damaged, recalibrate instruments.

Testing equipment: Provide Test equipment and tools to perform Documented Tests as follows:

- Special Testing equipment: If Documented, provide special equipment, tools and instruments required for Testing or calibration.
- Other Testing equipment: Provide standard Testing equipment.

Testing procedures

Verification: Verify Test procedures by:

- Manual Testing.
- Monitoring performance and analysing results using the control system trend logs.
- A combination of the above methods.

Sampling: Sampling may be used subject to the following:

- Use a sampling strategy only for multiple identical pieces of non-life-safety or otherwise non-critical equipment.
- If at any point, more than one identical item has failed, stop Testing, determine the cause, rectify and Document changes made to remaining units, before continuing with functional Testing of the remaining units.

Type Tests

Type Test reports: Required, as evidence of conformance of proprietary equipment.

Sound pressure level measurements

Requirement: Conform to the following:

- Correction for background noise: To AS/NZS 2107 (2016) Table B1.
- External: To AS 1055 (2018).
- Internal: To AS/NZS 2107 (2016).

Measurement positions: If a Test position is designated only by reference to a room or space, do not take measurements less than 1 m from the floor, ground or walls. For large equipment items including chillers, measure at 2 m and 7 m from the equipment item.

- Sound pressure level analysis: Measure the sound pressure level and the background sound pressure level over the full range of octave band centre frequencies from 31.5 Hz to 8 kHz at the designated positions.
- Sound pressure levels: Measure the A-weighted sound pressure levels and the A-weighted background sound pressure levels at the designated positions.

Test outcome

Requirement: Test as Documented and achieve the following:

- Pass the Documented Pass/Fail Test, and/or
- Values that meet Documented requirements, and/or
- Verification of manufacturer's claimed performance.

Failure of multiple items

Requirement: If 10% or 3, whichever is greater, of identical pieces (size does not constitute a difference) of equipment fail to perform as Documented for any reason, treat all identical units as having failed. Submit notice of failure and conform to the following:

- Within one week of notification, examine all other identical units and record the results. Submit a report of the findings within two weeks of the original failure notice.
- Within two weeks of the original failure notification, submit a signed and dated explanation of the problem, including the cause of failure, the proposed solution, full equipment details and any other information. Do not exceed the Documented requirements of the original installation with the proposed solution.

Rectification of failure under Test

Requirement: If an item fails a Documented Test, rectify the cause of failure and repeat the Test.

Submissions: If submission of Test results is Documented, submit results of both successful and unsuccessful Tests.

Test reports

Requirement: Include the following:

- Documented performance criteria including, if Documented, tolerances.
- Observations and results of Tests and conformance or non-conformance with Documented requirements.

Test validity period

Requirement: As Documented or, if no validity period is Documented, no older than 5 years.

Controls

General: Calibrate, set and adjust control instruments, control systems and safety controls.

Circuit protection

General: Confirm that circuit protective devices are sized and adjusted to protect installed circuits.

Certification

General: On satisfactory completion of the installation, Testing and commissioning and before the Date for Practical Completion, certify that each installation is operating correctly.

Integrated system Tests

Requirement: Conduct integrated system Tests as Documented. Tests: Provide the following:

- Test the integrated operation of the systems listed in each mode Documented.

Restoration of the systems to their pre-Test condition on completion of the Tests above.

Failure: If any of the systems fails to perform as Documented, including return to normal operation, rectify the cause and repeat the integrated system Test.

Deferred and seasonal Tests

Deferred Tests: If Documented Testing cannot be completed at the scheduled or Documented time, the Superintendent may direct that they be deferred to a later time but as soon as possible after the scheduled or Documented time.

Seasonal Tests: If Documented Tests are dependent on specific weather conditions, they may be deferred to a time when weather conditions are close to the Documented Test conditions. Complete seasonal Testing as soon as possible but no later than one month before the end of the Defects Liability Period.

Functional Tests

Function: Carry out functional and operational Tests on each energised equipment item and circuit.

COMMISSIONING

Standard

Requirement: Conform to SA TS 5342 (2021).

Static completion

Requirement: Systems, components and building elements are statically complete when:

- Their construction and installation is complete and as Documented, including completion of all systems, components and building elements on which they are dependent for commissioning.
- All pre-commissioning Tests have been successfully completed.
- They are safe and ready for commissioning.
- All cleaning that may adversely affect commissioning is complete.
- They have been inspected and all outstanding remedial work that may adversely affect commissioning is complete.

- All spaces required for access for commissioning are safe to use and cleared of obstructions that may adversely affect commissioning.

Commissioning plan

Requirement: Provide a commissioning plan to SA TS 5342 (2021) including the following:

- A summary of the Work covered by the commissioning plan.
- The parties responsible for this Work and any commissioning interrelationships.
- The basis of the design.
- General sequence of commissioning.
- Project specific commissioning methodologies for each system and building element to be commissioned.
- Pre-commissioning requirements.
- Project specific commissioning procedures for each commissioning activity including integrated system Tests, deferred and seasonal Tests.
- A project specific building tuning plan for all commissioned systems. Include building tuning procedures and tuning team members.
- Requirements for witnessing of Tests and Documented demonstrations of completion of commissioning.
- Commissioning program to **COMMISSIONING, Commissioning Program**.

Commissioning Program

Submissions: Submit a program consistent with, and forming part of, the construction Program as follows:

- Set out the proposed program for completion, commissioning, Testing and instruction.
- Identify related works and timing of the works prerequisite to successful and timely completion of the Works.

Revisions: Submit revisions of the program as the project proceeds.

Plant operation period: Include time in the program for the Documented plant operation period before the Date for Practical Completion.

Commissioning activities

Requirement: Provide the following to SA TS 5342 (2021):

- Manage the commissioning process.
- Establish and manage the completion process.
- Review design Documents for commissionability. Submit a report including any recommended changes.
- Review Documented commissioning requirements. Submit a report including any recommended changes.
- Review construction Documents for commissionability. Submit a report including any recommended changes.
- Develop, review and update the commissioning plan and commissioning Program.
- Develop, review and update commissioning methodologies.
- Develop, review and update commissioning procedures.
- Report on interdependencies between trades that may affect commissioning.
- Develop, review and update procedures for initial start-up of systems.
- Develop, review and update integrated system Test procedures.
- Carry out pre-commissioning activities. Record results and submit pre-commissioning records.
- Conduct commissioning activities to the commissioning methodologies and procedures. Record and submit commissioning records. Facilitate and conduct integrated system Tests and demonstrations. Record and submit integrated system Test records.
- Conduct Documented demonstrations of completion of commissioning.

- Report on the progress of commissioning work.
- Report on conformance to the commissioning plan and program.
- Report on commissioning Defects and issues and progress on their resolution.
- Develop, review and update commissioning report.
- Develop, review and update training materials, conduct training sessions to **TRAINING**.
- Develop, review and update operation and maintenance manuals to **OPERATION AND MAINTENANCE MANUALS**.
- Manage and report deferred and seasonal Testing activities to **TESTING - GENERALLY**.
- Management and reporting of building tuning process.
- Periodically review performance data.

Verification of commissioning

Requirement: On completion of commissioning of the equipment or system, provide additional Tests to verify that it is fully commissioned and operating to Documented requirements.

9.2 BUILDING TUNING

General

Standard: To SA TS 5342 (2021).

Frequency: Three monthly or more frequently.

Duration: Until the end of the maintenance period. Provide last building tuning in the month before the end of the maintenance period.

Requirement: Provide the following:

- Review data from all recording systems against Documented requirements.
- Review of building occupant feedback.
- If discrepancies are identified from the above, take corrective action to rectify them.
- Report on the findings of the reviews, corrective action and effect of corrective action.
- Recommend other action to improve the effectiveness, reliability and efficiency of systems.

10 PROJECT RECORDS

10.1 RECORD DRAWINGS

General

Requirement: Prepare record drawings showing the following:

- Installed locations of building elements, services, plant and equipment.
- Off-the-grid dimensions and depth if applicable.
- Any provisions for the future.

Recording, format and submission

Requirement: Record changes made during the progress of the WUC on a set of drawings kept on Site for that specific purpose.

Drawing layout: Use the same borders and title block as the Contract drawings. Quantity and format: Conform to **SUBMISSIONS**.

Endorsement: Sign and date all record drawings.

Accuracy: If errors in, or omissions from, the record drawings are found, amend the drawings and re-issue in the quantity and format Documented for **SUBMISSIONS**.

Date for submission: Not later than 2 weeks after the Date for Practical Completion.

Services record drawings

General: To **RECORD DRAWINGS, General** and **Recording, format and submission** and the following:

- Extensions and/or changes to existing: If a drawing shows extensions and/or alterations to existing installations, include sufficient detail of the existing installation to make the drawing comprehensible without reference to drawings of the original installation.
- Detention: If on-Site detention tanks or pondage are provided, include the volume required on the drawing and the permitted flow rate to the connected system.
- Domestic cold water or fire mains: Show the pressure available at the initial connection point and the pressure available at the most disadvantaged location on each major section of the Works.
- Stormwater: If storm water pipes are shown, include the pipe size and pipe grade together with the maximum acceptable flow and the actual design flow.

Diagrams: Provide diagrammatic drawings of each system including the following:

- Controls.
- Piping including all valves and valve identification tags.
- Principal items of equipment.
- Single line wiring diagrams.
- Acoustic and thermal insulation.
- Access provisions and space allowances.
- Fasteners.
- Fixtures.
- Switchgear and controlgear assembly circuit schedules including electrical service characteristics, controls and communications.
- Charts of valve tag numbers, with location and function of each valve, keyed to flow and control diagrams.

CAD base drawings: Refer consultant File Transfer Disclaimer

Subsurface services: Record information on underground or submerged services to the Documented quality level, conforming to AS 5488.1 (2022).

Subsurface services recording quality level: Level A

10.2 BASELINE DATA

General

Requirement: Provide baseline data to permit routine service of fire protection systems and equipment to AS 1668.1 (2015), the NCC cited AS 1670.1 (2018), AS 1851 (2012) and AS/NZS 2293.1 (2018).

Include baseline data for the following:

- Active fire and smoke systems including automatic fire sprinkler systems, fire pumpsets, fire hydrant systems and water storage tanks for fire protection systems.
- Fire detection and alarm systems.
- EWIS, exit signs and emergency lighting.
- Standby generator sets and batteries.
- Lay flat fire hose, fire hose reels, portable and wheeled fire extinguishers and fire blankets.
- Passive fire and smoke systems, including the following:
 - . Fire and smoke elements (vertical and horizontal) - walls, floors, ceilings, access panels and hatches.
 - . Structural fire-resistant elements - beams, columns, girders and trusses.

- . Fire-resisting doorsets - hinged, pivoted and horizontal sliding.
- . Smoke doors - hinged and pivoted.
- . Fire shutters.
- . Fire-resisting glazing.
- . Ducts.
- . Dampers.
 - Fire and smoke control features of mechanical services.
 - Emergency planning in facilities.

Revisions: Amend baseline data to include changes made to the installation during construction and maintenance. Correct errors in baseline data.

Format: Provide baseline data in a format that facilitates the carrying out and recording of routine service tasks including drawings showing the extent and location of items to be serviced, schedules of items and unique identification of each item.

10.3 OPERATION AND MAINTENANCE MANUALS

General

Standard: To SA TS 5342 (2021).

Authors and compilers: Personnel experienced in the maintenance and operation of equipment and systems installed, and with editorial ability.

Referenced Documents: If referenced Documents or worksections require submissions of manuals, include corresponding material in the operation and maintenance manuals.

Subdivision: By installation or system, depending on project size.

Revisions: Amend the operation and maintenance manuals to include changes made to the installation during construction and maintenance including changes to software and commissioning records.

Contents of manual

Table of contents: Include a table of contents in each volume. Title to match cover. Table of amendments: Include a table of amendments.

Directory: Names and contact details of principal consultant, subconsultants, Contractor, subcontractors and name of main contact.

Record drawings: Include complete set of record drawings, full size.

Drawings and technical data: Include as necessary for the efficient operation and maintenance of the installation.

Installation description: Include a general description of the installation.

Systems descriptions and performance: Include a technical description of the systems installed including the basis of design, the interrelation with other systems and the building and mode of operation, presented in a clear and concise format readily understandable by the Principal's staff. Identify function, normal operating characteristics, safety features and limiting conditions.

Baseline data: Include the baseline data to **BASELINE DATA**.

Commissioning records: Include commissioning records to SA TS 5342 (2021). Link commissioning records to item codes on the record drawings.

Training material: Include materials used to provide training, to **TRAINING**, in a form that can be used to train others.

Fire systems and equipment: Include Documentation to AS 1851 (2012), including the schedule of essential functionality and performance requirements.

Digital photographic and video records: Provide Documented digital photographic and video records, keyed to the drawings.

Equipment: Include schedules with the following details for installed equipment:

- Item code for use on record and diagrammatic drawings, and spare parts schedule.
- Equipment name plate data including serial number, if any.
- Name and contact details of the manufacturer and supplier.
- Catalogue list number(s).
- Location.
- Function.
- Performance figures and capacity data.
- Date of manufacture.
- Manufacturer's product data sheets including only relevant matter for the project. Mark each product data sheet to clearly identify specific products and component parts used in the installation, and data applicable to the installation.
- Additional information and commentary to illustrate relations of component parts. Certificates:
 - Certificates from authorities.
 - Product certification.
 - Test certificates for each service installation and all equipment.
 - Warranties.

Trends: 7 day record of all trends at commissioning. Operation procedures: Include for systems installed:

- Manufacturer's technical literature as appropriate.
- Safe starting up, running-in, operating and shutting down procedures. Include logical step-by-step instructions for each procedure.
- Control sequences and flow diagrams.
- Legend for colour-codes services.
- Schedules of fixed and variable equipment settings established during commissioning and maintenance.
- A list of special safety devices and their set points.
- Procedures for seasonal changeovers.
- Warnings to operators.
- Procedures for identifying and rectifying common faults.
- Recommendations for efficient plant operation.
- If the installation includes cooling towers, recommendations for water efficiency.
- Building tuning plan and procedure to **COMMISSIONING, Commissioning plan**.

Building occupants' guide: Include a concise guide written and illustrated for building occupants with no technical background. Include the following:

- Security provisions.
- Safety and access.
- Environmental features, including energy and water efficiency and waste management.
- Occupant relevant information on design and operation.
- Information for occupants on environmental systems that rely partially or wholly on local controls for heating, lighting, cooling, and ventilation.
- Contact details for faults, maintenance and emergencies. Maintenance procedures:
 - Detailed recommendations for periodic maintenance and procedures, including schedule of maintenance work with frequency and manufacturer's recommended Tests.
 - Manufacturer's technical literature as appropriate. Register with manufacturer as necessary. Retain copies delivered with equipment.
 - Safe trouble-shooting, disassembly, repair and reassembly, cleaning, alignment and adjustment, balancing and checking procedures. Provide logical step-by-step instructions for each procedure.

- Schedule of spares, recommended to be held on Site, for those items subject to wear or deterioration and that may involve the Principal in extended deliveries when replacements are required. Include complete nomenclature and model numbers, and local sources of supply.
- Schedule of normal consumable items, local sources of supply, and expected replacement intervals up to a running time of 40 000 hours. Include lubrication schedules for equipment.
- Instructions for use of tools and Testing equipment.
- Troubleshooting procedures.
- Emergency procedures, including telephone numbers for emergency services, and procedures for fault finding.
- Safety data sheets (SDS).
- Instructions and schedules conforming to AS 1851 (2012), AS/NZS 3666.2 (2011),

AS/NZS 3666.3 (2011) and AS/NZS 3666.4 (2011).

Maintenance records:

- Prototype routine service records conforming to AS 1851 (2012) prepared to include project specific details.
- Prototype periodic maintenance records and report to AS/NZS 3666.2 (2011),

AS/NZS 3666.3 (2011) and AS/NZS 3666.4 (2011) as appropriate, prepared to include project specific details.

- Hard copies: Binders to match the manuals, containing loose leaf logbook pages designed for recording completion activities including operational and maintenance procedures, materials used, Test results, comments for future maintenance actions and notes covering the condition of the installation. Include completed logbook pages recording the operational and maintenance activities performed up to the Date for Practical Completion.
- Number of pages: The greater of 100 pages or enough pages for the maintenance period and a further 12 months.

Emergency information: For each type of emergency, including fire, flood, gas leak, water leak, power failure, water failure, system or subsystem failure, chemical release or spill, include the following:

- Emergency instructions.
- Emergency procedures including:
 - . Instructions for stopping or isolating.
 - . Shutdown procedures and sequences.
 - . Instructions for actions outside the property.
 - . Special operating instructions relevant to the emergency.
 - . Contact details relevant to the emergency.

Emergency information manual

Form of emergency information: Provide one of the following:

- An index and coloured tabs identifying emergency information for each type of emergency within the operation and maintenance manual.
- A separate Emergency manual containing copies of emergency information from the main operation and maintenance manual.

Format – electronic copies

Scope: Provide the same material as Documented for hardcopy in electronic format. Quantity and format: Conform to **SUBMISSIONS, Electronic submissions.**

Printing: Except for drawings required in **RECORD DRAWINGS** provide material that can be legibly printed on A4 size paper.

Format – hard copies

General: A4 size loose leaf, in commercial quality, 4 ring binders with hard covers, each indexed, divided and titled. Include the following features:

- Cover: Identify each binder with typed or printed title *OPERATION AND MAINTENANCE MANUAL*, to spine. Identify title of project, volume number, volume subject matter, and date of issue.
- Dividers: Durable divider for each separate element, with typed description of system and major equipment components. Clearly print short titles under laminated plastic tabs.
- Drawings: Fold drawings to A4 size with title visible, insert in plastic sleeves (one per drawing) and accommodate them in the binders.
- Pagination: Number pages.
- Ring size: 50 mm maximum, with compressor bars.
- Text: Manufacturer's printed data, including associated diagrams, or typewritten, single-sided on bond paper, in clear concise English.

Number of copies: 3.

Date for submission

Draft submission: The earlier of the following:

- 4 weeks before the Date for Practical Completion.
- Commencement of training.

Final submission: Within 2 weeks after the Date of Practical Completion.

11. MAINTENANCE

11.1 PERIODIC MAINTENANCE

General

Requirement: Provide Documented maintenance so that the condition and performance of the maintained work throughout and at the end of the maintenance period is equal to or better than that at the beginning of the maintenance period including with respect to the following:

- Performance, service delivery.
- Service life and reliability.
- Compliance with statutory requirements.
- Compliance with building rating requirements.
- Energy and water efficiency.
- Environmental impact.
- Health and safety.
- Risk management. Inclusions: Include the following:
 - Periodic and statutory maintenance, cleaning and replacement of consumables.
 - Emergency repairs.
 - Condition reporting.

Duration: From the time systems and equipment are put into service to the end of the maintenance period.

Maintenance period: The greater of the Defects Liability Period and the period Documented. Faults: Rectify promptly.

Emergencies: Attend emergency calls promptly.

Annual maintenance: Carry out recommended annual maintenance procedures within the four weeks before the end of the maintenance period.

12 SELECTIONS

12.1 PERFORMANCE

Noise level schedule

	Whole Project
Upper limit of noise caused by services	Refer mechanical Documents

12.2 SUBMISSIONS AND INSPECTIONS

Notices schedule

Item	Minimum notice
On-Site or Off-Site Inspection	5 Business Days

12.3 MAINTENANCE

Maintenance requirements schedule

Refer to Design drawings, Hydraulic, Mechanical, and Electrical Documents.

0172 ENVIRONMENTAL MANAGEMENT

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide environmental management, as Documented.

Management and control plans

Implementation: To Documented management plans.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.

1.3 INTERPRETATION

Definitions

General: For the purposes of this worksection, the following definitions apply:

- Clearances: A formal certificate, approval or condition issued by a statutory authority allowing work in a particular area.
- Contamination of land: The presence of a substance in, on or under the land that is designated hazardous material and/or is at a concentration above that which is normally found in that locality, and presents a risk of harm to human health or to the environment.
- Environment: The physical factors of the surroundings of human beings including the land, waters, atmosphere, climate, sound, odours, tastes; and the biological factors of animals; and plants and the social factor of aesthetics.
- Environmental audit: A review of environmental management practices, in particular the evaluation of a site for environmental liability.
- Environmental impact assessment (EIA): A method for predicting environmental impacts of a proposed development including strategies for minimising identified impacts.
- Environmental management plan (EMP): A project or site specific plan describing the management of the environmental issues and considerations for the activity being undertaken. This applies to the design, construction and operation of the buildings, external works and infrastructure.
- Organic waste: Includes all food wastes, vegetative wastes from land clearing and pruning operations, biosolids produced from the treatment of liquid wastes, garden wastes and forestry waste (bark and saw dust) and paper and cardboard products.
- Pollution Incident: An Incident or set of circumstances during or as a consequence of which there is, or is likely to be, a leak, spill or other escape of a substance as a result of which pollution has occurred, is occurring or is likely to occur.
- Weed: An invasive plant that degrades natural areas, reduces the sustainability or affects the health of people and animals.

1.4 SUBMISSIONS

Control plans

Requirement: Submit the following:

- Environmental management plan.

Time for submissions: before any Work has commenced on Site.

1.5 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- Non-conforming items, such as ground contamination.
- Completed removal or rectification of non-conforming items.
- Unexpected finds.

2 ENVIRONMENTAL ADMINISTRATION, MONITORING AND REPORTING

2.1 PROCEDURAL AND PERSONNEL

Emergency response

Emergency response personnel: Provide staff names and contact details.

Complaints

Reporting: Within 1 working day of receiving a complaint about an environmental impact, including pollution Incidents, prepare a written report detailing the complaint and remedial action taken.

Register: Keep a register of all complaints and action taken.

Unexpected finds

Requirement: Give notice and close off affected Site area with barrier tapes and warning signs to prevent access.

Unexpected finds: Include asbestos and other hazardous or volatile contaminants, archaeological finds and items of heritage value.

2.2 MONITORING

Internal monitoring

Approval authority: the Superintendent Documentation: Provide details of the following:

- Environmental monitoring: Implementation and recording procedures.
- Control measures to be implemented: Non-conformance control and corrective action procedures.

Records: Maintain records of results of environmental monitoring, including the effectiveness of any remedial action taken.

Internal monitoring personnel: Provide staff names and contact details. Machinery and equipment: Provide details of proposed plant.

Review timing: Review the EMP or control measures at the following stages:

- When there is a change in the project, including the scope.
- Following significant environmental impacts or pollution Incidents.
- When improved performance is required to reduce a specific environmental impact.
- At completion of environmental audits.
- At the end of the project.

2.3 REPORTING

General

Requirement: Prepare environmental management reports that record the progress of the following:

- Performance against statutory requirements.
- Performance against the EMP, environmental objective and policy, and ecologically sustainable development outcomes and targets.
- Summary of monitoring, inspection and audits.
- Summary of reports required to meet the statutory requirements.
- Summary of environmental impacts, pollution incidents, non-conformance and complaints.
- Summary of corrective actions where required.
- Unexpected finds.

Reporting frequency: End of each month

3 ENVIRONMENTAL MANAGEMENT

3.1 ENVIRONMENTAL MANAGEMENT PLAN

Control plan

EMP: Prepare an environmental management plan with the following details:

- Project description, including Site location, construction activities and project schedule.
- EMP context, describing how the EMP fits into the overall project planning process.
- EMP objective and environmental policy.
- Risk assessment.
- Assignment of responsibility for environmental controls, including hierarchy of management.
- Conditions of approvals, licences and permits to meet statutory requirements.
- Reporting requirements.
- Environmental training plan and procedures: Including a program to familiarise staff with the EMP and/or management controls, environmentally sensitive areas and responsibilities.
- Environmental audit program and corrective action procedures.
- Emergency response procedures, including response time.
- Control plans, as required.
- Details of operational control measures to reduce risk of environmental impacts on the following:
 - . Heritage.
 - . Visual values.
 - . Endangered species.
 - . Habitat.
 - Locations of, and environmental controls for, environmentally sensitive areas.
 - Details of environmental protection for each activity.
 - Communication procedures.
 - Other items necessary to protect the surrounding environment. Activities staging: Address the phases of activity, as appropriate:
 - Before construction and Site establishment.
 - During construction.
 - After construction, including rehabilitation activities and Site and landscaping maintenance.

3.2 SOIL EROSION AND SEDIMENT CONTROL

Control plan

Plan: Prepare a soil erosion and sediment control plan to protect natural overland flow paths and stormwater drainage systems from sedimentation by minimising soil erosion and sediment transportation. Include the following details:

- Staging of operations and sequence of Works.
- Diversion of upstream water around the Site.
- Provision of temporary drains and catch drains.
- Erosion control measures.
- Sediment control measures.
- Stormwater control measures, including diversion, dispersal and/or retention to control and dissipate stormwater throughout the Site without damage.
- Maintenance procedures for control measures.
- Progressive restoration of disturbed areas.

Extent: All Site areas, access and haulage tracks, borrow pits, stockpile and storage areas and compound areas.

Control measures

Requirement: Protect natural overland flow paths and stormwater drainage systems from sedimentation by minimising soil erosion and sediment transportation. Refer also to Civil Engineering Documents.

3.3 WASTE MANAGEMENT

Control plan

Plan: Prepare a waste management control plan to minimise solid waste generated by construction activities and to dispose of the solid waste. Include the following details:

- Major waste streams.
- For each waste stream describe the following:
 - . Location.
 - . Disposal method.
 - . Transport method.
 - . Storage method.

3.4 GROUND CONTAMINATION

Control plan

Plan: Prepare a ground contamination control plan for land suspected of being contaminated or containing presence of acid sulfate soil, in conformance with the state or territory Environmental Protection Authority (EPA) and the relevant state or territory planning guidelines. Include the following details:

- Preliminary investigation.
- Detailed investigation.
- Site Remedial Action Plan (RAP).
- Stockpile sites.
- Site auditing and reporting procedures.
- Record of maintenance procedures, including record of remediation work, certificates issued and restrictions placed on the Site.

3.5 WEED MANAGEMENT

Control plan

Plan: Prepare a weed management control plan with the following details:

- Weed species.
- Weed infestation zones within the work Site and the investigation period.
- Treatment of infestation, including treatment type and application rate.
- Cleaning methods for vehicles and machinery, including cleaning bay location and treatment date.
- Subcontractors: Names and contact details for proposed subcontractors and evidence of experience in treatment of weed infestations and name and qualifications of chemical treatment handlers.

3.6 AIR QUALITY

Control plan

Plan: Prepare an air quality control plan to protect adjoining owners, residents and the public from emissions of dust and exhaust gases. Include the following details:

- Dust control.
- Exhaust gases emission control.

3.7 WATER QUALITY

Control plan

Plan: Prepare a water quality control plan to keep earthworks free of water and to reduce impacts on groundwater and surface water. Include the following details:

- Dewatering system.
- Washout prevention.
- Cross connection prevention to public sewerage system.
- Water disposal.

Control measures

Requirement: Keep earthworks free of water and reduce impacts on groundwater and surface water. Refer also to Civil Engineering Documents.

3.8 NOISE CONTROL AND VIBRATION

Control plan

Plan: Prepare a noise and vibration control plan to protect adjoining owners, residents and the public from noise and vibration impacts. Include the following details:

- Noise control and vibration: To the recommendations of AS 2436 (2010).
- Maximum noise level at the Site boundary.
- Noise control.
- Vibration assessment.
- Vibration control.

3.9 FLORA AND FAUNA

Control plan

Plan: Prepare a flora and fauna control plan to protect native flora and fauna on Site and prevent introduction of pest species. Include the following details:

- Native flora and fauna assessment.
- Pest species assessment.
- Tree pruning.
- Habitat provision.

3.10 CULTURAL AND ABORIGINAL AND/OR TORRES STRAIT ISLANDER HERITAGE

Control plan

Plan: Prepare a cultural heritage management plan to protect and prevent damage or loss of items of cultural heritage or Aboriginal and/or Torres Strait Islander sites and artefacts. Include the following details:

- Cultural heritage assessment.
- Aboriginal and/or Torres Strait Islander heritage assessment.
- Unexpected finds.
- Training.

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide adhesives, sealants and fasteners, as documented.

Performance

Requirements: Conform to the following:

- Fitness for purpose: Capable of transmitting imposed loads, sufficient to maintain the rigidity of the assembly, or integrity of the joint.
- Finished surface: That will not cause discolouration.
- Compatibility: Compatible with the products to which they are applied.
- Sealant replacement: Capable of safe removal without compromising the application of the replacement sealant for future refurbishment.
- Movement: If an adhered or sealed joint is subject to movement, select a system certified to accommodate the projected movement under the conditions of service.

1.2 PRECEDENCE General

Order of precedence:

- The requirements of other work sections of the specification override conflicting requirements of this work section.
- The requirements of work sections override conflicting requirements of their referenced documents. The requirements of the referenced documents are minimum requirements.

1.3 CROSS REFERENCES General

Requirement: Conform to the following:

- *0171 General requirements.*

1.4 SUBMISSIONS Products and materials

Sealants: Submit technical data sheets.

Adhesives and sealants: Submit safety data sheets (SDS).

Samples

Visible joint sealants: Submit colour samples.

Compatibility testing: Submit adhesion and compatibility testing data demonstrating that adhesive, sealant or fastener is compatible with materials to be fixed and is suitable for the project conditions.

Warranties

Manufacturer's warranty: Submit the manufacturer's published product warranties.

1.5 INSPECTION Notice

Inspection: Give notice so that inspection may be made of prepared joints and penetrations for each sealant application included in the **Installed sealant tests schedule**.

2 PRODUCTS

2.1 ADHESIVES Standards

Gypsum plaster adhesive: To AS 2753.

High strength adhesive tape

General description: A foam of cross linked polyethylene or closed cell acrylic coated both sides with a high performance acrylic adhesive system, encased in release liners of paper or polyester.

Product classification: Select tape to suit substrate as follows:

- Firm high strength foam tapes: For high energy surfaces including most bare metals such as stainless steel and aluminium.
- Conformable high strength foam: For the following:
 - . Medium energy surfaces including many plastics and paints, and bare metals.
 - . Lower energy surfaces including many plastics, most paints and powder coatings, and bare metals.

Thickness: Select the tape to make sure a mismatch between surfaces does not exceed half the tape thickness under the applied lamination pressure.

2.2 SEALANTS Standards

General: To ISO 11600.

Lightweight building element joints

Joints subject to rapid changes of movement: Provide sealants that accommodate the movement of the contact materials.

Floor control joints

General: Provide trafficable sealants.

Bond breaking backing:

- Bond breaking materials: Non-adhesive to sealant, or faced with a non-adhering material.
- Foamed materials: Closed-cell or impregnated, not water-absorbing.

2.3 FASTENERS General

Masonry anchors: Proprietary expansion or bonded type anchors conforming to **SELECTIONS, ANCHORS**.

Plain washers: To AS 1237.1.

- Provide washers to the heads and nuts of bolts, and the nuts of coach bolts.

Plugs: Proprietary purpose-made plastic.

Stainless steel fasteners: To ASTM A240/A240M.

Steel nails: To AS 2334.

- Length: At least 2.5 times the thickness of the member being secured, and at least 4 times the thickness if the member is plywood or building board less than 10 mm thick.

Unified hexagon bolts, screws and nuts: To AS/NZS 2465.

Fasteners in CCA treated timber: Epoxy coated or stainless steel.

Bolts

Coach bolts: To AS/NZS 1390.

Hexagon bolts Grades A and B: To AS 1110.1.

Hexagon bolts Grade C: To AS 1111.1.

Corrosion resistance

Atmospheric corrosivity category: To *0171 General requirements*.

Steel products: Conform to the **Corrosion resistance table** or provide proprietary products with metallic and/or organic coatings of equivalent corrosion-resistance.

Corrosion resistance table

Atmospheric corrosivity category to AS 4312	Threaded fasteners and anchors		Powder actuated fasteners
	Material	Minimum local metallic coating thickness (µm)	Material
C1 and C2	Electroplated zinc or Hotdip galvanized	30	Stainless steel Type 316
C3	Hot-dip galvanized	45	Stainless steel Type 316
C4	Stainless steel Type 316	-	Stainless steel Type 316
Note: For categories C5, CX and T to the AS/NZS 2312 series, seek specialist advice.			

Finishes

Electroplating:

- Metric thread: To AS 1897.
- Imperial thread: To AS 4397.

Galvanizing:

- Threaded fasteners: To AS/NZS 1214.
- Other fasteners: To AS/NZS 4680.

Mild steel fasteners: Galvanize if:

- Embedded in masonry.
- In external timbers.
- In contact with chemically treated timber other than CCA treated timber.

Epoxy coated: CCA treated timber.

Nuts

Hexagon chamfered thin nuts Grades A and B: To AS 1112.4.

Hexagon nuts Grade C: To AS 1112.3.

Hexagon nuts Style 1 Grades A and B: To AS 1112.1.

Hexagon nuts Style 2 Grades A and B: To AS 1112.2.

Screws

Coach screws: To AS/NZS 1393.

Hexagon screws Grades A and B: To AS 1110.2.

Hexagon screws Grade C: To AS 1111.2.

Hexagon socket screws: To AS 1420.

Self-drilling screws: To AS 3566.1.

Self-tapping screws:

- Crossed recessed countersunk (flat – common head style): To AS/NZS 4407.
- Crossed recessed pan: To AS/NZS 4406.
- Crossed recessed raised countersunk (oval): To AS/NZS 4408.
- Hexagon: To AS/NZS 4402.
- Hexagon flange: To AS/NZS 4410.
- Hexagon washer: To AS/NZS 4409.
- Slotted countersunk (flat – common head style): To AS/NZS 4404.
- Slotted pan: To AS/NZS 4403.
- Slotted raised countersunk (oval – common head style): To AS/NZS 4405.

Blind rivets

Description: Expanding end type with snap mandrel.

Type: Closed end for external application, open end for internal application.

End material:

- Aluminium base alloy for metallic-coated or prepainted steel.
- Stainless steel for stainless steel sheet.
- Copper for copper sheet.

Size:

- For sheet metal to sheet metal: 3 mm.
- For sheet metal to supports, brackets and rolled steel angles: 4.8 mm.

3 EXECUTION

3.1 ADHESIVES General

Requirement: Install to the manufacturer's recommendations.

Preparation

Substrates: Conform to the following:

- Remove any deposit or finish which may impair adhesion.
- If framed or discontinuous, provide support members in full lengths without splicing.
- If solid or continuous, remove excessive projections.
- If previously painted, remove cracked or flaking paint and lightly sand the surface.

Contact adhesive

Precautions: Do not use contact adhesive if:

- A substrate is polystyrene foam.
- A PVC substrate may allow plasticiser migration.
- The adhesive solvent can discolour the finished surface.

- Dispersal of the adhesive solvent is impaired.

Two-way method: Immediately after application, press firmly to transfer adhesive and then pull both surfaces apart. Allow to tack off and then reposition and press firmly together. Tap areas in contact with a hammer and padded block.

One-way method: Immediately after application, bring substrates together and maintain maximum surface contact for 24 hours by clamps, nails or screws as appropriate. If highly stressed, employ permanent mechanical fasteners.

High strength adhesive tape Preparation:

- Non-porous surfaces: Clean with surface cleaning solvents such as isopropyl alcohol/water, wash down and allow to dry.
- Porous surfaces: Prime the surface with a contact adhesive compatible with the tape adhesive system.

Application to copper, brass, plasticised vinyl and hydrophilic surfaces such as glass and ceramics in a high humidity environment: Conform to manufacturer's recommendations.

Applied lamination pressure: Make sure the tape experiences 100 kPa.

Application temperature: Generally above 10°C and to the manufacturer's recommendations.

Completion: Do not apply loads to the assembly for 72 hours at 21°C.

3.2 JOINT SEALING General

Requirement: Install to the manufacturer's recommendations.

Joint preparation

Cleaning: Cut flush joint surface protrusions and rectify if required. Mechanically clean joint surfaces free of any deposit or finish which may impair adhesion of the sealant. Immediately before sealant application, remove loose particles from the joint, using oil-free compressed air.

Bond breaking: Install bond breaking backing material.

Taping: Protect the surface on each side of the joint using 50 mm wide masking tape or equivalent means. On completion of sealant application, remove the tape and remove any stains or marks from adjacent surfaces.

Primer: Apply the recommended primer to the surfaces in contact with sealant materials.

Sealant joint proportions

General weatherproofing joints (width:depth):

- 1:1 for joint widths less than 12 mm.
- 2:1 for joint widths greater than 12 mm.

Sealant application

General: Apply the sealant to dry joint surfaces using a pneumatic applicator gun. Make sure the sealant completely fills the joint to the required depth, provides good contact with the full depth of the sides of the joint and traps no air in the joint. Do not apply the sealant outside the recommended working time for the material or the primer.

Weather conditions

Two pack polyurethanes: Do not apply the sealant if ambient conditions are outside the following:

- Temperature: Less than 5°C or greater than 40°C.
- Humidity: To the manufacturer's recommendations.

Joint finish

General: Force the sealant into the joint and finish with a smooth, slightly concave surface using a tool designed for the purpose.

Excess sealant: Remove from adjoining surfaces using cleaning material nominated by the sealant manufacturer.

Protection

General: Protect the joint from inclement weather during the setting or curing period of the material.

Rectification

General: Cut out and remove damaged portion of joint sealant and reinstall so repaired area is indistinguishable from undamaged portion.

3.3 FASTENERS General

Requirement: Install to the manufacturer's recommendations.

Fastening to wood and steel

Timber substrates: To AS 1720.1 Section 4.

Self-drilling screws: To AS 3566.1 for timber and steel substrates.

Masonry anchors

Installation: To the manufacturer's recommendations.

4 SELECTIONS

4.1 ADHESIVES Application schedule

Application	Product	Relevant work sections
wall panels or splashbacks		<i>0551 Joinery</i>
Drywall lining/wall panels		<i>0511 Lining, 0522 Partitions – framed and lined</i>
Joinery doors		<i>0453 Doors and access panels</i>
Timber joinery fitments		<i>0551 Joinery</i>
Trim, mouldings, skirtings and architraves		<i>0511 Lining</i>

4.2 ANCHORS Bonded anchor schedule – As per engineers drawings and schedules Expansion anchor schedule –

Application	Product	Relevant work sections
Metal flashings and rainwater goods		<i>0421 Roofing – combined</i>
Metal flashings and sealing nonporous substrates		<i>0431 Cladding – combined</i>
Window and external doors		<i>0451 Windows and glazed doors, 0453 Doors and access panels</i>
Hydraulic services		<i>0802 Hydraulics design and install</i>

Adhesives, sealants and fasteners combined function schedule

Application	Product	Relevant work sections
Fixing and sealing acoustic ceiling tiles		<i>0531 Suspended ceilings – combined</i>
Control joints, tile adhesives and wet area sealants		<i>0631 Ceramic tiling, 0632 Stone tiling</i>
Wet area sealants and lightweight detail items		<i>0525 Cubicle systems, 0551 Joinery</i>

1 GENERAL

1.1 RESPONSIBILITIES General

Requirements: Provide metal and prefinishes, as documented.

Performance

Requirement: Provide metals in sections of strength and stiffness suited to their required function, finish and method of fabrication.

1.2 PRECEDENCE General

Order of precedence:

- The requirements of other work sections of the specification override conflicting requirements of this work section.
- The requirements of work sections override conflicting requirements of their referenced documents. The requirements of the referenced documents are minimum requirements.

1.3 CROSS REFERENCES General

Requirement: Conform to the following:

- *0171 General requirements.*

1.4 SUBMISSIONS Samples

General: Submit samples of the following:

- Stainless steel: One sample of every documented surface finish.
- Anodising: One sample of every colour and finishing option.

2 PRODUCTS

2.1 METALS

Aluminium and aluminium alloys Drawn pipe: To AS/NZS 1867.

Drawn rod, bar and strip: To AS/NZS 1865.

Extrusions: To AS/NZS 1866.

Plate and sheets: To AS/NZS 1734.

Coated steel

Electrogalvanized (zinc) coating on ferrous hollow and open sections: To AS 4750.

Metallic-coated: Steel coated with zinc or aluminium-zinc alloy as follows:

- Ferrous open sections by an in-line process: To AS/NZS 4791.
- Ferrous hollow sections by a continuous or specialised process: To AS/NZS 4792.
- Metallic-coated steel sheet: To AS 1397. Metal thicknesses specified are base metal thicknesses.

Steel wire: To AS/NZS 4534.

Copper and copper alloys Casting: To AS 1565.

Plate, sheet and strip: To AS 1566.

Rods, bars and sections: To AS/NZS 1567.

Composition and designations: To AS 2738.

Stainless steel

Bars: To ASTM A276/A276M.

Plate, sheet and strip: To ASTM A240/A240M.

Welded pipe (plumbing applications): To AS 1769.

Welded pipe (round, square, rectangular): To ASTM A554.

Steel

Sheet: To AS/NZS 1595.

Structural bars and sections: To AS/NZS 3679.1.

Structural hollow sections: To AS/NZS 1163.

Steel for prefinishes

Cold rolled bar: To AS 1443 - Bright.

Cold rolled sheet: To AS/NZS 1595.

- Designation: CA2S-E.

Electric resistance welded tube: To AS 1450.

3 EXECUTION

3.1 GENERAL Metal separation

Incompatible sheet metals: Prevent direct contact between incompatible metals. Provide separation by one of the following:

- Apply an anti-corrosion low moisture transmission coating such as alkyd zinc phosphate primer or aluminium pigmented bituminous paint to contact surfaces.
- Insert a concealed, non-metallic separation layer such as polyethylene film, adhesive tape, neoprene, nylon or bituminous felt.

Incompatible fixings: Do not use.

Incompatible service pipes: Install lagging or grommets. Do not use absorbent, fibrous or paper products.

Brazing

General: Make sure brazed joints have sufficient lap to provide a mechanically sound joint.

Butt joints: Do not use butt jointing for joints subject to load. If butt joints are used, do not rely on the filler metal fillet only.

Filler metal: To AS/NZS 1167.1.

Finishing

Visible joints: Finish visible joints made by welding, brazing or soldering using methods appropriate to the class of work (including grinding or buffing) before further treatment such as painting, galvanizing or electroplating. Make sure self-finished metals are without surface colour variations after jointing.

Preparation

General: Before applying decorative or protective prefinishes to metal components, complete welding, cutting, drilling and other fabrication, and prepare the surface using a suitable method.

Standard: To AS 1627 series.

Priming steel surfaces: If site painting is documented to otherwise uncoated mild steel or similar surfaces, prime as follows:

- After fabrication and before delivery to the works.
- After installation, repair damaged priming and complete the coverage to unprimed surfaces.

Welding

Aluminium: To AS/NZS 1665.

Stainless steel: To AS/NZS 1554.6.

Steel: To AS/NZS 1554.1.

3.2 STAINLESS STEEL FINISHES General

Requirement: Provide a surface finish to match the approved sample.

Pre-assembly

Mechanically polished and brushed finishes: Apply grit faced belts or fibre brushes that achieve unidirectional finishes with buffing, as required to provide the following:

Bead blasted finish: Provide a uniform non-directional low reflective surface by bead blasting. Do not use sand, iron or carbon steel shot. Blast both sides of austenitic stainless steel to equalise induced stress.

Post-assembly pre-treatment

Heat discolouration: Remove by pickling.

Welds: Grind excess material, brush, and polish to match the pre-assembly finish.

Post-assembly finish

Electropolish finish for external installations: Provide an electro-chemical process to stainless steel type 316.

Brushed electropolish finish: Conform to the following:

- Pre-assembly finish: No. 4 brushed finish.
- Post-assembly finish: Provide an electro-chemical processed finish to achieve a No. 7 to No. 8 brushed finish.

Mirror electropolish finish:

- Pre-assembly finish: Mill finish 2B or mirror polished finish.
- Post-assembly finish: Provide an electro-chemical processed finish to achieve a No. 8 mirror like finish.

Completion

Cleaning: Clean and rinse to an acid free condition and allow to dry. Do not use carbon steel abrasives or materials containing chloride.

Protection: Secure packaging or strippable plastic sheet.

3.3 NON-FERROUS FINISHING Mechanical finishes

Bright finished copper alloy surfaces: For indoor applications, apply a clear lacquer protective coating.

3.4 ELECTROPLATING Electroplated coatings

Chromium on metals: To AS 1192.

- Service condition number: At least 2.

Nickel on metals: To AS 1192.

- Service condition number: At least 2.

Zinc on iron or steel: To AS 1789.

3.5 ANODISING General

Standard: To AS 1231.

Thickness grade: To AS 1231 Table H1.

3.6 METAL SPRAYING Metal spray

Standard: To ISO 2063.2.

Minimum thicknesses:

- Indoor applications: 125 µm.
- Outdoor applications: 175 µm.

Process: Electric arc.

Seal coat: Cover the metal spray finish with two coats of vinyl seal to a total dry film thickness of 80 µm.

3.7 PREPAINTING

Air-drying enamel Application: Spray or brush.

Finish: Full gloss.

General use:

- Primer: Two-pack epoxy primer to AS/NZS 3750.13.
- Top coats: 2 coats to AS 3730.6.

Oil resistant use:

- Primer: Two-pack epoxy primer to AS/NZS 3750.13.
- Top coats: 2 coats to AS/NZS 3750.22.

Equipment paint system

Description: Brush or spray application using paint as follows:

- Full gloss enamel finish coats, oil and petrol resistant: To AS/NZS 3750.22, two coats.
- Prime coat to metal surfaces generally: To AS/NZS 3750.19 or AS/NZS 3750.20.
- Prime coat to zinc-coated steel: To AS 3730.15 or AS/NZS 3750.16.
- Undercoat: To AS/NZS 3750.21.

Prepainted metal products Standard: To AS/NZS 2728.

Product type as noted in AS/NZS 2728: Not lower than the type appropriate to the documented atmospheric corrosivity category.

Stoving enamel

Application: Spray or dip.

Two-pack liquid coating Application: Spray.

Finish: Full gloss.

Primer: Two pack epoxy primer to AS/NZS 3750.13.

Topcoat:

- Internal use: Proprietary polyurethane or epoxy acrylic system.
- External use: Proprietary polyurethane system.

3.8 COMPLETION Damage

Damaged prefinishes: Remove and replace items, including damage caused by unauthorised site cutting or drilling.

Repair

Metallic-coated sheet: If repair is required to metallic-coated sheet or electro galvanizing on inline galvanized steel products, clean the affected area and apply a two-pack organic primer to AS/NZS 3750.9.

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide TERMIMESH termite management systems to new slab connections.

Confirm existing termite protect (reticulation or others) – contractor to confirm and provide to refurbished building..

Performance

Objective: Protection of the whole of the building from damage caused by termite attack.

1.2 COMPANY CONTACTS

TERMIMESH technical contacts Website: www.termimesh.com.au.

1.3 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.

1.4 STANDARDS

General

Termite management systems: To AS 3660.1.

1.5 MANUFACTURER'S DOCUMENTS

Technical manuals

Reference: *TERMIMESH System Training and Reference Manual*.

Technical information: www.termimesh.com.au/termimesh-specifications.

For builders: www.termimesh.com.au/for-builders.

1.6 SUBMISSIONS

Certification

Certificate of installation: Submit certificate to AS 3660.1 Appendix A3.

Operation and maintenance manuals

Maintenance regime: For systems requiring post construction monitoring, provide a maintenance manual with the details of the following:

- Inspection frequency.
- Instructions for inspection of termite activity.
- Contact details of manufacturer authorised suppliers for replacement parts/components.

Products and materials

Product data: For each product/material, submit manufacturer's data including the following:

- Construction details, material description and dimensions of individual components.

Type tests: Submit results, as follows:

- Termite management systems to AS 3660.3.

Records

Completion: Submit record drawings identifying the locations of the installed system.

Subcontractors

General: Submit names and contact details of TERMIMESH SYSTEM accredited applicators.

Warranties

Management system warranty: Submit the TERMIMESH SYSTEM and the installer's warranty of the material, workmanship and application.

1.7 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- Completed earthworks or substrate preparation before system application/installation.
- Set out for installation of the TERMIMESH SYSTEM to control joints and all penetrations to the slab.
- Retaining walls after application of any waterproofing membrane.
- Building perimeter before installation of external brickwork, cladding or applied finishes.
- The completed termite management system.

2 PRODUCTS

2.1 GENERAL

Product substitution

Other products: Conform to PRODUCTS, **GENERAL, Substitutions** in *0171 General requirements*.

Product identification

General: Marked to show the following:

- Manufacturer's identification.
- Product brand name.
- Product type.
- Quantity.
- Product reference code and batch number.
- Date of manufacture.

2.2 TERMIMESH SYSTEM

General

Description: A physical termite barrier comprising TMA725 stainless steel woven mesh.

Protection for concrete slabs: To AS 3660.1 Section 4.

Termite caps and sheeting: To AS 3660.1 Section 5.

Certification: CodeMark certificate CM30012.

TMA725 Stainless steel woven mesh

Description: Stainless steel mesh, woven from 0.18 mm TMA725 stainless steel diameter wire and tested to AS 3660.3.

Termiparge

Description: Specially formulated termite resistant adhesive cement used to bond the mesh to concrete, masonry or other termite resistant substrates and tested to AS 3660.3.

Termiflange

Description: Prefabricated stainless steel mesh with stainless steel clamps.

3 EXECUTION

3.1 TERMIMESH SYSTEM

Concrete slab

Standard: To AS 3660.1 Section 4.

Termite caps and sheeting Standard: To AS 3660.1 Section 5.

Installation

Requirement: Conform to the manufacturer's recommendations and the TERMIMESH SYSTEM recommendations using only TERMIMESH personnel or accredited installers.

Slab penetrations

Slab penetrations: Fit Termiflange to service penetrations through the slab before pouring the concrete slab.

Multiple or clustered penetrations: Fabricate Termiflanges on site to TERMIMESH's recommendations and to AS 3660.1.

3.2 COMPLETION

Termite management system notice

General: Permanently fix a durable notice in a prominent location to BCA B1.4(i)(ii) or BCA 3.1.4.4.

Waste materials

Progressive cleaning: Remove waste building materials, which could attract termites, from the site.

Warranties

Residential construction:

- Warranty form: TERMIMESH Pledge Residential Guarantee.
- Warranty period: 10 years on materials and 10 years on workmanship.
- Renewability: Renewable on a continuous basis without limit.
- Value of warranty: Unlimited for termite damage.

Commercial construction:

- Warranty form: TERMIMESH Pledge Commercial Guarantee.
- Warranty period: 10 years on materials and 10 year on workmanship.
- Renewability: Renewable on a continuous basis without limit.
- Value of warranty: Unlimited for termite damage.

Completion inspection

Report: At the end of the defects liability period, inspect the termite management system and prepare a report on its efficacy and status.

4 SELECTIONS

4.1 TERMIMESH SYSTEM SCHEDULE

Whole of building protection

Requirement: To the TERMIMESH SYSTEM or RECTIULATION schedule for whole building protection. This is a requirement for the three refurbishments and one new building.

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide timber products with finishes and treatments, as documented.

Performance

Requirements:

- Appropriate for durability and fire-resistance.
- Appropriate certification for the finishing applications.

1.2 PRECEDENCE

General

Order of precedence:

- The requirements of other work sections of the specification override conflicting requirements of this work section.
- The requirements of work sections override conflicting requirements of their referenced documents. The requirements of the referenced documents are minimum requirements.

1.3 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 *General requirements.*
- 0184 *Termite management.*
- 0671 *Painting.*

1.4 STANDARDS

General

Sawn and milled products:

- Hardwood: To AS 2796.1.
- Softwood: To AS 4785.1.

Reconstituted wood based panels:

- Particleboard: To AS/NZS 1859.1.
- Dry process fibreboard: To AS/NZS 1859.2.
- Decorative overlaid wood panels: To AS/NZS 1859.3.
- Wet process fibreboard: To AS/NZS 1859.4.

Plywood:

- Structural: To AS/NZS 2269.0.
- Interior: To AS/NZS 2270.
- Exterior: To AS/NZS 2271.
- Marine: To AS/NZS 2272.

Glued laminated timber: To AS/NZS 1328.1.

Laminated veneer lumber: To AS/NZS 4357.0.

1.5 INTERPRETATION

Abbreviations

General: For the purposes of this work section the following abbreviations apply:

- EWPA: Engineered Wood Products Association of Australasia.
- LVL: Laminated Veneer Lumber.
- MDF: Medium Density Fibreboard.

Definitions

General: For the purposes of this work section the definitions given in AS/NZS 4491 and the following apply:

- Dry process fibreboard (MDF): Panel material with a nominal thickness of 1.5 mm or greater, manufactured from lignocellulosic fibres (derived from wood or other materials) with application of heat and/or pressure, the bond of which is derived from a synthetic adhesive added to the fibres and the panels are manufactured with a forming moisture content of less than 20%.
- Particleboard: Panel material manufactured under pressure and heat from particles of wood (wood flakes, chips, shavings, sawdust and similar) and/or lignocellulosic material in particle form (flax shives, hemp hurds, bagasse fragments, rice hulls, wheat straw and similar) with the addition of an adhesive.
- Wet process fibreboard: Panel material with a nominated thickness of 1.5 mm or greater, manufactured from lignocellulosic fibres (derived from wood or other materials) with application of heat and/or pressure, the bond of which is derived from the felting of the fibres and the panels are manufactured with a forming moisture content greater than 20%.

1.6 SUBMISSIONS

Products and materials

Pressure preservative treatment: For timber required to be pressure treated, submit a certificate or other evidence showing that the timber has been treated.

Treated timber: Submit safety data sheets (SDS) for preservative treated timber.

2 PRODUCTS

2.1 GENERAL

Storage and handling

General: Deliver timber products to site in unbroken wrapping or containers and store so that the moisture content is not adversely affected.

Product identification

Preservative treated timber: Marking to include the following:

- A unique identifier for the treatment plant.
- A unique identifier for the preservative.
- Hazard class.

2.2 CERTIFICATION

Timber source certification

Requirement: Use timber products originating from sustainably managed forests. Application: Items requiring source certification:

- Timber framed doors and windows
- Timber wall panelling
- Timber door jambs and sills
- Timber veneers

Branding: Brand timber products under the authority of a certification scheme applicable to the product. Locate the brand on faces or edges that will be concealed in the Works.

Engineered timber product certification and identification

Branding: Brand timber products under the authority of a certification scheme applicable to the product. Locate the brand on faces or edges which will be concealed in the works.

Inspection: If neither branding nor certification is adopted, have an independent inspecting authority inspect the timber.

2.3 FIRE-RESISTANCE

General

Timber structures: To AS/NZS 1720.4.

Bushfire prone areas

Standard: To AS 3959 Appendix F.

2.4 DURABILITY

General

Requirement: Provide timbers with natural durability appropriate to the conditions of use, or preservative-treated timber of equivalent durability.

Natural durability class: To AS 5604.

Naturally termite-resistant timbers: To AS 3660.1 Appendix C.

Timber quality: Free of core wood (material within 50 mm of the tree's centre) and free of splits, checks, loose knots and cavities. Free of sapwood (lighter coloured wood found on the outer layer of the tree).

Lyctid susceptible timbers: Do not provide untreated timbers containing lyctid susceptible sapwood.

Untreated sapwood: If used, place to the outside of joints or in locations exposed to higher levels of ventilation.

Preservative treatment

Sawn and round timbers: To AS 1604.1.

Reconstituted wood-based products: To AS/NZS 1604.2.

Plywood: To AS/NZS 1604.3.

Laminated veneer lumber (LVL): To AS/NZS 1604.4.

Glued laminated timber products: To AS/NZS 1604.5.

Moisture content

Test: Methods as follows:

- Timber: To AS/NZS 1080.1.
- Plywood: To AS/NZS 2098.1.
- Reconstructed wood-based products: AS/NZS 4266.1.

Protection: Protect timber and timber products stored on site from moisture and weather. For milled, prefinished, prefabricated and similar elements that are to be protected in the final structure, provide temporary weather protection until the permanent covering is in place.

Termite management

Requirement: To *0184 Termite management*.

2.4 FINISHING

Production finish

Hardwood: To AS 2796.1 Table B1.

Softwood: To AS 4785.1 Table B1.

Surface coating

Painting and staining: To *0671 Painting*.

Application: To the manufacturer's specification.

Surface finish: Dressed for all visible surfaces.

Edge detail: Arrised edges unless noted otherwise. Abutting faces to have square edges.

Surface coating

Painting and staining: To *0671 Painting*.

3 EXECUTION

3.1 JOINTS

General

Joints and connections: Use hot-dipped galvanized or stainless steel fasteners, composite bolts, nails or nailed metal connectors.

Timber-to-timber interfaces: Provide a seal coating of preservative treatment and include inside bolt holes and the end grain of the timber.

Water retention: Avoid details that may trap water including housed, checked or birdsmouth joints.

Fasteners: To prevent chemical treatments reacting with fasteners, install to manufacturer's recommendations.

3.2 SHRINKAGE RESTRAINT

General

Requirement: Use seasoned timber, if possible, to avoid shrinkage restraint, particularly where timber elements are integrated with steel and/or concrete.

Moisture content: Maintain a timber moisture content near the anticipated in-service equilibrium moisture content. Use finishes and end-grain sealants to minimise moisture content changes.

Fasteners: Align fasteners along member axis and use single fasteners at joints.

Connections: Use connections that allow for movement without adversely affecting the performance of the connection.

Unseasoned timber: Provide as follows:

- Drill holes 10% oversize.
- Use species with similar shrinkage values to reduce movement and shrinkage.
- For framing provide adequate clearance at the top of masonry veneer and face fixed members to reduce vertical movement.

3.3 FINISHING

Ploughing

General: Back plough boards liable to warp (e.g. if exposed externally on one face). Make the width, depth and distribution of ploughs appropriate to the dimensions of the board and degree of exposure.

Painting

Edges: Chamfer edges of work to receive paint or similar coatings.

Priming: For woodwork to be painted, prime hidden surfaces before assembly.

Working with treated timber

Safety: Handle preservative treated timber to the manufacturer's recommendations and to NOHSC 2003 and the recommendations of NOHSC 3007.

3.4 COMPLETION

Warranties

Type: Manufacturer and applicator / installer interlocking warranty: Interlocking warranty to cover manufacturing Defects and Defects with products and materials delivered to Site, including their application and / or installation.

Period: As offered by the manufacturer and the installer.

0191 SUNDRY ITEMS

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide sundry items, as documented.

Performance

Requirements: Installation as follows:

- Undamaged and free of surface defects or distortions.
- Correctly located and aligned, plumb, level and straight.
- Fixed firmly in position.
- Connected to the nominated service(s).

1.2 CROSS REFERENCES General

Requirement: Conform to the following: - *0171 General requirements*.

1.3 SUBMISSIONS Certification

Sealant compatibility: Submit statements from all parties to the installation certifying the compatibility of sealants with items and substrates.

Operation and maintenance manuals

Requirement: Submit a maintenance manual and, if required, an operation manual with the technical specification and manufacturer's recommendations for the item to be installed. **Warranties**

Requirement: Submit a manufacturer's and/or installer's warranty.

1.4 INSPECTION Notice

Inspection: Give notice so inspection can be made of the following:

- Set-out of item locations before fixing.
- Completion of installation.

2 PRODUCTS

2.1 GENERAL Storage and handling

General: Deliver, unload and store components and accessories in unbroken manufacturer's packaging.

3 EXECUTION

3.1 PREPARATION Substrates

General: Prepare the substrate to receive the item.

Protection

General: Protect existing work from damage during the installation and rectify any damage. Provide temporary coverings if required.

3.2 INSTALLATION Accessories and trim

Requirement: Provide accessories and trim necessary to complete the installation.

3.3 COMPLETION Cleaning

Requirement: Remove packaging. Clean the completed assembly and surrounds. Wipe down appliances and fittings with a damp, soft, clean cloth.

Warranties

Requirement: Cover materials and workmanship in the terms of the warranty in the form of interlocking warranties from the supplier and the installer.

Form: Against failure of materials and execution under normal environment and use conditions.

4 SELECTIONS

4.1 SIGNAGE & GLASS TRANSFER WINDOW FILM

Allow the following provisional sums for supply and installation of signage and glass transfer window film. (excludes Braille & tactile signage)

4.2 PROVISIONAL SUM

TWENTY FIVE THOUSAND DOLLARS (\$25,000.00)

Contractor to allow for coordination with the selected AV contractor and incorporate into their works programme.

4.3 BRAILLE & TACTILE SIGNAGE

Male toilet and locker room Location: A1 – Registry A2 – Workshop A3 – Warehouse A4 – New Build	braillesignsupplies.com.au Order Code: Colour: Aluminum Grey Size: 250mm x 180mm x 1.6mm Mounting Height: Single line of tactile characters not less than 1250 mm and not higher than 1350 mm above the floor or ground surface. Mounting Position: TBC	 A rectangular sign with a dark background. It features white icons of a man, a woman, and a shower head. Below the icons, the text 'Male Toilet & Locker Room' is written in white. At the bottom, there is a line of Braille characters.
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Female toilet and locker room Location: A1 – Registry A2 – Workshop A3 – Warehouse A4 – New Build	braillesignsupplies.com.au Order Code: Colour: Aluminum Grey Size: 250mm x 180mm x 1.6mm Mounting Height: Single line of tactile characters not less than 1250 mm and not higher than 1350 mm above the floor or ground surface. Mounting Position: TBC	
Accessible Toilet LH Location: A3 – Warehouse A4 – New Build	braillesignsupplies.com.au Order Code: Aluminum Grey Colour: Blue/White Size: 250mm x 180mm x 1.6mm Mounting Height: Single line of tactile characters not less than 1250 mm and not higher than 1350 mm above the floor or ground surface. Mounting Position: On the wall on the latch side of the door with the leading edge of the sign located between 50 mm and 300 mm from the architrave.	
Unisex Toilet & Ambulant Toilet Location: A3 – Warehouse A4 – New Build	braillesignsupplies.com.au Order Code: Aluminum Grey Colour: Blue/White Size: 250mm x 180mm x 1.6mm Mounting Height: Single line of tactile characters not less than 1250 mm and not higher than 1350 mm above the floor or ground surface. Mounting Position: On the wall on the latch side of the door with the leading edge of the sign located between 50 mm and 300 mm from the architrave.	
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Confirm sizes of all Whitegoods prior to Joinery Manufacture at shop drawing preparation stage.

4.4 TELEVISIONS/ WALL MOUNTED (TV/W)

Client to provide TV units and mounting brackets. Contractor to install as directed. Refer to electrical drawings for location. Provide additional nogging and sheet plywood in stud framing as required to TV Locations.

4.5 MEETING ROOMS

Supply as per electrical drawings to provide. Contractor to install as directed. Provide additional nogging and sheet plywood in stud framing as required to projectors / TV

4.6 MIRRORS (MI)

Provide & install powder coated frameless pill shaped mirrors as indicated on drawings throughout. Sizes as indicated on drawings. Generally above all Hand Basins & as other locations shown on plans.

4.7 ACCESSIBLE AMENITIES

Mirror (MI) Provide and install framless oval shaped mirror as indicated on drawings throughout. Sizes as indicated on drawings

Hook (HK) - Provide 1 No. hat & coat hooks Legge 12841 per shower cubicle.

Grabrails to WC – Provide stainless steel grabrails to comply with AS1428.1 – 2009. Provide flange covers to all fixings. Provide padded back rest to ambulant and PWD toilets

Soap dispenser (SD) & Toilet roll Holder

4.8 AMBULANT AMENITIES (MALE & FEMALE)

Mirror (MI) Provide and install framless oval shaped mirror as indicated on drawings throughout. Sizes as indicated on drawings

Hook (HK) - Provide 1 No. hat & coat hooks Legge 12841 to each.

Grabrails to WC's – Provide stainless steel grabrails to comply with AS1428.1 – 2009. Provide flange covers to all fixings.

Soap dispensers (SD) & Toilet roll Holders

4.8 DRIP STRIPS (RDS)

Provide and install: over new windows and doors externally exposed to weather and as shown on plans provide Raven drip stripsRP67. Provide minimum 50mm overlap on each side of the doors and windows

0201 DEMOLITION

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Carry out demolition, as documented.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.

1.3 STANDARDS

General

Demolition: To AS 2601.

1.4 INTERPRETATION

Definitions

General: For the purposes of this work section, the following definitions apply:

- Demolition: The complete or partial dismantling of a building or structure, by pre-planned and controlled methods or procedures.
- Dilapidation record: The photographic or video and written record of the condition of the portion of the existing building retained, adjacent buildings, and other relevant structures or facilities, before the start of demolition work.
- Dismantle: The reduction of an item to its components in a manner to allow re-assembly.
- Recover: The disconnection and removal of an item in a manner to allow re-installation.

1.5 SUBMISSIONS

Authority approvals

Evidence of compliance: Before starting demolition, submit evidence of the following:

- Requirements of authorities relating to the work under the contract have been obtained.
- A permit to demolish has been obtained from the appropriate authority.
- A scaffold permit has been obtained from the appropriate authority (if scaffolding is proposed to be used).
- Certification that each person having access to the construction site has completed site-specific WHS induction training.
- Precautions necessary for protection of persons and property have been taken and suitable protective and safety devices have been provided to the approval of the relevant authority.
- Treatment for rodent infestation has been carried out and a certificate has been obtained from the appropriate authority.
- Fees and other costs have been paid.

Execution details

Requirement: Submit the following, as documented:

- Investigation and work plan.

Off-site disposal locations: Submit details of the proposed locations for the disposal of material required to be removed from the site, and evidence of conformance with the requirements of relevant authorities.

Records

Dilapidation record:

- Before demolition: Submit to each owner of each adjacent property, a copy of the part of the record relating to that property and obtain their written agreement to the contents.
- Rectification work: Submit written acceptance of rectification works from the owner of each adjoining property affected.

1.6 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- Adjacent structures before starting and at completion of demolition.
- Services before disconnection or diversion.
- Trees documented to be retained, before starting demolition.
- Contents of building before starting demolition.
- Structure after stripping and removal of roof coverings and external cladding.
- Underground structures after demolition above them.
- Excavations remaining after removal of underground work.
- Site after removal of demolished materials.
- Services after reconnection or diversion.
- Relocate existing containers (new build site) as directed on site,

2 PRODUCTS

2.1 DEMOLISHED MATERIALS

Demolished material classes table

Class	Requirement	Ownership
Demolish for removal	Remove from the site demolished materials identified in the Demolish for removal schedule . Do not burn or bury on site Transit: Prevent spillage of demolished materials in transit	Contractor

3 EXECUTION

3.1 INVESTIGATION AND WORK PLAN

General

Requirement: Before demolition or stripping work, prepare the work plan to AS 2601 Section 2.

Include the check list items appropriate to the project from AS 2601 Appendix A, and the following:

- Method of protection and support for adjacent property.
- Locations and details of service deviations and terminations.
- Sequence of work.
- If the demolition program results in components temporarily cantilevered, provide a certificate from a professional engineer.
- Proposals for the safe use of mobile plant on suspended structural members including provisions for the protection of lower floors in the event of structural failure.
- Wheel loads of tipping or loading vehicles.

3.2 SUPPORT

Temporary support

General: If temporary support is required, certification for its design and installation is required from a professional engineer engaged by the contractor.

Existing buildings: Until permanent support is provided, provide temporary support for sections of existing buildings which are to be altered and which normally rely for support on work to be demolished.

Ground support: Support excavations for demolition of underground structures.

Adjacent structures: Provide supports to adjacent structures where necessary, sufficient to prevent damage resulting from the works.

- Lateral supports: Provide lateral support equal to that given by the structure to be demolished.
- Vertical supports: Provide vertical support equal to that given by the structure to be demolished.

3.3 PROTECTION

Encroachment

General: Prevent the encroachment of demolished materials onto adjoining property, including public places.

Weather protection

General: If walls or roofs are opened for alterations and additions or the surfaces of adjoining buildings are exposed, provide temporary covers to prevent water penetration. Provide covers to protect existing plant, equipment and materials intended for re-use.

Dust protection

General: Provide dustproof screens, bulkheads and covers to protect existing finishes and the immediate environment from dust and debris.

Security

General: If walls or roofs are opened for alterations or additions, provide security against unauthorised entry to the building.

Temporary screens

General: Fill the whole of designated temporary openings or other spaces using dustproof and weatherproof temporary screens, fixed securely to the existing structure, and installed to shed water to avoid damage to retained existing elements or adjacent structures and contents.

Type: Timber framed screens sheeted with 12 mm plywood and painted. Seal the junctions between the screens and the openings.

Exposed surfaces

General: Where necessary, protect and weatherproof the surfaces of adjacent structures exposed by demolition.

Existing services

Location: Before starting demolition, locate and mark existing underground services in the areas which will be affected by the demolition operations.

Utility services: Contact DIAL BEFORE YOU DIG to identify location of underground utility services pipes and cables.

Excavation: Do not excavate by machine within 1 m of existing underground services.

Fixed items

Individual protection: Protect the following items in their existing positions:

- Existing Fibre line
- Existing sewer lines. Note where existing line under new building is rerouted. Refer Hydraulics plans.
- Existing Electrical lines
- Existing hydrant lines

3.4 DEMOLITION – BUILDING WORKS

Encroachment

General: If encroachments from adjacent structures are encountered and are not documented, give notice and obtain instructions.

Concrete slabs

General: Using a diamond saw, neatly cut back or trim to new alignment with a clean true face existing concrete slabs to be partially demolished or penetrated. Do not overcut at corners.

Material below grade

Extent: Demolish the following:

- partial demolitions/cut existing concrete driveway for new footings as noted on structural plans.

Explosives

General: Do not use explosives.

3.5 DEMOLITION – BUILDING SERVICES

General

Requirement: Decommission, isolate, demolish and remove from the site all equipment and associated components that become redundant as a result of the demolition.

Breaking down: Disassemble or cut up equipment where necessary to allow removal.

3.6 COMPLETION

Notice of completion

General: Give at least 5 working days' notice of completion of demolition so that adjacent structures may be inspected following completion of demolition.

Reinstatement

Assessment of damage: Use the dilapidation record to assess the damage and rectification work arising from the demolition work.

Rectification: Repair damage arising out of demolition work. Obtain written acceptance from the owner of each adjoining property of the completeness and standard of the rectification work.

Temporary support

General: Remove at completion of demolition.

4 SELECTIONS

Demolish for removal schedule

Item
Cut portions of concrete generally 150mm thick slabs for new footings, services, etc.
Existing services where noted for demolitions. Cap services.
Demolish internal cabinetry as noted on plans
Demolish walls / doors / windows as noted on plans
Demolish concrete where noted on site plans.
Demolish existing amenities fully strip tiles / floor covering
Demolish compactus units
Demolish ceiling tiles

0221 SITE PREPARATION

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide site preparation, as documented.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 General requirements.

1.3 INTERPRETATION Definitions

General: For the purposes of this work section the following definitions apply:

- Authority: Any organisation with statutory authority relating to the project, including clearances.
- Clearances: A formal certificate, approval or condition issued by a statutory authority allowing work in a particular area.
- Plant establishment period: The period between the date of practical completion and the end of the defects liability period.
- Utility service provider: Includes organisations providing power, water, sewerage, gas and telecommunications services.

1.4 SUBMISSIONS Certification

Vermin: Submit pest exterminator's certification as evidence that the completed site works are free from vermin.

Execution details

Requirement: Submit details of methods and equipment proposed for the following:

- Clearing and grubbing.
- Tree removal and transplanting.
- Protecting ground within and adjacent to tree driplines from compaction by proposed earthworks machinery.

1.5 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Enclosures around trees requiring protection.
- Trees requiring removal.
- Trees for transplanting to determine final orientation.

2 EXECUTION

2.1 COMMUNITY LIAISON Notification

General: Notify residents about construction activities which will affect access to, or disrupt the use of, their properties.

Notice: Minimum 5 working days, unless the work is of an urgent nature with safety implications.

Notification content:

- Description of the work.
- The reason for the work.
- The expected duration.
- Changes to traffic arrangements and property access.
- The 24-hour contact number of the representative responsible.

2.2 EXISTING SERVICES General

Requirement: Before starting earthworks, locate and mark existing underground services in the areas affected by the earthworks operations including clearing, excavating and trenching.

Utility services: Contact DIAL BEFORE YOU DIG to identify location of underground utility services pipes and cables.

Excavation: Do not machine excavate within 1000 mm of existing services.

Existing service lines: If required, divert services detected during excavation, clear of the building, and reconnect to the utility service provider's requirements.

2.3 SITE CLEARING Extent

Requirement: Clear only areas occupied by works such as structures, paving, excavation, regrading and landscaping or other areas documented for clearing.

Contractor's site areas: If not included within the areas documented above, clear only to the extent necessary for the performance of the works.

Clearing and grubbing

Clearing: Remove everything on or above the site surface, including rubbish, scrap, grass, vegetable matter and organic debris, scrub, trees, timber, stumps, boulders and rubble.

Grubbing: Grub out stumps and roots over 75 mm diameter to a minimum depth as follows:

- Below subgrade under buildings, embankments or paving: 500 mm.
- Below finished surface in unpaved areas: 300 mm.

Backfilling: Fill holes remaining after grubbing with sand material to prevent ponding of water. Compact the material to the relative density of the existing adjacent ground material.

Redundant/decommissioned works: Remove works no longer required, including slabs, foundations, paving, drain, and access chambers and covers within the works zone.

Surplus material

Topsoil and excavated material: Remove unwanted stripped soil and other material from the site as the work proceeds, including any material dropped on footpaths or roadways.

2.4 STORMWATER AND SEDIMENT CONTROL General

Waterways and drains

Waterways: If required, temporarily divert ditches, field drains and other waterways affected by excavation and reinstate on completion.

Stormwater drains: Divert drains detected during excavation, clear of the building, and reconnect as documented or obtain approval.

2.5 EXISTING WORKS TO REMAIN Marking

Requirement: Identify existing works to remain with 1000 mm high, 50 x 50 mm timber stakes connected by yellow plastic tape to prevent accidental damage.

2.6 COMPLETION Temporary works

Remove at completion: contractors site sheds, temporary fencing, etc

Site restoration

Requirement: Reinstate undeveloped ground surfaces to the condition existing at the commencement of the contract.

Any damage to the existing grounds needs to be rectified and returfed where contractor access may be required.

Clean up

Progressive cleaning: Keep the works clean and tidy, and regularly remove from the site, waste and surplus material arising from execution of the work.

Vermin management

Requirement: Employ a suitably qualified pest exterminator to remove vermin found during site preparation.

1 GENERAL**1.1 RESPONSIBILITIES General**

Requirement: Provide earthworks to the dimensions and tolerances, as documented. Refer to civil documents

1.2 DESIGN General

Designer: Patcol Group

Geotechnical report: refer attached appendix

Requirements

Design of footing or pier depths: as per Patcol Group structural engineers design documents.

Authority requirements: to be confirmed with building approvals.

1.3 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements*.

1.4 STANDARDS General

Earthworks: Conform to the recommendations of those parts of AS 3798 that are referenced in this work section.

Description and classification of soils: To AS 1726.

1.5 INTERPRETATION Abbreviations

General: For the purposes of this work section the following abbreviations apply:

- GITA: Geotechnical inspection and testing authority.
- GTA: Geotechnical testing authority.

Definitions

General: For the purposes of this work section the definitions given in AS 3798 and the following apply:

- Bad ground: Ground unsuitable for the works, including fill liable to subsidence, ground containing cavities, faults or fissures, ground contaminated by harmful substances and ground that is, or becomes, soft, wet or unstable.
- Rock: Monolithic material with volume greater than 0.3 m³ that cannot be removed until broken up by rippers or percussion tools.
- Site topsoil: Natural soil, excavated from the site, that contains organic matter, supports plant life, conforms generally to the fine-to-medium texture classification to AS 4419 and is free from the following:
 - o Stones more than 25 mm diameter.
 - o Clay lumps more than 50 mm diameter.
 - o Weeds and tree roots.
 - o Sticks and rubbish.
 - o Material toxic to plants.
- Subgrade: The trimmed or prepared earth material on which the pavement, footing or slab is constructed. Generally taken to relate to the upper line of the earth material.
- Zone of influence: A foundation zone bounded by planes extending downward and outward from the bottom edge of a footing, slab or pavement and defining the extent of foundation material having influence on the stability or support of the footings, slab or pavement.

1.6 TOLERANCES General

Finish: Finish the surface to the required level, grade and shape within the following tolerances:

- Under building slabs and load bearing elements: + 0, - 25 mm.
- Pavement subgrades: + 0, - 40 mm.
- Batters: No steeper than the slope shown on the drawings. Make sure flatter slopes do not impact on boundaries or required clearances to buildings, pavements or landscaping.
- Other ground surfaces: ± 50 mm, provided the area remains free draining and matches adjacent construction where required. Provide smoothness as normally produced by a scraper blade.

1.7 SUBMISSIONS Execution details

Report: Submit a time-based schedule detailing the methods and equipment proposed for the earthworks, including the following:

- Dewatering and groundwater control and disposal of surface water.
- Excavation methods, stages, clearances, batters and temporary supports.
- Stockpiles and borrow pits.
- Placing and compaction methods and stages.

Disposal location: Submit details of the locations and evidence of compliance with the appropriate authority requirements for the disposal of material requiring removal from site.

Records of measurement: Submit a certified copy of the agreed records of measurement.

Site records: Submit the following to AS 3798 clause 3.4 and Appendix B:

- Geotechnical site visit record.
- Earthworks summary report or daily geotechnical reports.

Products and materials

Imported fill: Submit certification or test results by a GTA registered laboratory of the imported fill as evidence of conformity with the contract, including the source.

Tests

Compaction: Submit certification and/or test results in conformance with the documented level of inspection and testing to AS 3798.

1.8 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Items to be measured as listed in **RECORDS OF MEASUREMENT**.
- Areas to be cleared and/or stripped of topsoil.
- Areas stripped of topsoil.
- Excavation completed to contract levels or founding material.
- Proof rolled subgrade before placing fill.
- Filling completed to contract levels.
- Stockpiled topsoil before spreading.

2 PRODUCTS

2.1 FILL MATERIALS General

Suitable material: To AS 3798 clause 4.4 including inorganic, non-perishable material suitably graded and capable of compaction to the documented density.

Unsuitable materials: To AS 3798 clause 4.3.

Sulfur content: Do not provide material with sulfur content exceeding 0.5% within 500 mm of cement bound elements (for example concrete structures or masonry) unless the elements are protected by impermeable membranes or equivalent means.

Re-use of excavated material: Only re-use suitable material to AS 3798 clause 4.4.

Stockpiles

General: Segregate the earth and rock material and stockpile for re-use in backfilling operations.

Location: Do not stockpile excavated material against tree trunks, buildings, fences or obstruct the free flow of water along drainage channels.

2.2 BORROW OR IMPORTED FILL General

Borrow or imported material: Use only when suitable excavated material from site is not available.

- Suitable material: To AS 3798 clause 4.4.

Material conforming to the following: as per PATCOL GROUP structural Engineers documents

2.3 GEOTEXTILE General

Material: UV stabilised, permeable, polymeric, woven or non-woven textile material used in contact with soil/rock material.

Identification and marking: To AS 3705.

3 EXECUTION

3.1 RECORDS OF MEASUREMENT Excavation and backfilling

Agreed quantities: If a schedule of rates applies, provisional quantities are documented, or there are variations to the contract levels or dimensions of excavations, do not backfill or place permanent works in the excavation until the following have been agreed and recorded:

- Depths of excavations related to the datum.
- Final plan dimensions of excavations.
- Quantities of excavations in rock.

Method of measurement: By registered surveyor.

Rock

Level and class: If rock is measured for payment purposes, either as extra over excavation of material other than rock or for adjustment of provisional measurements, do not remove the rock until the commencing levels and the classes of rock have been determined.

- stockpiles from contamination by other excavated material, weeds and building debris.

3.2 EXCAVATION Extent

Site surface: Excavate the site to the levels and profiles required for the documented structures, pavements, filling and landscaping. Make allowance for compaction, settlement or heaving.

Footings, pits, wells and shafts: Excavate to the required sizes and depths. Confirm that the foundation conditions meet the design bearing capacity.

Bearing surfaces

Requirement: Provide even plane bearing surfaces for loadbearing elements including footings. Step to accommodate level changes. If supporting masonry, make the steps appropriate to the courses.

Rock

General: Do not use explosives.

Existing footings

Requirement: If excavation is required within the zone of influence of an existing footing, provide supports to the footing sufficient to prevent damage arising from the works. Use methods including temporary shoring or underpinning.

Existing services

Location: Before starting earthworks, locate and mark existing underground services in the areas that will be affected by the earthworks operations including clearing, excavating and trenching.

Utility services: Contact DIAL BEFORE YOU DIG to identify location of underground utility services pipes and cables.

Excavation: Do not excavate by machine within 1000 mm of existing services.

Disposal of excess excavated material

General: Remove excess excavated material from site not required or unsuitable for fill.

Standard: To AS 3798 clause 6.1.8.

3.3 REINSTATEMENT Deterioration of bearing surfaces

Requirement: If the bearing surface deteriorates because of water or other cause, excavate to a sound surface before placing the loadbearing element.

Subgrades affected by moisture

Requirement: If, due to high moisture content the subgrade cannot support construction equipment or the overlying pavement cannot be compacted, perform one or more of the following: - Allow the subgrade to dry until it provides support for equipment and allows compaction.

- Scarify the subgrade to a depth of 150 mm, work as necessary to accelerate drying, and recompact when the moisture content is satisfactory.
- Excavate the wet material and remove to spoil, and backfill excavated areas.

Over excavation

Requirement: If excavation exceeds the required depths, reinstate to the correct depths, levels and bearing capacity.

Zone of influence: Within the zone of influence of footings, beams, or other structural elements, use concrete of strength equal to the structural element, minimum 15 MPa. Make sure that remedial concrete does not create differential bearing conditions.

Below slabs or pavements: Rectify the over excavation as follows:

- Generally: Provide selected fill compacted to the documented density.

- Less than 100 mm: Do not backfill. Increase the thickness of the layer above.

Rock depressions and subsoil drains: Backfill rock depressions and over excavation of subsoil drains using coarse subsoil filter.

3.4 FILL MOISTURE CONTROL General

Moisture content: Adjust the moisture content of fill during compaction within the range of 85% to 115% of the optimum moisture content determined by AS 1289.5.1.1 or AS 1289.5.2.1, as appropriate, to achieve the required density.

Note : Refer Structural Engineers notes and drawings for extent of earthworks scope. Generally limited earthworks over existing concrete carpark area.

0223 SERVICE TRENCHING

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide trenching for underground services, as documented. Refer also electrical, hydraulic and mechanical engineering documents.

1.2 DESIGN Requirements

Trench design: Design and coordinate all trenching required for proposed inground services, as documented.

Existing services: Where possible, design to avoid all existing services. Locate existing services to

EXECUTION, EXISTING SERVICES.

1.3 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements.*

1.4 STANDARDS Trenching

Earthworks: To AS 3798.

Electrical services: To AS/NZS 3000.

Hydraulic services: To AS/NZS 3500.1.

Communication services: AS/CA S009.

1.5 TOLERANCES Surface levels

Earthworks: Finish the surface to the required level, grade and shape within the following tolerances:

- Under building slabs and load bearing elements: + 0, - 25 mm.
- Pavement subgrades: + 0, - 40 mm.
- Batters: No steeper than the slope shown on the drawings. Make sure flatter slopes do not impact on boundaries or required clearances to buildings, pavements or landscaping.
- Other ground surfaces: ± 50 mm, provided the area remains free draining and matches adjacent construction where required. Provide smoothness as normally produced by a scraper blade.

Pavement base and subbase: Finish the surface to the required level, grade and shape within the following tolerances:

- Subbase: + 10 mm, - 25 mm.
- Base: + 10 mm, - 5 mm.

Finished pavement or paving surface: Conform to the documented level within the following tolerances:

- Asphalt: ± 10 mm.
- Concrete: + 10 mm, - 0 mm.
- Paving:
 - o Finished level: ± 8 mm.
 - o Height deviation between adjacent units (lippage): ± 2 mm.
- Granular surfaces: ± 10 mm.
- Lippage between restored surface and adjacent existing surface: ± 5 mm.

1.6 SUBMISSIONS Execution details

Excavation method: Submit details of proposed equipment and method of excavation, including the following:

- Service location and type: A plan of the trench works showing the location and type of service.
- Open excavation: Proposed duration.
- Shuttering and/or bracing of trench sides: If required for safety and stability, provide proposals.
- Geotechnical data: Geotechnical report supporting the procedures proposed for trenching and/or boring.
- Boring: Proposals for the following:
 - . Limits on length.
 - . Existence of other services and method of protection.
 - . Pressure grouting to voids.
 - . The effect of pressure grouting on other services, ground heave and proposals for minimising such effects.
 - . Access to properties outside the site.
 - . Council permits.
 - . Service interruptions including a plan for minimising unintended interruptions.
- Hazards: Identify WHS hazards that may be encountered with deep trenches including toxic gases and liquids.

Off-site disposal location: Submit details of the proposed disposal locations and evidence of conformance with the relevant authorities for the disposal of material required to be removed from the site.

Records

As-built location: Upon completion submit to the relevant authority, as-built documentation to show the location of the installed services.

1.7 INSPECTION Notice

Inspection: Give notice so that inspection may be made at the following stages:

- Items to be measured as listed in **GROUND CONDITIONS, Records of measurement**.
- Service trenches excavated before laying the service.
- Services laid in trenches and ready for backfilling.
- Completed surface restoration.

2 PRODUCTS

2.1 FILL MATERIALS General

Suitable material: To AS 3798 clause 4.4 including inorganic, non-perishable material suitably graded and capable of compaction to the documented density.

Unsuitable materials: Do not use unsuitable material for fill in conformance with AS 3798 clause 4.3.

Sulfur content: Do not provide filling with sulfur content exceeding 0.5% within 500 mm of cement bound elements (for example concrete structures or masonry) unless such elements are protected by impermeable membranes or equivalent means.

Re-use of excavated material: Only re-use suitable material in conformance with AS 3798 clause 4.4.

Material in reactive clay areas: In sites classified M, M-D, H1, H1-D, H2, H2-D, E or E-D to AS 2870, re-use excavated site material at a moisture content within $\pm 1\%$ of that of the adjoining in situ clay.

2.2 SURFACE RESTORATION MATERIALS General

Re-use: If possible re-use the existing surface materials that were removed during trench excavation, whilst conforming to the documented material requirements.

Subbase and base

Requirement: Provide crushed rock material configured in layers and depths to match existing and adjacent work, as follows:

- Base: 20 mm nominal size.
- Subbase: 40 mm nominal size.
-

Pathways and paved surfaces generally

Requirement: Provide materials consistent with those of the existing surface before service trenching works commenced.

Concrete surfaces

Material requirements: Normal-class to AS 1379.

Concrete strength grade: N25.

Slump: Maximum 100 mm.

Pavers

Concrete and clay pavers: To AS/NZS 4455.2.

Bedding and joint filling sand: Well-graded sand, free of deleterious material such as soluble salts which may cause efflorescence.

Stone pavers and setts: Provide sound stone pavers and setts of uniform quality. Reject any with defects liable to affect strength and durability.

Bedding mortar mix (cement:sand): Select from the range 1:3 to 1:4 to obtain satisfactory adhesion. Provide minimum water.

3 EXECUTION

3.1 EXISTING SERVICES Location

Requirement: Before commencing service trenching, locate and mark existing underground services in the areas which will be affected by the service trenching operations.

Utility services: Contact DIAL BEFORE YOU DIG to identify location of underground utility services pipes and cables.

Excavation

General: Do not excavate by machine within 1 m of existing underground services.

3.2 EXISTING SURFACES Concrete

Method: Sawcut trench set out lines for the full depths of the bound pavement layers except where the set out line is located along expansion joints.

Removal of concrete and asphalt: Break out concrete or asphalt pavement material between the trench set out lines, remove and dispose of off-site.

Paving

Removal: Take up paving units both full and cut by hand, between the trench set out lines, and neatly stack on wooden pallets.

Concrete edging: Break out, remove and dispose of off-site.

Concrete subbase: If present, sawcut along the trench set-out lines.

Grass

Removal method: Neatly cut grass turf between trench set-out lines into 300 mm squares.

Grass suitable for re-use: Take up and store the turf and water during the storage period.

Unsuitable grass: Remove and dispose of off-site.

Small plants, shrubs and trees

Remove for re-planting: Take up and store. Wrap the rootball in a hessian or plastic bag with drain holes and water during the storage period.

Unsuitable vegetation: Remove and dispose of off-site.

3.3 GROUND CONDITIONS As found site conditions

Unexpected conditions: If the following are encountered, give notice immediately and obtain instructions before carrying out any further work in the affected area:

- Bad ground.
- Discrepancies to expected ground conditions.
- Rock.
- Springs, seepages.
- Topsoil > 100 mm deep.

Records of measurement

Excavation and backfilling: If a schedule of rates applies, provisional quantities are specified, or there are variations to the contract levels or dimensions of excavations, do not commence backfilling or place permanent works in the excavation until the following have been agreed and recorded:

- Depths of excavations related to the datum.
- Final plan dimensions of excavations.
- Quantities of excavations in rock.

3.4 EXCAVATION General

Requirement: Excavate for underground services in conformance with the following:

- To required lines and levels, with uniform grades.
- Straight between access chambers, inspection points and junctions.
- With stable sides.
- Width tolerance: ± 50 mm, unless constrained by adjacent structures.
- Tree protection: To AS 4970.

Adjacent structures

Existing footings: If excavation is required within the zone of influence of an existing footing, use methods including (temporary) shoring or underpinning that maintain the support of the footing and make sure that the structure and finishes supported by the footing are not damaged.

Temporary supports: Provide supports to adjacent structures where necessary, sufficient to prevent damage arising from the works, as follows:

- Lateral supports: Provide lateral support using shoring.
- Vertical supports: Provide vertical support where necessary using piling or underpinning or both.

Permanent supports: If permanent supports for adjacent structures are necessary and are not described, give notice and obtain instructions.

Encroachments: If encroachments from adjacent structures are encountered and are not shown on the drawings, give notice and obtain instructions.

Trench widths

General: Keep trench widths to the minimum, consistent with the laying and bedding of the relevant service and construction of access chambers and pits.

Trench depths

General: As required by the relevant service and its bedding method.

Obstructions

General: Clear trenches of sharp projections. Cut back roots encountered in trenches to at least 600 mm clear of services. Remove other obstructions including stumps and boulders which may interfere with services or bedding.

Dewatering

General: Keep trenches free of water. Place bedding material, services and backfilling on firm ground, free of surface water.

Pumping: Provide pump-out from adjacent sumps or install well points.

Adjacent subsidence: Provide recharge points to isolate the dewatering zone.

Excess excavation

General: If trench excavation exceeds the correct depth, reinstate to the correct depth and bearing value using compacted bedding material or sand stabilised with 1 part of cement to 20 parts of sand by volume.

Stockpiles

Topsoil removal: Stockpile topsoil intended for re-use to a maximum height of 1500 mm.

Excavated material for backfill: If required, segregate the earth and rock material and stockpile, for reuse in backfilling operations.

Locations: Do not stockpile excavated material against tree trunks, buildings, fences or obstruct the free flow of water along gutters where stockpiling is permitted along the line of the trench excavation.

Disposal: If stockpiling is not permitted, dispose of excavated material off-site.

Unsuitable material

Disposal: Remove unsuitable material from the bottom of the trench or at foundation level and dispose of off-site. Replace with trench backfill material.

Boring

Subcontractor: If boring is required instead of trenches, engage a suitably qualified subcontractor to do the work.

3.5 TRENCH BACKFILL General

Place fill: To AS 3798 clauses 6.2.2 and 6.2.6.

Timing: Backfill service trenches as soon as possible after laying and bedding the service, if possible on the same working day.

Removal of supports: Remove temporary supports progressively as backfilling proceeds.

Marking services

Marking tape: Provide marking tape above service, with appropriate labelling, to AS/NZS 2648.1 and as follows:

- Non-metallic services: Provide tape capable of being detected by inground scanning devices.
- Location: Locate tape approximately half the depth of the service being marked, to a maximum depth of 200 mm below the finished ground level.

Boring: If boring techniques are used to install the service, provide permanent on site labelling at the start and end of the service run and record on the as-built documentation.

Bedding, haunch, side and overlay zones

Installation and material: To the particular utility authority or utility service requirements. Secure pipes against floatation.

Bedding of services: Surround pipes or conduits on all sides with a minimum of 75 mm compacted bedding sand, or as documented.

Overlay zone thickness: Maximum 300 mm immediately over the utility service.

3.6 SURFACE RESTORATION Subbase and base

Compaction: Uniformly compact each layer of the subbase and base courses over the full area and depth within the trench to a relative compaction of 100% when tested in conformance with AS 1289.5.4.1.

Compacted layer thickness:

- Maximum: 200 mm.
- Minimum: 100 mm.

Compaction test frequency: Minimum 1/every second layer/50 m² of restoration surface area.

Concrete surfaces

Construction: Conform to the following:

- Prime coat the cut edges of the existing surfaces with cement slurry. Lay and compact concrete so that the edges are flush and the centre is cambered 5 mm above the adjoining existing surfaces.
- Surface finish and pattern: Match existing adjoining work.
- Minimum thickness: 75 mm or the adjacent pavement thickness, whichever is thicker.
- Reinforcement and dowels: If required, provide steel reinforcement with dowels into the adjacent concrete.
- Expansion joints: 15 mm thick preformed jointing material of bituminous fibreboard placed in line with joints in existing concrete.
- Control joints:
 - . Form control joints strictly in line with the control joints in existing concrete.
 - . Around service poles: Terminate the concrete paving 200 mm from the pole and fill the resulting space with cold mix asphalt.

Weather: Do not place concrete in ambient temperatures above 30°C or below 10°C, without adequate precautions. Protect surface from rain damage, if required.

Compaction: Compact as follows:

- Thickness 100 mm or less: Compact by placing, screeding and finishing processes. If required use a hand-held vibrating screed at the surface. Do not use immersion vibrators.
- Thickness more than 100 mm and downturns: Use an immersion vibrator.

Curing: Cure by keeping continuously wet for 7 days.

Landscaped areas

In topsoil areas: Complete the backfilling with topsoil for at least the top 100 mm.

Lawn: Re-lay stockpiled turf. If existing turf is no longer viable, re-sow the lawn over the trench and other disturbed areas.

Planted areas: Overfill to allow for settlement.

3.7 COMPLETION General

As-built documentation: Upon completion, record the location of all installed services on the as-built documentation.

Note: Contractor to ensure all existing services are located and marked clearly on site when excavating and installing new services crossing these areas. Ensure all associated subcontractors are made aware of these services.

0224 STORMWATER – SITE

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide stormwater drainage, as documented. Refer also civil, hydraulic engineer drawings.

1.2 DESIGN General

Designer: as per Hydraulics & civil stormwater plans.

1.3 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements*.

1.4 STANDARDS

Stormwater drainage Standard: To AS/NZS 3500.3.

1.5 INTERPRETATION Definitions

General: For the purposes of this work section the following definition applies:

- Pipe surround: Includes pipe overlay, pipe side support, side zone and haunch zone.

1.6 SUBMISSIONS Certification

Requirement: Submit certificate stating that network is leak free upon completion.

Samples

General: Submit samples of the following:

- Each type of imported pipe bedding material.
- Each type of filter material.

Tests

Results: Submit results from pre-completion leak testing.

1.7 INSPECTION Notice

Inspection: Give notice so that inspection may be made at the following stages:

- Excavated surfaces prior to placing bedding material.
- Underground services, or services to be concealed, before being covered.
- Pipe joints before covering.
- Placing of concrete.
- Upon completion.

2 PRODUCTS

2.1 MATERIALS Concrete and mortar

Concrete: To AS 3500.3 clause 2.9 and the following:

- Grade: N15.
- Cement: To AS 3972.
. Type: GP, GL or GB.

Steel reinforcement:

- Bars and machine welded mesh: To AS/NZS 4671.

Joints

Solvent cement and priming fluid: To AS 3879.

Rubber ring joints/elastomeric seals: To AS 1646.

Pipes and fittings

Fibre reinforced cement (FRC): To AS 4139 and the following:

- ≤ 450 mm diameter: Rubber ring joints to AS 4139.
- > 450 mm diameter: With a purpose machined internal spigot and socket system within the pipe wall.

Glass-reinforced plastic (GRP): To AS 3571.1.

Access covers and grates: To AS 3996.

Polyvinyl chloride (PVC): To AS/NZS 1254, AS/NZS 1260 or AS 1273, as appropriate.

Polyethylene (PE): To AS/NZS 4129, AS/NZS 4130, ISO 8770 or AS/NZS 2033, as appropriate.

Precast concrete: To AS/NZS 4058.

Plastic pipe for subsoil drainage: To AS 2439.1.

Vitrified clay or ceramic: To AS 1741.

Bedding material

Bed and haunch zones: Provide granular material graded to the AS 1141 series. **Bedding material grading table**

Sieve size (mm)	Weight passing %	
	Bed and haunch	Side zones
75.0	-	100
19.0	100	-
9.5	-	50 - 100
2.36	50 - 100	30 - 100
0.60	20 - 90	15 - 50
0.30	10 - 60	-
0.15	0 - 25	-
0.075	0 - 10	0 - 25

Filter material

Requirement: Materials consisting of natural clean washed sands and gravels and screened crushed rock, to AS/NZS 3500.3 clause 2.13.1.

2.2 GEOTEXTILES General

Material: UV stabilised, permeable, polymeric, woven or non-woven textile material used in contact with soil/rock material.

Identification and marking: To AS 3705.

Subsoil drainage

Filter: Conform to AS/NZS 3500.3 clause 2.13.2.

2.3 PREFABRICATED PITS General

Requirement: Precast or prefabricated pits conforming to AS/NZS 3500.3 clauses 2.12.8 and 7.5.

Access covers and grates

Cover finish and load classification: Provide access cover and grate with the documented finish, slip resistance and load class.

3 EXECUTION

3.1 PIPING Installation

Laying: Lay lengths separately with the barrel bearing evenly on the prepared bedding.

Sockets: Lay with socket ends pointing upstream.

Cleaning: Clean pipe interior of dirt, debris, mortar and other foreign matter.

Protection: Provide temporary caps over the ends of incomplete sections to prevent the entry of foreign matter.

3.2 TOLERANCES Pipeline tolerances table

	Permissible angular deviation from the documented alignment	Permissible displacement from the documented positions
Horizontal	1:300	15 mm
Vertical	1:500	5 mm
Note: These tolerances are conditional on falls to outlets being maintained and no part of a pipeline having less than the documented gradient.		

3.3 STORMWATER DRAINS Location

Layout: Provide stormwater drains to connect surface drains, subsoil drains and drainage pits to the outlet point or point of connection. Make sure that location of piping will not interfere with other planned services and building elements not yet installed or built. Subject to the preceding and documented layouts, follow the most direct route with the least number of changes in direction.

Laying

Installation: Lay in straight lines between changes in direction or grade. If other pipes are adjacent, set each pipe true to line and complete each joint before laying the adjacent pipe.

Identification

Requirement: Lay a detectable strip or plastic tape in the trench after pipe laying, testing and initial backfilling.

Pipe underlay

Bedding: Bed piping on a continuous underlay of bedding material, minimum 75 mm, maximum 150 mm thick after compaction. Grade the underlay evenly to the gradient of the pipeline.

Chases: If necessary, form chases to prevent projections such as sockets and flanges from bearing on the trench bottom or underlay.

Pipe surrounds

Placement: Place the material in the pipe surround in layers, maximum 200 mm loose thickness, and compact without damaging or displacing the piping.

Trench backfill

General: Backfill the remainder of the trench to the underside of the subgrade with fill material placed in layers, maximum 200 mm loose thickness and compacted to minimum 90%, or 95% under pavements, of the standard maximum dry density.

Lifting holes

Sealing: Seal lifting holes in all pipes with plastic preformed plugs or 3:1 (sand:cement) mortar, before the commencement of backfilling.

Anchor blocks

Restraint: If necessary, to restrain lateral and axial movement of the stormwater pipes, provide anchor blocks at junctions and changes of grade or direction conforming to AS/NZS 3500.3 clause 7.9.

3.5 PITS Installation

Requirement: Prepare foundation, install pit and connect pipes, to manufacturer's recommendations.

Location: At junctions, changes of gradient and changes of direction of stormwater drains, as documented.

Finish to in situ exposed surfaces

General: Provide a smooth, seamless finish, using steel trowelled render or concrete cast in steel forms.

Corners: Cove or splay internal corners.

Metal access covers and grates

Cover levels: Top of cover or grate, including frame:

- In paved areas: Flush with the paving surface.
- In landscaped areas: 25 mm above finished surface.
- Gratings taking surface water runoff: Locate to receive runoff without ponding.

3.6 TESTING Pre-completion tests

General: Before backfilling or concealing, carry out an air or water pressure test to AS/NZS 3500.3 Section 9.

Leaks: If leaks are found, rectify and re-test.

3.7 COMPLETION Cleaning

General: Clean and flush the whole installation.

1 GENERAL**1.1 RESPONSIBILITIES General**

Requirement: Provide landscaping. Refer PC allowances for Landscaping & Irrigation only excluding topsoil & turf allowances.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements*.

1.3 INTERPRETATION Definitions

General: For the purposes of this work section the following definitions apply:

- Imported topsoil: Similar to local natural soil, suitable for the establishment and ongoing viability of the selected vegetation, free of weed propagules and of contaminants, and classified by texture to AS 4419 Appendix K Table K1, as follows:
 - . Fine: Clay loam, fine sandy clay loam, sandy clay loam, silty loam, loam.
 - . Medium: Sandy loam, fine sandy loam.
 - . Coarse: Sand, loamy sand.
- Plant establishment period: The period between the date of practical completion and the end of the defects liability period.
- Site rock: Rocks selected for salvage.
- Site topsoil: Natural soil, excavated from the site, that contains organic matter, supports plant life, conforms generally to the fine-to-medium texture classification to AS 4419 and is free from the following:
 - . Stones more than 25 mm diameter.
 - . Clay lumps more than 50 mm diameter.
 - . Weeds and tree roots.
 - . Sticks and rubbish.
 - . Material toxic to plants.
- Soil blend: A landscape soil derived from the blending of two or more of sand, natural soil material or organic materials, and with a bulk density and organic matter content to meet site specific requirements.
- Top dressing: A soil which is suitable for surface application to turf and lawns.
- Topsoil: Includes landscape soil, low density soils and soils for turf and lawns.

1.4 SUBMISSIONS Subcontractors

General: Submit names and contact details of proposed suppliers and evidence of the following, if appropriate:

2 PRODUCTS

2.1 TOPSOIL Standard

Site and imported topsoil: To AS 4419.

Potting mixes: To AS 3743.

Composts, soil conditioners and mulches: To AS 4454.

Source

General: If the topsoil of documented quality cannot be provided from material recovered from site, provide imported topsoil.

Imported topsoil

Requirement: Imported topsoil to AS 4419 Tables 1, 2 and 3, and as documented.

Imported topsoil particle size table (% passing by mass)

Sieve size (mm)	Soil textures		
	Fine	Medium	Coarse
2.36	100	100	100
1.18	90 – 100	90 – 100	90 – 100
0.60	75 – 100	75 – 100	70 – 90
0.30	57 – 90	55 – 85	30 – 46
0.15	45 – 70	38 – 55	10 – 22
0.075	35 – 55	25 – 35	5 – 10
0.002		2 – 15	2 – 8

Imported topsoil nutrient level table

Nutrient	Unit	Sufficiency range
Nitrate-N (NO ₃)	mg/kg	> 25
Phosphate-P (PO ₄) – P tolerant	mg/kg	43 - 63
Phosphate-P (PO ₄) – P sensitive	mg/kg	< 28
Phosphate-P (PO ₄) – P very sensitive	mg/kg	< 6
Potassium (K)	mg/kg	178 - 388
Sulfate-S (SO ₄)	mg/kg	39 - 68
Calcium (Ca)	mg/kg	1200 - 2400
Magnesium (Mg)	mg/kg	134 - 289
Iron (Fe)	mg/kg	279 - 552
Manganese (Mn)	mg/kg	18 - 44
Zinc (Zn)	mg/kg	2.6 - 5.1

Copper (Cu)	mg/kg	4.5 - 6.3
Boron (B)	mg/kg	1.4 - 2.7

Method References

pH in H₂O (1:5), pH in CaCl₂ (1:5) and Electrical Conductivity (EC) by Rayment & Higginson (1992) method 4A2, 4B2, 3A1.

Soluble Nitrate-N by APHA 4500.

Soluble Chloride by Rayment and Lyons 2011 modified method 5A2.

Extractable P by Mehlich 3 – ICP.

Exchangeable cations – Ca, Mg, K, Na by Mehlich 3 – ICP.

Extractable S by Mehlich 3 – ICP.

Extractable trace elements (Fe, Mn, Zn, Cu, B) by Mehlich 3 - ICP.

2.2 GRASS Turf

Description: Cultivated turf of even thickness, free from weeds and other foreign matter.

Supplier: A specialist grower of cultivated turf.

2.3 FERTILISER General

Description: Proprietary fertilisers, delivered to the site in sealed bags marked to show manufacturer or supplier, weight, fertiliser type, N:P:K ratio, recommended uses and application rates.

Application rate: Vary the application rate to allow for the plant-available immediate fertiliser equivalence value of the soil conditioning compost.

3 EXECUTION

3.1 PREPARATION Weed eradication

Herbicide: Eradicate weeds using environmentally acceptable methods, such as a non-residual glyphosate herbicide in any registered formulae, at the recommended maximum application rate.

Manual weeding: Regularly remove weed growth by hand throughout grassed, planted and mulched areas. Remove weed growth from an area of 750 mm diameter around the base of the trees in grassed areas. Continue weeding throughout the course of the works and during the planting establishment period.

Vegetative spoil

Disposal: Remove vegetative spoil from site. Do not burn.

3.2 TREATMENT General

Pest attack or disease: If evidence of pest attack or disease of plant material is discovered, immediately give notice.

Physical removal

General: Remove pest infestation and diseased plant material by hand if appropriate.

Pesticide

Product: Spray with insecticide, fungicide or both, as required.

3.3 ESTABLISHMENT Planting

Requirement: Make sure the general appearance and presentation of the landscape and the quality of plant material at date of practical completion is maintained for the planting establishment period.

Plant replacement: Replace failed, dead and/or damaged plants at maximum 3 weekly intervals as necessary throughout the plant establishment period.

Pruning: To AS 4373 and as documented.

Application of fertiliser: Apply either an all-purpose fertiliser or a 12 month slow release fertiliser, in two rows and cultivated into soil to a depth of 100 mm.

- Program: September and March according to seasonal growth requirement.

Weeding: Remove unwanted broadleaf plants and grasses considered invasive to the locality.

Remulching: Maintain the original ground levels around the base of plants.

Watering: Minimum 3 complete waterings, soaking to a depth of 150 mm at fortnightly intervals for the first 6 weeks of plant establishment irrespective of natural rainfall.

Grass surfaces

Preparation: Remove litter and fallen branches before mowing.

Mowing:

- Grass height: Consistent with the growth habit of the grass variety and maintained at 25 mm to 40 mm throughout the year. Do not remove more than one-third of the grass height at any one time.
- Program: Weekly during the mowing season, November to March, and at fortnightly intervals from April to October. Do not mow during wet conditions. Carry out last mowing not more than 7 days before end of plant establishment period.
- Clippings: Remove grass clippings from the site after each mowing.

Weeding: Remove unwanted broadleaf plants and grasses considered invasive to the locality.

- Program: Quarterly, and as required to maintain the general lawn condition.

Edge trimming: At the same time as mowing, trim lawn edges to plant beds, pathways, base of trees and other obstacles. Do not damage trees and shrubs.

Topdressing for established lawns: Weed-free imported sandy topsoil to a depth of 5 mm.

- Program: The spring following initial establishment.

Application of fertiliser: Apply lawn fertiliser at the completion of the first and last mowings of the plant establishment period, and at other times as required to maintain healthy grass cover.

3.4 COMPLETION Cleaning

Stakes and ties: Remove those no longer required at the end of the planting establishment period.

Temporary fences: Remove temporary protective fences at the end of the planting establishment period.

4 SELECTIONS

4.1 TURF

Allow a provisional quantity of **1,250m² of new turf**. Location to be directed on site.

Provide min 50thk top soil as sub grade to turf areas.

Provide and lay turf as directed. Grade site to provide falls to field inlets and surface drains. Roll turf with lawn roller and top dress all gaps.

Spread Fertilizer 'Grass master' to site before laying turf to quantities recommended by manufacture.

Turf Species to be confirmed.

Contractor to water before installing roll to provide even plane and fill and top dress throughout to provide even surface.

Location of turf to be directed on site

4.2 LANDSCAPING, PLANTING, IRRIGATION, TREES

Allow a provisional sum of **TWENTY THOUSAND DOLLARS (\$20,000.00)** for general new in-ground irrigation system (excluding topsoil to turf areas, equipment, fences)

0274 CONCRETE PAVEMENT

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide concrete pavement as documented. Refer to structural, civil, design drawings and documentation.

Performance Requirement:

- Free draining and evenly graded between level points.
- Even and smooth riding surfaces.

Conformance: Conform to the local authority requirements for levels, grades and minimum thickness, reinforcement and concrete strength for pavements within the kerb-and-gutter property boundaries.

1.2 DESIGN General

Coordination: Determine the local authority requirements that may affect grades, transitions and work zones, including the following:

- Drainage.
- Trees (due to settlement).
- Adjacent structures.

1.3 CROSS REFERENCES General

Requirement: Conform to the following:

- *0171 General requirements.*
- *0222 Earthwork.*
- *0271 Pavement base and subbase.*

1.4 STANDARDS Concrete

Specification and supply: To AS 1379.

Materials and construction: To AS 3600.

Residential pavements: To AS 3727.1.

Slip resistance

Classification: To AS 4586.

1.5 INTERPRETATION Definitions

General: For the purposes of this work section the following definitions apply:

- Ambient temperature: The air temperature at the time of mixing and placing of concrete.
- Concrete class – normal: Concrete that is specified primarily by a standard compressive strength grade up to 50 MPa and otherwise in conformance with AS 1379 clause 1.5.3.
- Concrete class – special: Concrete that is specified to have certain properties or characteristics different from, or additional to, those of normal-class concrete and otherwise in conformance with AS 1379 clause 1.5.4.
- Weather – cold: Ambient shade temperature less than 10°C.
- Weather – hot: Ambient shade temperature greater than 30°C.

1.6 TOLERANCES General

Surface abutting gutters: ± 5 mm from the level of the gutter edge.

Rigid pavement surface:

- From design level: + 10 mm, - 0 mm.
- From a 3 m straightedge placed anywhere on surface: 5 mm.

Horizontal position of outer concrete edge: 30 mm from documented position.

Joint locations in plan : 10 mm from documented position.

1.7 SUBMISSIONS Certification

Test certificates and records: Submit test certificates and also retain results on site.

Compliance certificate: If product testing is not proposed, submit a manufacturer's certificate together with results of recent tests undertaken by the manufacturer, showing compliance with test criteria.

Execution details

Concrete: Submit proposals for mixing, placing, finishing and curing concrete including the following:

- Addition of water on site.
- Changes to the concrete mix.
- Temperature control, curing and protection methods.
- Cutting or displacing reinforcement.
- Handling, placing, compaction and finishing methods and equipment, including pumping.
- Sequence, size and times of concrete pours, and construction joint locations and relocations.

Cores, fixings and embedded items: If required, submit shop drawings showing the proposed locations, clearances and cover, and indicate any proposed repositioning of reinforcement.

Cutting or coring: If cutting or coring of hardened concrete is proposed, provide details.

Sawn joints: Submit proposed methods, timing and sequence of sawing joints.

Damaged galvanizing: If repair is required, submit proposals to AS/NZS 4680 Section 8.

Splicing: If undocumented splicing is proposed, submit details.

Welding: If welding of reinforcement is proposed, provide details and give notice before welding reinforcement.

Joint sealants: Submit proposals for installation methods and sealant performance.

Crack assessment: If unplanned cracks occur in the finished pavement, submit proposals for investigation.

Surface repair method: If required, submit details of the proposed method before commencing repairs.

Trial section: Submit proposal for trial pavement.

Products and materials

Aggregate: Nominate the source for all aggregates.

Reinforcement: Submit the manufacturer's certificate of compliance with AS/NZS 4671, or submit test certificates from an Accredited Testing Laboratory.

Liquid curing compounds: Submit certified test results, including the application rate and the efficiency index to AS 3799 Appendix B.

Curing by covering: Submit details of the proposed covering material.

Repair materials: Submit proposals for epoxy resin/grout and elastomeric sealant.

Concrete: Submit the concrete supply delivery dockets.

Trial mix design report: Six weeks before commencing production, submit a report for each mix design containing the information required in AS 1012.2, the individual and combined aggregate particle size distribution, and the records and reports for the tests.

Subcontractors

Pre-mixed supply: Submit names and contact details of proposed pre-mixed concrete suppliers, and alternative source of supply in the event of breakdown of pre-mixed or site mixed supply.

Tests

Site tests: Submit results, as follows:

- Slip resistance test of completed installations.

1.8 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Base or subgrade before covering.
- Membrane or film underlay installed on the base or subgrade.
- Concrete formwork, reinforcement and dowels in position.
- Commencement of concrete placing.
- Completion of concrete placing.
- Evaluation of surface finish.

2 PRODUCTS

2.1 REINFORCEMENT

Steel reinforcement Standard: To AS/NZS 4671.

Reinforcement and joint requirements: as per Patcol Group Structural Engineers drawings and details

Surface condition: Free of loose mill scale, rust, oil, grease, mud or other material that may reduce the bond between the reinforcement and concrete.

Protective coatings

Requirement: For concrete containing protective coated reinforcement, provide the same coating type to all reinforcement and embedded ferrous metal items, including tie wires, stools, spacers, stirrups, plates and ferrules.

Epoxy coating: Provide high build, high solids chemically resistant coating to AS/NZS 3750.14.

- Thickness: 200 µm minimum.

Galvanizing: To AS/NZS 4680, as follows:

- Sequence: If fabricating after galvanizing, repair damaged galvanising and coat cut ends.
- Zinc-coating (minimum): 600 g/m².

Steel fibres

Fibre reinforcement: To CIA CPN35.

Steel fibre content: 45 to 75 kg/m³ and the steel fibre manufacturer's recommendations.

Accessories

Reinforcement supports: To AS/NZS 2425.

Tie wire: Galvanized annealed steel 1.25 mm diameter minimum.

Dowels

General: Provide each dowel in one piece, straight, cut accurately to length with ends square and free from burrs.

Standard: To AS/NZS 4671.

Grade: 250R steel bars 450 mm long.

Tie bars

Type: Deformed bar, 12 mm diameter, grade 500N, 1 m long.

2.2 AGGREGATE Characteristics

Standards: AS 2758.1.

Durability: Tested to AS 1141.22:

- Wet strength not less than 80 kN.
- 10% Fines Wet/Dry Variation not to exceed 35%.

Recycled concrete aggregate (RCA): If blending coarse RCA with natural aggregates, make sure substitution rates are below 30%.

2.3 CEMENT General

Standard: To AS 3972.

Type: GP – General Purpose Cement

Moisture: Protect from moisture until used. Do not use caked or lumpy cement.

Age: Less than 6 months old.

Storage: Store cement bags under cover and above ground.

Supplementary cementitious materials Fly ash: To AS/NZS 3582.1.

Slag: To AS 3582.2.

Amorphous silica: To AS/NZS 3582.3.

2.4 WATER General

Quality: Drinking water free from materials harmful to concrete or reinforcement, and not salty or brackish.

Limits: Not containing more than:

- 600 parts per million of chloride ion, tested to AS 3583.13.
- 400 parts per million of sulphate ion, tested to AS 1289.4.2.1.

2.5 ADMIXTURES General

Standard: Chemical admixtures to AS 1478.1, used to the manufacturer's recommendations.

Quality: Free from calcium chloride, calcium formate, or triethanolamine or any other accelerator. Do not use admixtures or combinations of admixtures without prior written approval.

Dosage: Vary the dosage of chemical admixture to account for factors such as air temperature, setting time and cement content to the manufacturer's recommendations.

Types of admixtures

Air entraining agent: Adjust mix for workability allowing up to 5% air entrainment.

Warm season retarder: During the warm season, (October to March inclusive), use a lignin or ligninbased (ligpol) set-retarding admixture (Type Re or Type WRR) if required to control slump within the limits stated in Concrete mix, properties.

Cool season accelerator: During the cool season, (April to September inclusive), use only a lignin or lignin based set-retarding admixture, if required, containing not more than 6% reducing sugars (Type Ac or Type WRAc to AS 1478.1).

2.6 CURING COMPOUNDS General

Curing compounds: To AS 3799 and AS 1160, Type 2.

Sheet material covering: To ASTM C171, white opaque or clear polyethylene film, or white burlappolyethylene sheet, or equivalent material.

2.7 OTHER MATERIALS

Tactile ground surface indicators Standard: To AS/NZS 1428.4.1.

3 EXECUTION

3.1 GENERAL Traffic control

Traffic restriction: Do not allow traffic or construction plant other than that associated with testing, saw cutting, cleaning or joint sealing on pavement for minimum 10 days after placing, or when the concrete has reached a compressive strength of at least 20 MPa, and joints have been completely sealed.

3.2 SUBGRADE Preparation

Conformance: Prepare subgrade to *0222 Earthwork*.

Extent: Prepare a uniform subgrade for the full pavement formation, extending at least to the back of kerbs or at least 300 mm beyond each side of the carriageway if kerbs are not proposed.

Reinstatement: Make sure of uniformity for backfilling of any utility trenches.

3.3 SUBBASE Thickness

Subbase thickness: as per Structural Engineers details

Width

Subbase width: Extend the subbase at its full depth to at least the back of kerbs or other edge stops before their installation.

No integral kerbs: Extend granular unbound subbase at least 300 mm beyond each side of the carriageway.

Tolerance

Subbase finished surface level: + 0 mm, - 10 mm.

Placement

Bound and unbound subbase materials and placement: To *0271 Pavement base and subbase*.

Friction reduction

Requirement: Provide 200 µm thick polyethylene sheeting with 200 mm taped minimum laps and/or a 20 mm thick layer of sand (silt and clay material less than 5%) directly beneath the concrete pavement.

3.4 CONCRETE MIX Standard

Concrete mix and supply: To AS 3600 clause 17.1 and AS 1379.

Properties

Concrete pavement thickness: as per structural engineers drawings.

Concrete pavement strength: as per structural engineers drawings.

Slump: Maximum 100 mm.

Drying shrinkage: Maximum 450 µε after 21 days of air drying.

Elapsed delivery time

General: Make sure that the elapsed time between the wetting of the mix and the discharge of the mix at the site is in conformance with the **Elapsed delivery time table**. Do not discharge at ambient temperature below 10°C or above 30°C unless approved heating or cooling measures are taken to deliver concrete within the range 5°C to 35°C.

Elapsed delivery time table

Concrete temperature at time of discharge (°C)	Maximum elapsed time (minutes)
5 – 24	120
24 – 27	90
27 – 30	60
30 – 35	45

Site mixed supply

Emergencies: If mixing by hand, provide details.

Plant: Mix concrete in a plant located on the construction site.

Pre-mixed supply

Addition of water: Do not add water.

Transport: Make sure the mode of transport prevents segregation, loss of material and contamination of the environment, and does not adversely affect placing or compaction.

Concrete delivery docket: For each batch, provide a docket listing the information required by AS 1379 clause 1.7.3, and the following information:

- Any binders or additives.
- Method of placement and climate conditions.
- Name of concrete delivery supervisor.
- The concrete element or part of the works for which the concrete was ordered, and where it was placed.

3.5 TESTING Standards

Sampling, identification, testing and recording: To the AS 1012 series.

Specimens: Sample the concrete on-site, at the point of discharge from the agitator.

Type and frequency: To AS 1379.

Testing authority: Concrete supplier or Accredited Testing Laboratory.

Concrete testing methods

Slump: To AS 1379 clause 5.2.

Compressive strength: Test to AS 1012.8.1 and AS 1012.9.

Drying shrinkage: Test to AS 1012.8.4 and AS 1012.13.

Flexural strength: Test to AS 1012.8.2 and AS 1012.11.

Acceptance criterion for strength: The average strength of any set of 3 consecutive project samples must be equal to or greater than the specified minimum value.

Sampling frequency: Provide a minimum of one sample from each 50 m³ of concrete.

3.6 INSTALLATION Junctions with existing pavements

Trimming: If new pavement is to be joined to an existing pavement, trim the edge of the existing pavement to create a neat vertical edge for its full depth before placing new pavement material.

Fixed formwork Description:

- Steel forms.
- Seasoned, dressed timber planks, free of warps, bends or kinks.

Depth: Equal to the edge thickness of the slab and in one piece.

Tolerances on position:

- Level of top of form: - 0 mm, + 10 mm from pavement surface design level.
- Horizontal tolerance: 10 mm (maximum departure from a plane surface).
- Verticality: 3 mm departure from vertical.

Staking: Stake forms in position using at least 3 steel stakes per form, not more than 1.5 m apart. Lock joints between form sections to prevent movement.

Release agent: Before placing reinforcement, apply a release agent compatible with the contact surfaces, to the interior of the formwork, except where the concrete is to receive an applied finish for which there is no compatible release agent.

Re-use: Clean and recoat the forms each time before placing concrete.

Keyways: Form the keyways of keyed construction joints using steel or timber form strips accurately located at the mid-depth of the slab and securely fastened flush against the formwork face.

Reinforcement

Tolerances in fabrication and fixing: To AS 3600.

Locate reinforcement: Place reinforcement in the top half of the pavement.

Minimum cover to reinforcement: 30 mm.

Splicing mesh: Overlap a minimum of 2 crosswires.

Supports: Provide reinforcement supports as follows:

- Able to withstand construction and traffic loads and maintain the concrete cover, as documented.
- With a protective coating if they are ferrous metal extending to the surface of the concrete.
- Use plastic or concrete supports with galvanized or zinc-coated reinforcement.
- Spacing:
 - . Bars: ≤ 60 diameters.
 - . Mesh: ≤ 600 mm.
- Supports over membranes: Prevent damage to waterproofing membranes or vapour barriers. If appropriate, place a metal or plastic plate under each support.

Projecting reinforcement: If starter or other bars extend beyond reinforcement mats or cages, through formwork or from cast concrete, provide a plastic protective cap to each bar until it is cast into later work.

Tying: Secure the reinforcement against displacement at intersections with either wire ties, or clips. Bend the ends of wire ties away from nearby faces of formwork or unformed faces to prevent the ties projecting into the concrete cover.

Mats: For bar reinforcement in the form of a mat, secure each bar at alternate intersections.

Cores, fixings and embedded items

Position: Fix cores and embedded items to prevent movement during concrete placing. In locating cores, fixings and embedded items, displace but do not cut reinforcement, and maintain cover to reinforcement.

Isolation: Isolate embedded items to prevent water tracking to concrete providing minimum cover to reinforcement.

3.7 CONCRETE PLACING AND COMPACTION Concrete placing

General: Place concrete uniformly over the width of the slab or lane and so that the face is generally vertical and normal to the direction of placement. Hand spread concrete using shovels, not rakes.

Ponding: Remove any water ponding on the base or subbase before starting placement.

Placing sequence: Commence from one corner (usually the lowest point) and proceed continuously out from that point.

Weather: Do not place concrete in ambient temperatures above 30°C or below 10°C, without adequate precautions.

Compaction

Thickness 100 mm or less: Compact by placing, screeding and finishing processes. If required use a hand-held vibrating screed at the surface. Do not use immersion vibrators.

Thickness more than 100 mm and downturns: Use an immersion vibrator.

Placing records

Log book: Keep on site and make available for inspection a log book recording each placement of concrete, including the following:

- Date.
- Specified grade and source of concrete.
- Slump measurements.
- The portion of WUC.
- Volume placed.

Rain

Protection: During placement and before setting, protect surface from damage.

Concrete placing in cold weather

Cement: Do not use high alumina cement.

Temperature limits: Maintain the following:

- Freshly mixed concrete: $\geq 5^{\circ}\text{C}$.
- Formwork and reinforcement before and during placing: $\geq 5^{\circ}\text{C}$.
- Water: Maximum 60°C when placed in the mixer.

High early strength cement: If deteriorating weather conditions are predicted, use high early strength cement.

Temperature control: Heat the concrete materials, other than cement, to the minimum temperature necessary so that the temperature of the placed concrete is $\geq 5^{\circ}\text{C}$.

Admixtures: Do not use calcium chloride, salts, chemicals or other material in the mix to lower the freezing point of the concrete.

Frozen materials: Do not allow frozen materials or materials containing ice to enter the mixer, and keep free of frost and ice any formwork, materials, and equipment coming in contact with the concrete.

Placed concrete: Prevent from freezing, without using salts or chemicals.

Concrete placing in hot weather

Handling: Prevent premature stiffening of the fresh mix and reduce water absorption and evaporation losses.

Hot weather placing: Maintain concrete at a temperature $\leq 35^{\circ}\text{C}$.

Formwork and reinforcement: Before and during placing maintain temperature $\leq 35^{\circ}\text{C}$.

Severe weather: If ambient shade temperature more than 38°C , do not mix concrete.

Temperature control: Select one or more of the following methods of maintaining the specified temperature of the placed concrete:

- Cool the concrete using liquid nitrogen injection before placing.
- Cover the container in which the concrete is transported to the forms.
- Spray the coarse aggregate using cold water before mixing.
- Use chilled mixing water or ice.

Evaporation control barriers: Erect barriers to protect freshly placed concrete from drying winds.

3.8 CONCRETE FINISH General

Commencement: Immediately after placement, spreading and compaction of the concrete, start initial finishing procedures to achieve the documented finish.

Final finishing: Do not commence final finishing until all bleed water has evaporated from the surface after initial finishing procedures.

Unformed surfaces

General: Strike off, screed and level slab surfaces to finished levels, to the tolerance class and finish documented.

Formed surfaces

Damage: Do not damage concrete works through premature removal of formwork.

Curing: If forms are stripped when concrete is at an age less than the minimum curing period, commence curing exposed faces as soon as the stripping is completed.

Finishing methods - primary finish Machine float finish:

- After levelling, consolidate the surface using a machine float.
- Cut and fill and refloat immediately to a uniform, smooth, granular texture.
- Hand float in locations inaccessible to the machine float.

Wood float finish: After machine floating use wood or plastic hand floats to produce the final consolidated finish free of float marks and uniform in texture and appearance.

Broom finish: After machine floating and steel trowelling draw a broom or hessian belt across the surface to produce a coarse even-textured transverse-scored surface.

Scored or scratch finish: After screeding, give the surface a coarse scored texture using a stiff brush or rake drawn across the surface before final set.

Sponge finish: After machine floating and steel trowelling, obtain an even textured sand finish by wiping the surface using a damp sponge.

Finishing methods - supplementary finish

Abrasive blast: After steel trowelling, abrasive blast the cured surface to provide texture or to form patterns without exposing the coarse aggregate, using hard, sharp graded abrasive particles.

Coloured applied finish: After machine floating, apply a proprietary liquid or dry shake material to the manufacturer's recommendations and trowel to achieve the required appearance.

Stamped and coloured pattern finish: Provide finishing system.

Surface repairs

Method: If surface repairs are required, detail proposals.

3.9 CONCRETE CURING General

Curing: Commence curing as soon as possible after finishing, when the concrete has set sufficiently not to be damaged by the curing process, and extend for a minimum period of 7 days.

End of curing period: Prevent rapid drying out at the end of the curing period.

Protection: Maintain at a reasonably constant temperature with minimum moisture loss, during the curing period.

Cold weather curing

General: Maintain concrete surface temperatures above 5°C for the duration of the curing period.

Hot weather curing

Requirement: If the concrete temperature exceeds 25°C, or the ambient shade temperature exceeds 30°C, protect from drying winds and sun by using an evaporative retarder until curing is commenced.

Curing methods

Covering sheet method: Cover concrete using damp hessian or cotton mats overlapped at least 150 mm and anchored against displacement by wind or other interference. Keep the mats continuously damp until covered by the covering sheet material. Repair tears immediately.

Moist curing method: Keep the concrete surface continuously damp by ponding or spraying constantly with water, fog, or mist, using suitable spraying equipment. Continue wetting for the curing period.

Curing compound: Provide a uniform continuous flexible coating to AS 3799, without visible breaks or pinholes. Make sure coating remains unbroken at least for the required curing period after application. Respray defective areas within 30 minutes. Respray within 3 hours after heavy rain.

Self-levelling toppings: If used also as curing compounds, conform to AS 3799.

Coloured concrete: Do not cure with plastic sheeting, damp sand or wet hessian. Use only chemical curing compounds compatible with the sealer or a sealer to the manufacturer's recommendations.

3.10 JOINTS General

Requirement: Construct expansion, contraction and construction joints straight and plumb. Make transverse joints normal to longitudinal joints. Extend transverse expansion and contraction joints continuously from edge to edge of the pavement through interconnected slabs.

Joint layout: Install joints as documented.

Joint spacings: as per Structural Engineers drawings

Joint widths: as per Structural Engineers drawings. Generally 10mm with compressible material and approved sealant.

Contraction joints

Installation: Construct transverse and longitudinal contraction joints by early power sawing at an appropriate time, tooling or by placing an insert in the fresh concrete.

Unplanned joints: If placement is interrupted for 30 minutes or longer, form a tied transverse construction joint within the middle third of the distance between planned joints but no closer than 1.5 m to the nearest planned joint. If necessary remove placed concrete back to the required location.

Expansion joints

Joint filling: Fill with jointing materials as documented. Finish visible jointing material neatly flush with adjoining surfaces.

Jointing materials: Provide jointing materials compatible with each other, and non-staining to concrete in visible locations.

Foamed materials (in compressible fillers): Closed-cell or impregnated, not water absorbing.

Isolation joints

Requirement: Provide formed full depth joints around structures and features which project through, into or against the pavement, and elsewhere as required.

Formed joints

Full depth joints: Form the edge of the concrete placed first to provide a smooth, vertical face. After stripping and cleaning fix the joint filler with a suitable waterproof adhesive to the face of the slab, and place the adjoining concrete after the adhesive has set.

Weakened plane joint: Cut a crack-inducing groove by using a suitable tool into the plastic concrete during finishing of the concrete surface. Compact and refinish the plastic concrete around the groove after forming the joint.

Rebated groove joints: Form the rebate by securely fixing removable steel or timber form strips to the form or forms on the slab which is placed first, so that the top of the strip is flush with the top of the form. After stripping and cleaning, fix the joint filler in the rebate after placing the adjoining concrete.

Sawn joints

Weakened plane joint: Saw the hardened concrete to depth at least $\frac{1}{4}$ to $\frac{1}{3}$ of the pavement thickness and to a uniform width in the range of 3 to 5 mm as follows:

- Timing: Commence sawing, regardless of time or weather conditions, as soon as the concrete has hardened sufficiently to permit cutting with only minor ravelling of the edges of the saw cut. Complete sawing no later than 24 hours after concrete placement.
- Sequence: If possible, saw every third transverse joint initially, then saw the intermediate joints. Start where concrete placement commenced.

- Cracking: If the concrete has already cracked near the location chosen for a joint, do not saw a joint in that location. If a crack develops ahead of the saw cut, discontinue sawing and submit proposals for extra sawn joints.
- Stand-by machines: Provide one stand-by sawing machine for each machine planned to be used.
- Cleaning and protection: Immediately after each joint is sawn, flush the saw cut and adjacent concrete surface using water, until the waste from sawing is removed from the joint.

Rebated groove joints: Saw straight, parallel sided grooves for joint seals on top of and centred on the sawn weakened plane joints.

- Timing: Commence sawing after the curing period has ended, immediately before joint sealing. Saw during daylight hours.

3.11 SURFACE SEALERS General

Sealer: same day sealer. 2 coats penetrating sealer. CCS Streetscape. Apply in accordance with Manufacturer's instructions.

Application: Apply surface sealer after the curing period and when concrete has dried to allow the sealer to penetrate into the concrete surface.

Curing sealer compound: If using the sealer as a curing compound, apply directly after finishing.

3.12 COMPLETION Completion tests

Slip resistance of completed installation: To AS 4663.

Rectification

Reinstating adjacent surfaces: Reinstall surfaces next to new pavements and associated elements. If an existing road pavement has been disturbed, trim back to a straight, neat and undisturbed edge, parallel to the new concrete for the full depth of the slab.

Concrete pavement: If pavement does not conform to the tolerances, submit rectification proposal.

Unplanned cracking:

- Maximum 0.4 mm wide crack is acceptable.
- > 1 mm must be assessed, detail a proposal for possible cause and rectification processes.

Cleaning

Excavated material: Remove from site.

4 SELECTIONS

4.1 SCHEDULES Concrete Pavement schedule

Code	BFC		
Location	External and mowing strips generally		
Primary finish Type	Broom Finished Concrete		
colour	Grey Concrete match existing (unless noted otherwise)		
Finishing method	Broom swept finish		
sealer	Clear Penetrating sealer 2 coats (refer to painting specification section)		

1 GENERAL**1.1 RESPONSIBILITIES****General**

Requirement: Provide cast in situ reinforced concrete, as documented. Refer to structural and civil engineering.

Performance

Requirements:

- Conforming to the design details and performance criteria.
- Satisfying quality and inspection requirements.
- Compatible with documented applied finishes.

1.2 DESIGN**General**

Formwork: The design of formwork, other than permanent composite form systems, is the contractor's responsibility. Allow for dimensional changes, deflections and cambers resulting from the following:

- Imposed actions.
- Concrete shrinkage and creep.
- Temperature changes.
- The application of prestressing forces (if any).

Structural design: To AS 3600.

Requirements

Authority requirements: As per Structural Engineers notes and drawings

1.3 CROSS REFERENCES**General**

Requirement: Conform to the following:

- *0171 General requirements.*

1.4 STANDARDS**General**

Formwork design and construction: To AS 3610.1.

Plywood formwork: To AS 6669.

Composite steel-concrete construction, including profiled steel sheeting and shear connectors: To AS/NZS 2327.

Reinforced concrete construction: To AS 3600.

Specification and supply of concrete: To AS 1379.

Strand, bar and wire: To AS 4672.1.

Design, installation and testing of post-installed and cast-in fastenings: To AS 5216.

Formed surfaces: To AS 3610.1.

Slip resistance

Classification: To AS 4586.

1.5 INTERPRETATION

Definitions

General: For the purposes of this work section the definitions given in AS/NZS 1314, AS 1379, AS 3600, AS 3610.1 and the following apply:

- Ambient temperature: The air temperature at the time of mixing and placing of concrete.
- Anti-burst reinforcement: Reinforcement cage surrounding anchorages to control the tensile bursting stresses.
- Early age strength: A mean compressive strength at 7 days exceeding the values shown in AS 1379 Table 1.2.
- Green concrete: Concrete which has recently set but has not achieved any design strength.
- Sample: A physical example that illustrates workmanship, materials or equipment, and establishes standards by which the work will be judged. It includes samples and sample panels.
- Specimen: A portion of a sample which is submitted for testing.
- Weather – cold: Ambient shade temperature less than 10°C.
- Weather – hot: Ambient shade temperature greater than 30°C.

1.6 TOLERANCES

Formwork

Plumb of elements > 8 m high: 1:1000.

Plumb of elements ≤ 8 m high: To AS 3610.1.

Position: Construct formwork so that finished concrete conforms to AS 3600 clauses 3.3 and 17.5 and as documented in the **Formwork dimensional deviation schedule**.

Reinforcement

Fabrication: To AS 3600 clause 17.2.

Reinforcement and tendon position: To AS 3600 clause 17.5.3.

Formed surfaces

Finish quality: To AS 3610.1 Table 3.3.3.1.

Unformed surfaces

Flatness: To the **Flatness tolerance class table**, using a straightedge placed anywhere on the surface in any direction, for the documented class of finish. **Flatness tolerance class table**

Class	Measurement	Maximum deviation (mm)
A	2 m straightedge	4
B	3 m straightedge	6
C	600 mm straightedge	6

1.7 SUBMISSIONS

Certification

Formwork design certification: For other than permanent composite form systems, submit certification by a professional engineer experienced in formwork design verifying conformance of the design.

Formwork execution certification: Submit certification by a professional engineer experienced in formwork design and construction, verifying conformance of the completed formwork, including the suitability of the formwork for the documented surface finish class.

Design documentation

Formwork calculations: Submit calculations by a professional engineer experienced in formwork design to show that allowable concrete stresses will not be exceeded and formwork capability will be maintained if the following is proposed:

- Formwork procedures or loadings that differ from those documented.
- Props above a floor that do not coincide with the props below.
- Undocumented formwork shoring or stripping procedures.
- Loadings from stacked materials.

Execution details

Reinforcement: Submit the following:

- General: Details of any proposed changes to documented reinforcement.
- Damaged galvanizing: Details of proposed repair to AS/NZS 4680 Section 8.
- Mechanical bar splices: Details and test certificates for each size and type of bar to be spliced.
- Provision for concrete placement: Details of spacing or cover to reinforcement that does not conform to AS 3600.
- Splicing: Details of any proposed changes to documented requirements.
- Welding: Details of any proposed welding of reinforcement to AS/NZS 1554.3.

Loading: Submit details of proposed construction systems, loads and procedures, including propping and re-shoring.

Concrete: Submit proposals for mixing, placing, finishing and curing concrete including the following:

- Changes to the concrete mix.
- Curing and protection methods.
- Curing period for low-pressure steam curing.
- Cutting or displacing reinforcement, or cutting or coring hardened concrete.
- Handling, placing, compaction and finishing methods and equipment, including pumping.
- Placing under water.
- Sequence and times for concrete placement, and construction joint locations and relocations. Include any proposed sequential placement of slab segments.
- Site storage, mixing and transport methods and equipment, if applicable.
- Temperature control methods.
- Sawn joints: Submit details of proposed methods, timing and sequence of sawing joints.

Pre-mixed supply delivery dockets: For each batch, submit a docket listing the information required by AS 1379, and the following:

- For special-class performance concrete: Documented performance and type of cement binder.
- For special-class prescription concrete: Details of mix, additives, and type of cement binder.
- Method of placement and climate conditions during pour.
- Name of concrete delivery supervisor.
- Project assessment carried out each day.
- The concrete element or part of the works for which the concrete was ordered, and where it was placed.
- The total amount of water added at the plant and the maximum amount permitted to be added at the site.

Surface repairs: If surface repairs are required, submit proposed methods.

Products and materials

Void formers: Submit type-test results as evidence of conformity to requirements of **PRODUCTS, FORMWORK, Void formers**.

Reinforcement strength and ductility: Submit type-test reports as evidence of conformity to AS 3600 Table 3.2.1 for each reinforcement type.

Product conformity: Submit evidence of conformity, as appropriate, as follows:

- Certification by a JAS-ANZ accredited third party.
- Report by an accredited testing laboratory describing tests and giving results which demonstrate that the product conforms.

Concrete mixes: Submit details, for each grade and type of concrete including any proposed use of special-purpose cement types.

Curing compounds: Submit details of any proposed curing compounds, including the following:

- Certified type-test results for water retention to AS 3799 Appendix B for liquid membrane-forming compounds.
- Evidence of compatibility with concrete, and with applied finishes including toppings and render, if any, including methods of obtaining the required adhesion.
- For visually important surfaces, evidence that an acceptable final surface colour will be obtained.

Admixtures: Submit details of any proposed admixtures, including the following:

- Brand name.
- Place of manufacture.
- Basic chemical composition.

Prototypes

Test panels: Provide test panels to AS 3610.1 clause 3.7 and as documented in the **Test panels schedule**.

Manufacture: Cast the panels using the form, concrete, compaction equipment, form release agents, curing and formwork removal methods which are to be used in the final work.

Storage: Once accepted, maintain the panels on site undamaged and protected from the weather, as reference prototypes for evaluation of completed work.

Surface treatment: Do not proceed with the related work until the acceptable range of surface treatments has been determined.

1.8 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- Used forms, after cleaning and before re-use.
- Base or subgrade before covering.
- Membrane or film underlay installed on the base or subgrade.
- Completed formwork and reinforcement, tendons, cores, fixings and embedded items fixed in place before placing concrete.
- Concealed surfaces or elements before covering.
- Commencement of concrete placing.
- Evaluation of the off-form finishes.
- Evaluation of surface finish.

2 PRODUCTS

2.1 CONCRETE

General

Stockpile: If uniform, consistent colour is documented, stockpile sand, cement and aggregates.

Properties

Concrete mix and supply: Conform to the following:

- Normal-class: To AS 1379 clause 1.5.3.
 - . Properties: As documented in the **Concrete properties schedule - performance**.
- Special-class: To AS 1379 clause 1.5.4.
 - . Performance properties: As documented in the **Concrete properties schedule - performance**.
 - . Prescription properties: As documented in the **Concrete properties schedule - prescription**.

Aggregates

Standard: To AS 2758.1.

Cement

Standard: To AS 3972.

Age: Less than 6 months old.

Storage: Store cement bags under cover and above ground.

Supplementary cementitious materials:

- Fly ash: To AS/NZS 3582.1.
- Slag: To AS 3582.2.
- Amorphous silica: To AS/NZS 3582.3.
-

Water

Standard: To AS 1379 clause 2.4.

Requirement: Clean, free from oil, acid, alkali, organic or vegetable matter and including not more than 500 mg/l of chloride ions.

Concrete colour

Standard: To AS 3610.1.

Chemical admixtures

Standard: To AS 1478.1, used to manufacturer's recommendations.

2.2 FORMWORK

General

Form face, linings and release agents: Compatible with documented concrete surface finish and any proposed applied finishes to concrete.

Trapped forms: Free of timber or chlorides and not to impair the structural performance of the concrete members.

Void formers

Requirement: Material capable of maintaining rigidity and shape until the concrete has set, capable of withstanding construction loads and non-collapsible on absorption of moisture.

Laboratory testing: Use void formers tested under laboratory conditions for conformance with the following:

- Deflection during placing and compaction of the concrete does not exceed beam or slab span/1000.
- Additional deflection between initial set and 7 days does not exceed span/400.

Test method: Place formers on damp sand and load with a mass of wet concrete at least equal to the mass of the beams or slabs to be supported.

Profiled steel sheeting composite forms

Material: Hot-dipped zinc-coated sheet steel to AS 1397.

Minimum steel grade: G550.

Accessories: Use materials and corrosion protection compatible with the profiled steel sheeting.

Plywood forms Material: To AS 6669.

Grade: Use appropriate grade for the documented design dimensions, loading and surface quality.

Joints: Seal the joints consistent with the documented surface finish class.

Tolerances: To AS 3610.1 Section 3.

2.3 REINFORCEMENT

Fibre reinforcement

Steel fibres: To AS 3600 clause 16.7.1.

Other fibres: To CIA CPN35.

Steel reinforcement

Standard: To AS/NZS 4671.

Shape: as nominated on Structural Engineers drawings and notes.

Strength grade and relative ductility class: as nominated on Structural Engineers drawings and notes.

Surface condition: Free of loose mill scale, rust, oil, grease, mud or other material which would reduce the bond between the reinforcement and concrete.

Protective coating

Standard: To AS 3600 clause 17.2.1.2.

Requirement: For concrete elements containing protective coated reinforcement, provide the same coating type to all that element's reinforcement and embedded ferrous metal items, including tie wires, stools, spacers, stirrups, plates and ferrules, and protect other embedded metals with a suitable coating.

Epoxy coating: High-build, high solids, chemically resistant coating to AS/NZS 3750.14.

- Thickness: 200 µm minimum.

Galvanizing: To AS/NZS 4680, as follows:

- Sequence: If fabricating after galvanizing, repair damaged galvanizing and coat cut ends.
- Zinc-coating (minimum): 600 g/m².

Tie wire

General: Annealed steel 1.25 mm diameter (minimum).

External and corrosive applications: Galvanized.

Supports

Standard: To AS/NZS 2425.

2.4 MISCELLANEOUS

Polymeric film underlay

Vapour barriers and damp-proofing membranes: To AS 2870 clause 5.3.3.

Curing compounds

Liquid membrane-forming compounds: To AS 3799.

Surface modifiers

Hardeners, sealants and protectors: If documented, proprietary products conforming to the manufacturer's recommendations.

Slip resistance treatment: If documented, proprietary products conforming to the manufacturer's recommendations.

3 EXECUTION

3.1 POLYMERIC FILM UNDERLAY

Location

Requirement: Under slabs on ground, including integral ground beams and footings, provide a vapour barrier or, in areas prone to rising damp or salt attack, a damp-proofing membrane.

Base preparation

Requirement: Conform to base type, as follows:

- Concrete working base: Remove projections above the plane surface, and any loose material.
- Graded prepared subgrade: Blind with sand to create a smooth surface free from hard projections. Lightly wet the sand just before laying the underlay.

Installation

Standard: To AS 2870 clause 5.3.3.

Requirement: Lay underlay over the base, as follows:

- Lap joints at least 200 mm and seal the laps and penetrations with waterproof adhesive tape.
- Face the laps away from the direction of concrete pour.
- Continue up vertical faces past the damp-proof course where applicable, and tape fix at the top.
- Patch or seal punctures or tears before placing concrete.
- Cut back as required after concrete has gained strength and formwork has been removed.

3.2 FORMWORK

General

Requirement: As documented in the **Formed surface finishes schedule**.

Substrates

Cleaning: Before placing concrete remove free water, dust, debris and stains from the form face and the formed space.

Bolt holes

Formwork tie bolts left in the concrete: Position more than 50 mm from the finished surface.

Corners

Work above ground: Bevel with a chamfer at re-entrant angles, and a fillet at corners.

Face of bevel: 25 mm.

Embedments

Fixing: Fix embedments through formwork to prevent movement, or loss of slurry or concrete, during concrete placement.

Openings

Requirement: In vertical forms provide openings or removable panels for inspection and cleaning, at the base of columns, walls and deep beams.

Access: For thin walls and columns, provide access panels for placing concrete.

Release agents

Application: Before placing reinforcement, apply a release agent to form face and linings.

Climbing formwork

Provision for inspection: Provide access below the movable formwork, from which surface treatment and inspection may be carried out.

Profiled steel sheeting composite formwork

Fixing: If sheeting cannot be fixed to structural steel supports with puddle welds, or with welded shear studs, provide details of proposed fixings.

Steel linings

Rust: Clean off any rust and apply rust inhibiting agent prior to re-use.

Visually important surfaces

Surface finish classes 1, 2 or 3: Set out the formwork to give a regular arrangement of panels, joints, bolt holes, and similar visible elements in the formed surface.

Void formers

Protection: Keep void formers dry until use, install on a firm level surface and place reinforcement and concrete with minimum delay.

3.3 REINFORCEMENT

Dowels

Fixing: If a dowel has an unpainted half, embed that half in the concrete placed first.

Tolerances:

- Alignment: 1:150.
- Location: $\pm$ half the diameter of the dowel.

Grade: 250 N.

Cover

Concrete cover generally: To AS 3600 clause 4.10.

Concrete cover for structures for retaining liquids: To AS 3735 clause 4.4.

Concrete cover for residential ground slabs and footings: To AS 2870.

Supports

Concrete, metal or plastic supports: Provide as follows:

- Able to withstand construction and traffic loads.
- With a protective coating if they are ferrous metal, located within the concrete cover zone, or are used with galvanized or zinc-coated reinforcement.

Spacing:

- Bars: ≤ 60 bar diameter.
- Mesh: ≤ 600 mm.

Supports over membranes: Prevent damage to waterproofing membranes or vapour barriers. If appropriate, place a metal or plastic plate under each support.

Projecting reinforcement

Protection: If starter or other bars extend beyond reinforcement mats or cages, through formwork or from cast concrete, provide a plastic protective cap to each bar until it is cast into later work.

Tying

Requirement: Secure the reinforcement against displacement at intersections with either wire ties, or clips. Bend the ends of wire ties away from nearby faces of formwork or unformed faces to prevent the ties projecting into the concrete cover.

Beams: Tie stirrups to bars in each corner of each stirrup. Fix other longitudinal bars to stirrups at 1 m maximum intervals.

Bundled bars: If required, tie bundled bars in closest possible contact. Provide tie wire at least 2.5 mm diameter and spaced not more than 24 times the diameter of the smallest bar in the bundle.

Columns: Secure longitudinal column reinforcement to all fitments (or helical reinforcement) at every intersection.

Mats: For bar reinforcement in the form of a mat, secure each bar at alternate intersections.

Fibre-reinforced concrete

Steel fibres: To AS 3600 Section 16.

3.4 CONCRETE SUPPLY

Elapsed delivery time

General: Make sure that the elapsed time between the wetting of the mix and the discharge of the mix at the site is in conformance with the **Elapsed delivery time table**. Do not discharge at ambient temperature below 10°C or above 30°C unless approved heating or cooling measures are taken to deliver concrete within the range 5°C to 35°C.

Elapsed delivery time table

Concrete temperature at time of discharge (°C)	Maximum elapsed time (minutes)
5 – 24	120
24 – 27	90
27 – 30	60
30 – 35	45

Pre-mixed supply

Addition of water: To AS 1379 clause 4.2.3.

Transport method: Select to prevent segregation, loss of material and contamination of the environment, and not to adversely affect placing or compaction.

Site mixed supply

Emergencies: If mixing by hand, provide details.

Plant: Mix concrete in a plant located on the construction site.

3.5 TESTING

General

Test authority: Concrete supplier or an Accredited Testing Laboratory.

Reports and records of test results: To the relevant parts of the AS 1012 series. Keep results on site.

Assessment process of test results Standard: To AS 1379.

Method of assessment: Project assessment.

Sampling

Method of sampling: AS 1012.1.

Sampling locations: To AS 1012.1 and the following:

- Slump tests: On site, at the point of discharge from the agitator.
- Compressive strength tests: Spread the site sampling evenly throughout the pour.

Frequency of sampling: To AS 1379 Sections 5 and 6 and the following:

- Slump tests: Take at least one sample from each batch.
- Compressive strength tests: To the **Project assessment strength grade sampling table**. **Project assessment strength grade sampling table**

Number of batches for each type and grade of concrete per day	Minimum number of samples: Columns and load bearing wall elements/batch	Minimum number of samples: Other elements/day
1	1	1
2-5	1	2
6-10	1	3
11-20	1	4
each additional 10	1	1 additional

Making and curing of specimens

General: To AS 1012.8.1 and AS 1012.8.2.

Specimens for compressive strength tests: Make and cure at least two specimens from the sample of each grade.

Specimen size:

- Aggregate size $\leq$ 20 mm: Nominally 200 x 100 mm diameter.
- Aggregate size $>$ 20 mm: Nominally 300 x 150 mm diameter.

Test methods

General: To the relevant parts of the AS 1012 series.

Acceptance criteria:

- General: As documented in the **Concrete properties schedule – performance**.
- Early age compressive strength: As documented in the **Control tests schedule**.

Drying shrinkage at 56 days: To AS 1012.8.4 and AS 1012.13.

Liquid retaining structures

Testing for liquid tightness: To AS 3735.

3.6 CORES, FIXINGS AND EMBEDDED ITEMS

General

Requirement: Install fasteners to manufacturer's recommendations and the assumptions of AS 5216 Appendix D.

Adjoining elements

Fixings: Provide fixings for adjoining elements. If required, provide temporary support to the adjoining elements during concreting, to prevent movement.

Protection

General: Grease threads. Protect embedded items against damage.

Compatibility: Provide inserts, fixings and embedded items that are compatible with each other, with the reinforcement and with the documented concrete mix and surface finish.

Corrosion: In external or exposed locations, galvanize anchor bolts and embedded fixings.

Structural integrity

Position: Fix cores and embedded items to prevent movement during concrete placement. In locating cores, fixings and embedded items, displace but do not cut reinforcement, and maintain documented cover to reinforcement.

Isolation: Isolate embedded items to prevent water tracking to concrete that provides minimum cover to reinforcement.

Tolerances

General: Maximum deviation from correct positions:

- Anchor bolt groups for structural steel: To AS/NZS 5131.
- Cores and embedded items generally: 10 mm.
- Other fixing bolts: 3 mm.

3.7 CONCRETE WORKING BASE

Finish

Membrane support: Wood float finish or equivalent.

Installation

General: Lay over the base or subgrade and screed to the required level.

Surface flatness tolerance

Maximum deviation: 6 mm from a 3 m straightedge.

3.8 PLACING AND COMPACTION

Placing

Horizontal transport: Use suitable conveyors, clean chutes, troughs, hoppers or pipes.

Methods: Avoid segregation and loss of concrete, and minimise plastic settlement. Maintain a nominally vertical and plastic concrete edge during placement.

Horizontal elements: Place concrete in layers not more than 300 mm thick. Compact the following layer into previous layer before previous layer has taken initial set.

Compaction

Methods: Use immersion and screed vibrators accompanied by hand methods as appropriate to remove entrapped air and to fully compact the mix.

Vibrators: Do not allow vibrators to contact set concrete, reinforcement or items including pipes and conduits embedded in concrete. Do not use vibrators to move concrete along the formwork. Avoid causing segregation by over-vibration.

Placing records

Log book: Keep on site and make available for inspection a log book recording each placement of concrete, including the following:

- Date.
- Specified grade and source of concrete.
- Slump measurements.
- The portion of work.
- Volume placed.

Rain

Protection: During placement and before setting, protect the surface from damage.

Time between adjacent placements

Minimum time delay: As documented in the **Minimum time delay schedule**.

Vertical elements

Placement: Limit the free fall of concrete to maximum of 2000 mm.

Placing in cold weather

Cement: Do not use high alumina cement.

Temperature limits: Maintain the following:

- Freshly mixed concrete: $\geq 5^{\circ}\text{C}$.
- Forms and reinforcement before and during placing: $\geq 5^{\circ}\text{C}$.
- Water: Maximum 60°C when placed in the mixer.

High early strength cement: If deteriorating weather conditions are predicted, use high early strength cement.

Temperature control: Heat the concrete materials, other than cement, to the minimum temperature necessary so that the temperature of the placed concrete is $\geq 5^{\circ}\text{C}$.

Admixtures: Do not use calcium chloride, salts, chemicals or other material in the mix to lower the freezing point of the concrete.

Frozen materials: Do not allow frozen materials or materials containing ice to enter the mixer, and keep free of frost and ice any forms, materials, and equipment coming in contact with the concrete.

Freezing: Prevent concrete from freezing.

Placing in hot weather

Handling: Prevent premature stiffening of the fresh mix and reduce water absorption and evaporation losses.

Temperature limits: Maintain the following:

- Normal concrete in footings, beams, columns, walls and slabs: $\leq 35^{\circ}\text{C}$. - For concrete strength grade less than 40 MPa with section thickness ≥ 1 m in all dimensions:
 $\leq 27^{\circ}\text{C}$. - For concrete strength grade 40 MPa or greater with section thickness ≥ 600 mm in all dimensions:
 $\leq 27^{\circ}\text{C}$.
- Forms and reinforcement before and during placing: $\leq 35^{\circ}\text{C}$.

Evaporation control barriers: Erect barriers to protect freshly placed concrete from drying winds.

Temperature control: Select one or more of the following methods of maintaining the temperature of the placed concrete at 35°C or less:

- Cool the concrete using liquid nitrogen injection before placing.

- Cover horizontal transport containers.
- Spray the coarse aggregate using cold water before mixing.
- Use chilled mixing water.

Placing under water

General: Do not place under water unless conditions prevent dewatering.

Minimum cement content for the mix: Increase by 25%.

3.9 JOINTS

Construction joints

Location: Do not relocate or eliminate construction joints, or form undocumented construction joints. If emergency construction joints are made necessary by unforeseen interruptions to the concrete pour, submit a report on the action taken.

Finish: Butt join the surfaces of adjoining pours. In visually important surfaces make the joint straight and true, and free from blemishes impermissible for its surface finish class.

Preparation: Roughen and clean the hardened concrete joint surface. Remove loose or soft material, free water, foreign matter and laitance. Dampen the surface just before placing the fresh concrete and coat with a neat cement slurry.

Expansion joints

Joint filling: Fill with jointing materials as documented. Finish visible jointing material neatly, flush with adjoining surfaces.

Preparation: Before filling, dry and clean the joint surfaces, and prime.

Watertightness: Apply the jointing material so that joints subject to ingress of water are made watertight.

Jointing materials: Provide jointing materials compatible with each other, and non-staining to concrete in visible locations.

Bond breaking: Provide back-up materials for sealants, including backing rods, which do not adhere to the sealant.

Foamed materials (in compressible fillers): Closed-cell or impregnated, not water absorbing.

Slip joints

Requirement: If concrete slabs are supported on masonry, provide proprietary slip joints.

3.10 FORMED SURFACES

General

Surface finish: As documented in the **Surface finish class schedule** and the **Formed surface finishes schedule**.

Damage: Do not damage concrete works through premature removal of formwork.

Curing

Requirement: If formwork is stripped before the minimum curing period for the concrete has elapsed, continue curing the exposed faces as soon as the stripping is completed.

Evaluation of formed surfaces

General: If evaluation of formed surface is required, complete the evaluation before surface treatment.

Finishing methods

Requirement: If soffits of horizontal concrete elements or faces of vertical concrete elements are to have a finish other than an off-form finish, provide finishes as documented.

Form removal: If vertical face formwork needs to be removed for finishing methods, while the concrete is green, make sure the concrete has sufficiently set to prevent slump.

Blasted finishes:

- Abrasive: Blast the cured surface using hard, sharp graded abrasive particles until the coarse aggregate is in uniform relief.
- Light abrasive: Blast the cured surface using hard, sharp graded abrasive particles to provide a uniform matt finish without exposing the coarse aggregate.

Bush hammered finish: Remove the minimum matrix using bush hammering to expose the coarse aggregate, recessing the matrix no deeper than half the aggregate size, to give a uniform texture.

Exposed aggregate finish: While the concrete is green, wet the surface and scrub with stiff fibre or wire brushes, flushing continuously with clean water, until the aggregate is uniformly exposed. Do not use acid etching. Rinse the surface with water.

3.11 CURING

General

Requirements: Taking into account the average ambient temperature at site over the relevant period affecting the curing, adopt procedures to make sure of the following:

- Curing: Cure continuously from completion of finishing, when the concrete has set sufficiently not to be damaged by the curing process, until the minimum total cumulative number of days or fractions of days, during which the air temperature in contact with the concrete is above 10°C, conforms to the following, unless accelerated curing is adopted:
 - . Fully enclosed internal surfaces/Early age strength concrete: 3 days.
 - . Other concrete surfaces: 7 days.
- End of curing period: Prevent rapid drying out at the end of the curing period.
- Protection: Maintain at a reasonably constant temperature with minimum moisture loss, during the curing period.

Curing method: as per structural engineers recommendations.

Curing compounds

Liquid membrane-forming compounds: Provide a uniform continuous flexible coating without visible breaks or pinholes, which remains unbroken for at least the required curing period after application.

Substrates: Do not use wax-based or chlorinated rubber-based curing compounds on surfaces forming substrates to applied finishes, concrete toppings and cement-based render.

Self-levelling toppings: If used also as curing compounds, conform to AS 3799.

Visually important surfaces: Apply curing compounds to produce uniform colour on adjacent surfaces.

Cold weather curing

Temperature: Maintain concrete surface temperatures above 5°C for the duration of the curing period.

Hot weather curing

Requirement: If the concrete temperature exceeds 25°C, or the ambient shade temperature exceeds 30°C, protect from drying winds and sun by using an evaporative retarder until curing is commenced.

Water curing

Method: Select a method of ponding or continuously sprinkling to prevent damage to the concrete surface during the required curing period.

3.12 COMPLETION

Formwork removal

Extent: Remove formwork, other than permanent forms and trapped forms, including formwork in concealed locations.

Timing: Do not disturb formwork until concrete has reached sufficient hardness to withstand formwork movements and removal without damage.

Stripping:

- General: To AS 3600 where it is more stringent than AS 3610.1.
- Vertical formwork: To AS 3610.1 Appendix C Table C2.
- Multi-storey work: Remove formwork without disturbing props supporting succeeding floors.
- Post-tensioned concrete: Remove formwork supporting post-tensioned concrete members to AS 3600 clause 17.6.2.7.

Removable bolts: Remove tie bolts without damaging the concrete.

Bolt hole filling: Provide material with durability and colour matching the concrete.

Recessed filling: Fill or plug the hole to 6 mm below the finished surface.

Curing: If formwork is stripped before the minimum curing period for the concrete has elapsed, continue curing the exposed faces as soon as the stripping is completed.

Protection

General: Protect the concrete from damage due to construction loads, physical and thermal shocks, and excessive vibrations, particularly during the curing period.

Surface protection: Protect finished concrete surfaces and applied finishes from damage.

Completion tests

Slip resistance of completed installation: To AS 4663.

4 SELECTIONS

4.1 CONCRETE SCHEDULES

Concrete schedules

Code / Property		
Type	Standard grey broom finished concrete	
Location	Refer to drawings	
Finish	Broom swept	
Sealer	CCS streetscape penetrating sealer or approved equal. 2 coats minimum	

0334 BLOCK CONSTRUCTION

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide block construction, as Documented. Refer to Design Drawings, and Structural Engineer's Documents.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 *General requirements.*

1.3 STANDARD

General

Materials and construction: To AS 3700 (2018).

1.4 INTERPRETATION

Definitions

General: For the purposes of this worksection, the definitions in AS 3700 (2018), AS/NZS 4455.1 (2008) and the following apply:

- Facework: Masonry intended to be exposed in a wall.
- Special mortar: Mortar with compositions not covered in AS 3700 (2018) Table 11.1.

1.5 TOLERANCES

General

Standard: To AS 3700 (2018) clause 12.5 and Table 12.1.

1.6 SUBMISSIONS

Products and materials

Type Tests: Submit Test results of the following:

- Characteristic unconfined compressive strength of masonry unit: To MATERIALS, Masonry units. Samples

Requirement: Submit samples to PRODUCTS, GENERAL, Samples.

Tests

Site Tests: Submit results for special mortar and masonry as follows:

- Characteristic compressive strength.
- Characteristic flexural tensile strength.
- Scratch index for mortar joints.

Warranties

Requirement: Submit warranties to COMPLETION, Warranties.

1.7 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- Set-out.
- Structural steelwork, including bolts and shelf angles, in position.
- Unit type, colour and texture.
- Bottoms of cavities, after cleaning out.
- Bottoms of core holes, before grouting.
- Reinforcement type and diameter.
- Positioning of reinforcement before grouting.

2 PRODUCTS

2.1 GENERAL

Samples

Requirement: Provide samples, as follows:

- Each type of face unit, showing the range of variation available, including colour, texture, surface irregularities, Defective arrises, and shape.

. Number of each type: 6.

- Each type of sealant, showing the finished colour.

. Quantity: Minimum two of each colour.

2.2 DURABILITY

General

Exposure environment: Moderate

Exposure locations: To AS 3700 (2018) clause 5.4.

2.3 MATERIALS

Masonry units

Selections: As Documented. Refer to Appendix B – Façade and Wall Schedule, and Structural Engineer's Documents.

Standard: To AS/NZS 4455.1 (2008) and AS/NZS 4455.3 (2008).

Salt attack resistance grade: To AS 3700 (2018) Table 5.1.

Characteristic unconfined compressive strength: To AS/NZS 4456.4 (2003).

Mortar materials

Mortar class: To AS 3700 (2018) Table 5.1.

Cement: To AS 3972 (2010).

White cement: With not more than 1% iron salts content. Lime: To AS 1672.1 (1997).

Sand: Fine aggregate with a low clay content, free from efflorescing salts and deleterious matter, selected for colour and grading.

Water: Clean and free from any deleterious matter. Admixtures: To AS 3700 (2018) clause 11.4.2.4.

Pigment: To EN 12878 (2014), and as follows:

- Integral pigment mix proportion: Not more than 10% by weight of cement.

Masonry cement mortar mix proportions table (cement:lime:sand), by volume

Mortar class to AS 3700 (2018)	Clay	Concrete	Calcium silicate	Water thickener
M3	1:0:4	1:0:4	N/A	Yes
M4	1:0:3	N/A	N/A	Yes

Cement (GP/GB) mortar mix proportions table (cement:lime:sand), by volume

Mortar class to AS 3700 (2018)	Clay	Concrete	Calcium silicate	Water thickener
M2	1:2:9	N/A	N/A	No
M3	1:1:6	1:1:6	N/A	Optional
M3	1:0:5	1:0:5	1:0:5	Yes
M4	1:0.5:4.5	1:0.5:4.5	N/A	Optional
M4	1:0:4	1:0:4	1:0:4	Yes
M4	1:0-0.25:3	1:0-0.25:3	N/A	Optional

Grout

Standard: To AS 3700 (2018) clause 11.7. Maximum aggregate size: 12 mm.

Minimum cement content: 300 kg/m³.

Characteristic compressive strength: Minimum 12 MPa. Nominal slump: 200 mm.

2.4 BUILT-IN COMPONENTS

General

Durability class of built-in components: To AS 3700 (2018) Table 5.1.

Reinforcement

Standard: To AS/NZS 4671 (2019).

Corrosion protection: To AS 3700 (2018) clause 5.9.

Minimum cover: To AS 3700 (2018) Table 5.1.

Connectors and accessories

Standard: To AS 2699.2 (2020).

Corrosion protection: To AS 2699.2 (2020).

3 EXECUTION

3.1 GENERAL

Mortar mixing

General: Measure volumes accurately to the Documented proportions. Machine mix for at least six minutes. If the initial set of the cement has taken place, discard the mortar. Do not retemper.

Storage and handling

Masonry units: Store above the surface of the ground and cover to prevent entry of rainwater and contaminants. Locate away from surface and ground water runoff.

Mortar materials: Protect from contamination and as follows:

- Sand: Store away from surface and ground water runoff and allow for free drainage of rainwater.
- Cement and lime: Store bags in a dry, under cover and above ground environment.

Bond

Type: Stretcher bond.

Mortar joints

General: Set out masonry with joints of uniform width and minimum cutting of masonry units. Solid and cored units: Lay on a full bed of mortar. Fill perpends solid. Cut mortar flush.

Hollow units: Face-shell bedded. Fill perpends solid. Cut mortar flush. Joint thickness: 10 mm.

Finish: Conform to the following:

- Externally: Tool to give a dense water-shedding finish.
- Internally: If wall is to be plastered, do not rake more than 10 mm to give a key.

Rate of construction

General: Regulate the rate of construction to eliminate joint deformation, slumping or instability.

Rods

Set-out: Construct masonry to the following rods:

- 190 mm high units: 3 courses to 600 mm.

Temporary support

General: If the final stability of the masonry is dependent on construction of (structural) elements after the masonry is completed, provide proposals for temporary support or bracing.

3.2 FACEWORK

Cleaning

General: Clean progressively as the WUC proceeds to remove mortar smears, stains and discolouration. Do not erode joints if using pressure spraying.

Colour mixing

Distribution: In facework, distribute the colour range of units evenly to prevent colour concentrations and banding.

Below ground

Facework: Commence facework at least one full course, below the adjacent finished surface level.

3.3 CONTROL JOINTS

General

Location and spacing: Provide control joints to AS 3700 (2018) clause 4.8.

Control joint filling

Filler material: Provide compatible sealant and bond breaking backing materials that are non-staining to masonry. Do not use bituminous materials with absorbent masonry units.

- Bond breaking materials: Non-adhesive to sealant, or faced with a non-adhering material.
- Foamed materials: Closed cell or impregnated, not water-absorbing.

Installation: Clean the joints thoroughly and insert an easily compressible backing material before filling with a gun-applied flexible sealant.

Sealant joint depth to width ratio (depth:width): 1:2. Minimum sealant depth: 6 mm.

3.4 BED JOINT REINFORCEMENT

Location

Stack bonded masonry: Conform to AS 3700 (2018) clause 4.12 and the following:

- Spaced vertically at centres not exceeding six times the thickness of the stack bonded leaf.
- In the first bed joint above or below an unrestrained horizontal edge of the masonry.
- One bed joint minimum, within 300 mm above or below a horizontal line of lateral support.

Installation

General: Lap 450 mm at splices. Fold and bend at corners so that the longitudinal wires are continuous. Stop 50 mm short of control joints. Extend 450 mm beyond each side of openings.

Reinforcement

Material: Galvanized welded wire mesh.

Width: Equal to the width of the leaf, less 15 mm cover from each exposed surface of the mortar joint.

3.5 REINFORCED AND GROUTED BLOCKWORK

Reinforcement

Cover: Maintain cover to vertical and horizontal steel reinforcement using plastic clips or wheels, as appropriate.

Vertical reinforcement: Tie vertical steel reinforcement to the starter bars through cleanout holes in each reinforced hollow masonry unit and fix in position at the top of the wall with plastic clips.

Horizontal: Lay horizontal steel reinforcement in contact with rebated webs. Hold in position using plastic clips if vertical steel is subsequently positioned to wall construction.

Cleaning core holes

General: Provide purpose-made cleanout blocks or machine cut a cleaning hole at the base of each grouted core.

Location: Locate on the side of the wall that is to be rendered or otherwise concealed.

Cleaning: Rod cores to dislodge mortar fins protruding from the blocks and mortar droppings from reinforcement. Remove through the cleanout blocks.

Grouting

Commencement: Do not commence until grout spaces have been cleaned out and the mortar joints have attained sufficient strength to resist blow-outs.

Height of lift: Limit the height of individual lifts in any pour to make sure that the grout can be thoroughly compacted to fill all voids.

Compaction: Compact by vibration or by rodding.

Topping up: On the completion of the last lift, top up the grout after 10 minutes and within 30 minutes, and vibrate or rod to mix with the previous pour.

3.6 TESTING

Special mortar

Durability: Scratch index Test to AS 3700 (2018) Appendix E. Compressive strength: To AS 3700 (2018) strength: To AS 3700 (2018)

3.7 COMPLETION

Warranties

Type: Interlocking material and installation warranty. Period: As provided by manufacturer and installer.

4 SELECTIONS

Refer to Appendix

0341 STRUCTURAL STEELWORK

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide structural steelwork, as documented. Refer to structural engineering drawings.

Performance

Construction category to AS/NZS 5131: as per structural engineers notes & recommendations.

Adjoining elements: Provide for the fixing of adjoining building elements that are to be connected to or supported on the structural steel.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.
- Structural Engineers drawings.
-

1.3 STANDARDS

General

Materials and design: To AS 4100.

Materials and design of cold-formed decking, purlins and girts: To AS/NZS 4600.

Composite steel-concrete construction including profiled steel sheeting and shear connectors: To AS/NZS 2327.

Fabrication and erection: To AS/NZS 5131.

1.4 INTERPRETATION

Abbreviations

General: For the purposes of this work section the following abbreviations apply:

- AECS: Architecturally Exposed Structural Steelwork.
- CC: Construction Category.
- NDE: Non-Destructive Examination.
-

Definitions

General: For the purposes of this work section the definitions given in AS/NZS 5131 apply.

1.5 TOLERANCES

General

Requirement: To AS/NZS 5131 Section 12 and Appendix F.

Tolerance class: 1.

AECS: To the AECS schedule.

1.6 SUBMISSIONS

Fabrication details

Distortions: Submit proposals for the following:

- Preventing or minimising distortion of galvanized components, welded components or welded and galvanized components.
- Restoration to the designed shape.

Identification marks: If members and/or connections will be exposed to view, submit details of proposed marking.

Program: Submit a fabrication program showing the proposed sequence of operations and time required.

Execution details

Anchor bolts: If anchor bolts do not meet documented location tolerances, submit proposals for rectification before proceeding.

Bolting connections: For connections not documented, submit proposals.

Bolt tensioning procedure: Submit details of procedure, equipment to be used and calibration of the process.

Site base plate holing: If hand cutting of bolt holes in column base plates is required, submit details.

Purlins and girts: If purlins and girts support components other than roofing or cladding, submit details.

Site modifications: Submit details of proposed on-site modifications or rectifications to any steel member, connection component, mechanical fastener, weld or corrosion protection.

Splices: If variations to documented splice locations or additional splices are proposed, submit details.

Temporary connections or attachments: If not documented, submit details.

Undocumented weld types: Submit proposals for weld type and electrodes.

Welding plan: Submit a welding plan to AS/NZS 5131 clause 7.2.

Work method statement: Before any erection work commences, submit a work method statement to AS/NZS 5131 clause 11.2.3.

Products and materials

Steel members and sections: Submit test reports or test certificates conforming to AS 4100 clause 2.2.2.

Bolts, nuts and washers: Submit test reports or test certificates conforming to AS/NZS 1252.1 Section 6.

Verification testing of bolt assemblies: Submit test reports or certificates conforming to AS/NZS 1252.2 Section 2, together with the Supplier Declaration of Conformity (SDoC).

Anchor bolts: If anchors, other than those documented, are required or proposed for supporting or fixing structural steel, submit evidence of the anchor capacity to carry the load.

Substitution: If alternative sections or connections are proposed, submit details.

Records

Survey: Submit survey of erected structural steel to verify components have been installed as documented.

Drawings: Submit as-built structural drawings, upon completion.

Shop detail documentation

General: Submit shop detail documentation to a scale that best describes the detail, conforming to AS/NZS 5131 clause 4.4.

Drawing format: 2D PDF's & 3D CAD Drawings, IFC Model, Electronic copy

Review of shop detail documentation: Structural Engineer – Patcol Group

Subcontractors

General: Submit names and contact details of proposed fabricator, detailer and installer.

Responsibilities: Submit names and contact details corresponding to the person/organisation assigned responsibility to the items listed in AS/NZS 5131 Table B3.

Tests

Site tests: Submit results, as follows:

- Bars and sections: Non-destructive tests.
- Plates: Ultrasonic tests.
- Welds: Non-destructive examinations.

1.7 INSPECTION

Notice – on-site

Inspection: Give notice so that inspection may be made of the following:

- Steelwork on-site before erection.
- Anchor bolts in position before casting in.
- Steelwork and column bases erected on site, before grouting, encasing, site protective coating or cladding.
- Tensioning of bolts in categories 8.8/TB, 8.8/TF, 10.9/TB and 10.9/TF.
- Reinforcement and formwork in place before any encasement.
- Completed grouting, encasement, fire protection or site applied protective coating.

2 PRODUCTS

2.1 GENERAL

Materials

Requirement: To AS/NZS 5131 Section 5.

Storage and handling

Requirement: Pack, support, transport and handle members and components without overstressing, deforming or damaging them or their protective coating.

Damaged items: Rectify or replace. Do not assemble into the structure without approval.

Protection: Wrap or otherwise protect members or components to prevent damage to surface finishes during handling and erection.

Storage: Store off the ground.

Lifting points: Do not allow steel slings to come into direct contact with coated steelwork.

Purchasing and traceability

Purchasing documentation and procedure: To AS/NZS 5131 clause 4.6.

Level of traceability: To AS/NZS 5131 clause 5.2.3 and the types defined in AS/NZS 5131 clause 4.7.

2.2 STRUCTURAL STEEL

Steel members and sections steel grade table

Type of steel	Minimum grade
Hot-rolled sections to AS/NZS 3679.1 and SA TS 102	300
Welded sections to AS/NZS 3679.2	300
Hot-rolled plates, floor plates and slabs to AS/NZS 3678 and SA TS 102	250
Hot-rolled flat products to AS/NZS 1594	HA300
Hollow sections to AS/NZS 1163 and SA TS 102: Circular sections less than 166 mm nominal outside diameter	C250
Hollow sections to AS/NZS 1163 and SA TS 102: Sections other than circular sections less than 166 mm nominal outside diameter	C350
Cold-formed purlins and girts to AS 1397	G450, Z350 or Z450

Certification

Steel: Minimum requirements for test and inspection certificates, to the following:

- Hot-rolled bars or sections: To AS/NZS 3679.1 clause 11.2.4.
- Welded I sections: To AS/NZS 3679.2 clause 11.2.4.
- Hot-rolled plate: To AS/NZS 3678 clause 11.2.4.
- Cold-formed hollow sections: To AS/NZS 1163 clause 11.2.4.

2.3 MECHANICAL FASTENERS

Standards

Bolts: To AS 1110.1, AS 1111.1 and AS/NZS 1252.1.

Nuts: To AS 1112.1, AS 1112.2, AS 1112.3, AS 1112.4 and AS/NZS 1252.1.

Bolting category

Requirement: To the **Bolting category schedule**.

Certification

High-strength bolt assemblies: Minimum requirements for test reports, to AS/NZS 1252.1 clause 6.4.2.

Finish

Bolts, nuts and washers: Hot-dip galvanized to AS/NZS 1214, corrosion-free, and in serviceable condition.

Anchor bolts

Hexagonal bolts: To AS 1111.1.

Hexagonal nuts: To AS 1112.3.

Plain washers: To AS 1237.1.

Requirement: Provide each anchor bolt with 2 nuts and 2 oversize washers with sufficient thread for the levelling nut and washer to set below the base plate.

Mechanical and chemical anchors: To AS 5216, installed to manufacturer's recommendations.

2.4 OTHER MATERIALS

Welding consumables

Requirement: To the relevant part of the AS/NZS 1554 series.

Studs and shear connectors

Requirement: To AS/NZS 5131 clause 5.6.

Grout

Requirement: To AS/NZS 5131 clause 5.8.

Type: non shrink cement based grout under all base plates as per structural engineers notes.

3 EXECUTION

3.1 PREPARATION, ASSEMBLY AND FABRICATION

Identification

Traceability: To AS/NZS 5131 clause 5.2.3.

Marking: Provide marks or other means of identifying each member compatible with the finish, for setting out, locating, erecting and connecting the steelwork to the marking plans.

High-strength bolting: If the work includes more than one bolting category, mark high-strength structural bolted connections with a 75 mm wide flash of colour, clear of holes.

Cold-formed members: Clearly mark material thickness.

Monorail beams: Identify and mark rated capacity in conformance with AS 1418.18 clause 5.12.6.

Natural beam camber

General: If steel beams have a natural camber, within the straightness tolerance, fabricate the steelwork element with the camber up.

Cutting

Shearing: Do not shear edges in areas designated as yielding regions for seismic design to AS 4100 earthquake design categories D and E.

Shaping

Requirement: Where forming, shaping or correcting distorted members, avoid damage and conform to AS/NZS 5131 clause 6.6.

Holing

Slotted holes: Do not use slotted holes for connections, other than those documented.

Tolerances

Measurement: Check tolerances by measurement after fabrication and application of corrosion protection.

3.2 WELDING

General

Requirements: To AS/NZS 5131 Section 7.

Standard: To AS/NZS 1554.1.

Weld category

Weld categories not documented: Category GP.

Weld type

Weld type not documented: Submit proposals for weld type and electrodes.

Non-destructive weld examination (NDE)

Requirement: To AS/NZS 5131 clause 13.6.2.

Non-visual NDE: By a third party testing authority.

Repairs: Repair welds revealed as faulty by NDE and repeat the examination.

Site welds

Completion: Weld only when correct alignment and preset or camber have been achieved.

3.3 MECHANICAL FASTENING

Connection contact surfaces

General: To AS/NZS 5131 clause 8.4.1.

Bolting categories 8.8/TF and 10.9/TF: Clean, as rolled and free from applied finishes.

Washers

Requirement: Place one washer under the part rotated during tightening process (nut or bolt head).

Method of tensioning TB and TF bolting categories

8.8/TB and 8.8/TF: Use part-turn method or a direct tension indicator device.

10.9/TB and 10.9/TF: Use a direct tension indicator device.

Permanent bolting

Completion: Bolt only when correct alignment and preset or camber has been achieved.

3.4 SURFACE PREPARATION AND TREATMENT

General

Requirement: Conform to *0345 Steel – protective paint coatings*, as appropriate. Also refer painting section.

3.5 ARCHITECTURALLY EXPOSED STRUCTURAL STEELWORK

General

Requirement: Provide AESS to AS/NZS 5131 Section 10 and the **AESS schedule**.

Fabrication

Additional requirements: To AS/NZS 5131 clause 10.4.

Corners and edges: Grind smooth sharp, marred, or roughened corners and edges.

Rough surfaces: Deburr and ground smooth.

Erection

Additional requirements: To AS/NZS 5131 clause 10.5.

3.6 ERECTION

General

Execution: Make sure every part of the structure has sufficient design capacity and is stable under construction loads produced by the construction procedure.

Temporary work

General: Provide all necessary temporary bracing or propping.

Temporary connections: Detail required cleats, if not shown on shop detail documentation.

Temporary members: If temporary members are required, fix so as not to weaken or deface permanent steelwork.

Anchor bolts

General: For each group of anchor bolts, provide a template with set-out lines clearly marked for positioning the bolts when casting in.

Beam camber

Requirement: If beam elements have a camber (natural or induced), erect them with the camber up.

Site work

General: Other than work shown on the shop detail documentation as site work, do not fabricate, modify or weld structural steel on-site.

Purlins

Trimming members: Provide to support edges of roof sheeting along hips, valleys and roof penetrations.

Movements

General: Allow for thermal movements during erection.

Grouting at supports

Preparation: Before grouting steelwork supported by concrete or masonry, set steelwork on packing or wedges.

- Permanent packing or wedges: Form with solid steel or grout of similar strength to the permanent grout.
- Temporary packing or wedges: Remove before completion of grouting.

Timing: Grout at supports before constructing supported floors, walls and roofing.

Temperature: Do not grout if the temperature of the base plate or the footing surface exceeds 35°C.

Drifting

Limitation: Use drifting only to bring members into position, without enlarging holes or distorting components.

3.7 REPAIRS

General

Requirement: Repair finishes to restore the full integrity of any coating.

3.8 COMPLETION

Tolerances

Conformance: After completing erection, verify conformance with AS/NZS 5131 Section 12 and Appendix F.

Temporary connections

General: Remove temporary cleats on completion and restore the surface.

0342 LIGHT STEEL FRAMING

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide light steel floor, wall, roof and truss framing, as documented. Floor, external wall, internal wall, and ceiling framing, as Documented. Refer to Design Drawings, and Structural Engineer's Documents.

Performance

Requirements:

Designer: RPEQ registered engineer.

- Suitable for having flooring, linings, cladding and roofing fixed to it.
- Conforming to the documented performance criteria.
- Conforming to the requirements of NASH-1 or NASH-2.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.

1.3 STANDARDS

General

Design, materials and protection: To AS/NZS 4600.

Residential and Low-rise steel framing: To NASH-1 (National Association of Steel Housing) and NASH-2.

1.4 INTERPRETATION

Definitions

General: For the purposes of this work section the definitions given in the NASH-1 and NASH-2 Standards apply.

1.5 TOLERANCES

General

Manufacturing, assembly and installation tolerances: To NASH-1 Appendix D and NASH-2 Appendix A.

1.6 SUBMISSIONS

Certification

Erected frame: Submit certification that the erected frame conforms to the documented project requirements.

Design documentation

General: Where the structural documentation defines performance criteria, submit, as follows:

- Design to AS/NZS 4600 or NASH-1: Independent design, documentation and certification from a professional engineer.
- To NASH-2: Certification of conformance to the requirements of NASH-2.

Reactions: Submit the location and magnitude of reactions that are to be accommodated by the support structure.

Floor and wall frame member sizes: Submit a schedule of proposed member sizes, certified as meeting stated project, and AS/NZS 4600 or NASH-2 requirements for span, spacings and loadings.

Warranties

Requirement: Submit warranties to **COMPLETION, Warranties**.

1.7 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- Damp-proof course installed before installation of steel framing.
- Steel framing erected on site before lining or cladding.

2 PRODUCTS

2.1 GENERAL

Storage and handling

Requirement: Transport all components to site and store, if required, so that components or their coating are not damaged or distorted.

Frames and trusses: If required, store on a flat even surface and do not load with other items.

Exposure: Minimise exposure of components to the weather, both during storage, handling and after erection.

2.2 COMPONENTS

Fascias and barge boards

Product: colorbond

Profile: 0.5 gauge purpose made colorbond formed fascia where indicated. For deeper fascia provide 0.5 gauge colorbond fabricated to suit appropriate depths.

Finish: colorbond

Colour: To match existing.

3 EXECUTION

3.1 GENERAL

Frame fabrication

Length: Cut members accurately to length so that they fit firmly against abutting members.

Service holes: If not pre-punched, form holes by drilling or punching, without compromising the structural integrity of the frame, located centrally within the centre third span of the element, conforming to the requirements of NASH-2.

Swarf: Immediately remove swarf and other debris from cold-formed steel framing.

Fastening

Prefabricated framing: Fasten framing elements using fasteners, as documented, to the fabricator's requirements.

Framing built in-situ: Use fasteners, as documented, from the following types:

- Bolting.
- Self-drilling, self-tapping screws.
- Blind rivets.
- Proprietary clinching system.
- Structural adhesives.
- Welding. On-site welded connections are not permitted.

Compatibility: Compatible with steel frame to prevent galvanic corrosion of dissimilar metals.

Welding

Burning: Avoid procedures that result in greater than localised burning of the sheets or framing members.

Prefabricated frames

General: Protect frames from damage or distortion during erection.

Unseasoned or CCA treated timber

General: Do not fix in contact with framing without fully painting the timber and/or the steel.

Earthing

Permanent earthing to AS/NZS 3000: Required.

Temporary earthing to AS/NZS 3000: Provide temporary earthing during erection until the permanent earthing is installed.

Protection

General: Restore coatings which have been damaged by welding or other causes. Thoroughly clean affected areas back to base metal and coat with a zinc rich organic primer.

Metal separation: Install lagging to separate non-ferrous service pipes and accessories from the framing.

Grommets: Provide grommets to isolate piping and wiring from cold-formed steel framing.

Site cut holes: Provide plastic bushes or grommets to site cut holes.

3.2 WALL FRAMING

Wall studs

General: Provide studs in single lengths without splices. Place a stud and a stiffened top plate under each structural load point from the roof or ceiling (except at openings). Provide multiple studs at points of concentrated load.

Maximum stud spacing: 600 mm.

Heads to openings

Requirement: Provide lintels appropriate to load and span.

Additional support

General: Provide additional support in the form of noggings, trimmers and studs for support and fixing of lining, cladding, hardware, accessories, fixtures and fittings.

Vermin barriers

Brick veneer barrier: Close nail steel wire mesh, with a maximum aperture of 10 mm, to the underside of the bottom plate of external stud walls, extending across the cavity for building into brickwork.

Damp-proof course

Requirement: Provide damp-proof courses under the bottom plate of stud walls built off slabs or masonry dwarf walls, as documented or as follows if not documented otherwise:

- External walls (not masonry veneer): Turn up a minimum of 75 mm on the inside and tack to studs. Project 10 mm beyond the external slab edge or dwarf wall and turn down at 45°.
- Walls of bathrooms, shower rooms and laundries: Turn up a minimum of 150 mm on the wet side and tack to studs.

Installation: Lay in long lengths. Lap full width at angles and intersections and at least 150 mm at joints.

Junctions: Preserve continuity of damp-proofing at junctions of damp-proof courses, sarking and waterproof membranes.

Flashings

Location: Provide flashings to external openings sufficient to prevent the entry of moisture. Form trays at the ends of sill flashings.

Masonry veneer construction: Extend across cavities and build into brickwork.

3.3 ROOF FRAMING

Beam framing

General: Construct framing for flat or pitched roofs where the ceiling follows the roof line, consisting of prefabricated roof beams, rafters or purlins supporting both ceiling and roof covering.

Battens

Requirement: Supply and fix battens suitable for span, spacing and proposed roofing material.

Anti-ponding boards Standard: To AS 4200.2.

3.4 ROOF TRIM

Fascia, valley and barge boards

Requirement: Fix fascia, valley gutter boards and barge boards in conformance with the manufacturer's recommendations. Colorbond finish

3.5 COMPLETION

Cleaning

General: On completion of framing remove debris from any gaps between members and make sure void between bottom chord of roof trusses and top of any non-supporting internal wall is clear.

Warranties

Requirement: Cover materials and workmanship in the terms of the warranty in the form of interlocking warranties from the supplier and the installer.

- Form: Against failure of materials and execution under normal environment and use conditions.

0345 STEEL – PROTECTIVE PAINT COATINGS

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide protective paint coatings for the protection of steel products and structural steelwork against interior and exterior atmospheric corrosion, as documented.

Performance

Requirement: Control atmospheric corrosion to structural steelwork and steel products until the first scheduled maintenance.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements.*
- 0341 *Structural steelwork*
- 0671 *Painting*

1.3 STANDARDS General

Surface preparation and coating: To AS/NZS 5131 Section 9 and the recommendations of AS 2312.1.

1.4 INTERPRETATION Abbreviations

General: For the purposes of this work section the following abbreviations apply:

- DFT: Dry Film Thickness.
- ITP: Inspection and Test Plan.
- MIO: Micaceous Iron Oxide.
- PDS: Product Data Sheet.
- SDS: Safety Data Sheet.

Definitions

General: For the purposes of this work section the definitions given in AS/NZS 2310 and the following apply:

- Coating contractor: The protective coatings application contractor conducting the on- or off-site coating application works.
- Coating manufacturer: The supplier and/or manufacturer of the protective coating materials used.
- Inspection and test plans (ITP): A series of formal inspection and test plans, prepared by the coating contractor to reflect the specific inspection and testing that will be carried out on the surface preparation, coating application and the record keeping tasks to be undertaken.

1.5 SUBMISSIONS Execution details

Detailing features: If design and fabrication features of the items to be coated may lead to difficulties, identify these and submit details for improvement.

Repair of damaged coating: If the protective coating is damaged, submit a coating repair proposal, based on the coating manufacturer's recommendations for reinstating the corrosion protection function of the system.

Reinstatement: If final coat varies from the submitted sample, submit proposals for reinstatement of the visible final coating system.

Maintenance painting

Existing steelwork: Identify, itemise and submit details of areas of corrosion, damage and other degradation.

Recoating systems: Submit details of coating systems for maintenance painting of previously coated items and structural elements, including surface preparation.

Products and materials

Multi-component coatings: If partial mixing of packs is proposed, submit details.

Quality

ITPs: Submit for each proposed coating system.

Quality supervisor: Submit the name and record of experience of the person responsible for the implementation of the ITPs.

Records

General: Prepare and maintain records of all surface preparation and coating application works, as follows:

- Standards: To AS 3894.10, AS 3894.11, AS 3894.12, AS 3894.13 and AS 3894.14.
- Reference the relevant parts of the ITP and record conformance.
-

Subcontractors

General: Submit names and contact details of proposed suppliers and applicators.

Requirement: Submit proof of currency of the applicator's environmental operating licence.

Substrate acceptance: Submit evidence of applicator's acceptance of the coating substrate before starting installation.

Warranties

General: Submit details of the proposed warranty terms, form and period.

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Items after fabrication, before commencing surface cleaning and preparation.
- Surfaces after preparation, before application of first coating.
- Coating stages:
 - . After application of primer or seal coats.
 - . After application of each subsequent coat.
- Repair of coating damage: Exposure of corrosion pitting or significant metal loss by blasting process.

2 PRODUCTS

2.1 GENERAL Storage and handling

Requirement: Handle, store, mix and apply all protective coatings in conformance with the manufacturer's recommendations.

Original containers: Deliver coating products to site in manufacturer's labelled and sealed containers.

Ambient temperature range for storage: 3°C to 30°C, or to manufacturer's recommendations.

Sunlight: Protect coating materials from direct sunlight before mixing or adding the converter (catalyst).

Use-by-date: Use products with limited shelf life before their use-by-date, unless written authorisation from the coating manufacturer's technical services section is provided.

Paint material

Requirement: To AS/NZS 5131 clause 9.9.3.

Proprietary products

Requirement: Provide all products from the one manufacturer's supply.

Product data sheets (PDS): Keep on site copies of all relevant manufacturer's PDS.

Safety data sheets (SDS): Keep on site copies of all relevant manufacturer's SDS.

Recording: To AS/NZS 5131 clause 9.9.5.

3 EXECUTION

3.1 GENERAL Product warnings

Requirement: Conform to the SDS.

Surroundings

Protection: Prevent the release of abrasives, overspray or paint waste debris into the air, ground or to any watercourse.

Prevent damage to other assets, services or equipment.

Reinstatement: Repair and/or clean affected surrounding areas.

Working area

General: Perform all painting under cover and/or protected from rain, condensation, dew, excessive wind, overspray or wind-blown dust.

Period: Continue protection where any of these conditions exist until the coating is no longer affected.

3.2 SURFACE PREPARATION General

Requirement: Conform to AS/NZS 5131 clauses 9.3, 9.4 and 9.5.

Treatment grade to AS/NZS 5131

Galvanized, aluminium and zinc primed surfaces

Requirement: Remove grease, oil and other solvent-soluble contaminants to AS 1627.1. Allow to dry and immediately proceed with the next operation.

Galvanized and aluminium surfaces: Abrade surfaces to a medium coarse type finish to provide an adhesion key.

Zinc primed surfaces: If present, remove zinc salts from zinc primers.

Treatment of welds

Requirement: Clean welds to remove roughness, using power tools to AS 1627.2. Remove filings by vacuuming or compressed air.

Temporary welds: Grind flush any temporary welds.

Porous, skip or stitch welds: Not permitted.

Site welding: If possible, avoid site welding. If on site welding is required, prepare and treat the weld to AS/NZS 5131 clause 9.12.2.

Shop priming

Requirement: Dust off and apply a coat of primer in conformance with the manufacturer's recommendations.

Site coating

General: High pressure wash down all surfaces with clean water. Lightly sand down primer/intermediate coats, which have been shop applied, before site application of next coat.

3.3 PREPARATION ASSESSMENT General

Conformance: Assess all surfaces of each steel member for conformance with the documented preparation requirements.

Abrasive blast cleaning

Assessment: To AS 1627.4 and AS 1627.9.

Mechanical cleaning Assessment: To AS 1627.9.

Surface profile

General: To AS 3894.5 Method A.

Surface dust from abrasion General: To AS 3894.6 Method C.

Chloride level testing Test: To AS 3894.6 Method A.

Maximum allowable chloride levels: 50 mg/m² for critical applications (heavy condensation, fresh water ponding or immersion) or to manufacturer's recommendations.

Conformance: If the maximum allowable chloride is exceeded, rewash the affected surface area until the chloride level is within the acceptable limits using clean water or chloride neutralising solutions. Jet-washing or steam cleaning is also acceptable before re-testing and re-abrasive blasting.

Timing of testing: Early in the blasting work so that removal procedures can be started before the blasting is completed.

3.4 MIXING General

Requirement: To AS/NZS 5131 clause 9.9.6.

Powered agitators: Mix package sizes larger than 4 litres using powered agitators driven by air motors.

Multi-component coatings: Combine as whole pack units before application.

Thinners: If addition of thinners is proposed, conform to the coating manufacturer's recommendations for the documented product.

Colour consistency: If colour consistency is required, pre-mix tinted products, before the addition of the curing agent or converter and before coating application.

3.5 COATING APPLICATION General

Requirement: Conform to AS/NZS 5131 clause 9.9 and the PDS.

Painting and coating colour: Verify all project finish colours with the retained samples.

Final surface preparation or coating application

Limits: If the environmental/climatic/substrate conditions listed in AS/NZS 5131 clause 9.9.10 and the following are present do not apply coating:

- Ambient air temperature below 5°C or above 40°C.
- Substrate temperature below 5°C or above 35°C.
- Full prime coat application cannot be carried out before the specified cleanliness of the surface deteriorates.
- Surface preparation standard has not been achieved.
- Time between final surface preparation and the commencement of coating has exceeded 4 hours.
- Visual tarnishing or black spots develop on the surface of the steel.

Exception: Preliminary blast or other surface preparations may be performed in conditions that are outside the limits, providing the final surface preparation and all coating applications are undertaken under the limit conditions.

Pre-coating: Before the spray application of each coating, stripe coat by brush method all edges, welds, seams, rivets, bolts, boltholes (including slots) and difficult to spray areas. Prime the underlying surfaces of replacement bolting, washers and nuts before installation.

Procedure: Conform to the coating order shown in **SELECTIONS, PROTECTIVE PAINT COATING SYSTEMS**.

Subsequent coats: Before applying any subsequent coating layer, make sure the surface condition of the preceding coat conforms to **SELECTIONS, PROTECTIVE PAINT COATING SYSTEMS** and is clean and free from defects.

Wet film thickness (WFT)

Method of measurement: To AS 3894.3 Appendix C using an approved wet film gauge continuously during application.

Dry film thickness (DFT)

Method of measurement: To AS 3894.3 clause 10.

Extent: Measure all surfaces at the completion of each prime, intermediate and finish coats, including areas of the element difficult to paint, masked by structure, or where double or light coating is likely.

Number of measurements: To AS 3894.3 clause 7.

Coatings with DFT 150 µm or less: If testing, deduct the effect of the measured surface profile from all DFT readings.

Single readings: Conform to the following:

- The average of 5 point readings for each 10 m² area of coating surface to be within the documented coating thickness range.
- No single point reading in any 10 m² to be less than 80% of the specified minimum coating thickness. If the average of three readings is used to produce a point reading, an individual reading may be less than 80% of the minimum coating thickness.
- Check any single reading that is greater than 150% of the documented maximum DFT with three additional readings within 50 mm of the original reading. If the average of these three readings is not greater than 150% of the specified DFT, take the average reading as the point reading. If greater than 150%, reject the DFT for that area. If no maximum limit for DFT is documented, consult manufacturer.

Rectification and defects

Rectification: Re-work areas rejected, using the same surface preparation, coatings and sequence as for the original work.

Defects (including under-thickness and over-thickness): Mark with dustless chalk, adhesive inspection labels or masking tape. Do not use crayon, paint or spirit based ink pens.

3.6 PROTECTION Contamination

Surfaces: Prevent contamination of coated surface, which are not yet dry, from blasting dust, abrasive or surface preparation debris and any other foreign matter.

Post application care

General: Protect the coating against physical, chemical, or atmospheric damage until all components are fully cured.

Care: Stack and handle all coated items using fabric slings or padded chains. Use soft packaging, carpet strips or other deformable materials between all coated items.

Water ponding: Stack coated items to prevent water ponding.

3.7 COATING REPAIR Repair of coating damage

Preparation: Feather back by hand or machine sanding all leading edges of intact coating adjacent to the repair, to remove any sharp edge.

Surface contamination: Remove by dusting or blowing down before applying the first coat of paint.

Sequence: Apply the repair coating in the same sequence and manner as the original coating.

Areas damaged without exposing the primer: Wash with a proprietary detergent solution, rinse with clean water and abrade so that edges of sound paint are feathered. Coat the area with the appropriate intermediate and finishing coat materials.

Areas damaged exposing the primer or steel surface: Blast clean to the original standard. Prepare at least 50 mm into the sound coating and to a further feathering zone of approximately 50 mm. Recoat with the documented system to restore the film thickness and integrity over the whole prepared surface including the feathered zone.

Aesthetic reinstatement: If required, repaint to a physical or discernible boundary line.

Defects: If corrosion pitting or areas of significant metal loss and defects are exposed by the blasting process, advise for inspection and have areas passed as being fit for service before proceeding with the coating system.

Timing: Apply the protective coating system within 4 hours of blast cleaning or in any case before visual tarnishing of the steel occurs.

3.8 COMPLETION General

Joints: On completion, seal all joints and mating surfaces with a compatible polyurethane sealant.

Warranties

Requirement: Cover materials and workmanship in the terms of the warranty in the form of interlocking warranties from the supplier and the applicator.

- Form: Against failure of materials and execution under normal environment and use conditions.
- Period: As offered by the supplier.

4 SELECTIONS

4.1 STRUCTURAL STEEL PROTECTIVE PAINT COATING SYSTEMS

Location	Primer	Second Coat	Third Coat
Interior non-visible	Dulux Zincanode 402, min 75 micron	Nil	Nil
Interior Visible	Dulux Zincanode 402, min 75 micron	Duremax GPE – Min 150 micron	Nil
Exterior non-visible	Dulux Zincanode 402, min 75 micron	Duremax GPE – Min 150 micron	Nil
Exterior Visible	Dulux Zincanode 402, min 75 micron	Duremax GPE – Min 150 micron	Weathermax HBR – Min. 100micron

Note: Paint thickness will be measured on site to ensure the required corrosion protection coating is achieved. All paint thickness nominated are for Dry Film Thickness

0411 WATERPROOFING – EXTERNAL AND TANKING

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide external waterproofing and tanking systems to substrates, as documented.

Performance

Requirements: Conform to the following:

- Graded to falls to dispose of stormwater without ponding above the depth of lapped seams.
- Able to accommodate anticipated building movements.
- Able to accommodate its own shrinkage over the warranty life of the roofing system.
- Able to resist water under hydrostatic pressure.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- *0171 General requirements.*

1.3 STANDARDS External waterproofing

Membrane materials: To AS 4654.1.

Membrane design and installation: To AS 4654.2.

Stormwater drainage Standard: To AS/NZS 3500.3.

1.4 INTERPRETATION Definitions

General: For the purposes of this work section the definitions given in AS 4654.1 and AS 4654.2 and the following apply:

- Bitumen: A viscous material from the distillation of crude oil comprising complex hydrocarbons, which is soluble in carbon disulphide, softens when it is heated, is waterproof and has good powers of adhesion. It is produced as a refined by-product of oil.
 - o APP Bitumen: Bitumen modified with atactic (meaning non-crystalline or amorphous) polypropylene wax to form a plastomeric sheet. The membrane is reinforced with fibreglass or non-woven polyester (NWP).
 - o SBS bitumen: Bitumen modified with Styrene Butadiene Styrene, a thermoplastic rubber that undergoes a phase inversion at elevated temperature and converts to an elastomeric material. The membrane is reinforced with fibreglass or non-woven polyester (NWP).
- Bond breaker: A system preventing a membrane bonding to the substrate, bedding or lining.
- Double detail joint: A joint formed by turning up and bonding the horizontal membrane to a vertical substrate and adding an over flashing of membrane material bonded to the vertical substrate and folded over and bonded to the horizontal membrane. In certain situations the double detail can be achieved by bonding an angle profile of membrane material to the junction prior to laying the membrane.
- Liquid applied: A water-based formulation which cures to form an elastomeric membrane.
- Polyurethane: Water or solvent based formulations which moisture cure to form an elastic rubber membrane.
- PVC membrane: Flexible plastic sheet membrane (vinyl).
- Slip sheet: A sheet used to isolate the membrane system from the supporting substrate or from the topping or mortar bedding. The most common material is polyethylene.
- Substrate: The surface to which a material or product is applied.

1.5 SUBMISSIONS Products and materials

Manufacturer's documentation: Submit copies of the following data:

- Product technical data sheets.

- Safety data sheets (SDS).
- Preventative maintenance procedures.
- Instructions and procedures for the repair of the membrane.
- Type test certificates verifying compliance with AS 4654.1 Section 2, Tables 2.1 to 2.3.

Records

Placing records: Photographically record the application of membranes and label with the following information:

- Date.
- Portion of work.
- Substrate preparation.
- Weather during application and curing.
- Protection provided from traffic and weather.

Liquid membrane applications:

- Record wet film thickness once every 10 m² and compare to the manufacturers requirements.
- On completion of every 100 m² of each coat compare the amount of membrane used with the manufacturer's application rate and record the result.

Subcontractors

General: Submit names and contact details of proposed suppliers and installers as recommended by the manufacturer.

Warranties

Requirement: Submit warranties to **COMPLETION, Warranties.**

2 PRODUCTS

2.1 GENERAL Storage and handling

General: Store and handle to the manufacturer's recommendations and as follows:

- Protect materials from damage.

2.2 MEMBRANES Membrane systems

Requirement: Provide a proprietary membrane systems suitable for the intended external waterproofing.

Tanking systems

Requirement: Provide a proprietary membrane system suitable for the intended below ground tanking.

2.3 PROTECTION Protection board

Description: Corflute or 4.5 FC Sheet (Polyethylene sheet or other flexible sheet is not acceptable)

3 EXECUTION

3.1 PREPARATION Substrates

General: Prepare substrates as follows:

- Fill all cracks in substrates wider than 1.5 mm with a filler compatible with the membrane system.
- Fill voids and hollows in concrete substrates with a concrete mix not stronger than the substrate.
- Remove projections.
- Remove deleterious and loose material.
- Remove all traces of a concrete curing compound if used.
- Leave the surface free of contaminants, clean and dust free.

Concrete substrates: Cure for more than 28 days.

Moisture content

Requirement: Verify that the moisture content of the substrate is compatible with the water vapour transmission rate of the membrane system by testing to AS 1884 Appendix A.

Falls

Requirement: Verify that falls in substrates are greater than 1:80.

Joints and fillets

Internal corners: 45 deg fillets

Fillet material: 45 deg fillets - plastic

External corners: Round or arris edges.

Control joints: Prepare all substrate joints to suit the membrane system.

Priming

Compatibility: If required, prime the substrates with compatible primers for adhesion of the membrane system.

3.2 APPLICATION Protection during installation

Damage: Protect membrane from damage during installation and for the period after installation until the membrane achieves its service characteristics that resist damage.

Drains

General: Prevent moisture from tracking under the membranes at drainage locations.

Drains and cages: Provide removable grates or cages to prevent blockage from debris. If the finished surface is above the level of the membrane, provide a slotted extension piece to bring the grate up to the level of the finished surface.

Overflows: Apply a bond breaker to the perimeter of the overflow outlet at its junction with the surface to which the membrane will be fixed. Turn the membranes into the overflow to prevent moisture from tracking behind the membrane.

Sheet membrane joints

Orientation of laps: Lap sheets on the upslope side of the roof fall over sheets on the downslope side.

End laps generally: Stagger end lap joints.

Bituminous sheet membranes:

- Side laps: 75 mm.

- End laps: 100 mm.
- Method: Heat welded.

Synthetic rubber membranes:

- Factory-vulcanized laps: More than 40 mm.
- Field side laps: More than 50 mm for side laps.
- Field end-laps: More than 100 mm for end laps.

PVC membranes:

- Factory welded laps: More than 30 mm.
- Field-welded laps:
 - . If used over insulation boards: More than 100 mm.
 - . Other instances: More than 75 mm overlaps.

Curing of liquid applied systems

General: To the manufacturers' instructions.

Control of movement

General: Provide control joints located over control joints in the substructure.

Fillets and bond breakers: Size to allow the membrane to accommodate movement.

Backing rod: closed cell polyethylene foam

Joint Sealant: refer manufacturer's recommendations

Control joint covers: Install after fixing hobs and membranes.

Bonded membranes: Carry control joints in the substrate through to and into the surface finish.

Membrane terminations

Membrane upturns: Provide upturns above the maximum water level expected from the exposure conditions of rainfall intensity and wind.

- Height: To AS 4654.2 Appendix A, Table A1.
- Anchoring: Secure sheet membranes along the top edge.
- Edge protection: Protect edges of the membrane.

Waterproofing above vertical terminations: Waterproof the structure above the termination to prevent moisture entry behind the membrane using cavity flashings, capping, waterproof membranes or waterproof coatings.

Horizontal terminations: Do not provide. Use vertical terminations.

Membrane vertical penetrations

Pipes, balustrades, ducts, and vents: Provide separate sleeves for all pipes, ducts and vents, and fix to the substrate.

Membrane horizontal penetrations

Sleeves: Protect rigid PVC-U conduits and pipes with a sleeve of SBS bitumen in order to seal to the membrane without burning the PVC-U. Do not use high density polyethylene (HDPE), polypropylene (PP) pipes or flexible PVC conduit.

Membrane at balcony doors and windows

Requirement: Install membrane before the fixing of door or window frames.

Hobless and flush thresholds: Install membrane before the fixing of door or window frames with a continuous grated drain abutting the external face of the door or window sill.

Membrane at gutters

Requirement: Terminate membrane over a corrosion resistant metal angle fixed to the gutter support substrate with the vertical leg of the angle turned down into the gutter at least 35 mm.

Membrane to below ground structures

Membrane: Externally apply membrane to all walls and return to horizontal surfaces to prevent water tracking around structure at joints and corners.

Protection board: Provide protection board to the full extent of the membrane.

Drainage cell: Provide geo-filter fabric wrapped drainage cell to vertical surfaces of the structure.

Reinforcement: Provide reinforcement to the membrane at junctions, corners and over joints to the manufacturer's recommendations.

3.3 COMPLETION Protection

General: Keep traffic off membrane surfaces until bonding has set or for 24 hours after laying, whichever period is the longer.

Reinstatement: Repair or replace faulty or damaged work. If the work cannot be repaired satisfactorily, replace the whole area affected.

Warranties

Waterproofing: Cover materials and workmanship in the terms of the warranty in the form of interlocking warranties from the supplier and the applicator.

- Form: Against failure of materials and execution under normal environment and use conditions.
- Period: As offered by the supplier.

0421 ROOFING – COMBINED

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide a roofing system and associated work, as documented. Refer to Design drawings and hydraulic documentation.

Ambient climatic conditions

Design rainfall intensity (mm/h) to AS/NZS 3500.3: 248mm/hr on 20 year average recurrence

Corrosion resistance

Material: To the manufacturer's recommendations for distance from marine influence. Distance from marine influence: by Colorbond.

Atmospheric corrosivity

Atmospheric corrosivity category: To 0171 *General requirements*.

Roof access

Type: normal roof maintenance

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements*.

1.3 INTERPRETATION Abbreviations

General: For the purposes of this work section, the following abbreviations apply:

- EPS: Expanded polystyrene.
- PF: Phenolic foam.
- PIR: Rigid polyisocyanurate.
- PUR: Rigid polyurethane. - XPS: Extruded polystyrene - MW: Mineral wool.

1.4 TOLERANCES Sheet metal roofing

Supporting members: To AS 1562.1 clause 4.2.3.

1.5 SUBMISSIONS Fire performance

Combustibility: Submit evidence of conformity to PRODUCTS, FIRE PERFORMANCE, Combustibility.

Fire hazard properties: Submit evidence of conformity to PRODUCTS, FIRE PERFORMANCE, Fire hazard properties.

Operation and maintenance manuals

On completion: Submit a manual of recommendations from the roofing manufacturer or supplier for the maintenance of the roofing system including, frequency of inspection and recommended methods of access, inspection, cleaning, repair and replacement.

Products and materials

Safety glazing materials: Submit evidence of conformity to AS/NZS 2208 Appendix A.

Thermal insulation performance: Submit evidence of performance to AS/NZS 4859.1 and AS/NZS 4859.2.

Type tests: As appropriate for the project, submit evidence of conformity to the following:

- Metal roofing generally: Roof sheeting and fastenings to AS 1562.1 clause 5.4 for resistance to concentrated load and AS 1562.1 clause 5.5 for resistance to wind pressure.
- Metal roofing in cyclonic regions to AS/NZS 1170.2: Roof sheeting and fastenings to AS 1562.1 clause 5.6.
-

Subcontractors

Installer experience: Submit evidence of experience.

Warranties

Roofing materials: Submit the manufacturer's published product warranties.

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Roof supports.
- The parts of the roofing, sarking, vapour barrier, insulation and roof plumbing installation before covering up or concealing.
- At completion of roof, prior to removal of edge protection

2 PRODUCTS

2.1 GENERAL Storage and handling

Storage: To the manufacturer's recommendations and the following:

- Metal roofing materials: Store away from uncured concrete and masonry, on a level base and not in contact with other materials that cause staining, denting or other surface damage.

Handling metal roofing materials: As follows:

- Use gloves when handling precoated metal roofing material.
- Use soft soled shoes when fixing or working on roofs.
- Protect edges and surfaces from damage. Do not drag sheets across each other or over other materials.
-

Safety mesh

Standard: To AS/NZS 4389.

2.2 FIRE PERFORMANCE Combustibility

Roofing: Tested to AS 1530.1.

Fire hazard properties

Insulated panels: Group number to AS 5637.1.

Insulation materials: Conform to the following, tested to AS/NZS 1530.3.

- Spread-of-Flame index: ≤ 9 .
- Smoke-Developed Index: ≤ 8 if spread of flame index > 5 .

Glazing plastics: Conform to the following, tested to AS/NZS 1530.3.

- Spread-of-Flame index: ≤ 9 .
- Smoke-Developed Index: ≤ 8 if spread of flame index > 5 .

2.3 PROFILED SHEET METAL ROOFING Standards

Design and materials: To AS 1562.1.

Fasteners

Prefinished exposed fasteners: Finish with an oven baked polymer coating to match the roofing material.

Fastenings to timber battens: Fastenings long enough to penetrate the thickness of the batten without piercing the underside.

Profiled fillers

Type: Purpose-made closed cell polyethylene foam profiled to match the roofing profile.

Location: Provide profiled fillers under flashings to the following:

- Ridges.
- Eaves.
- Lapped joints in roof sheeting.

Insulation spacers

Description: Proprietary spacer system to prevent excessive compression of insulation between roof sheeting and framing. Size spacers to suit the required insulation thickness documented and/or create an air space as required.

2.4 ROOF PLUMBING General

Description: Flashings, cappings, gutters, rainheads, outlets, downpipes and accessories necessary to complete the roofing system.

Flashing and capping: Notched to match profile of roof sheeting.

Matching fascia/barge capping: If the selected eaves gutter is a proprietary high front pattern forming part of a combined system of gutter, fascia and barge, provide matching proprietary fascias and barge cappings to roof verges and edges.

Standards

Roof drainage: To AS/NZS 3500.3.

Metal rainwater goods: To AS/NZS 2179.1.

Flashings and cappings: To AS/NZS 2904.

3 EXECUTION

3.1 INSTALLATION Protection

General: Keep the roofing and rainwater system free of debris and loose material during construction.

Thermal movement

Requirement: Allow for thermal movement in the roof installation and the structure, including movement in joints and fastenings.

Metal separation

Requirement: Prevent direct contact between incompatible metals, and between green hardwood or chemically treated timber and aluminium or coated steel, by one of the following methods:

- Applying an anti-corrosion, low moisture transmission coating to contact surfaces.
- Inserting a separation layer.

3.2 PROFILED SHEET METAL ROOFING Installation

Standard: To AS 1562.1.

Set-out point: ridge

Swarf: Remove swarf and other debris as soon as it is deposited.

Accessories: Provide accessories with the same finish as roofing sheets to complete the roofing installation.

Pan type sheets

Removal: Install sheets so that individual sheets can be removed without damage.

Ridges and eaves

Sheet ends: Treat as follows:

- Project sheets 50 mm into gutters.
- Close off ribs at bottom of sheets using mechanical means or with purpose-made fillers or end caps.
- Turn pans of sheets up at tops and down into gutters by mechanical means.
- Provide pre-cut notched eaves flashing and bird proofing if required.
- Close off ridges with purpose-made ridge fillers of closed cell polyethylene foam.

Ridge and barge

Capping: Finish off along ridge and verge lines with purpose-made ridge capping or barge rolls.

End laps

General: If end laps are unavoidable, and the sheet profile is not suitable for interlocking or contact end laps, construct a stepped type lap.

Length of lap (mm): minimum 250mm or as per manufacturers details.

3.3 ROOF PLUMBING Jointing sheet metal rainwater goods

Butt joints: Make joints over a backing strip of the same material.

Soldered joints: Do not solder aluminium or aluminium/zinc-coated steel.

Sealing: Seal fasteners and mechanically fastened joints. Fill the holes of blind rivets with silicone sealant.

Flashings

Installation: Flash roof junctions, upstands, abutments and projections through the roof. Preform to required shapes if possible. Notch, scribe, flute or dress down as necessary to follow the profile of adjacent surfaces. Mitre angles and lap joints 150 mm in running lengths. Provide matching expansion joints at 6 m maximum intervals.

Upstands: Flash projections above or through the roof with two part flashings, consisting of a base flashing and a cover flashing, with at least 100 mm vertical overlap. Provide for independent movement between the roof and the projection.

Large penetrations in low pitch roofs: Extend the base flashing over the roofing ribs to the ridge to prevent ponding behind the penetrating element.

Wall abutments: Where a roof abuts a wall, provide as follows:

- In masonry walls, planked cladding or concrete: Step in courses to the roof slope. Interleave with damp proof course, if any.
- Raking in masonry: Build into the full width of the outer leaf. Turn up within cavity, slope inward across the cavity and fix to or build into the inner leaf at least 75 mm above the roofing line.

Fixing to pipes: Solder or seal with neutral cured silicone rubber and secure with either of the following:

- Clamping ring.
- Proprietary flexible clamping shoe with attached metal surround flashing.

Gutters

Gutter and sump support: Provide framing and lining to support valley gutters, box gutters and sumps. Line the whole area under the gutters and sumps.

Expansion joints in guttering longer than 30 m: Provide as follows:

- Type: proprietary system

External downpipes

General: Prefabricate downpipes to the required section and shape where possible. Connect heads to gutter outlets and, if applicable, connect feet to rainwater drains.

Access cover: Provide a removable watertight access cover at the foot of each downpipe stack.

Downpipe support: Provide supports and fixings for downpipes.

3.4 COMPLETION Reinstatement

Extent: Repair or replace damage to the roofing and rainwater system. If the work cannot be repaired satisfactorily, replace the whole area affected.

Damage to pre-painted finish: Replace panels with scratches in the pre-painted finish greater than 2 mm in width visible from the ground.

Fasteners: Make sure weather tight and external panel facings are not distorted.

Cleaning

Roofing and rainwater drainage system: Remove debris, metal swarf, solder, sealants and unused materials.

Exposed metal surfaces: Clean surfaces of substances that interfere with uniform weathering or oxidation.

Roof plumbing: Clean out spoutings, gutters and rainwater pipes after completion of roof installation.

Protection: After completion, remove protective coatings using methods to the manufacturer's recommendations.

Warranties

Requirement: Provide warranties for materials and workmanship warranties in the form of interlocking warranties as follows:

- Form: Against failure of materials and execution under normal environment and use conditions.
- Period: As offered by the supplier/manufacturer.

4 SELECTIONS

4.1 PROFILED SHEET METAL ROOFING Profiled sheet metal roofing schedule

Property	MRS-1	MRS-2	
Location	Registry, Workshop, Warehouse	New Build	
Product	Stammit	Stammit	
Profile	Monoclad (TBC existing on site) To be confirmed	Monoclad (match other buildings)	
Material	Steel	Steel	
Base metal thickness (BMT) (mm)	0.48 (BMT)	0.48 (BMT)	
Finish	Colorbond	Colorbond	
Colour	TBC	TBC	
Fasteners	Pierced	Pierced	
Insulation spacer type	Air cell	Air ceill	

4.2 ROOF PLUMBING Flashing and capping schedule

Property	MAC	MAF	MBC
Type	Metal Apex Capping	Metal Apron Flashing	Metal Barge Capping
Material	Steel	Steel	Steel
Thickness and grade	0.55 (BMT)	0.55 (BMT)	0.55 (BMT)
Colour	Refer finishes schedule	Refer finishes schedule	Refer finishes schedule

Property	MRC	MF	
Type	Metal Ridge Capping	Metal Fascia	
Material	Steel	Steel	
Thickness and grade	0.55 (BMT)	0.55 (BMT)	
Colour	Refer finishes schedule	Refer finishes schedule	

Roof plumbing schedule

Item	Product	Material	Colour/Shape/Size
Eaves gutter – Type 1 A1, A2 & A3 Buildings	Custom colorbond gutter to match existing. Brackets as per manuf details or every 1200mm	Steel	Colorbond – refer finishes schedule
A4	Colorbond OG450 big one gutter	steel	Colorbond – refer finishes schedule

Provide kattsafe (or equal) roof safet anchor point to comply with AS5532:2013 to A4 – new building

Provide Kattsafe (or equal) roof mount support ladder bracket – refer to design plans for locations.

1 GENERAL**1.1 RESPONSIBILITIES General**

Requirement: Provide lightweight external wall cladding and associated work, as documented. Refer to design drawings.

Atmospheric corrosivity

Atmospheric corrosivity category: To 0171 *General requirements*.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements*.

1.3 INTERPRETATION Abbreviations

General: For the purposes of this work section the following abbreviations apply:

- AAC: Autoclaved aerated concrete.
- CCA: Copper chrome arsenate.
- CFC: Compressed fibre cement.
- EIFS: Exterior insulation and finish system.
- FC: Fibre cement.
- LOSP: Light organic solvent preservative.
- EPS: Expanded polystyrene.
- EPS-FR: Expanded polystyrene-fire retardant.
- MW: Mineral wool.
- PF: Phenolic foam.
- PIR: Rigid polyisocyanurate.
- PUR: Rigid polyurethane.
- XPS: Extruded polystyrene
-

1.4 TOLERANCES Permitted deviations

Flat sheet and panel cladding: To the manufacturer's recommendations.

Profiled metal sheet cladding: To AS 1562.1 clause 4.2.3.

Structural steelwork for wall cladding: ± 5 mm between bearing planes of adjacent supports.

1.5 SUBMISSIONS Fire performance

Combustibility: Submit evidence of conformity to PRODUCTS, FIRE PERFORMANCE, Combustibility.

Fire hazard properties: Submit evidence of conformity to PRODUCTS, FIRE PERFORMANCE, Fire hazard properties.

Fire-resistance level: Submit evidence of conformity to PRODUCTS, FIRE PERFORMANCE, Fire resistance of building elements.

Operation and maintenance manuals

General: Submit manufacturer's published use, care and maintenance requirements.

Products and materials

Type tests: As appropriate for the project, submit evidence of conformity to the following:

- Insulated panel systems:
 - . Air infiltration test to AS/NZS 4284 clause 8.4 for test pressures of ± 150 Pa or ± 300 Pa as documented.
 - . Water penetration test by static pressure to AS/NZS 4284 clause 8.5 at test pressure of 300 Pa.
 - . Water penetration test by cyclic pressure to AS/NZS 4284 clause 8.6 at test pressure of 600 Pa.
- Metal cladding generally: Cladding and fastenings to AS 1562.1 clause 5.5 for resistance to wind pressures.
- Metal cladding in cyclonic regions to AS/NZS 1170.2: Cladding and fastenings to AS 1562.1 clause 5.6.

Shop drawings

General: Submit shop drawings to a scale that best describes the detail, showing the following:

- Dimensioned elevations of all elements.
- Details of construction, connections and all support systems.
- Dimensions of all typical elements and of any special sizes and shapes.
- Provision for the exclusion and/or drainage of moisture.
- Jointing details and method of fixing between individual elements and between this installation and adjacent work, including adjustment.
- Sealant types and full size sections of all sealant-filled joints and backing rods.
- Provision for thermal movement.
- Provision for movement under seismic and wind loads.
- Sequence of installation.
- Co-ordination requirements with other work.
- Schedule of materials, finishes, componentry, hardware and fittings.

Subcontractors

General: Submit names and contact details of proposed suppliers and installers.

Warranties

Requirement: Submit warranties to **COMPLETION, Warranties**.

Cladding materials: Submit the manufacturer's product warranties.

Weatherproofing performance

Requirement: Submit evidence of conformity to **EXECUTION, GENERAL**, Weatherproofing.

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Framing, pliable membranes and insulation before covering up or concealing.

2 PRODUCTS

2.1 GENERAL Storage and handling

Requirement: Store and handle materials to the manufacturer's recommendations and the following:

- Protect materials including edges and surfaces from damage.
- Keep dry and unexposed to weather.
- Do not drag sheets or panels across each other or over other materials.
- Composite panels: Store unpacked panels by size in racks and protect from scratching, warping or bending.
- Sheeting: Stack flat and off the ground on at least 3 evenly placed bearers.
- Store metal materials away from uncured concrete and masonry on a level base.
- Do not store metal materials in contact with other materials which may cause staining, denting or other surface damage.
- Use gloves when handling precoated metal cladding material.

Components

Fasteners and ties: Type, size, corrosion resistance class and spacing to the cladding manufacturer's recommendations.

Flashings: To AS/NZS 2904.

2.2 FIRE PERFORMANCE Combustibility

Cladding: Tested to AS 1530.1.

Fire hazard properties

Group number: To AS 5637.1.

Bonded laminated materials: Tested to AS/NZS 1530.3. Fire hazard indices, as follows:

- Spread of Flame Index: 0.
- Smoke-Developed Index: ≤ 3 .

Insulation materials: Tested to AS/NZS 1530.3. Fire hazard indices as follows:

- Spread-of-Flame Index: ≤ 9 .
- Smoke-Developed Index: ≤ 8 if Spread-of-Flame Index > 5 .

Fire-resistance of building elements Fire-resistance level: Tested to AS 1530.4.

2.3 COMPRESSED FIBRE CEMENT (CFC) SHEETS General

Requirement: Proprietary compressed fibre cement sheets.

Standard: To AS/NZS 2908.2 and the following:

- Type A Category 5.

Quality: Smooth and even with factory sealed edges, free of imperfections such as chips.

Edge profile: Square.

Sealant and bond breaking tape: To the manufacturer's recommendations.

2.4 FIBRE CEMENT (FC) SHEETS General

Requirement: Proprietary single faced fibre cement sheets.

Standard: To AS/NZS 2908.2 and the following:

- Type A Category 3.

Sealant and bond breaking tape: To the manufacturer's recommendations.

2.5 PLYWOOD SHEETS General

Requirement: Proprietary plywood sheets.

Standard: To AS/NZS 2271.

Bond: Type A.

Presealed plywood: Sides and edges presealed with a machine applied sealer.

Visible surfaces with a clear finish: Veneer quality A.

Other visible surfaces: Veneer quality B.

Hidden surfaces: Veneer quality C or D.

Identification: Sheets labelled under the authority of a recognised certification scheme to *0185 Timber products, finishes and treatment*.

Fasteners

LOSP treated timber and non-corrosive timber cladding: Hot-dip galvanized steel.

CCA treated timber and corrosive timber cladding (including western red cedar or redwood): Stainless steel Type 316 or silicon bronze.

2.6 PROFILED SHEET METAL General

Requirement: Proprietary profiled sheet metal cladding. Parapet cladding longspan (confirm existing profile on site) selected Colorbond finish.

Design and installation: To AS 1562.1.

3 EXECUTION

3.1 GENERAL Preparation

Substrates or framing: Before fixing cladding, check the alignment of substrates or framing and adjust if required.

Flexible underlay: Check that the underlay or insulation is restrained.

Cladding: Make sure the cladding is clean and free of dust and loose particles.

Installation

Requirement: Install cladding as follows:

- Fix sheeting firmly against framing to the manufacturer's recommendations.
- Plumb, level, straight and to documented tolerances.
- Fixed or anchored to the building structure in conformance with the wind action loading recommendations.
- Isolated from any building loads, including loads caused by structural deflection or shortening.
- Allow for thermal movement.

Cladding layout: Cut/fabricate and install cladding to suit the layout as documented.

Protection: Protect surfaces and finishes, including the retention of protective coatings during installation.

Proprietary systems or products

Requirement: Use panels and components from a single proprietary system and install to the manufacturer's recommendations.

Accessories and trim

Requirement: Provide accessories and trim required to complete the installation, or as documented.

Corner flashing for profiled and seamed metal sheets: Finish off at corners with purpose-made folded flashing strips.

Weatherproofing

Requirement: Install cladding as documented. Conform to BCA FP1.4.

Metal separation

Requirement: Prevent direct contact between incompatible metals, and between green hardwood or chemically treated timber and aluminium or coated steel, by either of the following methods:

- Apply an anti-corrosion, low moisture transmission coating to contact surfaces.
- Insert a separation layer.

Incompatible metal fixings: Do not use.

Horizontal cladding Horizontal cladding surface:

- Minimum slope: 1:15.
- Staining: Slope away from visible vertical facade areas to prevent staining.

Defective and damaged parts

Defective components: Do not install component parts which are defective, including warped, bowed, dented, abraded or broken members.

Damaged parts: Remove and replace damaged parts during installation.

3.2 CFC SHEET CLADDING Joints

Control joint:

- Locate between the panel and fixing system and the supporting structure, as documented.
- Sheet edges: Square cut.
- Sealant: Do not apply finish coating over joint sealants.

Prefinished metal backing/jointing strip: Fix proprietary backing strip to the rear face of the panel with proprietary closed cell self-adhering foam and horizontal gasket.

- Seal the joint with a 3 mm epoxy fillet.

Vertical joints: Vertical gasket or prefinished jointing strip to framing member.

Arrangement: Set out as documented.

Fixing

General: Screw fix to proprietary framing supports at centres to the manufacturer's recommendations.

Concealed fixings:

- Predrill oversized holes.
- Countersink so that the top of the screw is 2 to 3 mm below the surface.
- Finish: Stop screw heads with epoxy filler. Smooth and level upon application and sand flush after curing.

3.3 FC SHEET CLADDING Joints

Control joints:

- Locate between the panel and fixing system and the supporting structure, as documented.
- Sheet edges: Square cut.
- Sealant: Do not apply finish coating over joint sealants.

Arrangement: Set out in even panels with joints coinciding with framing or as documented.

Fixing

General: Corrosion resistant nails or screws to the manufacturer's recommendations.

Eaves and soffit lining: Fix at 150 mm centres to soffit bearers at a maximum of 450 mm centres.

3.4 PROFILED SHEET METAL CLADDING General

Ground clearance: Maintain documented clearance.

Cutting sheets: Wherever possible, factory cut to length. Do not use an abrasion disc.

Accessories: Provide material with the same finish as cladding sheets.

Swarf: Remove swarf and other debris as soon as it is deposited.

Fixing

Fixing start location: As per manufacturer's recommendations

Joints

Expansion joints: as per manufacturer's recommendations.

3.5 COMPLETION Fasteners

Requirement: If required, adjust for weather tightness without distortion of external panel face.

Reinstatement

Extent: Repair or replace damage to the cladding. If the work cannot be repaired satisfactorily, replace the whole area affected.

Damage to pre-painted finish: Replace panels with scratches in the pre-painted finish greater than 2 mm in width visible from the ground.

Cleaning

Requirement: Remove excess debris, metal swarf, solder, sealants and unused materials.

Exposed metal surfaces: Clean surfaces of substances that interfere with uniform weathering or oxidation.

Protection: Remove protective coatings using methods required by the manufacturer after completion.

Panels: Clean surfaces with soft, clean cloths and clean water to the manufacturer's recommendations.

Warranties

Requirement: Cover materials and workmanship in the form of interlocking warranties from the supplier and installer.

- Form: Against failure of materials and execution under normal environment and use conditions.
- Period: As offered by the supplier.

4 SELECTIONS

Profiled sheet metal wall cladding schedule

Property	MWS-1		
Product	Strammit		
Location	A1, A2, A3, A4 buildings - parapet		
Profile	Longspan (match existing cladding)		
Material type	Colorbond steel		
Thickness (mm)	0.55mm BMT		
Colour	Refer Finishes schedule TBC		
Trims	Proprietary colorbond trims to match cladding colour		
Flashings and capping's	Colorbond to match cladding colour		
Fasteners	pierced		

Wall fixing Battens & spacing of battens to wall cladding as noted on Structural Engineers drawings and details. Confirm existing on site

Confirm existing cladding on site – contractor to advise.

0451 WINDOWS AND GLAZED DOORS

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide windows and glazed doors, as documented. Refer to design drawings

Performance

Product design: Provide windows with sashes capable of being opened to satisfy the maintenance requirements, as documented.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements*.

1.3 STANDARDS General

Selection and installation: To AS 2047.

Building classification: 5

Glazing

Glass type and thickness: To AS 1288, if no glass type or thickness is nominated.

Materials and installation: To AS 1288.

Quality requirements for cut-to-size and processed glass: To AS/NZS 4667.

1.4 INTERPRETATION Abbreviations

General: For the purposes of this work section the following abbreviations apply:

- AGWA: Australian Glass and Window Association (formerly Australian Window Association (AWA)).
- WERS: Window Energy Rating Scheme

Definitions

General: For the purposes of this work section the definitions given in AS/NZS 4668 and the following apply:

- Hardware: To AS 4145.1 Section 2.
- Louvres - continuous: Louvres that run continuously past, and are supported by, concealed framing or brackets.
- Louvres - horizontal: Louvres that span horizontally between frame stiles, mullions or vertical supports.
- Louvres - vertical: Louvres that span vertically between frame heads and sills, or horizontal supports.
- Total system SHGC: Solar heat gain coefficient as defined by the NCC and tested in conformance with NFRC 200.
- Total system U-Value: Thermal transmittance as defined by the NCC and tested in conformance with NFRC 100.

1.5 SUBMISSIONS Certification

Conformance: Submit evidence that window and door assemblies conform to AS 2047.

Sealant compatibility: Submit statements from all parties to the installation certifying the compatibility of sealants and glazing systems to all substrates.

Opacified glass: Submit a report, from the manufacturer, certifying that the proposed method of opacifying the glass will not be detrimental to the glass or affect the glass product warranty.

Toughened glass: For each batch of glass, submit certification from the manufacturer of heat soaking.

Protection of openable windows: Submit a certificate of on-site fall prevention testing.

Fire performance

Fire-resistance level: Submit evidence of conformity to PRODUCTS, **FIRE PERFORMANCE**, **Fire resistance of building elements**.

Operation and maintenance manual

Window and door assemblies: Submit the window and glazed door manufacturer's published instructions for operation, care and maintenance.

Hardware: Submit the manufacturer's published recommendations for use, care and maintenance.

Products and materials

Safety glazing materials: Submit evidence of conformity to AS/NZS 2208 Appendix A.

Type tests: Submit results, as follows:

- Acoustic performance of windows and doors.
- Protection of openable windows.

Samples in prototypes: Required samples may form part of prototypes.

Shop drawings

General: Submit shop drawings, to a scale that best describes the detail, showing the following:

- Full size sections of members.
- Hardware, fittings and accessories including fixing details.
- Junctions and trim to adjoining surfaces.
- Layout (sectional plan and elevation) of the window assembly.
- Lubrication requirements.
- Methods of assembly.
- Methods of installation, including fixing, caulking and flashing.
- Provision for vertical and horizontal expansion.
- Method of glazing, including the following:
 - . Rebate depth.
 - . Edge restraint.
 - . Clearances and tolerances.
 - . Glazing gaskets and sealant beads.

Subcontractors

General: Submit names and contact details of proposed manufacturers and installers.

Warranties

General: Submit warranties, as documented.

1.6 LOADINGS

Wind loads in accordance with AS 1170.2 2011. Also refer structural drawings notes for reference.

WIND LOADS	Parameters	
Structural importance level	3	
Region	A	

2 PRODUCTS

2.1 GENERAL Storage and handling

Storage: Store in a clean, dry area and unaffected by weather, to the manufacturer's recommendations. Protect from building materials and loose debris such as wet plaster, mortar, paint and welding splatter.

Handling: Handle frames to the manufacturer's recommendations and the following:

- Stack upright, off the ground and against a flat, vertical surface.
- Carry in the vertical position with sashes locked.
- Do not rack frames out of square.
- Do not remove any bands and corner bracing until after installation.

Acoustic performance

Windows and doors: Tested to AS 5218.

Protection of openable windows

Fall prevention: To BCA D2.24 and BCA 3.9.2.

Testing: To AS 5203.

Marking

Window assemblies: To AS 2047 Section 8.

2.2 FIRE PERFORMANCE

Fire-resistance of building elements Fire-resistance level: Tested to AS 1530.4.

2.3 GLASS Performance

Glass: Free from defects which detract from appearance or interfere with performance under normal conditions of use.

Glazing plastics: Free from surface abrasions, and warranted by the manufacturer for 10 years against yellowing or other colour change, loss of strength and impact resistance, and general deterioration.

Safety glass

Standard: To AS/NZS 2208.

Type: Grade A to AS 1288.

- Certification: Required.
- Certification provider: An organisation accredited by the Joint Accreditation System of Australia and New Zealand (JAS-ANZ).

Insulating glass units (IGUs)

Selection and installation: To AS 4666.

2.4 GLASS IDENTIFICATION

Safety glazing materials Identification: To AS 1288.

2.5 GLAZING MATERIALS General

Requirement: Glazing materials including putty, glazing compounds, sealants, gaskets, glazing tapes, spacing strips, spacing tapes, spacers, setting blocks and compression wedges appropriate for the conditions of application and required performance.

Jointing materials

Requirement: Provide jointing and pointing materials to manufacturer's recommendations that are compatible with each other and with the contact surfaces and non-staining to finished surfaces. Do not provide bituminous materials on absorbent surfaces.

Elastomeric sealants

Sealing compound (polyurethane, polysulfide, acrylic): To ASTM C920 or ISO 11600.

Sealing compound (silicone): To ASTM C920 or ISO 11600.

Sealing compound (butyl): To ASTM C1311.

Primer

Compatibility: Apply the manufacturer's recommended primer to the surfaces in contact with sealant materials.

Control joints

Depth of elastomeric sealant: One half the joint width or 6 mm, whichever is the greater.

Foamed materials (in compressible fillers and backing rods): Closed-cell or impregnated types that do not absorb water.

Bond breaking: Provide backing rods, and other back-up materials for sealants, that do not adhere to the sealant.

2.6 VENTILATING LOUVRE ASSEMBLIES General

Requirement: Metal louver blades, mounted in a metal surround frame or subframe, and able to withstand the permissible-stress-design wind pressure for that location, without failure or permanent distortion of members, and without blade flutter.

Expansion joints

Requirement: Provide for expansion and contraction in continuous sections (e.g. continuous louvers, interlocking mullions), at spacings not exceeding those recommended by the manufacturer, or 6 m, whichever is the lesser.

Adjustable louvers

General: Louver blades, clipped into blade holders pivoted to stiles or coupling mullions, linked together in banks, each bank operated by an operating handle incorporating a latching device, or by a locking bar.

Framed adjustable louvers

General: Louver blades, beaded into steel blade surround frames (sash), pivoted to pressed steel main frames, linked together in banks, each bank controlled by a proprietary sash operator.

Fixed metal louvers

General: Metal louver blades, mounted in a metal surround frame or subframe.

Screens

Requirement: Provide metallic-coated steel wire, stainless steel or PVC mesh screens behind louvers to prevent the entry of vermin, birds, rodents and wind-blown leaves and papers.

2.7 SECURITY WINDOW GRILLES General

Requirement: Proprietary metal security grilles, or operable screen and frames, fixed to the building structure with tamper resistant fastenings.

Security window grilles: To AS 5039.

2.8 ALUMINIUM FRAME FINISHES Powder coatings

Service condition category to AS 3715: category 4

Coating performance: to AAMA 2604

Coating type: Thermoset polyester powder coating.

Polyester coating grade: commercial

Colour: refer to finishes schedules

2.9 ANCILLARY COMPONENTS AND FITTINGS Trims

Timber: Solid timber at least 19 mm thick, mitred at corners.

Extruded gaskets and seals

General: Provide seals, as documented.

Materials: Non-cellular (solid) elastopressive seals as follows:

- Flexible polyvinyl chloride (PVC): To BS 2571, 100% solids with high consistency, ultraviolet stabilised.
- Rubber products (neoprene, ethylene propylene diene monomer (EPDM) or silicone rubber): To BS 4255-1.

Flashings

General: Corrosion resistant, compatible with the other materials in the installation, and coated with a non-staining compound where necessary.

Standard: To AS/NZS 2904.

Nylon brush seals

General: Dense nylon bristles locked into galvanized or stainless steel strips and fixed in a groove in the edge of the door or in purpose-made anodised aluminium holders fixed to the door with double sided PVC foam tape.

Pile weather strips

Standard: To AAMA 701/702.

Materials: Polypropylene or equivalent pile and backing, low friction silicone treated, ultraviolet stabilised.

Finned type: A pile weather seal with a central polypropylene fin bonded into the centre of the backing rod and raised above the pile level.

Weather bars

General: A weather bar for hinged external doors, located under the centres of closed doors.

2.10 HARDWARE Hardware documented generically

General: Provide hardware of sufficient strength and quality to perform its function, appropriate to the intended conditions of use, compatible with associated hardware, and fabricated with fixed parts firmly joined. Refer to appendix

Window locks and latches Standard: To AS 4145.3.

Window catches: Provide 2 catches per sash to manually latched awning or hopper sashes over 1000 mm wide.

Sash balances

Requirement: Match the spring strength of the balances to the sash weight they support.

Sash operators

Requirement: Provide sash operators, as documented.

2.11 KEYING Contractor's keys

Master key systems: Do not use any key under a master key system.

Identification

Labelling: Supply each key with a purpose-made plastic or stamped metal label legibly marked to identify the key, attached to the key by a metal ring.

Key material

Pin tumbler locks: Nickel alloy, not brass.

Lever locks: Malleable cast iron or mild steel.

Keying system

Requirement: Keying system, as documented.

Coding of locks: If window locks are included in building key code groups, provide cylinder or pin tumbler locks coded to match.

Number of keys table

Code	Key type	Minimum number of keys

Master key schedule to be confirmed with stanwell

3 EXECUTION

3.1 GLASS PROCESSING General

Processing: Perform required processes on glass, including cutting, obscuring, silvering and bending. Form necessary holes, including for fixings, equipment, access openings and speaking holes. Process exposed glass edges to a finish not inferior to ground arised.

3.2 INSTALLATION Glazing

General: Install the glass as follows:

- Permanently fix in place each piece of glass to withstand the normal loadings and ambient conditions at its location without distortion or damage to glass and glazing materials.
- No transfer of building movements to the glass.
- Watertight and airtight for external glass.

Temporary marking: Use a method which does not harm the glass. Remove marking on completion.

Toughened glass: Do not cut, drill, edge-work or permanently mark after toughening. Use installation methods which prevent the glass making direct contact with metals or other non-resilient materials.

Heat absorbing glass: In locations exposed to direct sunlight, provide wheel cut edges free from damage or blemishes, with minimum feather.

Preglazing

Window assemblies and glazed doors: Supply inclusive of glazing, shop preglazed.

External timber framed glazing: Glaze with putty.

Windows and glazed doors

General: Install windows and glazed doors frames as follows:

- Plumb, level, straight and true within building tolerances.
- Fixed or anchored to the building structure in conformance with the wind action loading requirements.
- Isolated from any building loads, including loads caused by structural deflection or shortening.
- Allow for thermal movement.
-

Weatherproofing

Flashing and weatherings: Install flashings, weather bars, drips, storm moulds, joint sealant and pointing to prevent water penetrating the building between the window frame and the building structure under the prevailing service conditions, including normal structural movement of the building.

Fixing

Fasteners and fastener spacing: Conform to the recommendations of the manufacturer.

Packing: Pack behind fixing points with durable full width packing.

Fasteners: Conceal fasteners.

Prepared masonry openings: If fixing of timber windows to prepared anchorages needs fastening from the frame face, sink the fastener heads below the surface and fill the sinking flush with a material compatible with the surface finish.

Joints

General: Make accurately fitted tight joints so that neither fasteners nor fixing devices such as pins, screws, adhesives and pressure indentations are visible on exposed surfaces.

Sealants:

- If priming is recommended, prime surfaces in contact with jointing materials.
- If frames are powder coated apply a neutral cure sealant.

Operation

General: Make sure moving parts operate freely and smoothly, without binding or sticking, at correct tensions or operating forces and are lubricated.

Protection

Removal: Remove temporary protection measures from the following:

- Contact mating surfaces before joining up.
- Exposed surfaces.

Trim

General: Provide mouldings, architraves, reveal linings, and other internal trim using materials and finishes matching the window frames. Install to make neat and clean junctions between frames and the adjoining building surfaces.

3.3 VENTILATING LOUVRE ASSEMBLIES Installation

General: Screw fix stiles and mullions to the building structure. Provide weather strips to heads and sills.

Framed adjustable louvres

Installation: Screw fix the main frame to the building structure with monel or stainless steel screws or masonry anchors of the type recommended by the louvre manufacturer.

Metal louvres

General: Install as for metal window installations.

3.4 HARDWARE Fasteners

Materials: Use materials compatible with the item being fixed and of sufficient strength, size and quality to perform their function.

- Concealed fasteners: Provide a corrosion-resistant finish.
- Exposed fasteners: Match exposed fasteners to the material being fixed.

Support: Provide appropriate back support (for example lock stiles, blocking, wall noggings and backing plates) for hardware fasteners.

- Hollow metal sections: Provide backing plates drilled and tapped for screw fixing, or provide threaded inserts (rivet nuts) with machine thread screws. Do not use self-tapping screws or pop rivets.

Proprietary window systems

Requirement: Provide the standard hardware and internal fixing points for personnel safety harness attachment, if required by and conforming to the governing regulations.

Operation

General: Make sure working parts are accurately fitted to smooth close bearings, without binding or sticking, free from rattle or excessive play, lubricated where appropriate.

Supply

Delivery: Deliver window hardware items, ready for installation, in individual complete sets for each window set, as follows:

- Clearly labelled with the intended location.
- In a separate dust and moisture proof package.
- Including the necessary templates, fixings and fixing instructions.

3.5 COMPLETION Hardware

Adjustment: Leave the hardware with working parts in working order, and clean, undamaged, properly adjusted, and lubricated where appropriate.

Keys

Contractor's keys: Immediately before the date for practical completion, replace cylinders to which the contractor has had key access during construction with new cylinders that exclude the contractor's keys.

Keys: For locks keyed to differ and locks keyed alike, verify quantities against key records, and deliver to the contract administrator at practical completion.

Key codes: Submit the lock manufacturer's record of the key coding system showing each lock type, number and type of key supplied, key number for re-ordering, and name of supplier.

Repair of finish

Polyester or fluoropolymer coatings: Contact supplier for approval to apply touch up products, otherwise replace damaged material.

Cleaning

Method: Clean with soft clean cloths and clean water, finishing with a clean squeegee. Do not use abrasive or alkaline materials.

Extent: All frames and glass surfaces inside and out.

Warranties

Window and door assemblies: Provide the manufacturer's published product warranties.

Hardware: Provide the manufacturer's published product warranties.

4 SELECTIONS

4.1 DOORS Door Schedule

- Refer to drawings for door schedule
- Refer to hardware schedule.

4.2 WINDOWS Window Schedule

- Refer to drawings for window schedule
- Glass type varies. Refer to window schedules

4.3 MASTER KEYING

Requirement: match Master key system throughout - TBC

Keying schedule to be confirmed. TBC

Coding of locks: TBC

To be designed by proposed subcontractor for client approval.

0455 DOOR HARDWARE

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide door hardware, as Documented. Refer to Appendix C – Door Hardware Schedule, prepared by formakaba

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.

1.3 INTERPRETATION

Abbreviations

General: For the purposes of this worksection, the abbreviations given in AS 4145.1 (2008) Appendix D apply.

Definitions

General: For the purposes of this worksection, the general definitions given in AS 4145.1 (2008) Section 2 and Appendix E apply.

1.4 SUBMISSIONS

Execution details

Door-by-door schedule: Submit a door-by-door hardware schedule.

- Information sources: This worksection and the Contract drawings. Key control system:
- Keyed hardware: Submit details to extend the existing key control security system for locks required to accept a group key.
- Electronic hardware: Submit details to extend the existing electronic control security system for locks required to accept a swipe card key.

Operation and maintenance manuals

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals. Records**

Door hardware schedule: Submit a schedule, prepared by the door hardware supplier, showing changes to the Contract door hardware schedule resulting from the following:

- Approval of a hardware sample.
- Acceptance of an equivalent to a specified proprietary item.
- A variation to a door hardware requirement.

Key coding system: Submit the lock manufacturer's record of the key coding system showing each lock type, number and type of key supplied, key number for re-ordering, and name of supplier.

Samples

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples. Subcontractors**

Automatic door operators: Submit names and contact details of proposed manufacturer and installer.

Pressure floor mat: Submit names and contact details of proposed manufacturer and installer.

Warranties

Requirement: Submit warranties to **COMPLETION, Warranties**.

2 PRODUCTS

2.1 GENERAL

Samples

Generic items: Provide samples of hardware items offered as meeting the description of items not specified as proprietary items.

Supply

Delivery: Deliver door hardware items, ready for installation, in individual complete sets for each door, as follows:

- Clearly labelled to show the intended location.
- In a separate dust and moisture proof package.
- Including the necessary templates, accessories, fastenings and fixing instructions.

Hardware specified generically: Hardware of the required strength and quality to perform its function, appropriate to the intended conditions of use, suitable for use with associated hardware, and fabricated with fixed parts firmly joined.

Storage and handling

General: Store and handle to protect materials from damage, to the manufacturer's recommendations.

2.2 LOCKS AND LATCHES

Standard

General: To AS 4145.2 (2008).

Padlocks

Standard: To AS 4145.4 (2002).

Lock and latch classification

Rating systems: To AS 4145.1 (2008) Section 3.

Performance requirements: To AS 4145.2 (2008) Section 3.

2.3 HINGES

Butt hinge materials

Timber doors in timber or steel frames:

- Material: refer to Appendix C – Door Hardware Schedule. Aluminium framed doors or timber doors in aluminium frames:
- Material: refer to Appendix C – Door Hardware Schedule. Doors fitted with closers: Provide low friction ball bearing hinges. Fire-resisting doors: To AS 1905.1 (2015).

Power transfer hinges: Provide hinges that allow wires to be transferred unbroken between the door and frame.

Lift-off doors: If sanitary compartments require lift-off doors, provide lift-off hinges and allow sufficient clearance at the head to allow door removal.

Timber solid core doors

Number of hinges: Determine the number of hinges required based on the nominated door leaf size and weight only. For other door leaf sizes or for doors with applied finishes, use the weight of the door to determine the number of hinges required. For a door leaf over 80 kg, use pivot hinges.

Size of hinges: Determine the size of the hinge based on the door leaf thickness:

- 35 to 43 mm: 100 x 75 mm butt hinges with a minimum thickness of 2.5 mm.
- 44 to 55 mm: 100 x 100 mm butt hinges with a minimum thickness of 2.5 mm.
- > 55 mm: As Documented.

Hinge pin: Supply fixed pins to hinges of doors opening out or designated as a security doors. For all other doors, provide loose pins.

Wide throw: If necessary, provide wide throw hinges to achieve the required door swings in the presence of obstacles such as nibs, deep reveals and architraves.

Hinges for timber doors table (unless noted otherwise in Appendix C – Door Hardware Schedule)

Nominal door leaf size (H x W x T) (mm)	Maximum door leaf weight (kg)	Number of hinges
2040 x 400 x 35	19	2
2040 x 600 x 35	29	2
2040 x 720 x 35	35	3
2040 x 820 x 35	39	3
2040 x 920 x 35	44	3
2040 x 1020 x 35	49	4
2040 x 720 x 40	37	3
2040 x 820 x 40	42	3
2040 x 920 x 40	48	3
2040 x 1020 x 40	52	4
2040 x 720 x 50	45	3
2040 x 820 x 50	50	3
2040 x 920 x 50	57	3
2040 x 1020 x 50	68	4
2400 x 720 x 40	50	4
2400 x 820 x 40	52	4
2400 x 920 x 40	55	4
2400 x 1020 x 40	60	4
2400 x 1220 x 50	72	5
2040 x 920 x 70	88	Pivot hinges

Aluminium doors

Application: Aluminium hinges for aluminium doors, or for doors of other materials in aluminium frames 40 kg or less.

Hinges for aluminium doors table (unless noted otherwise in Appendix C – Door Hardware Schedule)

Nominal hinge size (L x W x T) (mm)	Maximum door leaf weight (kg)	Knuckles (minimum)	Screws/hinge leaf (minimum)
100 x 70 x 3	30	3	3
100 x 80 x 3.5	50	5	4
130 x 50 x 3.4	75	Interfold	3
Length (L) is the dimension along the knuckles, not including hinge tips, if any, and width (W) is the dimension across both hinge leaves when opened flat.			

2.4 SLIDING DOOR HARDWARE

General

Requirement: Provide sliding door tracks and guides, as Documented. For Acoustic Sliding door refer to Appendix F – Acoustic Sliding Door Specification.

2.5 ANCILLARIES

Bolts

General: Barrel bolts, flush bolts and tower bolts with keepers, including lock plates, staples, ferrules or floor sockets.

Door seals

Acoustic applications: Tested to AS 1191 (2002) or EN ISO 10140-2 (2021) and rated to the NCC cited AS/NZS ISO 717.1 (2004).

Weather and energy saving seals: To AS 4420.1 (2016) Sections 5 and 6, and AS 2047 (2014).

Extruded gaskets and seals

General: Provide seals, as Documented.

Materials: Non-cellular (solid) elastomeric seals as follows:

- Rubber products: Neoprene, ethylene propylene diene monomer (EPDM) or silicone rubber.
- Flexible polyvinyl chloride (PVC): E type compounds, colourfastness grade B.

Nylon brush seals

General: Dense nylon bristles locked into holding strips and fixed in a groove in the edge of the door or in purpose-made anodised aluminium holders fixed to the door or frame to the manufacturer's recommendations.

Pile weatherstrips

General: Polypropylene or equivalent pile and backing, low friction silicone treated, ultraviolet stabilised, fixed to the door or frame to the manufacturer's recommendations.

Standard: To AAMA 701/702 (2023).

Rebated doors

General: For mortice locks or latches to rebated doors, provide purpose-made rebated pattern items.

Strike plates

General: Use strike plates supplied with the locks or latches. Do not provide universal strike plates.

Weather bars

General: Provide corrosion-resistant weather bars or threshold plates under hinged external doors, located under the centres of closed doors or to manufacturer's recommendations.

2.6 DOOR CONTROLLERS

Standard

General: To AS 4145.5 (2011).

Performance

Requirement: Door controllers, pivots, floor or overhead door closers, and automatic door operators, suitable for the door type, size, weight, sliding action and swings required and the operating conditions, including wind and air conditioning pressure.

Closers

Hinged or pivot doors used as fire-resisting doors: Closers Tested and certified for use as components of fire-resisting door assemblies to AS 1905.1 (2015).

2.7 ELECTRONIC CONTROL DEVICES

General

Requirement: Electric strikes, electric locks, drop bolts and/or similar devices to suit door construction and hardware.

Electromagnetic hold-open devices: To AS 1905.1 (2015) and the NCC cited AS 1670.1 (2018). Glass doors: Tumbler, drop bolts or magnetic holders.

Double leaf doors (solid frame): Electric strike or lock on the inactive leaf, connected to the door frame by concealed flexible wiring.

Activation

Activation device: Keypads, card readers or other activation devices located next to entry points. External: Weatherproof (IP56) hoods or housings for external units.

Mounting height: 900 to 1100 mm from floor level and not less than 500 mm from internal corners. Refer also to Architectural Drawings.

2.8 PANIC EXIT DEVICES

General

Standard: To EN 1125 (2008). Requirements:

- Field sizable.
- Keyed dogging.

2.9 KEYING

Keying requirements

Standard: To AS 4145.1 (2008) for keying security. Requirement: Provide door hardware and keys, as Documented. **Temporary construction keys and cylinders**

Requirement: Provide one of the following:

- Loan cylinder: Install for construction locks and replace at the Date for Practical Completion.
- Construction keyed master key cylinder: Keep up-to-date records of keys issued including Recipient's name, company and contact details, date issued and date returned.

Delivery of keys

Great grandmaster, grandmaster and master keys: Arrange for delivery direct to the Principal.

Locks keyed to differ and locks keyed alike: Check the quantity against key records, and deliver keys to the Superintendent at the Date of Practical Completion.

Group keying

Keying system: As Documented.

Existing system extension: Obtain the details of existing group or master key systems of the system to be extended.

Future extensions: Provide master and grandmaster group keying systems capable of accommodating future extensions.

Proprietary keying control security system: Provide for cylinder or pin-tumbler locks that accept a group key (e.g. master key, maison key).

Stamping: Stamp keys and lock cylinders to show the key codes and/or door number as scheduled.

Identification

Labelling: Supply each key with a purpose-made plastic or stamped metal label legibly marked to identify the key, attached to the key by a metal ring.

Key material

Lever locks: Malleable cast iron or mild steel. Pin tumbler locks: Nickel alloy, not brass.

Number of keys table

Key code	Key type	Minimum number of keys
GGMK		
GMK		
MK		
KD		
KA		

3 EXECUTION

3.1 INSTALLATION

General

Handing: Before supply, verify on Site, the correct handing of hardware items.

Operation: Make sure working parts are accurately fitted to smooth close bearings, without binding or sticking, free from rattle or excessive play, lubricated if appropriate.

Mounting height

Locks and latches: Centreline of the door knob or lever spindle above finished floor: 920mm AFFL.

Locks

Cylinders: Fix vertically and with consistent key alignment.

Door stops

Fixing location: Fix on the floor, skirting or wall, as appropriate, to prevent the door or door furniture striking the wall or other surface.

Fasteners

Materials: Provide materials compatible with the item being fixed, and of sufficient strength, size and quality to perform their function, and as follows:

- Concealed fasteners: Provide a corrosion-resistant finish to concealed fasteners.
- Exposed fasteners: Match exposed fasteners to the material being fixed.

Security: Locate exposed fasteners to lock furniture on the inside faces of external doors and on the inside faces of internal doors to lockable rooms.

Support:

- Hardware fasteners: Provide appropriate back support, such as lock stiles, blocking, wall noggings and backing plates.
- Hollow metal sections: Provide backing plates drilled and tapped for screw fasteners, or provide rivet nuts with machine thread screws. Do not use self-tapping screws or pop rivets.

Floor springs

General: Form a recess in the floor slab for the floor spring box, securely fix and grout the box in place so that the cover plate is flush with the finished floor.

Hinges

Setout:

- For two hinge doors:
 - . Top hinge: Top of hinge 150 mm down from top of door.
 - . Bottom hinge: Bottom of hinge 250 mm up from bottom of door.
- For three hinge doors:
 - . Top hinge: Top of hinge 150 mm down from top of door.
 - . Bottom hinge: Bottom of hinge 250 mm up from bottom of door.
 - . Centre hinge: Centre between top and bottom hinges.

Metal frames: Fix hinges using metal thread screws. Do not weld hinges to frames.

Timber doorsets: Install butt hinges in housings equal in depth to the thickness of the hinge leaf (except for hinges designed for mounting without housing), and fix with countersunk screws.

Seals

General: Provide the fastenings, rebates, grooves, and clearances required for installation and operation of the seals. Allow seals unwound from coils to settle before use. Install proprietary seals to manufacturer's recommendations and adjust correctly.

3.2 COMPLETION

Adjustment

General: Leave the hardware properly adjusted with working parts in working order, and clean, undamaged, properly adjusted, and lubricated if appropriate.

Opening force performance: To AS 1428.1 (2021).

Automatic door operators: Maintain and adjust the system throughout the Defects Liability Period.

Keys

Contractor's keys: Immediately before the Date of Practical Completion, replace or reset cylinders to which the Contractor has had key access during construction to exclude the Contractor's keys.

Operation and maintenance manuals

Requirement: Prepare a manual that includes the manufacturer's published recommendations for use, care and maintenance of the hardware provided.

Automatic door operators: Include the installer's proposal for continuing maintenance after completion on an annual renewal basis.

Warranties

Type: As provided by the manufacturers and installers. Period: As provided by the manufacturers and installers.

4 SELECTIONS

Refer to Appendix – Door Hardware Schedule and Architectural Drawings.

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide glazing, as Documented. Refer to Design Drawings, and Appendix – Façade and Wall Schedule.

Performance

Thermal qualities: U-Value and Solar heat gain coefficient (SHGC) as Documented.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 *General requirements*.

1.3 STANDARDS

Glazing

Glass type and thickness: To AS 1288 (2021), if no glass type or thickness is nominated. Materials and installation: To AS 1288 (2021).

Quality requirements for cut-to-size and processed glass: To AS/NZS 4667 (2000). Roof glazing: To AS 1288 (2021) Section 6.

1.4 INTERPRETATION

Abbreviations

General: For the purposes of this worksection, the following abbreviations apply:

- AGWA: Australian Glass and Window Association.

Definitions

General: For the purposes of this worksection, the definitions given in AS/NZS 4668 (2000) apply.

1.5 SUBMISSIONS

Certification

Design: Submit an engineer's certificate confirming conformance to AS 1288 (2021).

Opacified glass: Submit a report, from the manufacturer certifying that the proposed method of opacifying the glass will not be detrimental to the glass or affect the glass product warranty.

Toughened glass: Submit certification from the manufacturer for all toughened glass as evidence of heat soaking to **GLAZING, Heat soaking**.

Installation: Submit certification from the fabricator that the method of glazing, the selection of sealant systems and conditions next to the glass conform to the following:

- Compatible with the edge seal of insulating glass units (IGUs) and self-cleaning glass.
- Will not be detrimental to the long-term structural performance, weathering capabilities and visual qualities of the glass.

Glazier's data: Submit the glazing subcontractor's statement certifying the following:

- A satisfactory thermal safety assessment.
- The assembled frame provides the required glazing clearances and tolerances, and maximum and minimum joint configurations, based on the bow, warp and kink characteristics of the required glass types, and is ready for glazing.

Execution details

Site glazing: If Site glazing is intended, submit proposals.

Operation and maintenance manuals

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals. Products and materials**

Safety glazing materials: Submit evidence of conformity to AS 2208 (2023).

Samples

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples. Shop drawings**

Requirement: Submit shop drawings showing the following:

- Method of glazing.
- Rebate depth.
- Edge restraint.
- Clearances and tolerances.
- Glazing gaskets and sealant beads.

Warranties

Requirement: Submit warranties to **COMPLETION, Warranties.**

1.6 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- Glazing products before they are installed.

2 PRODUCTS

2.1 GENERAL

Samples

Requirement: Provide samples of glazing materials, each at least 200 x 200 mm, showing the visual properties and range of variation, if any, for each of the following:

- Tinted or coloured glass or plastics glazing.
- Surface modified or surface coated glass.
- Patterned or obscured glass or plastics glazing.
- Ceramic-coated glass.
- Wired glass.
- Insulating glass units.
- Mirror glass.

Storage and handling

Storage: Store glass and glazing materials in a clean, dry area unaffected by weather, to the manufacturer's recommendations. Protect from building materials and loose debris such as wet plaster, mortar, paint and welding splatter.

Handling: Handle glass to the manufacturer's recommendations.

2.2 GLAZING

Performance

Glazing: Free from Defects that detract from appearance or interfere with performance under normal conditions of use.

Plastics glazing: Free from surface abrasions and resistant to yellowing or other colour change. Capable of maintaining physical properties including strength and impact resistance for its design life.

Heat soaking

Requirement: Heat soak glass to AS 1288 (2021) clause 3.8. Standard: To EN 14179-1 (2016).

Marking: To EN 14179-1 (2016) or certified by the manufacturer to AS 1288 (2021) clause 3.8.2.

Safety glazing materials Standard: To AS 2208

(2023). Type: Grade A.

Marking: To AS 1288 (2021) clause 5.23.

Heat-strengthened glass

Standard: To ASTM C1048 (2025).

Insulating glass units (IGUs)

Requirement: Provide insulating glass units, as Documented. Manufacture,

Testing and installation: To AS 4666 (2012).

2.3 GLAZING MATERIALS

General

Requirement: Putty, glazing compounds, sealants, gaskets, glazing tapes, spacing strips, spacing tapes, spacers, setting blocks, shims and compression wedges appropriate for the conditions of application and required performance.

Primer

Compatibility: Apply the manufacturer's recommended primer to the surfaces in contact with sealant materials.

2.4 ANCILLARY COMPONENTS AND FITTINGS

Extruded gaskets and seals

General: Provide seals, as Documented.

Materials: Non-cellular (solid) elastomeric seals as follows:

- Rubber products: Neoprene, ethylene propylene diene monomer (EPDM) or silicone rubber.
- Flexible polyvinyl chloride (PVC): E type compounds, colourfastness grade B.

Pile weatherstrips

Standard: To AAMA 701/702 (2023).

Material: Polypropylene or equivalent pile and backing, low friction silicone treated, ultraviolet stabilised.

Finned type: A pile weatherseal with a central polypropylene fin bonded into the centre of the backing rod and raised above the pile level.

3 EXECUTION

3.1 GLAZING PROCESSING

General

Processing: Perform required processes on glazing, including cutting, obscuring, silvering and bending. Form necessary holes, including for fastenings, equipment, access openings and speaking holes. Process exposed glass edges to a finish not inferior to ground arised.

3.2 INSTALLATION

Glazing

Requirement: Install the glazing as follows:

- Permanently fix in place each piece of glass to withstand the normal loadings and ambient conditions at its location without distortion or damage to glazing materials.
- No transfer of building movements to the glazing.
- Watertight and airtight for external glazing.

Temporary marking: Use a method that does not damage the glazing. Remove marking only after certification and acceptance of the installation.

Toughened glass: Do not cut, work, or permanently mark after toughening. Use installation methods that prevent the glass making direct contact with metals or other non-resilient materials.

Heat-absorbing glass: In locations exposed to direct sunlight, provide wheel cut edges free from damage or blemishes, with minimum feather.

Preglazing

Window assemblies and glazed doors: Supply inclusive of glazing, shop preglazed. Curtain walls: Supply inclusive of glazing, shop preglazed.

Site glazing

Minimum dimensional requirements: Edge clearance, edge cover, front clearance and back clearance to AS 1288 (2021).

Vision panels: Install so that they are to be removable from: interior. External timber

framed glazing: Glaze with putty.

3.3 COMPLETION

Replacement

Requirement: After replacing damaged glass, leave the Work clean, polished, free from Defects, and in good condition.

Cleaning

Method: Clean with soft clean cloths and clean water, finishing with a clean squeegee. Do not use abrasive, acidic or alkaline materials.

Extent: All frames and glass surfaces internally and externally.

Operation and maintenance manuals

Requirement: Prepare a manual that includes the manufacturers' published recommendations for in-service use.

Warranties

Type: Interlocking manufacturer and installer warranty. Period: As offered by manufacturer and installer.

Glazing subcontractor's warranty: Provide an undertaking conditional only on compliance with the manufacturers' recommendations for maintenance, to repair or replace glass and glazing materials that become Defective or prove unsuitable for the nominated application; during the warranty period.

Glass manufacturer's warranty: Provide an undertaking, conditional only on compliance with the manufacturer's recommendation for installation and maintenance, to supply replacement glass units to the Site for replacement of Defective units defined as follows:

- IGU: Units in which the hermetic seal has failed as evidenced by intrusion of foreign matter, or internal condensation at temperature above 2°C.
- Coated glass units (including coated super insulating glass units): Units in which the metallic coating shows evidence of manufacturing Defects, including but not necessarily limited to cracking or peeling, as determined in conformance with ASTM C1048 (2025).

Toughened glass warranty: Provide a manufacturer's warranty that toughened glass supplied for use in curtain walls has been subjected to a heat soaking process that has converted at least 95% of the nickel sulfide content to the stable beta-phase.

4 SELECTIONS

Refer to drawings & Appendix A

0471 THERMAL INSULATION AND PLIABLE MEMBRANES

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide thermal insulation and pliable membrane systems, as documented.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.

1.3 INTERPRETATION

Definitions

General: For the purposes of this work section the following definitions apply:

- Bio-soluble: A product that dissolves in bodily fluids and is quickly cleared from the lungs.
- Fibre batts: Flexible insulation supplied as factory cut pieces and composed of mineral wool (glass and rock fibre) or polyester fibre.
- Fibre blankets: Flexible insulation supplied as factory cut rolls and composed of mineral wool (glass and rock fibre) or polyester fibre, and may be combined with reflective facings.
- Fire hazard properties: Terminology to BCA A5.5.
- Pliable building membrane: To AS/NZS 4200.1 and equivalent to sarking-type materials as defined in the NCC.
- Thermal insulation terminology: To AS/NZS 4859.1.
- Vapour permeable (breathable) membrane: A flexible membrane material, normally used for secondary waterproofing that allows for the transmission of water vapour.

1.4 SUBMISSIONS

Products and materials

Thermal insulation properties: Submit evidence of conformity to AS/NZS 4859.1 and AS/NZS 4859.2.

Warranties

Manufacturer's published product warranties: Submit on completion.

1.5 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- Insulation or pliable membrane materials after installation and before concealment.

2 PRODUCTS

2.1 GENERAL

Storage and handling

Labelling: Deliver mineral wool products to site in packaging with third party mark of conformity indicating product is bio-soluble and not listed as hazardous material in the Safe Work Australia *Hazardous Chemical Information System* (HCIS).

2.2 FIRE PERFORMANCE

Fire hazard properties

Insulation materials: Tested to AS/NZS 1530.3. Fire hazard indices as follows:

- Spread-of-Flame Index: ≤ 9 .
- Smoke-Developed Index: ≤ 8 if Spread-of-Flame Index > 5 .

Materials with reflective facing: Tested to AS/NZS 1530.3 and the recommendations of Appendix A6.

Pliable membranes: Flammability Index ≤ 5 tested to AS 1530.2.

2.3 MATERIALS

Thermal insulation

Standard: To AS/NZS 4859.1.

Wet processed fibreboard (softboard): To AS/NZS 1859.4.

Mineral wool insulation: Bio-soluble and not listed as a hazardous material in the Safe Work Australia *Hazardous Chemical Information System* (HCIS).

Pliable building membranes Standard: To AS/NZS 4200.1.

Vapour control membranes:

- Vapour barrier:
 - . Vapour control classification: Class 1 or Class 2, as documented.
- Vapour permeable (breathable) membrane:
 - . Vapour control classification: Class 3 or Class 4, as documented.

Water control (sarking) membrane (other than walls and gables):

- Water control classification: Water barrier.

2.4 COMPONENTS

Fasteners and supports

General: Metallic-coated steel.

Mesh support to roof insulation Welded safety mesh: To AS/NZS 4389.

Thermal break strips Product: Proprietary item.

R-Value ($\text{m}^2\cdot\text{K}/\text{W}$): ≥ 0.2 .

3 EXECUTION

3.1 GENERAL

Thermal insulation

Requirement: To AS 3999 and BCA J1.2 or BCA 3.12.1.1, as appropriate.

Installation: Firmly butt together with no gaps except as follows:

- Access openings and vents: Do not obstruct.
- Light fittings: To AS/NZS 3000 clause 4.5.
- Electrical cables: To AS 3999 clause 2.6.

Glass wool and rock wool insulation: Conform to the

ICANZ Industry code of practice for the safe use of glass wool and rock wool insulation.

Pliable building membrane

Installation: To AS 4200.2 and BCA J1.2 or BCA 3.12.1.1, as appropriate.

3.2 WALLS

Framed walls

Fibre batts: Friction fit between framing members. If other support is not provided, staple nylon twine to the framing and stretch tight.

Thermal break strips: Provide to steel framing with lightweight external cladding:

- Screw fixing: Button head screws at 1 m centres.
- Adhesive fixing: Wallboard adhesive walnuts at 1 m centres.

Vapour permeable (breathable) membrane

Requirement: Provide a vapour permeable membrane behind external facing material that does not provide permanent weatherproofing or that may be subject to condensation forming on the internal face, including the following:

- Boards or planks fixed vertically or diagonally.
- Boards or planks fixed in exposed locations where wind driven rain can penetrate the joints.
- Unpainted or unsealed cladding.
- Masonry veneer.

Installation: Run the vapour permeable membrane horizontally on the outer face of external wall framing, over the flashing, from the bottom plate up. Pull taut over the framing and fix to framing members. Seal across the wall cavity at the top.

Horizontal laps: At least 150 mm wide, lapped to make sure water is shed to the outer face of the membrane.

End or vertical overlaps laps: At least 150 mm wide made over framing.

Openings: Run the vapour permeable membrane over the openings and leave covered until windows and doors are installed.

Cut the membrane on a 45° diagonal from each corner of the opening, fold the flaps inside and fix to the inside frame of the opening. If the membrane is used to provide a continuous air tight layer, seal all joints with pressure sensitive adhesive tape.

Fixing: Install as follows:

- Timber frames: Metallic-coated clouts, 20 mm long 6 to 8 mm staples or punched multi-point metallic-coated steel brads.
- Steel or aluminium frames: Hex head screws, with either 20 mm diameter washers or through hardboard strips.
- Plywood: Alternatives:
 - o Metallic-coated clouts, 20 mm long 6 to 8 mm staples or punched multi-point metallic-coated steel brads at minimum 300 mm centres.
 - o Water based contact adhesive with a 50% adhesive cover.

3.3 CEILINGS

Framed ceiling

Suspended ceiling

Fibre batts and blankets: Lay batts/blankets over the ceiling system close butted to each other and to the suspension rods.

3.4 ROOFS

General

Requirement: Provide insulation to the whole of the roof area including skylight shaft walls, except the following:

- Eaves, overhangs, skylights, vents and openings.
- Roofs to outbuildings, garages, and semi-enclosed spaces such as verandahs, porches and carports.

Mesh support to roof insulation

Requirement: Provide support to the following:

- Water control (sarking), vapour barrier or reflective thermal insulation membranes laid over roof framing members that are spaced at more than 900 mm centres.
- Blanket type thermal insulation laid over roof framing members as sound insulation to metal roofing.

Installing welded safety mesh: To AS/NZS 4389.

Metal roofs

Fibre batts: Fit tightly between framing members.

Fibre blanket for sound insulation: Install over the roof framing, reflective thermal insulation (if any), and mesh support, so that the blanket is in continuous contact with the underside of the metal roofing sheets.

Combined fibre blanket and reflective insulation: Lay facing reflective insulation face downwards over safety mesh.

Thermal break strips: Provide to steel framing supporting metal sheet roofing.

- Screw fixing: Button head screws at 1 m centres.
- Adhesive fixing: Wallboard adhesive walnuts at 1 m centres.

Pliable building membranes

Vapour barrier: Lay over the roof framing with sufficient sag to allow the bulk insulation to achieve its full thickness. Overlap all edges 150 mm and seal all joints with pressure sensitive adhesive tape.

Water control (sarking) membrane: Provide sarking under tile and shingle roofing.

4 SELECTIONS

4.1 INSULATION

Wall insulation schedule

- Refer to section 0472 Acoustic insulation for extent and scope of wall insulation.

Ceiling insulation schedule

- Refer to section 0472 Acoustic insulation for extent and scope of ceiling insulation.

Roof Insulation Schedule

Property	Walls	Ceilings	
Application	165mm bulk insulation	165mm bulk insulation	
location	Stud walls A1 registry, A2 workshop, A3 main warehouse, A4 new build	Stud walls A1 registry, A2 workshop, A3 main warehouse, A4 New build	
Type/Product	Bradford Gold hi performance wall	Bradford Gold ceiling	
R-Value (m ² .K/W)	2.0	3.0	
Thickness (mm)	90 mm	165mm	

0472 ACOUSTIC INSULATION

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide acoustic insulation, as documented. Also refer copy of the Acoustic report in the appendix to these specifications.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.

1.3 INTERPRETATION

Definitions

General: For the purposes of this work section the following definitions apply:

- Acoustic insulation: Materials or methods of construction to reduce the transmission of airborne and structure-borne sound through floors, walls and ceilings or other enclosing elements in buildings.
- Acoustic underlay: A resilient material laid between the structural floor and the flooring material to provide sound isolation.
- Airborne sound: Sound radiated directly from a source, such as a loudspeaker or machine, into the surrounding air.
- Bio-soluble: A product that dissolves in bodily fluids and is quickly cleared from the lungs.
- Fibre batts: Flexible insulation supplied as factory cut pieces and composed of mineral wool (glass and rock fibre) or polyester fibre.
- Fibre blankets: Flexible insulation supplied as factory cut rolls and composed of mineral wool (glass and rock fibre) or polyester fibre, and may be combined with reflective facings.
- Fire hazard properties: Terminology to BCA A5.5.
- Impact sound: Sound caused by impacts on building structure. Typical sources include footsteps, dropped objects on horizontal surfaces and the slamming of doors.
- Sound insulation (isolation): Reduction of sound energy passing through building elements.
- Structure-borne sound: Sound waves transmitted within the building structure and re-radiated into other spaces as airborne sound. Typical sources include direct impact from dropped objects and vibrating machinery.

1.4 SUBMISSIONS

Fire performance

Fire hazard properties: Submit evidence of conformity to PRODUCTS, FIRE PERFORMANCE, Fire hazard properties.

Products and materials

Acoustic insulation properties: Submit evidence of conformity to documented requirements for insulation.

Warranties

Manufacturer's published product warranties: Submit on completion.

1.5 INSPECTION

Notice

Inspection: Give notice so inspection may be made of the following:

- Insulation after installation and before concealment.

2 PRODUCTS

2.1 GENERAL

Storage and handling

Labelling: Deliver mineral wool products to site in packaging with third party mark of conformity indicating product is bio-soluble and not listed as hazardous material in the Safe Work Australia Hazardous Chemical Information System (HCIS).

2.2 FIRE PERFORMANCE

Fire hazard properties

Insulation materials: Tested to AS/NZS 1530.3. Fire hazard indices as follows:

- Spread-of-Flame Index: ≤ 9 .
- Smoke-Developed Index: ≤ 8 if Spread-of-Flame Index > 5 .

Facing materials: Tested to AS 1530.2: Flammability Index ≤ 5 .

2.3 MATERIALS

General

Mineral wool insulation: Bio-soluble and not listed as a hazardous material in the Safe Work Australia *Hazardous Chemical Information System* (HCIS).

Bulk insulation

Mineral wool blankets and batts: Glass wool or rock wool bonded with thermosetting resin.

Polyester blankets and batts: Thermally bonded polyester fibres.

Board insulation

Mineral wool panels: High density glass wool or rock wool bonded with thermosetting resin.

Wet process fibreboard (including softboard): To AS/NZS 1859.4.

Composite plasterboard panels: Proprietary items.

Flexible sheet insulation

Impregnated vinyl: Lead impregnated vinyl sheeting.

Recycled rubber/cork: Recycled rubber granules and/or cork bound with polymers.

2.4 COMPONENTS

Fasteners and supports

General: Metallic-coated steel.

Resilient mounts: Proprietary fixing clips with rubber or acrylic pads.

Adhesives

General: Compatible with the substrate and the insulation and conforming to the insulation manufacturer's recommendations.

Sealants

Acoustic sealant: Non-hardening sealant compatible with the substrate materials.

Fire-resisting sealant: Non-hardening sealant compatible with the substrate materials and having a fire-resistance rating equal to that of the building element it seals.

Sealant strips: Closed cell resilient foam.

3 EXECUTION

3.1 GENERAL

Bulk insulation

General: Firmly butt together with no gaps except as follows:

- Access openings and vents: Do not obstruct.
- Light fittings: To AS/NZS 3000 clause 4.5.
- Electrical cables: To AS 3999 clause 2.6.

Glass wool and rock wool insulation: Conform to the

ICANZ Industry code of practice for the safe use of glass wool and rock wool insulation.

3.2 WALLS

Framed walls and partitions

Fibre batts: Friction fit between framing members. If other support is not provided, staple nylon twine to the framing and stretch tight.

Mineral wool panels: Fix to face of studs with adhesive and temporarily fasten with single screw until plasterboard installed.

3.3 CEILINGS

Framed ceiling

Fibre batts: Fit tightly between framing members. If support is not otherwise provided, staple nylon twine to the framing and stretch tight.

Suspended ceiling

Fibre batts and blankets: Lay batts/blankets over the ceiling system close butted to each other and to the suspension rods.

4 SELECTIONS

4.1 SCHEDULES

Acoustic insulation schedule - walls

Property	A	B	C
Application	Wall		
Product/Material	Bradford Gold hi performance wall		
Thickness (mm)	90mm		
Density (kg/m ³)			
Surface weight (kg/m ²)			
R Value	2.0		

Acoustic insulation schedule - Ceilings

Property	D	E	F
Application	Ceiling Tiles		
Product/Material	Bradford gold ceiling		
Thickness (mm)	165mm		
Density (kg/m ³)	8kg/m ³		
Surface weight (kg/m ²)			
R Value	3.0		

The insulation material shall be Bradford Gold partition rolls having a Material R Value, thickness and density as noted in tables above, and shall be deemed non-combustible when tested to AS1530.1. For installation specifications refer to the relevant Bradford product selector.

0511 LINING

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide internal lining systems, as documented. Refer to design drawings

Performance

Requirement: Provide lining system with a surface that is:

- Resistant to impacts expected in use.
- Resistant to moisture encountered under expected environmental conditions.
- Free of irregularities.
- A suitable substrate for the nominated final finish.

1.2 CROSS REFERENCES General

Requirement: Conform to the following: - *0171 General requirements.*

1.3 INTERPRETATION Definitions

General: For the purposes of this work section the definitions given in AS/NZS 4491 and the following apply:

- Decorative overlaid wood panels: Particleboard or fibreboard with a bonded decorative finishing surface such as thermosetting resin (low pressure melamine), PVC film, paper foils or wood veneer.
- Dry process fibreboard (MDF): Panel material with a nominal thickness of 1.5 mm or greater, manufactured from lignocellulosic fibres (derived from wood or other materials) with application of heat and/or pressure, the bond of which is derived from a synthetic adhesive added to the fibres and the panels are manufactured with a forming moisture content of less than 20%.
- Fibre cement sheet linings: Treated cellulose fibre in a matrix of cement and sand autoclaved sheet, sealed on one side.
- High pressure decorative laminates (HPDL):
 - o Panels consisting of core layers impregnated with phenolic and/or aminoplastic resins and a surface layer(s) impregnated with aminoplastic resins (mainly melamine resins).
 - o Sheets consisting of a decorative face and layers of fibrous sheet material (e.g. paper) impregnated with thermosetting resins and bonded together under heat and pressure of at least 5 MPa.
- Particleboard: Panel material manufactured under pressure and heat from particles of wood (wood flakes, chips, shavings, sawdust and similar) and/or lignocellulosic material in particle form (flax shives, hemp hurds, bagasse fragments, rice hulls, wheat straw and similar) with the addition of an adhesive.
- Wet process fibreboard: Panel material with a nominated thickness of 1.5 mm or greater, manufactured from lignocellulosic fibres (derived from wood or other materials) with application of heat and/or pressure, the bond of which is derived from the felting of the fibres and the panels are manufactured with a forming moisture content greater than 20%.

1.4 TOLERANCES Permitted deviations

Bearing surface of finished framing:

- Gypsum lining: To AS/NZS 2589 clause 4.2.2. - Other lining: 4 mm from a 1.8 m straightedge.

1.5 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Substrate or framing before installation of linings.
- Finished surface of installation before applying:
 - o Sealer.

2 PRODUCTS

2.1 GENERAL Storage and handling

Requirement: Dry and undamaged lining stacked in pallets horizontally on a smooth, level surface. Prevent distortion or moisture ingress.

Timber or fibreboard panels: Store off the ground in a well-ventilated area.

Handling: Do not drag sheets across each other or across other materials. Protect edges, corners and surface from damage.

Acclimatisation

Timber panels: Store on-site in final interior conditions for 2 to 3 weeks before installing. Do not install until the air conditioning system of the installation area is operating.

Certification

Timber based products: Label panels under the authority of a recognised certification scheme to *0185 Timber products, finishes and treatment*, as applicable to the product. Locate the label on faces or edges which will be concealed in the works.

2.2 FIRE PERFORMANCE

Fire hazard properties Group number: To AS5637.1.

2.3 PLASTERBOARD General

Standard: To AS/NZS 2588.

Location: refer to drawings

Grade: standard grade unless noted otherwise

Thickness (mm): Generally 10mm plasterboard;

Edge finish: recessed edges for a flush set joint finish.

2.4 FIBRE CEMENT General

Standard: To AS/NZS 2908.2.

Wall and ceiling linings: Type B category 2.

Minimum thickness: 6 mm.

Location: generally to wet areas, external soffits, external cladding as noted on drawings. Type: Type A for cladding and soffit linings.

Category: 3 for cladding and soffit linings

Thickness (mm): generally 6mm unless noted otherwise.

2.5 DRY PROCESS FIBREBOARD (INCLUDING MEDIUM DENSITY FIBREBOARD) General

Standard: To AS/NZS 1859.2.

Melamine overlaid medium density fibreboard: Medium density fibreboard (STD MDF) overlaid on both sides with low pressure melamine.

2.6 ADHESIVES, SEALANTS AND FASTENERS Adhesives

For wallboards: Gunnable synthetic rubber/resin based mastic contact adhesive formulated for bonding flooring and wallboards to a variety of substrates.

Sealants

Fire-resisting sealant: Non-hardening sealant, compatible with the materials to be sealed and having a fire-resistance rating equal to that of the building element it seals.

Acoustic sealant: Non-hardening sealant compatible with the materials to be sealed.

3 EXECUTION

3.1 CONSTRUCTION GENERALLY Conditions

Commencement: Do not start lining work until the building or installation area is enclosed and weathertight, and all wet trades have been completed.

Substrates

Requirement: Plumb, level, in true alignment and to the lining manufacturer's recommendations.

Timber, steel framing and battened masonry: To AS/NZS 2589 clause 4.2.

Preparation: Before fixing linings, check and adjust the alignment of substrates or framing, if necessary.

Battens

General: Fix at each crossing with structural framing members, to solid walls or ceiling support. Provide wall plugs in solid substrates.

Ceiling linings

General: Do not install until the timber roof structure is fully loaded for at least 14 days.

Accessories and trim

General: Provide accessories and trim as necessary to complete the installation.

Adhesives

General: Provide adhesive types appropriate for the purpose, and apply them so they transmit the loads imposed without causing discolouration of the finished surfaces.

Fire-resisting and acoustic rated installations

Sealing: Apply sealant to the manufacturer's recommendations and as follows: - Around services pipes and penetrations.

- Electrical outlets and recessed lights: Line back and sides of fixture with plasterboard and seal around fixture junction with sealant.
- Around perimeter of lining panels: Provide continuous runs of sealant.

3.2 PLASTERBOARD LINING Installation

Gypsum plasterboard and fibre reinforced gypsum lining: To AS/NZS 2589. Level of finish and jointing: To AS/NZS 2589 clause 3.1.

Supports

General: Install timber battens or proprietary cold-formed galvanized steel furring channels as follows:

- Where framing member spacing exceed the recommended spacing.
- Where direct fixing of plasterboard is not possible, due to the arrangement or alignment of the framing or substrate.
- Where the lining is the substrate for tiled finishes.
- If required for penetrations for services, including mechanical grilles and lighting fixtures.
- If required to support fixtures.

3.3 FIBRE CEMENT LINING Installation

Joints and layout: Run sheets across the framing members. In flush jointed applications, stagger end joints in a brick pattern and locate them on framing members, away from the corners of large openings. Provide supports at edges and joints.

Supports

General: Install timber battens or proprietary cold-formed galvanized steel furring channels as follows:

- Where framing member spacing exceed the recommended spacing.
- Where direct fixing of fibre cement is not possible, due to the arrangement or alignment of the framing or substrate.
- Where the lining is the substrate for tiled finishes.
- If required for penetrations for services, including mechanical grilles and lighting fixtures.
- If required to support fixtures.

Fixing

Timber framed construction: Nail only or combine with adhesive.

Steel framed wall construction: Screw only or combine with adhesive.

Wall framing: Conform to the following:

- Do not fix to top and bottom plates or noggings.
- In tiled areas: Provide an extra row of noggings immediately above wall-to-floor flashings. Fix sheet at 150 mm centres to each stud and around the perimeter of the sheet.

Ceilings: Fix using nail and/or adhesive to timber ceiling furring members. Do not fix sheets directly to the bottom chords of trusses. **Do not use countersunk screws for any ceiling/soffit fixings**

- Ceiling battens: Fix at 350 or 400 mm maximum centres – to be confirmed with project wind classification

Wet areas: Do not use adhesive fixing alone.

Joints

Joint width:

- Butt joints: 1 to 2 mm.
- Expressed joints: 10 mm maximum.

Joint backing for expressed joints: Black self-adhesive polyurethane tape.

Flush joints: Provide recessed edge sheets and finish flush using perforated paper reinforcing tape.

External corner joints: Make joints over metallic-coated steel corner beads.

Dry joints: Provide square edged sheet and finish with a PVC-U joining section.

Control joints: Provide control joints to coincide with structural control joints and as follows:

- Walls: ≤ 7.2 m centres.
- Ceilings: To divide into bays not larger than 10.8×7.2 m.
- Soffit linings: To divide into bays not larger than 4.2×4.2 m or 5.6×3.6 m.
- Control joint beads: Purpose-made metallic-coated.
- Support: Provide framing parallel to the joint on each side. Do not fix the lining to abutting building surfaces.

Wet areas: Provide additional supports, flashings, trim and sealants as required.

Joints in tiled areas: Bed perforated paper tape in bedding compound. Do not apply a topping coat.

- Control joints: Not more than 4.2 m centres and space to suit joints required in tiling.
- Internal corners: Reinforce with metallic-coated steel angles. In corners subject to continuous moisture, flash over the angle and under the sheeting with continuous bitumen coated aluminium flashing.

3.4 TRIM AND ACCESSORIES General

Requirement: Provide trim such as beads, mouldings and stops to make neat junctions between lining components, finishes and adjacent surfaces.

Proprietary items: Provide complete with installation accessories.

Timber and MDF trim: Fix using full length so that trim is secure and without movement. Where nail or screw fixings are used, make sure fastener finishes sufficiently below face of trim so that stopping piece finishes flush with the face.

3.5 COMPLETION General

Damaged or marked lining and components: Replace.

Exposed surfaces: Touch up shop applied finishes and restore damaged or marked areas.

Timber panels: If appearance is not uniform, replace panels.

Cleaning: Clean completed surfaces to remove irregularities and leave panels smooth and clean, to the manufacturer's recommendations. If required, sand with fine paper to remove irregularities and refinish panel surface.

- Debris and unused material: Remove from site.

Warranties

Requirement: At practical completion, submit warranties against defective materials and installation.

4 SELECTIONS

4.1 SHEET LINING Wall sheet lining schedule (also refer partition plans for reference)

Property / wall type	PB Plasterboard	VB Villaboard	
Location	As noted on plans	Wet Area	
Material	Plasterboard	Villaboard	
Thickness (mm)	10mm	9mm	
Configuration			
Edge type	Recessed edge	Recessed edge	
Joint type	flush	flush	
Fixing			
Level of finish	4	4	
Fire hazard properties: Group number	1		
Battens: Size (mm)			
Battens: Spacing	450 ctrs max	450 ctrs max	

Ceiling sheet lining schedule

Property	FC Joint fibre cement	Insulated Panelling	
Location	Typical external soffits	Verandahs to new Build Fineline profile under	
Material	Fibre cement	Colorbond	
Thickness (mm)	6mm nail fixed to 70x45 timber batten framing	Prefinished – selected colorbond finish	
Edge type	square		
Slot type	n/a	n/a	
Joint type	PVC straight joiner	n/a	
Level of finish	4		
Fire hazard properties: Group number	1		
Battens: Size (mm)	70x45 timber batten framing	-	
Battens: Spacing	450 ctrs max	-	

Note: Countersunk screws are not permitted for use on FC soffit sheeting. All soffits to have timber framing.

Joinery lining schedule

Property	Laminate		
Location	Refer Joinery plans		
Material	Laminated sheets		
Grade			
colour	Refer finishes schedule. TBC		
Thickness (mm)	18mm		
Width (mm) coverage			
Orientation of boards			
Joint type	As noted on plans – butt joints & expressed joints		

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide cubicles, as Documented. Refer to Design Drawings and Appendix A

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.

1.3 INTERPRETATION

Abbreviations

General: For the purposes of this worksection, the following abbreviations apply:

- ABS: Acrylonitrile butadiene styrene.

Definitions

General: For the purposes of this worksection, the following definitions apply:

- Acrylic polymer resin panels: Thermoplastic or thermosetting acrylic resin panels derived from acrylic acid, methacrylic acid or other related compounds, with mineral fillers and pigments.
- High pressure decorative laminates (HPDL):
 - . Panels consisting of core layers impregnated with phenolic and/or aminoplastic resins and a surface layer(s) impregnated with aminoplastic resins (mainly melamine resins).
 - . Sheets consisting of a decorative face and layers of fibrous sheet material (e.g. Paper) impregnated with thermosetting resins and bonded together under heat and pressure of at least 5 MPa.

1.4 TOLERANCES

General

Deviation from true grid lines and planes: 1:1000 to a maximum of 3 mm. Misalignment of adjoining surfaces at panel junctions: 1 mm.

Panel thickness: ± 0.5 mm.

Length and width: 0.1% of the dimension or 0.5 mm, whichever is the greater.

Flatness, twist, winding and bow: 1 mm deviation from a 2.4 m straightedge placed in any position. Maximum deviation of edges from the intended true line: ± 1 mm.

1.5 SUBMISSIONS

Operation and maintenance manuals

Requirement: Submit manual to COMPLETION, Operation and maintenance manuals. Products and materials

Manufacturer's data: Submit manufacturer's standard product literature for each system type.

Samples

Requirement: Submit samples to PRODUCTS, GENERAL, Samples. Shop drawings

General: Submit shop drawings to a scale that best describes the detail, including the following:

- Overall layout and dimensions.
- Materials, thicknesses and finishes of elements including doors, divisions, fronts, pedestals and top rail.
- Assembly components, including suspension beam and fixing hardware.
- Door hardware type and location.
- Relationship of assembly to adjacent building elements.
- Junctions and trim to adjoining surfaces.
- Methods of fixing partitions.

Subcontractors

General: Submit names and contact details of proposed suppliers and installers. Evidence of experience: QBCC Licence and CV.

Substrate acceptance

Requirement: Submit evidence of the installer's acceptance of the wall, floor and ceiling substrate before starting installation.

Warranties

Requirement: Submit warranties to COMPLETION, Warranties.

1.6 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- Set-out before installation.
- Completion of installation.

2 PRODUCTS

2.1 GENERAL

Samples

Requirement: Provide samples of the following:

- Each selected panel and door finish, at least 300 x 300 mm, with associated selected edge strips and trim.
- Sample panel Tested for graffiti removal:

. Nominate the cleaning agent.

. Mark the sample with spray paint and permanent markers.

. Clean half of each marked area.

. Repeat with alternate cleaning agents until removal has been achieved.

- All hardware and metal components in the selected finish.

Storage and handling

Requirement: Store in a clean, dry area, unaffected by weather and to the manufacturer's recommendations. Protect from building materials and loose debris, such as wet plaster, mortar, paint and welding spatter.

2.2 PRODUCT SYSTEMS

Acrylic polymer resin panels

Requirement: Chip, crack and scratch resistant acrylic polymer resin panels.

High pressure decorative laminate (HPDL) panels and doors

Material: Compact high pressure decorative laminate panels with an integral surface finish and edges sealed by the manufacturer.

Standard: To AS/NZS 2924.1 (2024).

- Classification: Compact general purpose standard (CGS).

Panel edge: Factory prefinished square cut, ground smooth and arrised. Cleaned and oiled by the manufacturer.

2.3 COMPONENTS

Suspension beam

General: For suspended systems, provide a suspension beam consisting of a galvanized mild steel channel, located immediately above the ceiling framing along the line of the partition fronts.

Hardware

Fixing hardware: Bolts, dowels, brackets, standards, cappings and stabilising bars supplied to complete the cubicle assembly.

Door furniture: As Documented.

3 EXECUTION

3.1 GENERAL

Pre-conditioning

General: Acclimatise wood-based system components in the in-service conditions for a minimum period of two weeks before assembly.

Control of movement

Assembly: Accommodate thermal expansion of panels.

3.2 PANELS

Manufactured cubicle system installation

Assembly: Attach divisions and nibs to walls and fronts with purpose-made proprietary fastenings. Cut nibs and divisions that abut walls, as required, so that assembly is plumb. Seal edges as recommended by the manufacturer.

Floor mounted/overhead braced type: Fix fronts to the floor with proprietary fittings and at the top to a metal channel head rail, supplied as part of the system. Run head rail across the fronts and fix to the walls at each end. Form the channel into a box section over doorways by snapping in a mating channel insert.

Heads of openings: Fix stabilising head channels by screwing to the top of the partitions. Provide an infill strip to the channel across the opening.

Ceiling-hung type: Hang the fronts from a suspension beam with attachments incorporating a means of height adjustment, supplied as part of the system.

Freestanding type: Fix fronts to the floor with proprietary fittings.

Stone panels: Avoid on-site cutting and mechanical processing of panels, if possible.

3.3 COMPONENTS

Shower seats

Fixing: Fix to structural elements using one of the following methods:

- Anodised aluminium channel to exposed edge, secured to walls at each end.
- Product assembly detail.
- Proprietary wall bracket.

Suspension beam

Installation: Install suspension beam as follows:

- Build the ends into masonry structure or provide end fastenings to the structure, as necessary, to transfer the load.
- Drill the bottom flange of the channel for the partition fasteners.

3.4 COMPLETION

Operation and maintenance manuals

Requirement: Prepare a manual that includes the following:

- Full product information for each system, including product designations, components list, colours and finishes, and accessories.
- Information on all doors and hardware supplied as part of the system including door type, size, finishes, and hardware details.
- Maintenance recommendations.

Warranties

Type: Interlocking manufacturer and installer warranty. Period: as offered by the manufacturer and installer.

4 SELECTIONS

Refer to Appendix

0531 SUSPENDED CEILINGS – COMBINED

1 GENERAL

1.1 RESPONSIBILITIES

General

Requirement: Provide suspended ceilings, as documented.

1.2 CROSS REFERENCES

General

Requirement: Conform to the following:

- 0171 General requirements.

1.3 STANDARDS

General

Suspended ceilings: To AS/NZS 2785.

1.4 INTERPRETATION

Definitions

General: For the purposes of this work section the definitions given in AS/NZS 2785 and the following apply:

- Ceiling unit: Tile, panel, plank, strip or open grid supported within a ceiling suspended system.

1.5 TOLERANCES

Suspension system

Flatness, twist, winding and bow: 1.5 mm deviation from a 1.5 m straightedge placed in any position.

Sheeted or flush ceiling system

Suspension system bearing surface for flush lined ceiling: To AS/NZS 2589 Table 4.2.2.

Deflection: To AS/NZS 2785 Table 2.4.5.

1.6 SUBMISSIONS

Operation and maintenance manuals

General: On completion, submit manufacturer's recommendations for the care and maintenance of the ceiling, and operating instructions for demounting, if applicable.

Shop drawings

Set-out drawings: Submit proposed set-out, indicating cut ceiling units if any, before installation.

Coordinate with plenum services layouts, building structure and other factors affecting the layout.

Warranties

Requirement: Submit manufacturer warranties with maintenance manuals.

1.7 INSPECTION

Notice

Inspection: Give notice so that inspection may be made of the following:

- The suspension system before the installation of ceiling units or lining.
- The ceiling assembly before the installation of fittings and site painting, if applicable.
- The completed ceiling.

2 PRODUCTS

2.1 GENERAL

Storage and handling

Requirement: Store suspended ceiling system and components in a dry and secure storage area, unaffected by weather.

2.2 CEILING SYSTEMS

Proprietary systems

Consistency: Provide suspended ceilings as complete proprietary systems, fabricated by one manufacturer.

Installation: Specialist installers recommended by the ceiling system manufacturer.

Ceiling units: As documented.

Support: Fix proprietary suspension system to the structural soffit.

2.3 SUSPENSION SYSTEM

Proprietary suspension system General: As documented.

Protective coatings for steel components: To AS/NZS 2785 Appendix F.

2.4 CEILING UNITS

General

Ceiling units: As documented.

2.5 LINING

Plasterboard

Standard: To AS/NZS 2588.

Fibre cement

Standard: To AS/NZS 2908.2.

Wall and ceiling linings: Type B category 2.

Minimum thickness: 4.5 mm.

Sealants

Fire-resisting sealant: Non-hardening sealant compatible with the ceiling materials and to the documented fire-resistance level.

Acoustic sealant: Non-hardening sealant compatible with the ceiling materials and rated to R_w 65.

3 EXECUTION

3.1 GENERAL

Working environment

General: Do not start work before the building is enclosed, wet work is complete and dry, and all work above the ceiling, including services, is complete.

Protection

General: Protect existing work from damage during the installation.

Partitions

General: If partitions are attached to the underside of the ceiling systems, include the partition mass in the seismic mass of the ceiling.

Stability

General: Install the ceilings level and fix to prevent looseness or rattling of ceiling components under normal conditions.

Structure-borne sound

General: Provide a ceiling system which does not amplify structure-borne sound. Provide suitable proprietary products or systems for reducing contact vibrations between structure and ceiling.

Control of movement

Abutments: Install the ceiling to allow for differential movement at abutting surfaces.

Alignment: Align ceiling control joints with structural control joints. Do not bridge structural control joints.

Prefinishes

General: Repair damaged prefinishes by recoating.

3.2 SUSPENSION SYSTEM

Ceiling grid

Set-out: Align ceiling unit joints and centrelines of visible suspension members with documented grid lines. If not documented, set out with equal margins.

Suspension system

Support members: Install support members as follows:

- Space as required by the loads on the system and the type of ceiling.
- Allow for the installation of services and accessories, including ductwork, light fittings and diffusers.
- Provide additional back support or suspension members for the fixing of services and accessories to prevent distortion, overloading or excessive vertical deflection.
- Allow for access for maintenance of services.

Failure: Provide a ceiling system where failure of any one suspension point does not cause a progressive failure of the ceiling.

Height adjustment: Provide height adjustment with a length adjustment device at each suspension point, permitting length variation of at least 50 mm.

Grid members: If required, notch grid members at the junction with the perimeter trim to make sure the ceiling units lay flat on the perimeter trim.

Restriction: Do not attach the suspension system to the lip or flange of purlins.

Services

Support: Conform to the following:

- If the service has not been designed to accept the ceiling load, do not fix suspension members to services (e.g. ductwork).
- If services obstruct the ceiling supports, provide bridging and suspension on each side of the services.
- Do not support services terminals on ceiling units.
-

Bracing

General: Provide bracing to prevent lateral movement and to resist the imposed horizontal seismic force.

Bulkheads

General: Integrate bulkheads with the ceiling structure and brace to prevent lateral movement. If the ceiling is terminated at a bulkhead, provide for seismic requirements.

External suspended soffits

General: Support external suspended soffits on rigid members capable of carrying the loads from imposed actions. Install members to minimise any eccentricity, and carry the upward and downward loads from wind actions through to the supporting structure.

Fasteners

General: Provide concealed fasteners. If material supporting hangers is less than 3 mm thick, do not use screw fasteners.

3.3 CEILING UNITS

Installation

Fitting: Fit ceiling units accurately and neatly, without distortion, and free from air leakage and staining.

Lock clips: If ceiling units are exposed to loads from wind actions or if required for security, insert lock clips at the junction of carrier rails and units.

Pattern and texture: Set out patterned or heavily textured materials with a consistent direction of pattern or texture, or as documented.

Service penetrations

General: Provide openings for all services elements, including light fittings, ventilation outlets, detectors, sprinklers and loudspeakers. If services pass through ceiling grid members, provide additional grid members and support or relocate service.

Cut ceiling unit edges

General: Conceal, or finish to match prefinished edges, including at openings for services elements.

3.4 PLASTERBOARD LINING

Installation

Gypsum plasterboard and fibre reinforced gypsum plaster: To AS/NZS 2589.

Suspended flush ceilings: Fix using screws or screws and adhesive to ceiling members or support frame.

Multiple sheet layers

Application: Fire-resisting and acoustic rated ceilings.

Joints: Fill and flush up all joints and fixings in each layer and caulk up perimeters and penetrations before installing following layers. Stagger all sheet joints by minimum 200 mm.

Joints

Flush joints: Provide recessed edge sheets and finish flush using perforated paper reinforcing tape.

Butt joints: Make joints over framing members or otherwise provide back blocking.

External corner joints: Make joints over metallic-coated steel corner beads.

Control joints: Align lining control joints with structural control joints and as follows:

- Ceilings: At maximum 12 m centres.
- Control joint beads: Purpose-made metallic-coated.
- Location: If possible, position joints to intersect light fixtures, vents or air diffusers.

Wet areas: Install additional supports, flashings, trim and sealants, as required.

3.5 FIBRE CEMENT LINING

Installation

General: Run sheets across the framing members. In flush jointed applications, stagger end joints in a brick pattern and locate them on framing members, away from the corners of large openings. Provide supports at edges and joints.

Suspended flush ceilings: Fix using screws or screws and adhesive to ceiling members or support frame.

Multiple sheet layers

Application: Fire-resisting and acoustic rated ceilings.

Joints: Fill and flush up all joints and fixings in each layer and caulk up perimeters and penetrations before installing following layers. Stagger all sheet joints by minimum 200 mm.

Joints

Flush joints: Provide recessed edge sheets and finish flush using perforated paper reinforcing tape.

External corner joints: Make joints over metallic-coated steel corner beads.

Dry joints: Provide square edged sheet and finish with a PVC-U joining section.

Control joints: Align lining control joints with structural control joints and as follows:

- Ceilings: To divide into bays not larger than 10.8 x 7.2 m.
- Soffit linings: To divide into bays not larger than 4.2 x 4.2 m or 5.6 x 3.6 m.
- Control joint beads: Purpose-made metallic-coated.
- Support: Provide framing parallel to the joint on each side. Do not fix the lining to abutting building surfaces.
- Location: If possible, position joints to intersect light fixtures, vents or air diffusers.

Wet areas: Install additional supports, flashings, trim and sealants, as required.

3.6 ACCESS PANELS

Finish

General: Match the access panels to the ceiling in appearance and performance.

Identification

General: Provide each access panel with an identification mark.

Non-demountable ceilings

General: Provide access panels supported and anchored to permit ready removal and refixing.

Reinforcement

General: Reinforce the back of the access panel to prevent warping and facilitate handling.

3.7 TRIM

General

Trim: Provide trim at junctions with other building elements and surfaces, including walls, beams and penetrations, consistent with the materials and finishes of the ceiling system.

Accessories

General: Provide accessories as part of the proprietary ceiling system necessary to complete the installation.

Plasterboard cornices

Fixing: Mitre at corners and adhesive fix with cornice cement. Pin in place at cornice edges until adhesive sets, remove pins and fill holes.

Fibrous plaster cornices and roses

Fixing: Pin or prop in place and fix with wet gypsum plaster and scrim straps over framing members.

Fire-resisting walls

Requirement: Seal to soffit with sealant with an equivalent fire-resistance level before fixing decorative cornices, if any.

3.8 COMPLETION

Spares

General: Provide spare matching ceiling components, as follows, and store the spare materials on site where directed:

- Supporting system: One spare supporting member (hanger or framework member) for every 100 members or part thereof of the same type installed in the ceiling.
- Ceiling units: One spare unit for every 50 units or part thereof installed in the ceiling.
- Accessories: One spare of each type for every 50 units or part thereof installed in the ceiling.
-

Warranties

Requirement: Provide warranties for materials and workmanship in the form of interlocking warranties from the supplier and the installer.

Form: Against failure of materials and execution under normal environment and conditions of use.

Warranty terms: as per manufacturers terms

4 SELECTIONS

4.1 SUSPENSION SYSTEM

Proprietary suspension system schedule

Property	SG/		
Product	Apex acoustic Ceiling Tiles (or equal)		
Property	SG/		
Material	Steel		
Basic grid (l x b) (mm)	1200x600		
Wall angle trim	10mm shadowline		
Finish/colour	White		

- to the Suspension System in podcast room 4.3 provide additional WHIG (green) suspension rod hangar Acoustic isolators installed with the suspended ceiling system.

4.2 LINING

Refer to section 0511 Lining and also cross reference with Drawings.

0551 JOINERY

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide joinery, as documented.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements.*
- 0185 *Timber products, finishes and treatment.*
-

1.3 STANDARDS General

Tactile indicators to stairs: To AS/NZS 1428.4.1.

1.4 TOLERANCES General

Requirement: Fabricate and install joinery items to substrates undamaged, plumb, level, straight and free of distortion.

Tolerances table

Property	Tolerance
Plumb and level	1:800
Offsets in flush adjoining surfaces	0.5 mm
Offsets in revealed adjoining surfaces	2 mm
Alignment of adjoining doors	0.5 mm
Difference in scribe thickness for joinery items centred between walls	2 mm
Doors centred in openings	0
Joints in finished surfaces	0

1.5 SUBMISSIONS Operation and maintenance manuals

General: Submit manufacturer's published recommendations for service use.

Products and materials

Manufacturer's data: Submit manufacturer's product data.

Proprietary items: Submit the manufacturer's standard drawings and details showing:

- Methods of construction.
- Assembly and fixing, with dimensions and tolerances.

Shop drawings

General: Submit shop drawings to a scale that best describes the detail, showing the following:

- Overall dimensions.
- Materials, thicknesses and finishes of elements including doors, divisions, shelves and benches.
- Type of construction including mitre joints and junctions of members.
- Hardware type and location.
- Temporary bracing, if required.
- Procedures for shop and site assembly and fixing.
- Locations of benchtop joints.
- Stone benchtop layout including joint arrangement and penetrations.
- Locations of sanitary fixtures, stoves, ovens, sinks, and other items to be installed in the units.
- Relationship of fixture to adjacent building elements.
- Details of fabrication involving other trades or components.
- Proposals for the break-up of large items as required for delivery to the site.
- Proposed method of joining the modules of large items.

Timing: Before fabrication.

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Shop fabricated or assembled items ready for delivery to the site.
- Openings prepared to receive assemblies.
- Site erected assemblies on completion of erection, before covering up by cladding and encasing.
- Surfaces prepared for, and immediately before, site applied finishes.
- Completion of installation.

2 PRODUCTS

2.1 DELIVERY AND STORAGE **General**

Requirement: Deliver joinery units to site in unbroken wrapping or containers and store so that its moisture content is not adversely affected. Do not store in areas of wet plaster. Store in an adequately ventilated space away from heat and direct sunlight. Keep storage time to a minimum by delivering items only when required for installation.

2.2 JOINERY MATERIALS AND COMPONENTS **Visible work**

Clear finished timber and veneer: Make sure all visible surfaces are free of branding, crayon or chalk marks and of blemishes caused by handling.

Joinery timber

Hardwood for trim: To AS 2796.1.

Hardwood for furniture: To AS 2796.3.

Seasoned cypress pine: To AS 1810.

Softwood for trim: To AS 4785.1.

Softwood for furniture: To AS 4785.3.

Finished sizes of milled timbers: Not less than the documented dimensions unless qualified by a term such as nominal, out of or ex to which industry standards for finished sizes apply.

Plywood

Interior use generally: To AS/NZS 2270.

Interior use, exposed to moisture: To AS/NZS 2271.

Visible surface with a clear finish: Veneer quality A.

Other visible surfaces: Veneer quality B.

Particleboard

Standard: To AS/NZS 1859.1.

Classification: HMR – High Moisture resistance

Melamine overlaid particleboard: Particleboard overlaid on both sides with low pressure melamine.

Dry process fibreboard (including medium density fibreboard) Standard: To AS/NZS 1859.2.

Melamine overlaid medium density fibreboard: Medium density fibreboard (STD MDF) overlaid on both sides with low pressure melamine.

Decorative overlaid wood panels Standard: To AS/NZS 1859.3.

High-pressure decorative laminate (HPDL) sheets Standard: To AS/NZS 2924.1.

Classification: CGS – Compact general purpose standard Minimum thickness: Conform to the following:

- For horizontal surfaces fixed to a continuous substrate: 1.2 mm.
- For vertical surfaces fixed to a continuous substrate: 0.8 mm.
- For post formed laminate fixed to a continuous substrate: 0.8 mm.
- For vertical surfaces fixed intermittently (e.g. to studs): 3.0 mm.
- For edge strips: 0.4 mm.

Stone facings

General: Provide stone or engineered stone slabs within the visual range of the approved samples. Repair mud veins or lines of separation that are integral to the selected pattern with resin fillers and back lining.

2.3 JOINERY ASSEMBLIES General

Standard: To AS 4386.

Plinths

Material: Select from the following:

- Exterior general purpose plywood.
- High moisture resistant particleboard.
- High moisture resistant medium density fibreboard.

Thickness: 16 mm. per laminated board.

Fabrication: Form up with front and back members and full height cross members at not more than 900 mm centres.

Fasteners: Conceal with finish.

Installation: Scribe to floor and secure to wall to provide level platform for carcasses.

Carcasses

Material: Select from the following:

- Overlaid high moisture resistant particleboard.
- Overlaid high moisture resistant medium density fibreboard.

Thickness: 16 mm.

Joints: Select from the following:

- Proprietary mechanical connections.
- Dowels and glue.
- Screws and glue.
- Proprietary joining plates and glue.

Adjustable shelves: Support on proprietary pins in holes bored at equal centres vertically.

- Spacing: 32 mm.

Finish: generally white laminate board to carcass. Prelaminated board (colour selected from Full Laminex range to visible areas and splashbacks) Fasteners: Conceal with finish.

Installation: Secure to walls at not more than 600 mm centres.

Drawer fronts and doors

Material: Select from the following:

- Melamine overlaid high moisture resistant particleboard.
- Melamine overlaid high moisture resistant medium density fibreboard.
- Soft closers

Thickness: 16 mm.

Door size: Not exceeding 1.5 m² on face, with 2400 mm maximum height and 900 mm maximum width.

Drawer fronts: Rout for drawer bottoms.

Finish: Refer colour schedule

Colour: Refer colour schedule

Drawer backs and sides

Material: PVC film wrapped particleboard.

Thickness: 12 mm.

Colour: Refer colour schedule

Installation: Mitre corners leaving outer skin of foil intact, finish with butt joints, glue to form carcass and screw to drawer front. Rout for drawer bottoms.

Drawer bottoms

Material: PVC film laminated hardboard.

Thickness: 3 mm.

Drawer and door hardware

Hinge types: 170° Blum hinges or equal. Concealed metal hinges with the following features:

- Nickel plated.
- Adjustable for height, side and depth location of door.
- Integrated self-closing action.
- Hold open function.
- Soft closers

Hold open angle: 170 deg

Piano hinges: stainless steel, extending full height of doors.

Slides: Blum Metabox full extension runners or equal. Metal runners and plastic rollers with the following features:

- 30 kg loading capacity.
- Integrated self-closing action.
- Full extension.
- Closure retention.
- White thermoset powder coating or nickel plated.

Pulls: handles – TBC

Stops: rubber

Material: 16mm prelaminated selected colour board.

2.4 WORKING SURFACES Laminated benchtops

Material: generally, laminate on high moisture resistant board Finish: High-pressure decorative laminated sheet:

Exposed edges: Extend laminate over shaped nosing, finishing more than 50 mm back on underside. Splay outside corners at 45°.

Balance underside: Extend laminate to the undersides of benchtops.

Installation: Scribe to walls. Fix to carcass at least twice per 600 mm length of benchtop.

Joint sealing: Fill joint with sealant matching finish and clamp with proprietary mechanical connectors.

Balance underside: Laminate undersides of benchtops.

3 EXECUTION

3.1 JOINERY General

Joints: Provide materials in single lengths whenever possible. If joints are necessary, make them over supports.

Framing: Frame and trim where necessary for openings, including those required by other trades.

Concealed surfaces: Prime surfaces concealed by substrates.

Deficiencies: Examine joinery units for completeness and remedy deficiencies.

Substrate: Damp clean and vacuum substrate surfaces that will be permanently concealed.

Openings: Provide openings for the following: generally for equipment via cable plugs, sinks, tubs, etc

Acclimatisation

General: Acclimatise the joinery items by stacking in the in-service conditions with air circulation to all surfaces after the following are complete:

- Air conditioning operational.
- Lighting operational.
- Site drainage and stormwater works are complete.
- Space fully enclosed and secure.
- Wet work complete and dry.

Accessories and trim

General: Provide accessories and trim necessary to complete the installation.

Fasteners

Visibility: Do not provide visible fasteners except in the following locations:

- Inside cupboards and drawer units.
- Inside open units, in which case provide proprietary caps to conceal fixings.

Visible fasteners: Where fasteners are unavoidable on visible joinery faces, sink the heads below the surface and fill the sinking flush with a material compatible with the surface finish. In surfaces which are to have clear or tinted finish, provide matching wood plugs showing face (not end) grain. In surfaces which are to have melamine finish, provide proprietary screws and caps finished to match.

Fix joinery units to substrates as follows:

- Floor mounted units: 600 mm centres maximum.
- Wall mounted units: To each nogging and/or stud stiffener.

Fasteners: Screws with washers into timber or steel framing, or masonry anchors.

Adhesives

General: Provide adhesives to transmit the loads imposed and for the rigidity of the assembly, without causing discolouration of finished surfaces.

Finishing

Junctions with structure: Scribe, plinths, benchtops, splashbacks, ends of cupboards, kickboards and returns to follow the line of structure.

Joints: Scribe internal and mitre external joints.

Edge strips: Finish exposed edges of sheets with edge strips which match sheet faces.

Matching: For surfaces which are to have clear or tinted finish, arrange adjacent pieces to match the grain and colour.

Hygiene requirements: To all food handling areas and voids at the backs of units in all areas, seal all carcass and junctions wall/floor, and cable and pipe entries with silicone beads for vermin proofing. Apply water resistant sealants around all plumbing fixtures and make sure sealants are fit for purpose.

Benchtops

Installation: Fix to carcass at least twice per 600 mm length of benchtop.

Joint sealing: Fill joints with sealant matching the finish colour and clamp with proprietary mechanical connectors.

Edge sealing: Seal to walls and carcasses with a sealant, which matches the finish colour.

3.2 TRIMS General

Requirement: Provide trims such as architraves, beads, mouldings, stops and skirtings to make neat junctions to openings and between lining components, finishes and adjacent surfaces.

Fixing

To masonry walls: Screw with wall plugs at 600 mm centres maximum.

To stud walls: Nail to plate or framing at 600 mm centres maximum.

3.3 COMPLETION Protection

Timber treads: Provide full timber or plywood casing.

Cleaning

Temporary coatings: On or before completion of the works, or before joining up to other surfaces, remove all traces of temporary protective coatings.

Requirement: Remove all dust, marks and rubbish from all surfaces and internal spaces. Clean and polish all self-finished surfaces such as anodised and powder coated metals, sanitary ware, glass, tiles and laminates.

4 SELECTIONS

Refer to appendix A for colour selection

1 GENERAL**1.1 RESPONSIBILITIES General**

Requirement: Provide metal fixtures, as documented.

Performance

Requirements:

- Undamaged, plumb, level and straight or as documented.
- Free of surface defects or distortions or as documented.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- *0171 General requirements.*

1.3 STANDARDS General

Access for maintenance: To AS 1657.

Tactile indicators: To AS/NZS 1428.4.1.

1.4 TOLERANCES General

Requirement: ± 2 mm from design dimensions.

1.5 SUBMISSIONS Operation and maintenance manuals

General: Submit manufacturer's published recommendations for service use.

Products and materials

Proprietary items: Submit the manufacturer's product data standard drawings and details showing:

- Methods of construction.
- Assembly and fixing, with dimensions and tolerances.

Stainless steel: For each batch of stainless steel supplied to the works, submit the certificate of conformance or test certificate to the applicable standard, as documented.

Stainless steel welding: Before fabrication commences, submit evidence of qualification of the welding procedure by testing to AS/NZS 1554.6 clause 4.7 or evidence of prequalification to AS/NZS 1554.6 clause 4.12.

Shop drawings

General: Submit shop drawings to a scale that best describes the detail, showing the following information:

- Overall and detail dimensions.
- Details of fabrication and components.
- Details of fabrication involving other trades or components.
- Information necessary for site assembly.
- Proposals for the break-up of large items as required for delivery to the site.
- Proposed method of joining the modules of large items.

Subcontractors

General: Submit names and contact details of proposed suppliers, fabricators and installers.

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Site erected assemblies on completion of erection, before covering up by cladding and encasing.
- Steel surfaces prepared for, and immediately before, site applied finishes.

2 PRODUCTS

2.1 GENERAL Storage and handling

Requirement: Store and handle fabricated metalwork, as follows:

- Deliver to site in unbroken wrapping or packing.
- Store on a level base, away from uncured concrete and masonry and areas of wet plaster.
- Do not store in contact with other materials that may cause staining, denting or other surface damage.
- Use gloves when handling precoated finishes.
- Keep storage time to minimum by delivering items only when required for installation.

Marking

General: Provide suitable and sufficient marks or other means for identifying each member of selected assemblies, and for their correct setting out, location, erection and connection. Mark bolted connections to show the bolting category. Do not mark stainless steel by notching.

2.2 MATERIALS Metals and components

Performance: Provide metals and components in quantity, lengths and cross-sections of strength and stiffness suited to their required function, finish, fabrication and method of installation.

Fasteners

Performance: Provide fasteners to resist galvanic corrosion in materials of structural and mechanical strengths and corrosion resistance at least equal to that of the lowest resistant metal in the connection.

Materials: Provide fasteners as follows:

- To copper and copper alloys: Copper or copper-alloy fixing devices only.
- To aluminium and aluminium alloys: Aluminium alloy or non-magnetic stainless steel fixing devices only.
- To stainless steel: Appropriate stainless steel materials only.

3 EXECUTION

3.1 CONSTRUCTION GENERALLY Aluminium structures

Standard: To AS/NZS 1664.1 or AS/NZS 1664.2.

Metals

Incompatible metals: Separate using concealed layers of suitable materials in appropriate thicknesses.

Fabrication

Workshop: Fabricate and pre-assemble items in the workshop wherever practicable.

Edges and surfaces: Keep clean, neat and free from burrs and indentations. Remove sharp edges without excessive radiusing.

Tube bends: Form bends in tube without deforming the cross section and the material thickness.

Colour finished work: Match colours of sheets, extrusions and heads of fasteners.

Thermal movement: Accommodate thermal movement in joints and fastenings.

Joints

General: Fit joints to an accuracy appropriate to the class of work. Finish visible joints made by cutting, drilling, welding, brazing or soldering using grinding, buffing or other methods appropriate to the class of work, before further treatment.

Self-finished metals: Free of surface colour variations, after jointing.

Joints: Fit accurately to a fine hairline or as documented.

Splicing

General: Provide structural members in single lengths.

3.2 WELDING AND BRAZING Welding

Quality: Provide finished welds which are free of surface and internal cracks, welding slag, and porosity.

Site welds: Avoid site welding wherever possible. If required, locate site welds in positions for down hand welding.

Butt weld quality level: Not inferior to the appropriate level recommended in AS/NZS 1554.1 Section 6, AS/NZS 1554.6 Section 6 or AS/NZS 1665 Section 6, as appropriate.

Brazing

General: Make sure brazed joints have sufficient lap to provide a mechanically sound joint.

Butt joints: Do not use butt joints for joints subject to load. If butt joints are used, do not rely on the filler material only.

3.3 STAINLESS STEEL FABRICATION Welding stainless steel

Certification of welders: To AS 1796.

Riveting

General: Use only to join stainless steel sheet or strip less than 1 mm thick. Drill (not punch) the rivet hole, and drive the rivet cold. On completion, clean and passivate the riveted assembly.

Soldering

General: Do not solder stainless steel.

3.5 COMPLETION Cleaning

Temporary coatings: On or before completion of the works, or before joining up to other surfaces, remove all traces of coatings used as temporary protection.

4 SELECTIONS

4.1 STEEL FRAMED JOINERY

Refer to details on drawings

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide plaster finishes as documented. Refer to Design drawings.

Performance

Requirements:

- Resistant to impacts expected in use.
- Free of irregularities.
- Consistent in texture and finish.
- Firmly bonded to substrates for the expected life of the application.
- Without obvious shrinkage cracks.
- As a suitable substrate for the nominated final finish.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- *0171 General requirements.*

1.3 INTERPRETATION Abbreviations

General: For the purposes of this worksection the following abbreviations apply:

- CRF: Cement render – finish.
- CRM: Cement render – medium.
- CRS: Cement render – stronger.
- CRW: Cement render – weaker.
- GPF: Gypsum plaster – finish.

Definitions

General: For the purposes of this work section the following definitions apply:

- Base coat: A plaster coat applied before the application of the finish coat.
- Bonding treatment: A treatment of a substrate which improves adhesion of a plaster system.
- Finish coat (rendering and plastering): The final coat of a coating system.
- Finishing treatment (plastering): The treatment applied to a finish coat which may include processes and results.
- Plaster: A mixture of binders, aggregate and water which is applied to substrates in a plastic state and dries and cures to a hard surface which may subsequently be decorated:
 - o Cement plaster: Contains Portland cement as the principal binder.
 - o Gypsum plaster: Contains hydrated or anhydrous calcium sulfate as the principal binder.
- Plastering: The process of coating the framing or solid surfaces of a building with a plastic material which hardens and then may be decorated or remain self-finished.

- Plastering system: One or more coats of plaster and associated treatments comprising some or all of the following in sequence:
 - o Base coat 1 or 2.
 - o Bonding treatment.
 - o Finish coat.
 - o Finishing treatment.
- Render, rendering: Plaster, plastering, usually single coat and usually cement:lime:sand.
- Substrate: The surface to which a material or product is applied.

1.4 TOLERANCES Tolerances table

Description	Alignment	Tolerance
Walls and other vertical structures	Vertical	6 mm in 2400 mm
Reveals sides	Vertical	3 mm in 1800 mm
Reveals head up to 1800 mm	Horizontal	3 mm in 1800 mm
Reveals head over 1800 mm	Horizontal	5 mm max
Reveals, piers, beams, wall stop ends up to 300 mm	Square	3 mm max
Reveals, piers, beams, wall stop ends over 300 mm	Square	5 mm max
Radius of corners	Round	Should not vary by more than $\pm 10\%$ over the length of the arris.

1.5 SUBMISSIONS Prototype

Plaster systems: Prepare prototypes of each plaster system complete with beads and other embedded items:

- Size: 1200 x 2400 mm.

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Prototypes ready for inspection.
- Substrates immediately before applying base coats.
- Finish treatments before decoration.

2 PRODUCTS

2.1 GENERAL Storage and handling

General: Store materials in a dry, well-ventilated and secure storage area, unaffected by weather.

2.2 MATERIALS AND COMPONENTS Accessories

Beads: Provide metal proprietary sections manufactured for fixing to substrates and/or embedding in the plaster to form and protect plaster edges and junctions.

Metal lath: Provide a proprietary product manufactured from raised expanded metal for use with plaster:

- Mass/unit area: 1.84 kg/m² or greater.
- Material thickness: 0.70 mm or greater.
- Mesh size: 9.5 x 28.6 mm.

Metallic-coatings to AS 1397: For beads or lath in cement plaster: To the **Corrosion resistance and durability table**.

Admixtures

Plasticisers or workability agents: Do not use in cement plasters.

Aggregates

Sand: Fine, sharp, well-graded sand with a clay content between 1% and 5% tested to AS 1141.12, and free from efflorescing salts.

Sand grading for base coat plaster table

Sieve size	Percent passing	
Sieve size	Percent passing	
	Minimum	Maximum
4.75 mm	100	100
2.36 mm	90	100
1.18 mm	60	90
600 µm	35	70
300 µm	10	30
150 µm	0	5
75 µm	0	3

Bonding products

General: Proprietary products manufactured for bonding cement-based plaster to solid substrates.

Cement

Standard: To AS 3972.

Type: GP.

Colouring products

General: Provide proprietary products manufactured for colouring cement plaster.

Integral pigment proportion: 5% maximum weight of cement.

Cornice cement

General: Provide a proprietary product manufactured for use with the cornice.

Cornices

Cast plaster: Proprietary item.

Corrosion resistance and durability

Compliance: To the **Corrosion resistance and durability table** or provide proprietary products with metallic and/or organic coatings of equivalent corrosion resistance and as follows:

- Galvanize: To AS/NZS 4680.

Corrosion resistance and durability table

Atmospheric corrosivity category to AS 4312	Metal lath, beads and embedded items	Minimum cement content (mix type) above damp-proof course
C1 and C2	Galvanize after fabrication 300 g/m ² stainless Type 316	CRW
	Powder coated aluminium	CRM
C3	Stainless Type 316 Powder coated aluminium	CRM
C4	Stainless Type 316 Powder coated aluminium	CRS
Note: For categories C5 and CX seek specialist advice.		

Curing products

General: Provide proprietary products manufactured for use with the plaster system.

Gypsum plaster

General: Provide a proprietary product containing calcium sulfate hemihydrate with additives to modify setting.

Lime

Limes for building: To AS 1672.1.

Lime putty

General: Prepare lime putty as follows:

- Stand dry hydrate of lime to AS 1672.1 and water for 24 hours or more without drying out.
- Stand quicklime and water for 14 days or more without drying out.

Mixes

General: Select a mix proportion to suit the conditions of application.

Measurement: Measure binders and sand by volume using buckets or boxes. Do not allow sand to bulk by absorption of water.

Plaster mixing: Machine mix for 3 to 6 minutes.

Strength of successive coats: Make sure successive coats are no richer in binder than the coat to which they are applied.

Mix proportion table – Cement render, by volume

Mix type		Substrate	Upper and lower limits of proportions by volume		
			Cement	Lime	Sand
Single or multi-coat systems with integral finishing treatments Base coats in multicoat systems with cement or gypsum finishes	CRS	Dense and smooth concrete and masonry	1	0	3
			1	0.5	4.5
	CRM	Regular clay or concrete masonry	1	0.5	4.5
			1	1	6
	CRW	Lightweight concrete masonry and other weak substrates	1	1	6
			1	2	9
Second coat - Internal	CRF	Cement render base coats	1	1	6
			1	2	9
Second coat - External	CRF	Cement render base coats	1	1	5
			1	2	6

Mix proportion table – Gypsum finish coat, by volume

Mix type		Substrate	Upper and lower limits of proportions by volume			
			Gypsum	Cement	Lime putty	Sand
Gypsum finish coats	GPF	Cement render base coats	1	-	1.5	-
			1	-	2	-

Mix proportion table – Gypsum finish coat, by weight

Gypsum plaster (kg)	Lime putty (kg)
17	25
34	50
51	75

Control joint products

General: Provide proprietary products manufactured for use with the plastering system and to accommodate the anticipated movement of the substrates and/or the plaster.

Water

General: Clean and free from any deleterious matter.

3 EXECUTION

3.1 PREPARATION Substrates

General: Provide substrates as follows:

- Clean and free from any deposit or finish which may impair adhesion of plaster.
- If framed or discontinuous, support members in full lengths without splicing.
- If solid or continuous, remove excessive projections and fill voids and hollows with plaster stronger than the first coat and not weaker than the substrate.

Absorbent substrates: If suction is excessive, control it by dampening without over-wetting, and do not plaster substrates showing surface moisture.

Dense concrete: If not sufficiently rough to provide a mechanical key, roughen by scabbling or the like to remove 2 mm of the laitance and expose the aggregate before applying a bonding treatment.

Painted surfaces: Remove paint and hack the surface at close intervals.

Untrue substrates: If the substrate is not sufficiently true for conformity with the thickness limits for the plaster system, or has excessively uneven suction resulting from variations in the composition of the substrate, apply additional coats without exceeding the thickness limits for the substrate or system.

Beads

Location: Fix beads as follows:

- Angle beads: At all external corners.
- Drip beads: At all lower terminations of external plaster.
- Beads for control of movement: At all control joints.
- Stop beads: At all terminations of plaster and junctions with other materials or plaster systems.

Joints in beads: Provide dowels to maintain alignment.

Mechanical fixing to substrate: ≤ 300 mm centres.

Bonding treatment

General: If bonding treatment is required, throw a wet mix onto the background. Mix proportions to the following:

- Cement plaster (cement:sand): 1:2.
- Gypsum plaster (gypsum:sand): 1:2.

Curing: Keep continuously moist for 5 days or more and allow to dry before applying plaster coats.

Thickness: Greater than or equal to 3 mm and less than 6 mm.

Embedded items

General: To the **Corrosion resistance and durability table**. If there are water pipes and other embedded items, sheath them to permit thermal movement.

Lath

Location: Provide lath as follows:

- Chases: If chases or recesses are 50 mm wide or greater, fix metal lath extending 75 mm or more beyond each side of the chase or recess.
- Masonry and concrete substrates: If mechanical key cannot be attained by scabbling and bonding, fix metal lath.
- Metal and other non-porous substrates: Fix metal lath to provide a key.

Installation: Fix lath as follows:

- General: Run the long way of the mesh across supports with strands sloping downwards and inwards from the intended face of the plaster.
- Fixing: Mechanically fix at centres of 150 mm or less.
- Laps: Tie with 1.25 mm galvanized wire at centres of 150 mm or less. Do not stop edges of sheets at corners but bend around.
- On solid substrates: Space the lath 5 mm or more clear of the substrate.
- Support spacing: ≤ 400 mm.

3.2 APPLICATION Plastering

Base coats: Scratch-comb each base coat in two directions when it has stiffened.

Metal lath: Press the plaster through the apertures of expanded metal lath and wings of beads.

Incidental work

General: Return plaster into reveals, beads, sills, recesses and niches. Plaster faces, ends, and soffits of projections in the substrate, such as string courses, sills, pilasters and corbels. Run neatly finished throating on soffits of external projections. Trim around openings. Plaster exposed internal surfaces of built-in cupboards.

Joining up

General: If joining up is required, make sure joints are imperceptible in the finished work after decoration.

Control joints

General: Provide joints in the finish to coincide with control joints in the substrate. Make sure the joint in the substrate is not bridged during plastering.

Size:

- Depth: Extend the joint right through the plaster and reinforcement to the substrate.
- Width: 3 mm, or the same width as the substrate joint, whichever is greater.

Damp-proof courses: Do not continue plaster across damp-proof courses.

Plastering on metal lath: Provide control joints to divide the plastering area into rectangular panels of 10 m² or less.

V-joints: Provide V-joints, cut right through the plaster to the substrate, at the following locations:

- Abutments with metal door frames.
- Abutments with other finishes.
- Junctions between different substrates.

Cornices

General: Accurately cut and mitre corners. Match and align ornament. Do not make butt joints in the length of a cornice unless required, or if full lengths are not available.

Installation: Butter edges, mitres and joins for the full length of the cornice with adhesive.

Mechanical fixing: If cornice projects across a ceiling 400 mm or more, provide additional mechanical fixing as follows:

- Fixing centres: ≤ 600 mm. **Plaster thickness table**

Substrate	Cement render, total thickness of single or multi-coat work (mm)	Gypsum/lime plaster (mm)
Dense concrete walls	15 max	3 max
Dense concrete ceilings	9 max	3 max
Brickwork and blockwork	12 min	3 max
Lightweight concrete and blocks	12 min	3 max
Metal lath measured from the face of the lath.	18 min	3 max

Temperature

General: If the ambient temperature is less than 10°C or more than 30°C, make sure the temperature of mixes, substrates and reinforcement at the time of application are between 5°C and 35°C.

3.3 FINISHES Finishing treatments

Plain even surfaces: Work the hardening plaster as follows:

- Bag: Rub the finish coat when set firm with a hessian bag or similar.
- Carborundum stone: Rub the finish coat when set hard with a carborundum stone to achieve a finish free from sand.
- Foam float: Float finish coat on application with a wood or plastic float to an even surface and finish with a foam float to achieve a fine sand textured finish.
- Steel trowel: Steel trowel finish coat to a smooth dense surface which is not glass-like and is free from shrinkage cracks and crazing.
- Wood or plastic float: Float the finish coat on application to an even surface with a wood or plastic float.

Sprayed textured surfaces: Spray plaster onto a substrate using a purpose-designed machine.

Stippled textured surfaces: Work the hardening plaster with a stiff brush.

Specialist plaster finishes Polymer modified render:

- Basecoat render: Proprietary polymer modified cementitious render supplied as a complete plastering system.
- Finish coats: Proprietary trowelled on coloured and textured polymer modified finish coats.

3.4 COMPLETION Curing

General: Prevent premature or uneven drying out and protect from the sun and wind.

Keeping moist: If a proprietary curing agent is not used, keep the plaster moist as follows:

- Base coats and single coat systems: Keep continuously moist for 2 days and allow to dry for 5 days before applying further plaster coats.
- Finish coats: Keep continuously moist for 2 days.

4 SELECTIONS

4.1 SCHEDULES Plastering construction schedule

Refer to section "0671 Painting" for specification requirements

0621 WATERPROOFING – WET AREAS

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide waterproofing systems to all new and refurbished wet areas

Performance

Requirements:

- Grade to floor wastes, to dispose of water without ponding.
- Prevent moisture entering the substrate or adjacent areas.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- *0171 General requirements.*

1.3 STANDARDS

Waterproofing wet areas Standard: To AS 3740.

1.4 INTERPRETATION Definitions

General: For the purposes of this work section the definitions given in AS 3740 and the following apply:

- Bond breaker: A system preventing a membrane bonding to the substrate, bedding or lining.
- Membranes (waterproof): Impervious barriers to liquid water which may be:
 - o Installed below floor finishes.
 - o Installed behind the wall sheeting or render and termed External.
 - o Installed to the face of the wall sheeting or render and termed Internal.
 - o Applied in liquid or gel form and air cured to form a seamless film.
 - o Applied in sheet form with joints lapped and sealed.
- Preformed shower base: A preformed, prefabricated vessel (including integral upstands) installed as the finished floor of a shower compartment, and provided with a connection point to a sanitary drainage system.
- Shower tray: An internal or external liquid applied or sheet membrane system used to waterproof the floor and the wall/floor junctions of a shower area.
- Substrate: The surface to which a material or product is applied.
- Waterproof (WP): The property of a material that does not allow moisture to penetrate through it.
- Waterproofing systems: Combinations of membranes, flashings, drainage and accessories which form waterproof barriers and which may be:
 - o Loose-laid.
 - o Bonded to substrates.
- Water resistant (WR): The property of material that restricts moisture movement and will not degrade under conditions of moisture.
- Wet area: An area within a building supplied with a floor waste.

1.5 SUBMISSIONS Products and materials

Documentation: Submit copies of product manufacturer's:

- Product technical data sheets.
- Safety data sheets (SDS).
- Type tests certificates verifying conformance to AS/NZS 4858 Table 8.1.

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Substrate preparation completed.
- Secondary layers preparation completed.
- Before membranes are covered up or concealed.
- After flood testing, if applicable.

2 PRODUCTS

2.1 GENERAL Storage and handling

General: Store and handle to the manufacturer's recommendations and as follows:

- Protect materials from damage.

2.2 MEMBRANES Standards

Standard: To AS/NZS 4858.

Membrane systems

Requirement: Provide a proprietary membrane systems suitable for the intended internal wet area waterproofing.

2.3 ACCESSORIES Water stop angles

Material: Rigid, corrosion resistant angles compatible with the waterproof membrane system.

Bond breakers

Requirement: Compatible with the extensibility class of the membrane to be used.

Material: Purpose made bond breaker tapes and closed cell foam backing rods or fillets of sealant.

Flashings

Requirement: Flexible waterproof flashings compatible with the waterproof membrane system.

Liquid membrane reinforcement

Requirement: Flexible fabric compatible with the waterproof membrane system.

Sealants

Requirement: Waterproof, flexible, mould-resistant and compatible with host materials.

Adhesives

Requirement: Waterproof and compatible with host materials.

3 EXECUTION

3.1 PREPARATION Substrates

General: Make sure substrates are as follows:

- Clean and free of any deposit or finish which may impair adhesion of membranes.
- If walls are plastered, remove loose sand.
- If walls or floors are framed or discontinuous, support members are in full lengths without splicing.
- If floors are solid or continuous:
 - . Excessive projections are removed.
 - . Voids and hollows greater than 10 mm with abrupt edges are filled with a cement:sand mix not stronger than the substrate nor weaker than the bedding.
 - . Depressions less than 10 mm are filled with a latex modified cementitious product with feathering eliminated by scabbling the edges.
 - . Cracks in substrates wider than 1.5 mm are filled with a filler compatible with the membrane system.

Concrete substrates: Cure for more than 28 days.

External corners: Round or arris edges.

Moisture content

Requirement: Verify that the moisture content of the substrate is compatible with the water vapour transmission rate of the membrane system by testing to AS 1884 Appendix A.

Falls

Membrane directly under the floor finish: Make sure the fall in the substrate conforms to the fall documented for the finish.

Sheet substrate fastening

Requirement: Fasten or adequately fix to the supporting structure.

Control joints

Finishes: Align control joints in finishes and bedding with control joints or changes in materials in the substrate.

Water stop angles

Requirement: Provide water stop angles at door thresholds and shower enclosures to support the waterproof membrane at junctions between waterproofed and non-waterproofed areas. Sizing: Size the vertical leg of the water stop angle to conform to the requirements of AS 3740.

Corners: Cut the horizontal leg and bend the vertical leg at corners instead of forming vertical joints between separate lengths of angle.

Fixing: Fix water stop angles to the substrate with compatible sealant or adhesive and corrosionresistant countersunk or wafer head screws.

Priming

Compatibility: If required, prime the substrates with compatible primer for adhesion of the membrane system.

Bond breakers

Requirement: After the priming of surfaces, provide bond breakers at all wall/floor, hob/wall junctions and at control joints where the membrane is bonded to the substrate.

Sealant fillet bond breakers:

- Application: Form a triangular fillet or cove of sealant to internal corners within the period recommended by the membrane manufacturer after the application of the primer.
- Widths: 5 mm x 5 mm to vertical corners. 6 mm x 6 mm to 9 mm x 9 mm to horizontal corners.

Backing rod bond breakers: Retain in position with continuous length of tape pressed firmly in place against the surfaces on each side of the rod.

3.2 APPLICATION Protection

Damage: Protect membrane from damage during installation and for the period after installation until the membrane achieves its service characteristics that resist damage.

Extent of waterproofing

Waterproof or water resistant surfaces: To the requirements of BCA F1.7 for Class 2, 3 and 4 buildings, or BCA 3.8.1.2 for Class 1 buildings.

Vertical membrane terminations

Upstands: At least 150 mm above the finished tile level of the floor or 25 mm above the maximum retained water level, whichever is the greater.

Anchoring: Secure sheet membranes along the top edge.

Edge protection: Protect edges of the membrane.

Flashings

Junctions between waterproof surfaces: Provide a bond breaker at internal corners behind flashings.

Junctions between waterproof surfaces and other surfaces: Provide a bead of sealant at the following junctions:

- Waterproof and water-resistant surfaces.
- Water-resistant and water-resistant surfaces.
- Water-resistant and non water-resistant surfaces.

Perimeter flashings: Provide continuous flashings to the full perimeter of waterproof areas at wall/floor junctions and to water stop angles.

Vertical flashings: Provide vertical corner flashings continuous across wall/wall junctions to at least 2000 mm above finished floor level.

Vertical liquid applied flashings:

- Return legs at least 40 mm on each wall.
- Overlap the vertical termination of the floor waterproofing membrane at least 20 mm.

Vertical sheet flashings:

- Return legs at least 50 mm on each wall.
- Overlap shower tray upstands at least 50 mm.
- Do not penetrate flashing with wall lining fasteners.

Reinforcement: At coves, corners and wall/floor junctions with gaps greater than 3 mm reinforce liquid applied membranes with reinforcement fabric tape recommended by the membrane manufacturer. Fold the tape in half lengthways and imbed it in the first flashing coat of membrane with one half of the tape on each side of the corner or joint. Apply a second coat of liquid membrane to seal the fabric.

Door jambs and architraves

Requirement: If the bottom of doorjambs and architraves do not finish above the floor tiling, waterproof their surfaces below tile level to provide a continuous seal between the perimeter flashing to the wall/floor junction and the water stop angle.

Drainage connections

Floor wastes: Provide floor wastes of sufficient height to accommodate the thickness of floor finishes and bedding at the outlet position. Position drainage flange to drain at membrane level. Turn membrane down 50 mm minimum into the floor waste drainage flanges, and adhere to form a waterproof connection.

Floor wastes in shower trays: Provide drainage of the tile bed and a waterproof connection between the tray and the drain.

Preformed drainage channels:

- With continuous drainage flanges: Provide a continuous waterproof connection between the membrane and the channel.
- Without drainage flanges: Provide continuous waterproofing under the channel and terminate the membrane at a floor waste with a recessed drainage flange.

Taps and spouts

Requirement: Waterproof penetrations for taps and spouts with proprietary flange systems or a sealant.

Provision for servicing: Install taps in a manner that allows tap washers or ceramic discs to be serviced without damaging the waterproofing seal.

Curing of liquid applied systems

General: To the manufacturer's instructions.

Curing: Allow membrane to fully cure before tiling.

Overlaying finishes on membranes

Requirement: Protect waterproof membranes with compatible water-resistant surface materials that do not cause damage to the membrane. Suitable materials: Conform to AS 3740.

Bonded or partially bonded systems: If the topping or bedding mortar is required to be bonded to the membrane, provide sufficient control joints in the topping or bedding mortar to reduce the movement over the membrane.

3.3 TESTING 3.4 COMPLETION Protection

General: Keep traffic off membrane surfaces until bonding has set or for 24 hours after laying, whichever period is the longer.

Reinstatement: Repair or replace faulty or damaged work.

Warranties

Requirement: Cover materials and workmanship in the terms of the warranty in the form of interlocking warranties from the supplier and the applicator.

- Form: Against failure of materials and execution under normal environment and use conditions.

0651 RESILIENT FINISHES

1 GENERAL

1.1 RESPONSIBILITIES General

Requirement: Provide resilient floor finishes to substrates, as documented.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- 0171 *General requirements*.

1.3 STANDARDS General

Installation: To AS 1884.

Slip resistance

Classification: To AS 4586.

1.4 INTERPRETATION Definitions

General: For the purposes of this work section the definitions given in AS 1884 and the following apply:

- Resilient floor coverings classification: To EN ISO 10874.
- Substrate: The surface to which a material or product is applied.
- Underlay: A non-structural layer of sheet material or in situ levelling material on the substrate to provide a smooth and level surface.

1.5 SUBMISSIONS Certification

General: Submit a certificate of conformity for static dissipative and static conductive floor installations.

Operation and maintenance manuals

General: Submit manufacturer's instructions for care and maintenance for each type of finish.

Products and materials

Manufacturer's data: Submit the manufacturer's product data sheets for each type of finish, and the manufacturer's recommendations for its application including the following, as appropriate:

- Thickness and width of sheet, or size of tile or plank.
- Adhesive and jointing method.
- Resistance to wear, indentation, chemicals, light and fire.
- Flexibility and bending strength.

Type tests: Submit results, as follows:

- Slip resistance to AS 4586.

Subcontractors

General: Submit names and contact details of proposed suppliers and installers.

Substrate acceptance: Submit evidence of installer's acceptance of the flooring substrate before starting installation.

Tests

Site tests: Submit results, as follows:

- Site slip resistance test of completed installations.
- Surface pH test.
- Moisture content test.
-

Warranties

Requirement: For each type of resilient finish specified, submit the manufacturer and installer's warranty of the material, workmanship and application.

Warranty items: provide manufacturer's warranty and include in Operation and Maintenance manuals.

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Substrate immediately before fixing resilient finishes or underlay.
- Completed installation.

2 PRODUCTS

2.1 FIRE PERFORMANCE Fire hazard properties

Critical radiant flux: Tested to AS ISO 9239.1.

2.2 UNDERLAYS Cementitious

General: Polymer modified cementitious smoothing and self-levelling compound.

Thickness: 3 mm minimum.

2.3 ADHESIVES General

Requirement: To the resilient finishes manufacturer's recommendations.

2.4 SHEETS, TILES AND PLANKS Polyvinyl chloride (PVC)

Resilient floor covering, homogeneous: To EN ISO 10581.

Resilient floor covering, heterogeneous: To EN ISO 10582.

Resilient floor covering, jute or polyester felt backing: To EN 650.

Resilient floor covering, with foam layer: To EN 651.

Resilient floor covering, with particle based enhanced slip resistance: To EN 13845.

Resilient floor covering, semi-flexible polyvinyl chloride tiles: To EN ISO 10595.

2.5 OTHER MATERIALS

Tactile ground surface indicators Standard: To AS/NZS 1428.4.1.

3 EXECUTION

3.1 SUBCONTRACTORS General

Requirement: Use specialist installers recommended by the material manufacturers.

3.2 PREPARATION Substrates

General: To AS 1884 Section 3. **Substrate tolerance table**

Property	Length of straightedge laid in any direction	Max. deviation under the straightedge
Planeness	2000 mm	4 mm
Abrupt deviation tolerance	150 mm	0.5 mm

Concrete substrates

Requirement: Do not start installation of the resilient finishes until the concrete substrate conforms to AS 1884 clause 3.1 and the adhesive and resilient finish manufacturers' recommendations.

Substrate rectification: Conform to the following:

- Surface treatments: Mechanically remove any incompatible surface treatments, including the following:
 - . Sealers and hardeners.
 - . Curing compounds.
 - . Waterproofing additives.
 - . Surface coatings and contamination.
- Surface quality: Remove projections and fill voids and hollows with a smoothing and self-levelling compound compatible with the adhesive. Allow filling or levelling compound to dry to manufacturer's recommendations.

Cleaning: Remove loose materials or dust.

Working environment

General: Do not start work before the building is enclosed, wet work is complete and dry, overhead work is complete and good lighting is available. Protect adjoining surfaces.

Conditioning

Conditioning of floor covering and subfloor: To AS 1884 clause 4.1 and manufacturer's recommendations.

3.3 INSTALLATION General

Requirement: To AS 1884 Section 5 and the manufacturer's recommendations.

Sheet set-out

General: Set out sheets to give the minimum number of joints. Position joints away from areas of high stress. Run sheet joints parallel with the long sides of floor areas, vertically on non-horizontal surfaces.

Plank set-out

General: Set out planks from centre of room. Align patterns, texture and grain in one direction.

Edges

General: Make sure edges are firm, unchipped and machine-cut accurately to size and square to the face, and that edges are square to each other before installation.

Joints

Non-welded: Butt edges together to form tight neat joints showing no visible open seams.

Doorways: Where changes of floor finish occur at doorways, locate the joint on the centreline of the door leaf in the closed position.

Expansion joints

General: To the manufacturer's recommendations for joint widths, and area and length limitations.

Junctions

General: Scribe neatly up to returns, edges, fixtures and fittings. Finish flush with adjoining surfaces.

Rolling

General: If rolling is required, roll the finish in multiple directions before the adhesive sets.

Change of finish

General: Maintain finished floor level across changes of floor finish including carpet.

Cleaning

General: Keep the surface clean as the work proceeds.

3.4 SHEETING Welded joints

Thermal welding: After fixing, groove the seams using a grooving tool and weld the joints with matching filler rod, using a hot air welding gun. When the weld rod has cooled, trim off flush.

Chemical welding: Apply seaming compound 100 mm wide to the substrate centrally under the seam. Roll the seam until the compound is forced up into the joint. Clean off flush with a damp cloth.

Epoxy jointing: Join seams with epoxy adhesive.

3.5 JOINTS AND ACCESSORIES Accessories

General: Provide purpose-made matching moulded accessories for nosings, coves, skirtings, edge cover strips and finishes at junctions, margins, and angles, if available. Otherwise, form accessories from the sheet material. Provide solid backing for radiused coves and nosings.

Edge strips

General: Provide aluminium angle at junction with different floor finishes. Butt floor finishes to the vertical edge of the angle. Provide Ardit floor leveller, or equal (feather out 600), to maintain consistent finished surface level across materials

Control joints

Location: Provide control joints as follows:

- Over structural control joints.
- At junctions between different substrates.

Depth of joint: Right through to the substrate.

Sealant width: 6 to 25 mm.

Depth of elastomeric sealant: One half the joint width, or 6 mm, whichever is the greater.

Control joint materials – sheet flooring

Proprietary slide plate divider strip: Provide interlocking metal plates grouted into pockets formed in the concrete joint edges to finish flush with the flooring surface.

Coved skirtings

Site formed coving: Carry the flooring material up over a profiled coving section to form the skirting and mitre and weld all joints. Make sure the radius of the coving section conforms to the floor finish manufacturer's recommendations for sheeting material and thickness.

3.6 TESTING Substrate tests

Moisture content: Test substrate for suitability for the installation of resilient floor coverings to AS 1884 Appendix A.

- Maximum relative humidity of concrete: To AS 1884 Appendix A3.2.
- Moisture content of timber, plywood and particleboard subfloors: To AS 1884 Appendix A3.3.

Surface pH: Test concrete subfloor for suitability for the installation of resilient floor coverings to AS 1884 Appendix C.

- Maximum pH: 10.

-

Completion tests

Slip resistance testing of completed installation: To AS 4663.

3.7 COMPLETION Protection

Finished floor surface: Keep traffic off floors for a minimum of 24 hours after laying or until bonding has set, whichever period is the longer. Avoid contact with water for minimum 7 days after laying.

Reinstatement

Extent: Repair or replace faulty or damaged work. If the work cannot be repaired satisfactorily, replace the whole area affected.

Cleaning

General: Clean the finished surface. Buff and polish. Before the date for practical completion, mop and leave the finished surface clean and undamaged on completion.

Spare materials

General: Supply spare matching resilient finishes and accessories of each type for future replacement purposes. Store the spare materials on site where directed.

Quantity: At least 1% of the quantity installed.

4 SELECTIONS

4.1 PRODUCTS Sheet planks

NOTE : vinyl to finish flush with adjoining floor finish.

Provide aluminium angle trim to butt carpet and Vinyl to.

A1 - PROPOSED REGISTRY BUILDING:

Property	VYL-1	VYL-2	VYL-3	VYL-4
Location	Refer to floor plan (Colour 1)	Refer to floor plan (Colour 2)	Amenities	
Type	Ref			
Form	Sheet vinyl	Sheet vinyl	Sheet vinyl	
Colour	Refer colour schedule			
Sheet width (mm)	Rolls	Rolls	Rolls	
Thickness (mm)	2mm	2mm	2mm	
Slip resistance classification	R11	R11	R11	
Skirting	Feather edge skirting	Feather edge skirting	Coved vinyl	

A2 - PROPOSED WORKSHOP BUILDING:

Property	VYL-1	VYL-2	VYL-3	VYL-4
Location	Refer to floor plan (Colour 1)	Refer to floor plan (Colour 2)	Amenities	
Type	Ref			
Form	Sheet vinyl	Sheet vinyl	Sheet vinyl	
Colour	Refer colour schedule			
Sheet width (mm)	Rolls	Rolls	Rolls	
Thickness (mm)	2mm	2mm	2mm	
Slip resistance classification	R11	R11	R11	
Skirting	Feather edge skirting	Feather edge skirting	Coved vinyl	

A3 - PROPOSED WAREHOUSE BUILDING:

Property	VYL-1	VYL-2	VYL-3	VYL-4
Location	Refer to floor plan (Colour 1)	Refer to floor plan (Colour 2)	Amenities	
Type	Ref			
Form	Sheet vinyl	Sheet vinyl	Sheet vinyl	
Colour	Refer colour schedule			
Sheet width (mm)	Rolls	Rolls	Rolls	
Thickness (mm)	2mm	2mm	2mm	
Slip resistance classification	R11	R11	R11	
Skirting	Feather edge skirting	Feather edge skirting	Coved vinyl	

A4 - PROPOSED NEW BUILDING:

Property	VYL-1	VYL-2	VYL-3	VYL-4
Location	Refer to floor plan (Colour 1)	Refer to floor plan (Colour 2)	Amenities	
Type	Ref			
Form	Sheet vinyl	Sheet vinyl	Sheet vinyl	
Colour	Refer colour schedule			
Sheet width (mm)	Rolls	Rolls	Rolls	
Thickness (mm)	2mm	2mm	2mm	
Slip resistance classification	R11	R11	R11	
Skirting	Feather edge skirting	Feather edge skirting	Coved vinyl	

1 GENERAL**1.1 RESPONSIBILITIES General**

Requirement: Provide coating systems to substrates, as documented.

Performance Requirement:

- Consistent in colour, gloss level, texture and dry film thickness.
- Free of runs, sags, blisters, or other discontinuities.
- Paint systems which are fully opaque or at the documented level of opacity.
- Clear finishes at the level of transparency consistent with the product.
- Fully adhered.
- Resistant to environmental degradation within the manufacturer's stated life span.
-

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- *0171 General requirements.*

1.3 STANDARDS Painting

General: To the recommendations of those parts of AS/NZS 2311 referenced in this work section.

1.4 INTERPRETATION Definitions

General: For the purposes of this work section the definitions given in AS/NZS 2310 and the following apply:

- Gloss: The optical property of a surface, characterised by its ability to reflect light specularly.
- Gloss unit: Numerical value for the amount of specular reflection relative to that of a standard surface under the same geometric conditions.
- Levels of gloss finish: When the specular direction is 60 degrees, surfaces with the following specular gloss reading is defined as follows:
 - o Full gloss: Over 85 gloss units.
 - o Gloss: Over 50 and up to 85 gloss units.
 - o Semi-gloss (satin): Over 20 and up to 50 gloss units.
 - o Low gloss (low sheen): Over 5 and up to 20 gloss units.
 - o Flat finish (matt): Up to 5 gloss units.
- Opacity: The ability of a paint or textured and membrane coating to obliterate the colour difference of a substrate.
- Paint or coating system: A product in liquid form, which when applied to a surface, forms a dry film having protective, decorative or other specific technical properties.
- Primer, prime coat: The first coat of a painting system that helps bind subsequent coats to the substrate and which may inhibit its deterioration.
- Sealer: A product used to seal substrates to prevent the following:
 - o Materials from bleeding through to the surface.
 - o Reaction of the substrate with incompatible top coats.
 - o Undue absorption of the following coat into the substrate.
- Substrate: The surface to which a material or product is applied.
- Undercoat: An intermediate coat formulated to prepare a primed surface or other prepared surface for the finishing coat.

1.5 SUBMISSIONS Products and materials

General: Submit the following at least 3 weeks before the paint is required:

- Paint brand name and product range quality statement.
- Safety data sheets (SDS) showing the health and safety precautions to be taken during application.
- The published recommendations for maintenance.

Subcontractors

Specialist applicators: Submit name and contact details of proposed specialist applicators.

Warranties

General: Submit the coating manufacturer's warranties .

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Opaque finishing stages:
 - o Completion of surface preparation.
 - o After application of final coat.
- Clear finishing stages:
 - o Before surface preparation of timber.
 - o Completion of surface preparation.
 - o After application of final coat.

2 PRODUCTS

2.1 GENERAL Storage and handling

General: Store materials not in use in tightly covered containers in well-ventilated areas with temperatures maintained at the manufacturer's recommendations.

Delivery: Deliver paints to the site in the manufacturer's labelled and unopened containers.

2.2 PAINTING MATERIALS Standards

Paint types: To AS/NZS 2311 Table 4.2 and the following:

- Metal primer - general purpose for iron and steel: To AS/NZS 3750.19.
- Metal primer - latex for metallic zinc surfaces: To AS 3730.15.
- Metal primer - solvent borne for ferrous metallic surfaces: To AS 3730.21.
- Metal primer - zinc-rich organic for iron and steel: To AS/NZS 3750.9.

Combinations

General: Do not combine products from different manufacturers in a paint system.

Clear timber finish systems: Provide only the combinations of putty, stain and sealer recommended by the manufacturer of the topcoat.

Putty and fillers

Material: To the recommendations of the paint system manufacturer, suitable for the substrate and compatible with the primer.

Tinting

General: Provide only products which are colour tinted by the manufacturer or supplier.

Toxic ingredients

General: To the *Poisons Standard (SUSMP)* Part 2 Section 7.

3 EXECUTION

3.1 PREPARATION Order of work

Other trades: Before painting, complete the work of other trades as far as practicable within the area to be painted, except for the installation of fittings, floor sanding and laying flooring materials.

Clear finishes: Complete clear timber finishes before commencing opaque paint finishes in the same area.

Protection

General: Before painting, clean the area and protect it from dust contamination. Use drop sheets and masking agents to protect surfaces, including finished surfaces and adjacent finishes, during painting.

Fixtures and furniture: Remove door furniture, switch plates, light fittings and other fixtures before painting, and conform to the following:

- Labelling and storage: Attach labels or mark fixtures using a non-permanent method, identifying location and refixing instructions, if required. Store and protect against damage.

Difficult to remove fixtures: Where removal is impractical or difficult, apply surface protection before substrate preparation and painting.

Substrate preparation – generally

General: Prepare substrates to receive the documented paint system.

Cleaning: Clean down the substrate surface. Do not cause damage to the substrate or the surroundings.

Filling: Fill cracks and holes with fillers, sealants, putties or grouting cements as appropriate for the finishing system and substrate, and sand smooth: - Clear finish: Provide filler tinted to match the substrate.

Clear timber finish systems: Prepare the surface so that its attributes will show through the clear finish without blemishes, using methods including the following:

- Removal of bruises.
- Removal of discolourations, including staining by oil, grease and nailheads.
- Bleaching where necessary to match the timber colour sample.
- Puttying.
- Fine sanding, with the last abrasive no coarser than 220 grit, so that there are no scratches across the grain.

Treated surfaces: If surfaces have been treated with preservatives or fire retardants, make sure the paint system is compatible with the treatment and does not adversely affect its performance.

Unpainted surfaces

Standard: To AS/NZS 2311 Section 3.

Previously painted surfaces

Preparation of a substrate in good condition: To AS/NZS 2311 clause 7.4.

Preparation of a substrate in poor condition: To AS/NZS 2311 clause 7.5.

Preparation of steel substrates with protective coatings: To AS 2312.1 Section 8 and AS 1627.1.

PVC-U: Clean with methylated spirit and a nylon scouring pad.

Wallcovering: Remove wallcovering and residual paste with clean water. Patch and repair substrate to a uniform surface before painting.

Lime wash paints: Remove by brushing with warm water.

Reconditioned damaged surfaces in galvanized steel: To AS/NZS 4680 clause 8.

Cleaning external surfaces

Sound external surfaces other than timber: Remove dirt, grease, loose and foreign matter, efflorescence and mould by water blasting or steam cleaning without damaging the surface. Remove remaining loose material with hand tools. Use sanding blocks to preserve the arrises of masonry and stone details.

Efflorescence: Eliminate the source of salt and water before cleaning. Allow surface to dry for 15 to 30 days before repainting.

New masonry: Allow 30 days for the masonry to cure and pH level to stabilise before painting.

3.2 PAINTING SYSTEMS General

Number of coats: Except where one or two coat systems are documented, each paint system consists of at least 3 coats.

Low VOC emitting paints

General: Provide the VOC limits as documented.

New unpainted interior surfaces Standard: To AS/NZS 2311 Table 5.1.

New unpainted exterior surfaces Standard: To AS/NZS 2311 Table 5.2.

Previously painted surfaces

Interior surfaces: To AS/NZS 2311 Table 8.2.

Exterior surfaces: To AS/NZS 2311 Table 8.3.

Specialised painting systems

Standard: To AS/NZS 2311 clause 5.2. Provide the following final coats:

- High build textured or membrane finishes for concrete and masonry: Paint reference number B38 using products conforming to the AS 4548 series.
- Two-pack gloss pigmented polyurethane: Paint reference number B44.
- Two-pack epoxy: Paint reference number B29.
- Two-pack water based epoxy: Paint reference number B29A.

3.3 APPLICATION

Light levels General: ≥ 400 lux.

Substrate moisture content

Requirement: Use a moisture meter to demonstrate that the moisture content of the substrate is at or below the recommended maximum level for the type of paint and the substrate material.

Paint application

Standard: To AS/NZS 2311 Section 6.

Timing: Apply the first coat immediately after substrate preparation and before contamination of the substrate can occur. Apply subsequent coats after the manufacturer's recommended drying period has elapsed.

Painting conditions

General: Unless the paint is recommended for such conditions, do not paint under the following conditions:

- Dusty conditions.

- Relative humidity: > 85%.
- Surface temperature: < 10°C or > 35°C.

Priming timber before fixing

General: Apply one coat of wood primer, and 2 coats to end grain, to the back of the following before fixing in position:

- External fascia boards.
- Timber door and window frames.
- Bottoms of external doors.
- Associated trims and glazing beads.
- Timber board cladding.

Spraying

General: If the paint application is by spraying, use conventional or airless equipment which conforms to the following:

- Satisfactorily atomises paint being applied.
- Does not require paint to be thinned beyond the maximum amount recommended by the manufacturer.
- Does not introduce oil, water or other contaminants into the applied paint.

Paint with known health hazards: Provide personal protection, masking, ventilating and screening facilities to AS/NZS 4114.

Sanding

Clear finishes: Sand the sealer using abrasives no coarser than 320 grit without cutting through the colour. Take special care with round surfaces and edges.

Repair

Requirement: Clean off marks, paint spots and stains progressively and restore damaged surfaces to their original condition.

Maintenance painting: To AS/NZS 2311 Section 8.

Repair of galvanizing

Cleaning: For galvanized surfaces which have been subsequently welded, power tool grind to remove all surface contaminants, including rust and weld splatter. Prime affected area immediately after cleaning.

Primer: Type 2 organic zinc-rich coating for the protection of steel to AS/NZS 3750.9.

Tinting

General: Tint each coat of an opaque coating system so that each has a noticeably different tint from the preceding coat, except for top coats in systems with more than one top coat.

Services

General: Paint all new services and equipment, including those in plant rooms, if not embedded, except chromium, anodised aluminium, GRP, PVC-U, stainless steel, non-metallic flexible materials and normally lubricated machined surfaces.

Proprietary items: Repaint only if damaged.

Windows

Operation: Make sure opening windows function correctly before and after painting.

Doors

Drying: Maintain door leaf in the open position during drying. Do not allow door hardware or accessories to damage the door finish during the drying process.

Wet paint warning

Notices: Place in a conspicuous location and do not remove until the paint is dry.

3.4 COMPLETION General

Protection and masking: Remove masking and protection coverings before paint has dried.

Cleaning: On completion of painting, remove splatters from adjacent finished surfaces by washing, scraping or other methods which do not scratch or damage the surface.

Reinstatement: Repair, replace or refinish any damage, including works of other trades. Touch up new damaged paintwork or misses only with the paint batch used in the original application.

Fixtures: Refix removed and undamaged fixture in the original location, make sure they are properly fitted and in proper working order.

Disposal of paint and waste materials

Requirement: Conform to requirements of the local government authority.

4 SELECTIONS

PAINT SCHEUDLE

4.1 INTERIOR PAINTING SCHEDULE

Item/Location	Coat/Method	Notes
Ceiling	- Dulux one step primer - 2 coats Dulux wash and wear matt	- Last coat to be rolled
Woodwork and trim (including doors)	- Dulux professional Enviro2 acrylic seal - 2 coats Dulux Aquanamel gloss	- Note all timber is LOSP treated. Provide primer recommended for LOSP
		treated timber
Villaboard	- Dulux one step primer - 2 coats Dulux wash and wear low sheen	-
Plasterboard	- Dulux one step primer - 2 coats Dulux wash and wear low sheen	-
Metalwork	- Dulux MetalShield all surface primer - 2 coats Aquanamel semi-gloss acrylic	-
Structural Steel, rigging bars, etc	Primer Dulux Zincode 402 , 75 micron DFT (refer structural steel spec) - 2nd coat - Duremax GPE – Min. 150micron DFT - 3rd coat – Weathermax HBR – Min. 100micron DFT	

Refer finishes schedule for paint colours.

4.2 EXTERIOR PAINTING SCHEDULE

Item/Location	Coat/Method	Notes
Downpipes, PVC, Vent pipes etc. (including above roof)	- 1 coat etch primer - 2 coats gloss acrylic	Paint vent pipes above roof.
FC – walls, soffits, exotec cladding	- 1 coat Dulux professional primer - Minimum 3 coats: Dulux Weathershield Low Sheen	-
Structural Steel work Metal work / posts Brackets, beams etc.	- Dulux Zincode 402, 75 micron (refer Structural Steel Spec (75 micron) - 2nd coat - Duremax GPE – Min. 150micron - 3rd coat – Weathermax HBR – Min. 100micron	All visible steelwork to be painted. Total minimum DFT 325 micron. NOTE: paint thickness will be measured on site to ensure the required corrosion protection coating is achieved. - All existing downpipes
Concrete Pavement	- 2 coats CCS Streetscape penetrating	-

Refer finishes schedule for paint colours.

1 HYDRAULIC SYSTEMS

1.1 RESPONSIBILITIES General

Requirement: Provide hydraulic services, as documented.

Design for durability and maintainability

Design for durability: Systems to achieve the documented performance, reliability, service life, energy efficiency and safety requirements, and are easily maintainable.

Access for maintenance: systems to conform to **ACCESS FOR MAINTENANCE** in *0171 General requirements*.

Operating environment

Requirement: Provide equipment suitable for the environment in which it operates.

Hydraulic system design

Requirement: Provide as per Hydraulics design documents.

Design life

Requirement: To *0171 General requirements*.

Design life:

- Hot water heaters: 10 years.
- All other components and systems: 25 years.

Energy efficiency Requirement: To BCA J5.

Noise levels

Ambient noise emitted: Lower than the level that can be heard within a habitable room in any neighbouring premises, regardless of whether any door or window to that room is open.

Noise levels in occupied spaces: To *0171 General requirements*, **Noise levels**.

Acoustic treatment of hydraulic services: Provide acoustic treatment to achieve the documented space and ambient noise levels and acoustic separation.

Seismic restraint

Requirement: To *0171 General requirements*, **SEISMIC RESTRAINT OF NON-STRUCTURAL COMPONENTS**.

Fire separation

Requirement: As documented.

1.2 CROSS REFERENCES General

Requirement: Conform to the following:

- *0171 General requirements*.
- *0223 Service trenching*.
- *0421 Roofing*

1.3 STANDARDS General

Plumbing and drainage: To AS/NZS 3500.0, AS/NZS 3500.1, AS/NZS 3500.2, AS/NZS 3500.3, AS/NZS 3500.4 and the PCA.

Copper pipe and fittings-installation and commissioning: To AS 4809.

Gas: To AS/NZS 5601.1.

Microbial control: To AS/NZS 3666.1, AS/NZS 3666.2 and the recommendations of SA/SNZ HB 32.

Authorised products

Authorised products: Listed in the WaterMark Product Database, unless otherwise required by the Network Utility Operator.

Labelling

Water efficiency labelling: Provide products conforming to and labelled to the Water Efficiency Labelling Scheme (WELS).

1.4 INTERPRETATION Abbreviations

General: For the purposes of this worksection the following abbreviation applies:

- LPG: Liquefied petroleum gas.

Definitions

General: For the purposes of this worksection the following definitions apply:

- Hot-dip galvanized: Zinc coated to AS/NZS 4680 after fabrication with coating thickness and mass to AS/NZS 4680 Table 1.
- Network Utility Operator: The entity undertaking the piped distribution of drinking water or natural gas for supply or is the operator of a sewerage system or external stormwater drainage system.

1.5 SUBMISSIONS Authority approvals

Authority submissions: Submit evidence of approval (including notices) from authorities relating to the works.

Certification

Completion: Submit certificate as verification that the design and installation conforms to all contractual and statutory requirements.

Operation and maintenance manuals

Requirement: Conform to *0171 General requirements*, **OPERATION AND MAINTENANCE MANUALS**.

Products and materials

Data: Submit technical data for all items of plant and equipment, including the following:

- Assumptions.
- Calculations.
- Model name, designation and number.
- Capacity of all system elements.
- Country of origin and manufacture.
- Materials used in the construction.
- Size, including required clearances for installation.
- Certification of conformance to the applicable code or standard.
- Technical data schedules corresponding to the equipment schedules in the contract documents. If there is a discrepancy between the two, substantiate the change.
- Manufacturers' technical literature.
- Type test reports.

Subcontractors

General: Submit names, contact details, licence numbers and type of licence of proposed suppliers and installers.

1.6 INSPECTION Notice

Inspection: Give notice so that inspection may be made of the following:

- Excavated surfaces.
- Concealed or underground services.

1.7 INSTALLATION Connections to Network Utility Operator mains

General: Excavate to locate and expose the connection points and connect to the Network Utility Operator mains. On completion, backfill and compact the excavation and reinstate surfaces and elements which have been disturbed such as roads, pavements, kerbs, footpaths and nature strips to *0223 Service trenching*.

Service trenching

Requirement: To *0223 Service trenching*.

Accessories

General: Provide the accessories and fittings necessary for the proper functioning of the systems, including taps, valves, outlets, pressure and temperature control devices, strainers, gauges and pumps.

Isolating valves: In addition to valves required to meet statutory requirements, provide valves to allow safe isolation of parts of the system, with minimum inconvenience to the building occupants, in event of leaks or maintenance.

Movement compensation

Compensation: Arrange piping and wiring crossing building expansion joints so that moment in the joint does not cause damage.

1.8 PIPING Finishes

Exposed piping: Finish exposed piping, including fittings and supports, as follows:

- In internal locations such as toilet and kitchen areas: Chrome plate copper piping to AS 1192 service condition 2, bright.
- Externally and steel piping and iron fittings internally: Paint.
- In concealed but accessible spaces (including cupboards and non-habitable enclosed spaces): Leave copper and plastic unpainted except for identification marking. Prime steel piping and iron fittings.

Valves: Finish valves to match connected piping.

2 SANITARY FIXTURES

2.1 STANDARDS General

Design for access and mobility: To AS 1428.1 and AS 1428.2.

3 TAPWARE

3.1 STANDARDS General

Design for access and mobility: To AS 1428.1 and AS 1428.2.

3.2 TAPS General

Material: Brass or bronze.

Taps other than hose taps: Provide anti-splash aerator nozzles.

Construction: Provide the following:

- Ceramic disk valve.
- O-ring seals.
- Vandal resistant handle.
- Vandal resistant aerator nozzles.

Hose taps

Construction: Provide hose taps as follows:

- Type: DN20 diameter brass finish hose tap.
- Riser: DN20 diameter copper riser with a back plated elbow.
- Heads: Provide anti-vandal heads where documented.

Backflow prevention: Provide vacuum breaker backflow prevention device and isolation valve.

Installation: Conform to the following:

- Fix hose tap 450 mm above the floor.
- Install against a masonry or concrete wall. Fix the back plated elbow to the wall with 3, 20 mm brass screws, screwed into 20 x 6 mm expanding plastic plugs.

Taps and valve heads

Metal heads and handles: Provide brass fittings or a suitable bush to prevent electrolysis and growth.

Plastic heads and handles: Provide compact fittings designed to prevent fracture and exposure of jagged or rough edges.

Vandal-proof heads: Provide vandal-proof or anti-tampering devices for the designated types.

4 WATER HEATERS

4.1 ELECTRIC STORAGE WATER HEATERS Description

General: Proprietary automatic electrically heated water heater including connections, controls and fittings.

Standards

General: To AS/NZS 4692.1.

Energy performance: To AS/NZS 4692.2.

Tariff

General: Install so that the heating system qualifies for the tariff concession or subsidy offered by the electricity distributor.

4.2 INSTANTANEOUS WATER HEATERS Gas instantaneous heaters

Standard: To AS/NZS 5601.1.

Energy rating: Minimum 6 stars.

Power supply: Provide gas and electrical supply with isolation at the heater.

Free air flow: Make sure air flow around and above the heater is not obstructed and discharge air does not short circuit to the air intake.

Electric instantaneous heaters Standard: To AS/NZS 60335.2.35.

4.3 INSTALLATION General

Standard: Install to AS/NZS 3500.4.

Gas water heaters

Standard: Install to AS/NZS 5601.1.

Solar water heaters Collectors: Install collectors:

- Facing $\pm 20^\circ$ of north.
- Inclined at the angle for the nearest location listed in AS/NZS 3500.4 Table G2.
- In a location not subject to shading at any time of year especially in winter when the sun is low.

Adjustment: If installing at other angles and/or subject to shading, increase the size of the collectors to that required to give the performance of an unshaded collector oriented as specified above.

Manifolds

General: If multiple heaters are installed in banks use the manufacturer's standard manifold arrangement to provide equal flow thorough each heater in the bank.

5 STORMWATER – BUILDINGS

5.1 DESIGN General

Requirement: Design the roof drainage system, including sumps, rainwater outlets, overflow outlets and downpipes.

Roof drainage system design, application and calculations Standards: Conform to the following:

- AS/NZS 3500.3.
- National Construction Code Series Volume 3: Plumbing Code of Australia (PCA).

Handbooks: Conform to the recommendations of:

- SA HB 39.

Rainfall intensity for design:

- Eaves gutters: 1 in 20 years storm.
- Box gutters: 1 in 100 years storm.

Box gutter overflow

Overflow: Design for 100% overflow.

Methods of calculation

Requirement: Manual or software that employs the data and methods in the applicable standard.

Design documentation

Drawings: Show the following on the drawings:

- General layout of the system.
- Calculated capacities.
- Details of components including internal and external metal gutters, downpipes, sumps and rainheads.
- Spatial relationship to other services and building structure.
- Means of accommodating thermal and building movement.
- Details of penetrations.
- Coordination and interfaces with other trades.
- Connection points to site stormwater system.

5.2 STORMWATER DRAINS Location

General: Provide stormwater drains to connect downpipes, surface drains, subsoil drains and drainage pits to the outlet point or point of connection. Make sure location of piping will not interfere with other services and building elements not yet installed or built. Subject to the preceding and documented layouts, follow the most direct route with the least number of changes in direction.

Downpipe connections

Termination: Select from the following:

- Termination over pit: Stop downpipe 100 mm above the ground level and discharged into grated pit. Do not connect directly into stormwater pipes.
- Direct connection: Bring downpipes out from the building at a suitable angle and level so the downpipe enters the underground drain at the finished level of the surrounding area. Turn up branch pipelines with bends to meet the downpipe, finishing horizontally 50 mm (nominal) above finished ground or pavement level. Seal joints between downpipes and drains.

Laying

General: Lay in straight lines between changes in direction or grade with sockets pointing up hill. If other pipes are adjacent, set each pipe true to line and complete each joint before laying the next pipe. If work is not continuous, cap open ends to prevent entry of foreign matter.

Pipe underlay (bedding)

General: Bed piping on a continuous underlay of bedding material, minimum 75 mm and maximum 150 mm thick after compaction. Grade the underlay evenly to the gradient of the pipeline.

Chases: If required, form chases to prevent projections such as sockets and flanges from bearing on the trench bottom or underlay.

Pipe surrounds

General: Place the material in the pipe surround in layers, maximum 200 mm loose thickness, and compact without damaging or displacing the piping.

Anchor blocks

General: If required to restrain lateral and axial movement of the stormwater pipes, provide reinforced concrete anchor blocks at junctions and changes of grade or direction conforming to AS/NZS 3500.3 clause 7.9.

Thermal movement

General: Arrange piping to accommodate thermal expansion. Provide proprietary expansion joints in copper and plastic pipes where pipe flexibility does not allow movement. Make sure movement does not strain branch connections.

5.3 SUBSOIL DRAINS General

Requirement: Provide subsoil drains to intercept groundwater seepage and prevent water build-up behind walls and under floors and pavements. Connect subsoil drains to surface drains or to the stormwater drainage system as applicable.

Trench width: Minimum 450 mm.

Trench floor: Grade the trench floor evenly to the gradient of the pipeline. If the trench floor is rock, correct any irregularities with compacted bedding material.

Pipe depth: Provide the following minimum clear depths, measured to the crown of the pipe, below the following elements:

- Formation level of the pavement, kerb or channel: 100 mm.
- Average gradient of the bottom of footings: 100 mm.
- Finished surface of unpaved ground: 450 mm.

Jointing

General: At junctions of subsoil pipes provide tees, couplings or adaptors to AS 2439.1.

Pipe underlay (bedding)

General: Bed piping on a continuous underlay of bedding material, at least 75 mm thick after compaction. Lay the pipe with one line of perforations at the bottom.

Bedding thickness: Minimum 75 mm, maximum 150 mm, after compaction.

Chases: If necessary, form chases to prevent projections such as sockets and flanges from bearing on the trench bottom or underlay.

Pipe surrounds

General: Place the material in the pipe surround in layers, of a maximum 200 mm loose thickness, and compact without damaging or displacing the piping.

Depth of overlay:

- To the underside of the bases of overlying structures such as pavements, slabs and channels.
- To within 150 mm of the finished surface of unpaved or landscaped areas.

Geotextile

Requirement: Provide polymeric fabric formed from plastic yarn composed of at least 85% by weight propylene, ethylene amide or vinylidene chloride and containing stabilisers or inhibitors which provide resistance to deterioration due to ultraviolet light.

Marking: To AS 3705.

Protection: Provide heavy duty protective covering. Store clear of the ground and out of direct sunlight. During installation, do not expose the filter fabric to sunlight for more than 14 days.

Filter socks

General: Provide permeable polyester socks, capable of retaining particles 0.25 mm and greater. Securely fit or join the sock at each joint.

5.4 PITS Finish to in situ exposed surfaces

General: Provide a smooth, seamless finish, using steel trowelled render or concrete cast in steel forms.

Location: At junctions, changes of gradient and changes of direction of stormwater drains.

Metal access covers and grates Standard: To AS 3996.

Cover levels: Top of cover or grate, including frame:

- In paved areas: Flush with the paving surface.
- In landscaped areas: 25 mm above finished surface.
- Gratings taking surface water runoff: Locate to receive runoff without ponding.

5.5 TESTING Pre-completion tests

General: Before backfilling or concealing, carry out the following tests to AS/NZS 3500.3 Section 9:

- Downpipes within buildings: Air or water pressure test.
- Site stormwater drains and main internal drains: Air or water pressure test.
- Rising mains from pumped discharge: Water pressure test .

Leaks: If leaks are found, rectify and re-test.

5.6 COMPLETION Cleaning

General: Clean and flush the whole installation.

6 WASTEWATER

6.1 PRODUCTS Material selection

Environmental conditions: Provide materials capable of withstanding the operational environmental conditions. Select and install to manufacturers' recommendations.

Dissimilar materials: Connect dissimilar materials using adapters to Network Utility Operator requirements and manufacturer's recommendations.

Rubber banded sleeves: Do not provide.

6.2 FLOOR WASTES General

Requirement: Provide each floor waste with a trap constructed of the material specified for the sanitary plumbing system. Fit off each riser with a chromium plated brass grating finished flush with the surrounding floor finish. If the floor surfaces are vinyl, provide gratings and outlets designed to permit the vinyl to be turned down into the outlet and the grating clamped down onto the surface.

Waterproofing: Make sure all penetrations through floors and finishes up to the edge of grates are fully waterproof.

Priming: Provide priming of floor wastes. If floor wastes cannot be primed via fixture, provide priming valves to maintain the water seal to AS/NZS 3500.2.

6.3 SANITARY PLUMBING Location

General: Verify location and invert level of piping before commencing installation.

Layout: Arrange piping in conformance with the following:

- Avoid interference with other services and building elements.
- Follow the most direct route with the least number of changes of direction.

Ducts: If installed in ducts, locate and fix stacks, wastes and pipes independently of other services. Arrange so they are easily accessible and removable throughout their entire length.

Order of work

Requirement: Start drain laying at the downstream end of the drainage system (at the connection point to site infrastructure), not the upstream end. Confirm invert levels with building elements before starting to lay drains.

Discharge from air handling systems

Trays, sumps and plumbing: To AS/NZS 3666.1.

Expansion joints

Location: Provide expansion joints where pipes cross seismic or movement joints in the building, and from the building to below ground outside the building.

Inspection openings

Location: Provide inspection openings at each upstream end of branch and main drains, change of direction, entry to stacks and to AS/NZS 3500.2. Provide inspection openings complete with access riser brought up to finished floor levels. If access risers are located in tiled floor areas or surfaces with similar finishes, provide anti-slip inspection covers with neoprene gas tight sealing rings.

Size: Provide inspection openings that allow full access to the waste pipe.

Vertical stacks: Provide a removable access gate opening of size equal to the diameter of the pipe approximately 600 mm above finished floor level. If the stack is concealed behind a wall or duct, provide a hinged access panel in the wall or duct with finish to match the surface in which it is installed.

Thermal movement

General: Arrange piping to accommodate thermal expansion. Provide proprietary expansion joints in copper and plastic pipes where pipe flexibility does not allow for movement. Make sure that movement does not strain branch connections.

Tundishes

Location: Provide suitably sized, trapped tundishes to collect condensate wastes from mechanical equipment. Connect tundishes to nearest waste or floor drain. Connect tundish waste to floor wastes, wastes or drains and provide traps and vents where necessary.

Charging: If tundishes are not provided with a constant discharge from equipment and are connected directly to the sanitary plumbing system or drainage system, provide a trap seal primer valve to ensure that the trap of the tundish is charged at all times.

Vent pipes

Requirement: Provide upstream and downstream vents to AS/NZS 3500.2.

Location: Locate vents at least 6 m from any air intake or grille and at least 3 m from exhaust discharges.

Staying to roof: If fixings for stays penetrate the roof covering, seal the penetrations and make watertight.

Terminations: Provide vent cowls of the same material as the vent pipe.

Wet area floors

General: Where drainage connections pass through wet area floors, terminate 4 mm below the substrate surface.

6.4 SANITARY DRAINAGE Laying

General: Lay in straight lines between changes in direction or grade with sockets pointing up hill. If other pipes are adjacent, set each pipe true to line and complete each joint before laying the next pipe. If work is not continuous, cap open ends to prevent entry of foreign matter.

6.5 PIPING Supports

Adverse soil conditions under structures: If soil under structures is unable to support piping without movement, suspend piping from the structure above on type 316 stainless steel hangers, fasteners and saddles at 600 mm maximum centres. Backfill with non-compacted light weight material and size supports to carry the load of the piping and backfill and conform to the **Pipeline tolerances table**. Cover metal components with petroleum based mastic then wrap with petroleum based tape to provide a continuous barrier to prevent ingress of water. Provide 50% minimum overlap of tape.

Differential movement: If the geotechnical site investigation report predicts differential movements between buildings and the ground, conform to **Differential movement** in *0171 General requirements*. **Pipeline tolerances table**

	Permissible angular deviation from alignment	Permissible displacement from alignment
Horizontal	1:300	15 mm
Vertical	1:500	5 mm

6.6 PITS Installation

General: Prepare foundation, install pit and connect pipes, to manufacturer's recommendations.

Location: At junctions, changes of gradient and changes of direction of stormwater drains.

Metal access covers and grates Standard: To AS 3996.

Cover levels: Top of cover or grate, including frame:

- In paved areas: Flush with the paving surface.
- In landscaped areas: 25 mm above finished surface.
- Gratings taking surface water runoff: Locate to receive runoff without ponding.

6.7 TESTING Pre-completion tests

Requirement: Test to AS/NZS 3500.2 Section 15, before backfilling or concealing.

Leaks: If leaks are found, rectify and re-test.

6.8 COMPLETION Cleaning

General: On completion clean and flush the whole installation.

7 COLD AND HEATED WATER

7.1 DESIGN General

Requirement: Design the cold water and heated water systems.

Design parameters

Standards: Conform to the following:

- AS/NZS 3500.1.
- National Construction Code Series Volume 3: Plumbing Code of Australia (PCA).
- Pressure and velocity: Design drinking water and recycled water systems to achieve the following:
 - o Minimum outlet pressure at any point: 250 kPa.
 - o Maximum outlet pressure at any point: 500 kPa.
 - o Maximum velocity at any point: 1.6 m/s.

7.2 PRODUCTS Backflow prevention devices

Standard: To AS/NZS 2845.1 and AS 2845.2.

Pressure drop: Select for lowest pressure drop compatible with the required functions.

Thermostatic mixing valves Standard: To AS 4032.1.

Requirement: Provide thermostatic mixing valves that automatically control the temperature at the mixed outlet to a preselected temperature and suitable for the number of outlets served by the individual valve.

Controls: Include the following:

- A temperature sensitive automatic control that maintains temperature at the preselected setting and rapidly shuts down the flow if either the supply system fails, or if the normal discharge water temperature is exceeded.
- Hot water flush facility.

Wall box: If documented, house the thermostatic mixing valve in a stainless steel recessed wall box with a hinged door and keyed lock.

Water meters

Standard: To AS 3565.4.

Installation: To the requirements of the Network Utility Operator.

7.3 PIPING Mains connection

Requirement: Connect the cold water supply system to the Network Utility Operator's main through a stop valve and meter.

Cold water system: Provide the cold water supply system, installed from the meter to the draw-off points or connections to other services.

Heated water system: Provide the heated water system, installed from the cold water connection points to the draw-off points or connections to other services.

Fittings and accessories

General: Provide the fittings required for the proper functioning of the water supply system, including taps, valves, backflow prevention devices, pressure and temperature control devices, strainers, gauges and automatic controls and alarms.

Provision for dismantling: Arrange piping by the provision of unions or similar so that valves, taps and other maintainable components can be removed for maintenance without disturbing or cutting adjacent piping.

Press-fit fittings

Type: Permanent and inseparable after pressing.

Fitting material: Copper, stainless steel or gunmetal.

Leak path: Incorporate a positive leak path in the form of a channel in the metallic body of the fitting that indicates leakage when dry pressure tested is over the range 2.2 kPa to 300 kPa or wet pressure tested is over the range 100 kPa to 650 kPa.

Seals: EPDM suitable for solar applications and incorporating a cylindrical pipe guide in the front of the seal.

Pipes under pressure embedded in concrete

Prohibition: Do not embed or cast water service pipes into concrete structures.

Sleeves

Requirement: Provide sleeves at penetrations to *0171 General requirements*.

Provision for expansion

Requirement: Provide for thermal expansion of piping to AS/NZS 3500.4 clause 4.12.3 using either:

- Inherent flexibility of the piping.
- Proprietary expansion compensators consisting of a corrugated stainless flexible hose inside a reinforced metallic braid and fitted with stainless steel flanges. Install in bending mode and provide guides and anchors to manufacturer's recommendations.

Expansion loops: If expansion loops are used, install isolation valves either side of the expansion loops within ceiling spaces and riser ducts.

Venting

Location: Provide 15 mm minimum size air release vents at the following locations:

- High points of the system.
- Sections of the piping in which air may collect.
- Upstream from each item of heat exchange equipment.

Risers: Provide a 150 mm high riser set vertically from the pipe and fabricated from the same diameter and material as the pipe. Provide an automatic air vent at the top of the riser.

7.4 BACKFLOW PREVENTION Location

Requirement: Provide backflow prevention devices in the following locations:

- On main incoming domestic cold water supplies, downstream of meters.
- On all mechanical plant, upstream of the plant.
- On all irrigation systems.

In other locations required by the Network Utility Operator and AS/NZS 3500.1.

Installation

Location: Arrange to be readily accessible and easily removal.

External valve locations: Protect from damage and vandalism.

Arrangement: Provide each backflow prevention device with the following:

- Provide unions if $\leq$ DN50, flanges for larger sizes.
- Isolating valves upstream and downstream of each backflow prevention devices.
- Dual check valves to AS/NZS 3500.1 clause 4.6.2.3.
- Line strainer upstream of the each backflow prevention device.
- Tundish and drain with connection to waste drain to AS/NZS 3500.1.

Registration: Register valves to Network Utility Operator requirements.

In-wall mounted backflow prevention valves

Arrangement: Provide pre-plumbed proprietary high hazard valve assemblies as follows:

- Dual hot and cold valve trains.
- Flush mount, lockable, recessed Type 304 stainless steel cabinet.
- Slide-in front panel.
- DN50 waste connection.

Vacuum breaker valves

Requirement: Provide vacuum breaker valves where required to prevent cross-connection of the cold water service.

7.5 PIPING INSULATION General

Requirement: Insulate the following:

- Non-chrome plated heated water piping, fittings and valves.
- Non-chrome plated cold water piping in spaces subject to condensation.
- Cold water piping subject to freezing.

Application: Fit insulation tightly to piping surfaces without gaps. Close butt ends of insulation sections. Minimise number of joints. If the insulation is in half-sections, make only half-circumferential joints at any one place. Seal longitudinal seams in foil laminate and fix insulation at maximum 500 mm centres with polypropylene, zinc-coated steel or aluminium straps.

Unions and other items requiring service: Install the insulation so that it is readily removable.

Fittings: Provide insulation with insulation R-Value at least equal to that of the adjacent piping insulation.

7.6 MARKING Notice plate

General: Provide a notice plate containing condensed emergency instructions, legibly printed or engraved on durable material resistant to defacement, at least 3 mm thick or mounted on board at least 3 mm thick, permanently fixed in a convenient position at the control valves.

Buried services

Requirement: Provide a detectable marker tape with trace wire to identify the route of buried piping.

Marker tape: Provide a minimum 100 mm durable plastic in colour to AS 1345 continuously printed with the words DANGER – BURIED DRINKING WATER SERVICE BELOW.

Location: Lay in backfill 150 mm above the pipe.

Trace wire: Terminate with 1000 mm coil in a readily accessible location. Tag to match the record drawings.

7.7 TESTING Pre-completion tests

Pressure tests: Before insulation is applied to joints, pressure test piping to AS/NZS 3500.1 and AS/NZS 3500.4 as appropriate.

Equipment: Before testing, disconnect any equipment not rated for the test pressure.

Leaks: If found, rectify and re-test.

Cross connections: Isolate systems individually and check for cross connections.

Backflow prevention: To AS/NZS 3500.1.

Tapware: Check for leaks.

Completion test

General: Provide a full operational test to verify conformance.

7.8 COMPLETION Commissioning

Strainers: Remove, clean and replace strainer baskets.

Cleaning and disinfection: To AS/NZS 3500.1 Appendix I.

Cold water systems: Test and commission to AS/NZS 3500.1 Section 18.

Heated water systems: Test and commission to AS/NZS 3500.4 Section 9.

Testable backflow prevention devices: Test and commission to AS/NZS 2845.3 by a licensed plumber with backflow device accreditation. Tag and certify to the requirements of the Network Utility Operator.

Charging

Completion: On completion of installation, commissioning, testing and disinfection, fill the system with water, turn on control and isolating valves and the energy supply and leave the water supply system in full operational condition.

Thermostatic mixing valves Field testing: To AS 4032.3.

8 HYDRAULIC MAINTENANCE

8.1 GENERAL Objective

Requirement: Maintain the hydraulic systems for the documented maintenance period so that the performance, reliability, service life, energy efficiency and safety of the system is equal to or better than that at the beginning of the maintenance period, in parallel with and including:

- Periodic and statutory maintenance, cleaning and replacement of consumables.
- Emergency repairs.

Maintenance period: To *0171 General requirements*.

8.2 MAINTENANCE Cold and heated water

Maintenance of tanks and piping for drinking water: To AS/NZS 3500.1 and AS/NZS 3500.4.

Ball float valves: Check and adjust for no overflow.

Heated water systems:

- Conform to the recommendations of AS/NZS 3500.4 Appendix M.
- Inspection and maintenance: To AS/NZS 3666.2.
- Provide service tags recording inspections and tests.

Leaks: Inspect cold and heated water systems at least annually for the following:

- Leaks, including leaks from cisterns.
- Other defects.
- Safe condition.
- Conformance to the PCA and Network Utility Operator requirements.

Leaks and defects: Report if found and rectify.

Strainers: Inspect and clean at least annually.

Sanitary fixtures schedule

A1 - Registry

Office					
Location	count	Fixture code	Fixture Type	Selection	Accessories & Tapware
Kitchen	1	SK	Sink	Vera Double Sink 790mm - Black Granite	Elysian Commercial Pull-Out Kitchen Mixer - brushed nickel
	1	ZIP	tap	Zip HydroTap G5 BC Classic Plus Tap - brushed nickel	Cold water
Female	1	BSN	Wall mounted Basin - Corian	Custom Design washplane with 2 basins	Hot & cold Posh Solus Basin Mixer Tap brushed nickel (5 Star) Lead Free
	2	WC	Toilet	Caroma Forma Close Coupled Back to Wall Bottom Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Cold water
	1	SHR	Sower		<u>Roca Ona Shower Mixer Tap</u> brushed nickel Roca Ona Twin Rail Shower brushed nickel (3 Star)
Male	1	BSN	Wall mounted Basin - Corian	Custom Design washplane with 3 basins	Hot & cold Posh Solus Basin Mixer Tap brushed nickel (5 Star) Lead Free
	3	WC	Toilet	Caroma Forma Close Coupled Back to Wall Bottom Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Hot & cold
	2	UR	Urinal	Caroma Leda Wall Hung Urinal P Trap White (3 Star)	Cold water
	1	SHR	Sower		<u>Roca Ona Shower Mixer Tap</u> brushed nickel Roca Ona Twin Rail Shower brushed nickel (3 Star)

Office					
Location	count	Fixture code	Fixture Type	Selection	Accessories & Tapware
Kitchen	1	SK	Sink	Vera Double Sink 790mm - Black Granite	Elysian Commercial Pull-Out Kitchen Mixer - brushed nickel
	1	ZIP	tap	Zip HydroTap G5 BC Classic Plus Tap - brushed nickel	
Walkway	1	Trough	Trough	Britex - Pre-plumbed Hand Wash Trough (TPWDPH) 1800mm	Cold
Female	1	BSN	Wall mounted Basin - Corian	Custom Design washplane with 2 basins	Hot & cold Posh Solus Basin Mixer Tap brushed nickel (5 Star) Lead Free
	2	WC	Toilet	Caroma Forma Close Coupled Back to Wall Bottom Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Cold water
	1	SHR	Sower		<u>Roca Ona Shower Mixer Tap</u> brushed nickel Roca Ona Twin Rail Shower brushed nickel (3 Star)
Male	1	BSN	Wall mounted Basin - Corian	Custom Design washplane with 3 basins	Hot & cold Posh Solus Basin Mixer Tap brushed nickel (5 Star) Lead Free
	4	WC	Toilet	Caroma Forma Close Coupled Back to Wall Bottom Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Hot & cold
	1	UR	Urinal	Britex regency urinal – custom length 3100mm – Stainless steel	Cold water
	3	SHR	Sower		<u>Roca Ona Shower Mixer Tap</u> brushed nickel Roca Ona Twin Rail Shower brushed nickel (3 Star)

Office					
Location	count	Fixture code	Fixture Type	Selection	Accessories & Tapware
Kitchen	1	SK	Sink	Vera Double Sink 790mm - Black Granite	Elysian Commercial Pull-Out Kitchen Mixer - brushed nickel
	1	ZIP	tap	Zip HydroTap G5 BC Classic Plus Tap - brushed nickel	
Female	1	BSN	Wall mounted Basin - Corian	Custom Design washplane with 1 basins	Hot & cold Posh Solus Basin Mixer Tap brushed nickel (5 Star) Lead Free
	1	WC	Toilet	Caroma Forma Close Coupled Back to Wall Bottom Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Cold water
	1	WC ambulant	Toilet	Caroma Forma Close Coupled Back to Wall Back Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Cold water Ambulent grab rails
	1	SHR	Sower		<u>Roca Ona Shower Mixer Tap</u> brushed nickel <u>Roca Ona Twin Rail Shower</u> brushed nickel (3 Star)
Male	1	BSN	Wall mounted Basin - Corian	Custom Design washplane with 2 basins	Hot & cold Posh Solus Basin Mixer Tap brushed nickel (5 Star) Lead Free
	1	WC ambulant	Toilet	Caroma Forma Close Coupled Back to Wall Back Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Cold water Ambulant grab rails
	1	WC	Toilet	Caroma Forma Close Coupled Back to Wall Bottom Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Hot & cold
	1	UR	Urinal	Britex regency urinal – custom length 3100mm – Stainless steel	Cold water
	1	SHR	Sower		<u>Roca Ona Shower Mixer Tap</u> brushed nickel <u>Roca Ona Twin Rail Shower</u> brushed nickel (3 Star)

PWD	1	BSN	Basin	Tribute Rectangle 600 Right Hand Shelf WB 1TH OF	Hot & cold Care Plus Optional Basin Care Handle H/C
	1	WC	Toilet	Care 610 Cleanflush Connector S Trap Suite with Backrest and Caravelle Single Flap Seat	Cold water Care Support Grab Rail - 140 Degree Angled 870x700 RH - Stainless Steel
	1	SHR	Shower		Hot & cold Care Support Shower Set with Inverted T Rail - RH- Chrome Care Support Shower Seat - White (GermGard®)

Office					
Location	count	Fixture code	Fixture Type	Selection	Accessories & Tapware
Kitchen	2	SK	Sink	Vera Double Sink 790mm - Black Granite	Elysian Commercial Pull-Out Kitchen Mixer - brushed nickel
	2	ZIP	tap	Zip HydroTap G5 BC Classic Plus Tap - brushed nickel	
Female	1	BSN	Wall mounted Basin - Corian	Custom Design washplane with 3 basins	Hot & cold Posh Solus Basin Mixer Tap brushed nickel (5 Star) Lead Free
	2	WC	Toilet	Caroma Forma Close Coupled Back to Wall Bottom Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Cold water
	1	WC ambulant	Toilet	Caroma Forma Close Coupled Back to Wall Back Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Cold water Ambulant grab rails
	1	SHR	Sower		<u>Roca Ona Shower Mixer Tap</u> brushed nickel <u>Roca Ona Twin Rail Shower</u> brushed nickel (3 Star)
Male	1	BSN	Wall mounted Basin - Corian	Custom Design washplane with 4 basins	Hot & cold Posh Solus Basin Mixer Tap brushed nickel (5 Star) Lead Free
	1	WC ambulant	Toilet	Caroma Forma Close Coupled Back to Wall Back Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Cold water Ambulant grab rails
	3	WC	Toilet	Caroma Forma Close Coupled Back to Wall Bottom Inlet Over Height Rimless Toilet Suite with Soft Close Quick Release Seat White (4 Star)	Hot & cold
	2	UR	Urinal	Britex regency urinal – custom length 3100mm – Stainless steel	Cold water
	1	SHR	Sower		<u>Roca Ona Shower Mixer Tap</u> brushed nickel <u>Roca Ona Twin Rail Shower</u> brushed nickel (3 Star)

PWD	1	BSN	Basin	Tribute Rectangle 600 Right Hand Shelf WB 1TH OF	Hot & cold Care Plus Optional Basin Care Handle H/C
	1	WC	Toilet	Care 610 Cleanflush Connector S Trap Suite with Backrest and Caravelle Single Flap Seat	Cold water Care Support Grab Rail - 140 Degree Angled 870x700 RH - Stainless Steel
	1	SHR	Shower		Hot & cold Care Support Shower Set with Inverted T Rail - RH- Chrome Care Support Shower Seat - White (GermGard®)
Externally	1	Trough	Trough	Britex - Pre-plumbed Hand Wash Trough (TPWDPH) 1800mm	Cold

Just

1 FIRE SERVICES SYSTEMS

1.1 RESPONSIBILITIES

General

Requirement: Design systems and provide fire services, as documented.

1.2 DESIGN

Design for durability and maintainability

Design for durability: Develop the design so the systems achieve the documented performance, reliability, service life, energy efficiency and safety requirements, and are easily maintainable.

Access for maintenance: Develop the design so the systems conform to **ACCESS FOR MAINTENANCE** in *0171 General requirements*.

Operating environment

Requirement: Provide equipment suitable for the environment in which it operates.

Fire services system design

Requirement: Provide engineering design that:

- Maximises the functionality, performance, safety, flexibility and reliability of the fire services.
- Is technically sound.
- Can be constructed using currently accepted methods.
- That provide the lowest combined owning and operating cost over the design life of the systems.

Fire extinguishers and blankets: To *0573 Fire extinguishers and blankets*.

Electrical systems:

- Electrical design: To AS/NZS 3000.
- Selection of cables: To AS/NZS 3008.1.1.
- Fire-resisting protection: Provide for switchboards and associated electrical conductors to BCA C2.13.

Authority submissions: Make submissions (including notices) to authorities relating to the works.

Design life

Requirement: To *0171 General requirements*.

Design life: All components and systems: 25 years.

Seismic restraint

Requirement: To *0171 General requirements*, **SEISMIC RESTRAINT OF NON-STRUCTURAL COMPONENTS**.

Fire separation

Requirement: As documented.

1.3 CROSS REFERENCES

General

Requirement: Conform to the following:

- *0171 General requirements.*
- *0223 Service trenching.*
- *0573 Fire extinguishers and blankets.*

1.4 STANDARDS

Hydraulic fire services

Plumbing and water supply: To AS/NZS 3500.1.

Sprinklers: To AS 2118.1.

Hydrants: To AS 2419.1.

Hose reels: To AS/NZS 1221 and AS 2441

Combined wet fire suppression systems: To AS 2118.6.

Tanks for fire protection systems: To AS 2304.

Pumps for fire services: To AS 2941.

Electrical fire services

Requirement: To AS/NZS 3000, unless otherwise documented.

Fire detection and alarms: To AS 1670.1, AS 1670.5, AS 4428.16, AS 7240.2, AS 7240.4 and AS 7240.13.

Emergency warning and intercommunication: To AS 1428.5, AS 1670.4, AS 1670.5, AS 4428.4, AS 4428.16, AS ISO 7240.24 and AS 60849.

Wiring fire and mechanical performance classification: To AS/NZS 3013.

Degrees of protection (IP code): To AS 60529.

Electromagnetic compatibility (EMC): To the AS/NZS 61000 series.

Communications systems: To AS/CA S008, AS/CA S009, AS/NZS 11801.1 and AS/NZS 14763.2.

1.5 SUBMISSIONS

General

Requirement: Conform to *0171 General requirements.*

Authority approvals

Authority submissions: Submit evidence of approval (including notices) from authorities relating to the works.

Certification

Completion: Submit certificate as verification that the design and installation conforms to all contractual and statutory requirements.

Design documentation

Hydrants: Submit the following:

- Hydraulic calculations: Submit hydraulic calculations of the hydrant systems to AS 2419.1.
- Calculation method: Use commercially available software written and validated to conform to AS 2419.1.
- Drawings: Based on the calculations, submit drawings showing the most hydraulically advantaged and disadvantaged hydrants to AS 2419.1.

Sprinklers: Submit the following:

- Hydraulic calculations: Submit hydraulic calculations of the sprinkler systems to AS 2118.1 Section 14.
- Calculation method: Use commercially available software written and validated to conform to AS 2118.1.
- Drawings: Based on the calculations, submit drawings showing the most favourable and unfavourable areas to AS 2118.1.
-

Operation and maintenance manuals

Requirement: Conform to *0171 General requirements*, **OPERATION AND MAINTENANCE MANUALS**.

Products and materials

Data: Submit technical data for all items of plant and equipment, including the following:

- Assumptions.
- Calculations.
- Model name, designation and number.
- Capacity of all system elements.
- Country of origin and manufacture.
- Materials used in the construction.
- Size, including required clearances for installation.
- Certification of compliance with the applicable code or standard.
- Technical data schedules corresponding to the equipment schedules in the contract documents. If there is a discrepancy between the two, substantiate the change.
- Manufacturers' technical literature.

Shop drawings

Requirement: Submit detail drawings at minimum 1:100 scale, showing the following:

- Pipework and equipment layout and sections showing the work to be installed on the level that the services are installed.
Do not submit glass floor drawings.
- Riser layouts and sections.
- Piping and other schematic drawings including numbering of each valve to correspond to valve tags notation.
- Inclusions: Include the following on the drawings:
 - o Details of control panels including indicating equipment, control and power diagrams.
 - o Location, type, grade and finish of piping, fittings, valves and pipe supports.
 - o Provision of a temporary fire hydrant service in the construction period.
 - o Provision of site treatment and fire vehicle parking as required adjacent to the fire hydrant booster inlet valve.
 - o Tank stands and supporting structures.
 - o Penetrations and associated building work. If penetrations are through external walls, detail flashing and weatherproofing at 1:10 scale.
 - o Fire hose reel and extinguisher layout.
 - o Fire hydrant site plan for the entire site.
 - o Fire fan control schematics and layouts.
 - o Location of detector system components.
 - o Circuit identification.
 - o Labelling details.

Subcontractors

General: Submit names, contact details, licence numbers and type of licence of proposed suppliers and installers.

2 HYDRANTS

2.1 PIPING

General

Piping material: To AS 2419.1.

2.2 VALVES

General

Valves: To AS 2419.1.

Existing to be maintained

Provide new as documented on hydraulic design

2.3 WATER SUPPLY

General

Supply: Provide water supply, as documented.

Backflow prevention: Provide backflow prevention to AS/NZS 3500.1 Section 4.

Service trenches: To *0223 Service trenching*.

2.4 INSTALLATION

General

System: To AS 2419.1.

Valves: Locate valves to permit satisfactory operation and maintenance.

Isolating valves: Provide isolating valves to AS 2419.1 clause 8.7.

Ring mains: To AS 2419.1 clause 8.6.

Pressure gauges: Provide pressure gauges at the hydraulically most disadvantaged fire hydrant in the installation and each pressure zone.

Proving test facilities

Requirement: Provide facilities to verify that the system flows and pressures meet AS 2419.1.

Flow sensor: Low loss pitot type averaging sensor, with 2 flared isolating valves for connection of pressure lines and stainless steel wetted parts.

Installation: Install to the manufacturer's recommendation for installation, connection and valving. Provide manufacturers recommended straight lengths of pipe upstream and downstream of tapping point. Mount in the piping using an adaptor bushing and welding boss.

Performance:

- Accuracy: Within $\pm 1.5\%$ over the range of flow anticipated.
- Stability: Within $\pm 0.125\%$ over five years.
- Repeatability: $\pm 0.1\%$.

2.5 COMPLETION

General

System: To AS 2419.1.

Flush: Before testing, flush the piping system with clean water to AS 2419.1. Flush until the piping has been thoroughly cleaned out. Operate the system until all foreign matter has been removed.

Hydrostatic test: Test the piping system to AS 2419.1. Maintain the test pressure for the minimum time required to AS 2419.1 or longer if necessary to complete the inspection of the system under test.

Baseline data

Requirement: Provide baseline data to AS 1851 and AS 2419.1.

3 HOSE REELS

3.1 FIRE HOSE REELS

General

Standard: To AS/NZS 1221.

Product certification: Required, to AS/NZS 1221 clause A3.

Certification provider: An organisation accredited by the Joint Accreditation System of Australia and New Zealand (JAS-ANZ).

Type: Swivel hose guide.

Existing to be maintained

4 SUPPLIED

General

Standard: To AS/NZS 2444.

Allow to install below and required signage

5x Fire Blankets

10 x 9.0kg ABE Fire Extinguisher (4 in new build, 2 per renovated offices)

1 x 9.0 dry chemical fire extinguisher (1 in lab refurbished office)

Locations to be confirmed